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AVALON RARE METALS INC.

TECHNICAL REPORT
DISCLOSING THE RESULTS OF THE
FEASIBILITY STUDY
ON THE
NECHALACHO RARE EARTH ELEMENTS PROJECT

Report Date: 31 May, 2013
Effective Date: 17 April, 2013

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1.0 SUMMARY

1.1 INTRODUCTION

Micon International Limited (Micon) has been retained by Avalon Rare Metals Inc. (Avalon) to compile the Technical Report under Canadian National Instrument 43-101 (NI 43-101) which discloses the results of the feasibility study prepared by SNC-Lavalin (SLI) on Avalon's Nechalacho rare earth elements project.

SLI was engaged by Avalon to complete the feasibility study for development of the Nechalacho project under an agreement dated 14 December, 2011. The study was completed in April, 2013 and the results summarized in a press release dated 17 April, 2013.

1.1.1 Scope of the Feasibility Study

The Nechalacho mine and the three processing plants comprise the following principal components of the project:

- The Nechalacho site on the north shore of Great Slave Lake, Northwest Territories, which includes:
    - Underground mine and paste backfill plant.
    - Concentrator and tailings management facility (TMF).
    - Site utilities and infrastructure including power plant, accommodation, airstrip and dock facilities.
- The Pine Point Site on the south side of Great Slave Lake, Northwest Territories, which includes:
    - Hydrometallurgical plant, including sulphuric acid plant, hydrometallurgical tailings facility (HTF) and dock facility.

The Geismar site near Baton Rouge, Louisiana, United States, which includes:

- Rare earth refinery, including administration and maintenance facilities, and gypsum impoundment.

The flotation concentrate produced at the Nechalacho site will be transported to the hydrometallurgical plant at Pine Point where a mixed rare earth precipitate and an enriched zirconium concentrate (EZC) will be produced. The EZC will be sold to one or more third parties. The mixed rare earth concentrate will be transported to the leaching and rare earth separation facility at Geismar where a range of separated rare earth products will be recovered for sale.

The Nechalacho project is 100% owned by Avalon.

In addition to SLI, the principal contributors and consultants providing input to the feasibility study are as follow:

- Avalon.
- Mine Ventilation Services Inc. (MVS).
- Roscoe Postle Associates Inc. (RPA).
- EBA Engineering Consultants (a Tetra Tech Company) (EBA).
- Knight Piésold Ltd. (KPL).
- FLSmidth (FLS).
- Golder Associates Ltd. (Golder).
- URS Corporation (URS).
- Hwozdyk Inc. (mining consulting services).

Other technical consultants retained by Avalon include:

- SGS Canada Inc. (metallurgical testwork).
- Xstrata Process Support (XPS).
- J.R. Goode and Associates (metallurgical consultant).
- Jerald Tabor (independent metallurgical consultant).
- PRS & Associates (independent project management consultant).

The feasibility study was completed over a period of two years at a cost of approximately \$60 million. Capital cost estimates were completed to an intended level of accuracy of +15%, consistent with an AACEI Class 3 estimate, based on prices as of second quarter 2012 and excluding escalation. Operating cost estimates were completed to an intended level of accuracy of +15%, based on prices as of fourth quarter 2012.

It should be noted that rare earth and yttrium oxides have been calculated and expressed herein as RE_2O_3 although oxides of cerium, praseodymium and terbium are often given using the formulae CeO_2, Pr_6O_{11} and Tb_4O_7, respectively. The latter convention is used for oxide pricing.

1.2 GEOLOGY

The rare metals deposits which are the basis for the Nechalacho project occur within the Blatchford Lake Complex (BLC), which includes Archaean Yellowknife Supergroup metasedimentary rocks of the southern Slave geologic province. The BLC has an alkaline character and intrusive phases vary successively from early pyroxenite and gabbro through to leuco-ferrodiorite, quartz syenite and granite, to peralkaline granite and a late syenite.

The Nechalacho deposit is hosted by the Nechalacho Nepheline Syenite and consists of a layered series of rocks with increasing peralkaline characteristics at depth. It is interpreted that the Nechalacho deposit is virtually undeformed where most of the features observed were generated by magmatic and hydrothermal processes. The distribution of most of the rock units and the mineralization generally follow a sub-horizontal pattern that can be traced for several hundreds of metres.

1.2.1 Mineralization

Mineralization in the Nechalacho deposit includes light rare earth elements (LREE) found principally in allanite, monazite, bastnaesite and synchysite; yttrium, heavy rare earth elements (HREE), niobium and tantalum found in fergusonite; niobium in ferrocolumbite; HREE, niobium, tantalum and zirconium in zircon; and gallium in biotite, chlorite and feldspar in albitized feldspathic rocks.

There are some trends in chemical and mineralogical characteristics in the Nechalacho deposit that have both geological and metallurgical significance. These trends are most apparent in vertical zoning of the deposit and include the following:

- The proportion of HREE to LREE gradually increases from top to bottom of the deposit. The proportion of HREE within total rare earth elements (TREE) at the top of the deposit may be 6% and at the bottom of the deposit it may exceed 30%.

- Some five types of zircon textures have been identified, which show imperfect vertical zoning, with “cumulate” textures being more common towards the bottom of the Basal Zone and “wavy banded zircon” textures more common in the upper parts of the Basal Zone and in Upper Zones.

- Liberation data from QEMSCAN® studies have shown that liberation decreases downwards in the deposit – that is at specific grain sizes, the proportion of free and liberated zircon decreases downwards.

1.3 EXPLORATION

The Nechalacho property has been systematically explored for several different metals at different deposits within the property over a period of 30 years. On taking over the property in 2005, Avalon sampled archived drill core from the Nechalacho deposit to extend the area of known yttrium and rare earth element (REE) mineralization. Starting in August, 2007, Avalon has conducted continuous drilling campaigns and an airborne magnetic survey was completed in winter 2009 to aid in mapping the local geology and structures.

1.4 DRILLING

Since 1977, diamond drilling has been carried out intermittently by various operators over five separate mineralized zones at Thor Lake. A total of 51 holes had been completed on the Nechalacho deposit through 1988.

Avalon commenced diamond drilling on the Nechalacho deposit in July, 2007 the goal of which was to further delineate zones of REE and tantalum mineralization. Eleven tightly-spaced inclined holes were drilled to obtain a mini-bulk sample for continued metallurgical work on the REE-enriched zones.

From February, 2009, the drill spacing was reduced and was focused on the southeastern part of the deposit where the Basal Zone has higher TREE grades but, also, higher HREE grades, along with thicker intercepts.

Until the end of 2011, a custom designed database for geological, geotechnical and assay data was managed by a consultant, with regular back-ups off-site. Since the beginning of 2012, the drill hole database has been rebuilt into DataShed of Maxwell Geoservices and is maintained by Avalon geologists.

Core recoveries are generally high at the Nechalacho deposit, due to the exceptionally competent nature of the rock, with an average of 97% in the mineralization and with no bias in the sampling due to incomplete sample recovery.

A comprehensive core logging and sampling protocol was established at the beginning of the July 2007 drilling program. This protocol has been strictly applied for all of the subsequent drilling programs. A comprehensive geotechnical logging protocol was introduced at the start of the summer 2009 drill program.

1.4.1 Sampling Method and Approach

On average, and except for PQ drilling, assay samples were 2 m long except where, in the geologist's opinion, it was advisable to follow lithological boundaries. In 2010, when PQ drilling commenced at the project, the sample interval was reduced to typically 1 m due to health and safety concerns with the weight of PQ core. A 1-m sample of PQ core weighs about 18 kg.

1.5 SAMPLE PREPARATION, ANALYSES AND SECURITY

All sample preparation, from identification of sample intervals to bagging of drill core, was completed by employees of Avalon. Subsequent preparation, such as crushing, grinding and further steps, were all completed by commercial laboratories, ALS Limited (ALS), Acme Analytical Laboratories (Vancouver) Limited (Acme) and Activation Laboratories Limited (Actlabs).

Internally developed standards were inserted routinely. Blanks, composed of split drill core of unaltered and unveined diabase dyke intersected in drilling beneath Thor Lake, were inserted every 40th sample.

Avalon has developed a number of standards for the project to date and all were subject to Round Robin analysis for certified values.

1.5.1 Specific Gravity

Specific gravity (SG) is measured on core samples taken at 5 m intervals within the hole. Each sample is approximately 10 cm long.

In 2010, 32 samples of drill core were submitted to ALS for an independent check of the SG values. In 2012, a further 10 samples of drill core were submitted for independent check of SG.

1.6 DATA VERIFICATION

The database used by Avalon for the resource estimate contained drill hole collar location, downhole deviation surveys, lithology, density, magnetic susceptibility, rock quality designation, sample numbers, sample intervals, and analytical data.

Over the years, a number of steps have been taken to ensure the accuracy and validity of the database. Cross-referencing was routinely done over time to help validate the values of economic interest in the assay table by comparison with MS81, MS81h and XRF input tables, as well as various comparisons between sample duplicates, check samples, and comparison between laboratory results.

A total of 248 electronic assay certificates from 2007 to 2012, representing approximately a quarter of the assay database, were used for validation of rare earth and rare metal elements, as well as yttrium, uranium and thorium. Approximately 10% of the assays in the database were verified. Additional assay spot checks were done for samples from five historical drill holes, by comparison with scans of old paper records, either analysis certificates or drill hole logs containing a variable mix of cerium, yttrium, gallium, zirconium and niobium values, in element parts per million or percent oxide.

Scatter plots were performed for pairs of rare metal, LREE and HREE pairs of elements.

Collar locations of drill holes were compared with older printed records and GPS survey files. Downhole deviation surveys were visually inspected and spot checks were done for drill hole azimuth and dip.

1.6.1 Quality Assurance/Quality Control

In 2007 and 2009, Avalon commissioned CDN Resource Laboratories Ltd. (CDN) in British Columbia to generate standards for insertion into the assay stream. Dr. Barry Smee was retained to review the Round Robin procedure and assess the quality of the data. A further two standards were developed in 2010 and these were also subject to Round Robin analysis and certification by Charles Beaudry.

Quality control procedures used during 2011 were continued through 2012 with the addition of core duplicate sampling for the summer 2012 drilling campaign. New rare earth element standard samples at various grades have been certified for use starting in 2013.

The rate of insertion of standards was varied according to whether the samples were from a mineralized zone or not with a standard every 10 samples in mineralization and every 40 samples outside mineralization. The results of the standard analyses were checked against the certified or provisional means and tolerances listed in the standard certificates, as well as against the laboratory precision tolerance level of $\pm$10%. The three rare earth elements with the potential highest value (neodymium, terbium and dysprosium) were routinely monitored along with the overall values for the TREE and HREE.

Precision results of the QA/QC program for all laboratories, as measured by relative standard deviation (standard deviation/sample population mean), are measured for the individual payable elements in the deposit.

Avalon monitors the results of its internal standards during routine analysis of drill core and all core is analyzed.

1.7 MINERAL PROCESSING AND METALLURGICAL TESTING

Extensive metallurgical testwork has been completed at a number of different laboratories over the last four years and a large number of testwork reports have been issued to summarize this work.

Much of the pertinent metallurgical and mineralogical development studies have been undertaken using bulk composite samples that represent the Nechalacho deposit mineralization spatially and in terms of lithology. These selected composite samples tended to be selected to represent mineralization at different depths in the deposit in terms of elevation. The composites designated UZ were from Upper Zone mineralization and BZ were from Basal Zone mineralization.

Since 2010, Avalon has completed four flotation pilot plant tests at Xstrata Process Support (XPS) and SGS Minerals, Lakefield, Ontario (SGS-M). All of these pilot plants were conducted using bulk samples sourced from drill core.

1.7.1 Mineralogy

The mineralogy of the Nechalacho deposit has been studied by SGS-M using QEMSCAN®, a scanning electron microscope (SEM) and an electron microprobe (EMP). A second major body of work continues to be conducted by a team under Professor Anthony Williams-Jones of McGill University, Montreal.

Nechalacho mineralization is hosted in nepheline syenite that has been extensively hydrothermally altered in areas of mineralization. The payable elements of the Nechalacho deposit are typically hosted in a number of minerals, summarized as follows:

- LREEs dominantly occur in bastnaesite, synchisite, monazite and allanite.
- HREEs dominantly occur in zircon, fergusonite and rare xenotime.
- Zirconium (Zr), along with HREE, niobium and tantalum occurs in zircon and other zircono-silicates (eudialyte).
- Niobium and tantalum occur in columbite and ferrocolumbite, fergusonite and zircon.

The mineralogy of the Nechalacho ore is complex and guides metallurgical development and performance.

1.7.2 Hydrometallurgical Testwork

Six hydrometallurgical pilot plant campaigns were conducted at SGS-M between June and October, 2012. The main objectives of these campaigns were to:

- Test a continuous version of the hydrometallurgical flow-sheet.
- Optimize REE extraction in the pregnant solution.
- Remove target contaminants (iron, uranium and thorium).
- Ensure the final mixed rare earth precipitate product had an acceptable grade of REE while reducing the uranium and thorium contents below 500 ppm.
- Ensure the concentrations of species in the filtrate from the tailings circuit met target environmental levels.

The final pilot plant campaign, which operated between September 24 and October 5, 2012, demonstrated the technical viability of the process and provided crucial input for the final hydrometallurgical flowsheet, process design criteria and process engineering adopted for the feasibility study.

1.7.3 Refinery

The refinery to be constructed at Geismar comprises two plants, the leaching and the separation plants. The leaching plant removes impurities from the hydrometallurgical precipitate in order to attain a purified feed to the separation plant where the individual rare earth products will be produced.

A large number of testwork reports have been issued to summarize the testwork that has been undertaken at a number of different laboratories over the last few years. All relevant testwork has been completed using the rare earth precipitate produced during the hydrometallurgical pilot plant testwork program.

1.8 MINERAL RESOURCE ESTIMATE

The mineral resource estimate for the Nechalacho project presented in the feasibility study based on the block model prepared by Avalon was audited originally by RPA on 21 November, 2012. Subsequent to this, Avalon updated the database and re-estimated the resource as of 3 May, 2013. The update included correction of some minor assay data entry errors and drill hole locations. The net effect of these changes is considered immaterial as the resource change was less than 1% in most individual parameters. The largest changes were for ZrO_2 grade, and the effect was an increase in grade in Measured and Indicated resources of between 0.1% and 3.2% of the overall grade in the various categories.

The current mineral resource estimated by Avalon and accepted by RPA for the Nechalacho deposit is summarized in Table 1.1. The mineral resource is reported at a cut-off value of US$320/t. The effective date of the mineral resource estimate is 3 May, 2013.

Table 1.1
Nechalacho Deposit Mineral Resource Estimate as at 3 May, 2013

Category	Zone	Tonnes (million)	TREO (%)	HREO (%)	ZrO_2 (%)	Nb_2O_5 (%)	Ta_2O_5 (%)
Measured	Basal	10.86	1.67	0.38	3.23	0.40	0.04
	Upper	-	-	-	-	-	-
Total Measured		**10.86**	**1.67**	**0.38**	**3.23**	**0.40**	**0.04**
Indicated	Basal	55.81	1.55	0.33	3.01	0.40	0.04
	Upper	54.59	1.42	0.14	1.96	0.28	0.02
Total Indicated		**110.40**	**1.49**	**0.24**	**2.49**	**0.34**	**0.03**
Measured and Indicated	Basal	66.67	1.57	0.34	3.05	0.40	0.04
	Upper	54.59	1.42	0.14	1.96	0.28	0.02
Total Measured and Indicated		**121.26**	**1.50**	**0.25**	**2.56**	**0.34**	**0.03**
Inferred	Basal	61.09	1.29	0.25	2.69	0.36	0.03
	Upper	122.28	1.26	0.12	2.21	0.32	0.02
Total Inferred		**183.37**	**1.27**	**0.17**	**2.37**	**0.33**	**0.02**

1. CIM definitions were followed for Mineral Resources.
2. Mineral Resources are estimated at a NMR cut-off value of US$320/t. NMR is defined as "Net Metal Return" or the in situ value of all payable metals, net of estimated metallurgical recoveries and off-site processing costs.
3. An exchange rate of US$1=CAD1.05 was used.
4. Heavy rare earth oxides (HREO) is the total concentration of: Y_2O_3, Eu_2O_3, Gd_2O_3, Tb_2O_3, Dy_2O_3, Ho_2O_3, Er_2O_3, Tm_2O_3, Yb_2O_3 and Lu_2O_3.
5. Total rare earth oxides (TREO) is HREO plus light rare earth oxides (LREO): La_2O_3, Ce_2O_3, Pr_2O_3, Nd_2O_3 and Sm_2O_3.
6. Rare earths were valued at an average net price of US$62.91/kg, ZrO_2 at US$3.77/kg, Nb_2O_5 at US$56/kg, and Ta_2O_5 at US$256/kg. Average REO price is net of metallurgical recovery and payable assumptions for

contained rare earths, and will vary according to the proportions of individual rare earth elements present. In this case, the proportions of REO as final products were used to calculate the average price.

7. ZrO_2 refers to zirconium oxide, Nb_2O_5 refers to niobium oxide and Ta_2O_5 refers to tantalum oxide.

1.8.1 Cut-off Grade

Both rare metals and rare earths contribute to the total revenue of the Nechalacho deposit.

An economic model was created, using metal prices that were updated from those used in the prefeasibility study, flotation and hydrometallurgical recoveries, the effects of payable percentages, and any payable Net Smelter Return (NSR) royalties. The payable percentages of elements contained within the EZC were also included. The net revenue generated by this model is termed the NMR.

The mineral resource estimate is based on the minimum NMR value being equal to an operating cost of US$320/t, a break-even cut-off value.

1.9 MINERAL RESERVE ESTIMATE

The mineral reserve estimate for the Nechalacho project presented in the feasibility study was estimated from the block model prepared by Avalon and audited originally by RPA on 21 November, 2012. As noted above, subsequent to this, Avalon updated the database and re-estimated the resource as of 3 May, 2013. The mineral reserve estimate is derived from this block model by applying the appropriate technical and economic parameters to extraction of the REE with proven underground mining methods.

The mineral reserve has been estimated based on conversion of the high grade mineral resources at a cut-off value greater than US$320/t NMR. Payable elements include the REE, zirconium, niobium and tantalum. No Inferred mineral resources were converted to mineral reserves. The high grade mineral resources are 34.7% and 14.7% of the total Measured and Indicated mineral resources, respectively.

1.9.1 Mine Design Parameters

The key design criteria set for the Nechalacho mine are:

- Initial design based on a 20-year life-of-mine (LOM) of high grade material.
- Mechanized cut and fill and long hole mining methods with paste backfill.
- Minimum mining thickness of 5 m.
- Extraction ratio of 94.2%.
- Internal dilution of 8.5%.
- External dilution of 5% applied to secondary stopes.
- Estimated total average dilution for the life of mine of approximately 11%.
- Production rate of 2,000 t/d ore (730,000 t/y).
- Ore bulk density of 2.91 t/m^3.

The mineral reserve estimate for the Nechalacho project shown in Table 1.2 has an effective date of 3 May, 2013.

The figures in the table are rounded to reflect that the numbers are estimates. The conversion of mineral resources to mineral reserves includes technical information that requires subsequent calculations or estimates to derive sub-totals, totals and weighted averages. Such calculations or estimations inherently involve a degree of rounding and consequently introduce a margin of error. Where these occur, Micon does not consider them to be material.

To the best of Micon's knowledge there are no legal, political, environmental, or other issues which would materially affect development of the estimated mineral reserve.

Table 1.2
Mineral Reserve Estimate as at 3 May, 2013

Description	Mineral Reserve Category		
	Proven	Probable	Proven and Probable
Tonnage (Mt)	**3.68**	**10.93**	**14.61**
TREO (%)	**1.7160**	**1.6923**	**1.6980**
HREO (%)	**0.4681**	**0.4503**	**0.4548**
HREO/TREO	27.28%	26.61%	26.78%
La_2O_3	0.256%	0.256%	0.256%
Ce_2O_3	0.570%	0.567%	0.568%
Pr_2O_3	0.072%	0.071%	0.071%
Nd_2O_3	0.284%	0.283%	0.283%
Sm_2O_3	0.065%	0.065%	0.065%
Eu_2O_3	0.008%	0.008%	0.008%
Gd_2O_3	0.062%	0.061%	0.061%
Tb_2O_3	0.010%	0.010%	0.010%
Dy_2O_3	0.058%	0.056%	0.056%
Ho_2O_3	0.011%	0.010%	0.010%
Er_2O_3	0.029%	0.027%	0.028%
Tm_2O_3	0.004%	0.004%	0.004%
Yb_2O_3	0.023%	0.022%	0.022%
Lu_2O_3	0.003%	0.003%	0.003%
Y_2O_3	0.259%	0.249%	0.251%
ZrO_2	3.440%	3.309%	3.342%
Nb_2O_5	0.425%	0.413%	0.416%
Ta_2O_5	0.046%	0.045%	0.045%

1. CIM definitions were followed for Mineral Reserves.
2. Mineral Reserves are based on Mineral Resources published by Avalon in News Release dated November 26th, 2012 and audited by Roscoe Postle Associates Inc., and modified as of 3 May, 2013.
3. Mineral Reserves are estimated using price forecasts for 2016 for rare earth oxides given below.
4. HREO grade comprises Y_2O_3, Eu_2O_3, Gd_2O_3, Tb_2O_3, Dy_2O_3, Ho_2O_3, Er_2O_3, Tm_2O_3, Yb_2O_3, and Lu_2O_3. TREO grade comprises all HREO and La_2O_3, Ce_2O_3, Nd_2O_3, Pr_2O_3, and Sm_2O_3.
5. Mineral Reserves are estimated using a NMR cash cost cut-off value of US$320/t.
6. Rare earths were valued at an average net price of US$62.91/kg, ZrO2 at US$3.77/kg, Nb_2O_5 at US$56/kg, and Ta_2O_5 at US$256/kg. Average REO price is net of metallurgical recovery and payable

assumptions for contained rare earths, and will vary according to the proportions of individual rare earth elements present. In this case, the proportions of REO as final products were used to calculate the average price.

7. Mineral reserves calculation includes an average internal dilution of 8.5% and external dilution of 5% on secondary stopes.

Micon believes the key assumptions, parameters and methods used to convert mineral resource to mineral reserve are appropriate. To the best of Micon's knowledge there are no known mining, metallurgical, infrastructure, permitting or other relevant factors that may materially affect the mineral reserve estimate.

1.10 MINING METHODS

Underground mining of the Measured and Indicated mineral resource of the Basal Zone was investigated for the feasibility study. The majority of the mineral resource of the Basal Zone lies directly beneath and to the north of Long Lake, approximately 200 m below surface. Thus, the deposit is to be mined using underground mining methods.

The mine production rate is 2,000 t/d (730,000 t/y) of ore and the mine life is 20 years.

Geotechnical information for the mine design was based on geotechnical data collection completed in conjunction with Avalon's on-going exploration drill program. The stope dimensioning and crown pillar stability analysis was performed by KPL. The analysis indicated that excavations 15 m wide, 5 m high and 100 m in length will be stable with the proper installation of ground support and mitigation strategies.

The deposit at the Nechalacho project is relatively flat lying and will be mined with a combination of longhole stoping, and cut and fill methods. The mine will be accessed through a mine portal located near the concentrator. The dimensions of the 1,600 m main ramp will accommodate the overhead conveyor system and access for men and equipment.

Zones less than 10 m thick will be mined by cut and fill methods in a primary and secondary mining sequence. Zones over 10 m thick will be mined with longhole stoping. Secondary stopes will be mined after the adjoining primary stopes have been filled. The mining of the secondary stopes will be the same as the mining of the primary stope.

Blasted material will be mucked and transported by rubber tyred equipment to the crusher station. The crushed ore will be transported to the surface by conveyor.

Paste backfill will be used to improve the overall mine stability, reduce the surface footprint for the Nechalacho TMF, and enable the extraction of secondary stopes for increased mining recovery.

The overall mine layout is shown in Figure 1.1.

Figure 1.1
Isometric View of Overall Mine Layout

The Nechalacho production schedule is designed to extract high grade material located closest to the underground crusher station in the initial mining years to maximize present value.

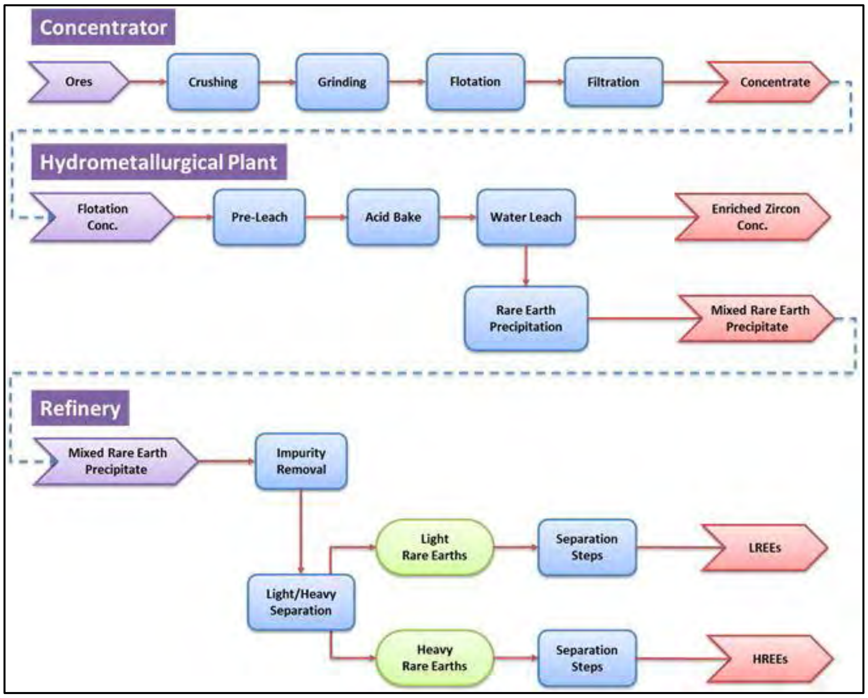
Steady production at an average rate of 2,000 t/d ore is anticipated by the sixth month of operation.

1.11 RECOVERY METHODS

The processing facilities comprise three separate plants, the concentrator located at the Nechalacho site, the hydrometallurgical plant at the Pine Point site and the rare earth refinery at the Geismar site.

The flowsheets and the process design criteria developed for the three processing facilities is based on extensive metallurgical testwork, as described above. A simplified flowsheet showing the three processes is presented in Figure 1.2.

Figure 1.2
Simplified Flowsheet for the Concentrator, the Hydrometallurgical Plant and the Refinery



1.11.1 Nechalacho Concentrator

Ore will be fed to the primary crusher from the underground ROM storage pocket and fine crushed ore product will be delivered by conveyor to a surface fine ore bin which will feed the grinding circuit. The crushing circuit has been designed to accommodate an expanded capacity of 4,000 t/d.

The grinding circuit will be a conventional rod mill/ball mill operation. The rod mill will be operated in open circuit, and the ball mill in closed circuit with classifying hydrocyclones.
The cyclone overflow will gravitate to two stages of magnetic separation, followed by a desliming circuit. The magnetics from the magnetic separation circuit and the fines from desliming will be routed to tailings. The deslimed slurry will feed the flotation circuit.

This flotation circuit design comprises three stages of bulk flotation, four stages of cleaner flotation and a single cleaner scavenger stage. Flotation concentrate will be pumped to a gravity separation circuit for further enrichment before being thickened and filtered to final product concentrate. The light material (gravity tailings) will be recycled to the bulk rougher flotation circuit.

The tailings will be thickened, the overflow from which will be pumped to the process water tank although a portion will be fed to a water treatment plant to remove impurities. The tailings thickener underflow will be directed to the either the TMF or the paste backfill plant.

The paste backfill plant has been designed to produce 1,738 t/d of backfill using concentrator tailings.

1.11.2 Pine Point Hydrometallurgical Plant

The hydrometallurgical plant at Pine Point comprises the following process sections:

- Pre-leach.
- Acid bake.
- Water leach.
- Neutralization and impurity removal.
- Impurity removal re-dissolution.
- Rare earth precipitation.
- Waste water treatment.
- Tailings neutralization.

The concentrate from Nechalacho will be fed to the pre-leach section of the plant where excess sulphuric acid produced in the water leach section will be used to neutralize the base materials. The product from the pre-leach circuit will be filtered and fed to the acid bake system while the filtrate will feed the iron reduction circuit.

The filter cake from the pre-leach circuit will be mixed with concentrated sulphuric acid and fed into the acid baking rotary kiln where the REE in the concentrate will be converted to sulphates. The discharge from the acid bake kiln will be leached in water to recover approximately 80% of the LREEs and 50% of the HREEs. The solids containing the balance of the REEs, along with most of the zirconium, niobium and tantalum, will be filtered, washed, neutralized and dried to approximately 8% moisture. This dried product will be packaged and shipped to customers as EZC.

The rare earth filtrate from the water leach process will be cleaned through several neutralization and impurity removal steps. The resulting slurry will be filtered and washed and the final rare earth precipitate will be dried to approximately 8% moisture.

The filtered solids from the waste water treatment circuit will be washed and transferred to the Pine Point HTF.

1.11.3 Geismar Rare Earth Refinery

The rare earth refinery facilities at Geismar consists of two key sections, the leaching plant to remove impurities, and the separation plant where products are separated and refined to the quality required for the customers. The refinery design is based on experience and widely-accepted, well proven standard rare earth sulphate and chloride processing methods and equipment selections.

The leaching plant comprises a series of processes, including re-dissolution, precipitation, solvent extraction and selective precipitation. Impurities, principally uranium and thorium, will be removed in a series of dissolution, selective precipitation, filtration and solvent extraction steps.

Rare earth separation will be achieved using a kerosene mixture as the extracting agent for most separations, with hydrochloric acid as the stripping agent. The purified strip solution from the respective solvent extraction stage will feed their atmospheric precipitation tanks where soda ash or oxalic acid will be added in order to precipitate the pure REEs as carbonates or oxalates, respectively. The slurry streams containing the precipitates will be thickened, filtered, dried and calcined to produce pure rare earth oxides. The filtrate will be forwarded to the water treatment facility.

1.12 INFRASTRUCTURE

Project infrastructure comprises items required for access and operation of the three sites at Nechalacho, Pine Point and Geismar, and communications between them.

The flotation concentrate produced at the Nechalacho site will be barged along the eastern side of Great Slave Lake to Pine Point where the flotation concentrate will be upgraded to a mixed rare earth precipitate and EZC. The rare earth precipitate will be shipped by rail to the Geismar site for leaching and separation of rare earths. The EZC will be shipped for sale to one or more third parties.

1.12.1 Nechalacho Site

There is no permanent road connecting the site with nearby communities and there are no plans to construct one. The site is accessible by barge during the summer months and year-round by aircraft using an existing airstrip.

An existing 5 km long road connects the concentrator area with the planned dock site on the north shore of Great Slave Lake and will be upgraded during project construction to allow year-round operations.

The shoreline topography at Great Slave Lake south of the Nechalacho site is generally rocky and steep. The design of the permanent dock facility comprises a barge loading-unloading dock and a large marshalling yard. All supplies, except diesel fuel, will arrive in standard shipping containers (those used for concentrate) or in full size containers.

The existing airstrip will be extended to a total length of 1,000 m for operations to allow year-round access for light freight and personnel movement. An airstrip depot, located on the south side of the airstrip and west of the access road, will serve as communication centre, passenger waiting area and for security check in/check out.

Power will be generated using diesel generators on the basis of a build-own-operate arrangement where a third party will provide complete operational facilities and provide the power to the Avalon facilities at an agreed cost. The power plant will consist of five diesel generators. Any one diesel generator unit will be utilized to supply backup power to critical process, electrical heat tracing and other loads during outages of the power plant.

Fresh water for the potable water treatment plant will be provided via a floating pump barge located in the southwestern portion of Thor Lake connected by pipeline to the concentrator site. Fresh water for the potable water treatment, firewater and the concentrator will be stored in a single tank. A portion of the water from the fresh water storage tank will be treated for potable use and stored in a separate tank. Grey water and sewage from the concentrator site facilities will be collected and pumped to a rotating biological contactor (RBC) treatment plant. Waste water from the flotation process at will be treated prior to release.

The administration office will comprise open office space, private offices, lunch rooms, meeting/conference rooms, toilets and service rooms.

The metallurgical laboratory will be equipped with analytical facilities, offices and service rooms.

Dry facilities will include lockers and change facilities for both men and women.

The maintenance shop will be located on the ground floor at the northeast corner of the concentrator building.

Diesel fuel for the Nechalacho site will be transported by barges in the summer months and temporarily stored in three holding tanks located close to the dock. Fuel will be pumped to the main fuel storage tank farm. No gasoline will be stored or dispensed at site.

Waste handling facilities will comprise a sorting station and incinerator. The waste will be sorted on site and organic waste will be directed to the incinerator. Hazardous and recyclable wastes will be stored in sealed containers and shipped off-site on a regular and as-required basis to existing municipal handling facilities.

The accommodations complex is designed to provide accommodation, medical, dining and recreational requirements for 120 persons. The accommodations complex will be built under a build-own-operate arrangement where a third party will provide complete operational facilities and provide accommodation and catering services during operations.

Proven, reliable, and state-of-the-art telecommunications systems will be provided for permanent operations and maintenance at the Nechalacho site facilities. The project facilities, encompassing the Nechalacho and Pine Point sites in Canada, and the Geismar site in the United States will be connected via a combination of satellite land-station terminals and terrestrial communication links to provide access to voice, data and video services.

1.12.2 Pine Point Site

The hydrometallurgical plant will treat concentrate from the Nechalacho site to produce a rare earth precipitate and EZC.

The site is accessible year-round from Hay River via Highways 5 and 6 and an access road from Highway 6 to the site. A network of local gravel roads was developed for past mining operations and provides alternative access to the site. An existing 13.8 km long gravel road connecting the site with the dock located on the south shore of Great Slave Lake will be used for concentrate shipping during the summer months.

At the location of the Pine Point docking facility, Great Slave Lake is very shallow with water depths of +1 m within 150-200 m of the shoreline. Due to the shallow depth, dredging is planned. Access to the docked barge will be via steel ramp or suitable gravel pushout ramp from the Great Slave Lake yard. The dock yard will be used as a lay down area for concentrate containers that will be shipped from the concentrator at the Nechalacho site, as well as for returning empty containers. Approximately 3,300 containers will be shipped during a shipping season.

Primary power for Pine Point site will be provided via the existing NWT Hydro Corporation (NTHC) substation located at the former Pine Point mine site. Standby power requirements for critical loads will be supplied by a diesel generator.

Two concentrate storage buildings are sized to store 11 months of production.

Prilled-sulphur feedstock for the sulphuric acid plant will be trucked from Hay River. Sulphuric acid will be stored in four product acid tanks with dedicated heating and circulating systems. It will be built as a build-own-operate arrangement with a third party. In addition to prilled sulphur, the principal bulk reagents will be dolomite, magnesium hydroxide, slaked lime, hydrogen peroxide and flocculants, all of which will be delivered by truck.

Raw water for process operations will be pumped via a raw/fresh water pipeline over a distance of 2.2 km from the J-44 historic open pit, to a fresh water storage tank. Raw/fresh water will then be distributed from this storage tank to the potable water treatment plant, the fire water system distribution system, the boiler water treatment, the gland water storage tank and the process water storage tank. Sewage will be treated using RBC treatment plants.

The main fuel storage and dispensing facility will consist of one storage tank. A fuel line will be installed from the storage tank to the sulphuric acid plant, the kiln and process boilers at the hydrometallurgical plant, and the maintenance shop.

Solid waste will be collected at the point of origin in colour-coded containers which will then be transported via a contractor to an appropriate municipal or private collection/treatment facility.

A two-storey administration building will be connected to the plant building.

A maintenance area will be provided within the hydrometallurgical plant building with service bays, offices and maintenance equipment.

Proven, reliable and state-of-the-art telecommunications systems will be provided for permanent operations and maintenance at the Pine Point site facilities.

1.12.3 Geismar Site

A two-lane access road connecting the Geismar site to the existing Ashland Road will be constructed.

A new rail spur, approximately 1.2 km in length, will be constructed from the existing CN railway line, located northeast of the Geismar site, to the product storage area.

Fresh water for the packaged potable water treatment plant will be provided from a well equipped with a submersible pump. Fresh water for process and fire water will be provided via a HDPE pipeline from the adjacent chlor-alkali plant. A waste water treatment plant will be provided to treat run-off water from the contaminated areas of the site and gypsum impoundment, as well as the process waste water stream prior to discharge to the Mississippi River. Sewage will be treated using RBC treatment plants.

Administration, laboratory and shops/warehousing will be located in stand-alone buildings.

The reagents required for the process include sulphuric acid, hydrochloric acid, sodium hydroxide, soda ash, oxalic acid, slaked lime, naphthenic acid, P507, DEPHA, and kerosene. Reagents will be transported to site by rail, ship or pipeline and stored on-site. Sodium hydroxide and hydrochloric acid will be supplied by the adjacent chlor-alkali plant and waste brine from Avalon's operation will be returned to the chlor-alkali plant for disposal.

A new 34.5 kV overhead line will be installed by Entergy Corporation to supply the main step-down substation. For identified critical loads, an emergency diesel generator will provide power for short periods in the case of interruption to utility power supply.

Product will be stored in sealed containers and placed in shipping containers that will be used as storage facilities. Both road and railroad access will be available for product off-site shipping.

Proven, reliable and state-of-the-art telecommunications systems will be provided for permanent operations and maintenance at the Geismar site facilities.

1.13 MARKET STUDIES AND CONTRACTS

In preparing its analysis of markets for the Nechalacho project feasibility study, Avalon has collected historical price information, supply/demand analysis, and forecasts from a number of different published and proprietary sources.

Based on the predominance of LREE in the anticipated product mix of new supply sources outside China, Avalon and its consultants consider it likely that a supply shortfall in many of the more critical rare earths is likely to emerge by 2016. This is expected to result in firmer prices for neodymium, europium, gadolinium, terbium, erbium and yttrium.

1.13.1 Product Pricing

The market for rare earths and yttrium products is small, and reliable, publicly available pricing information, forecasts and refining terms are not readily accessible. The prices used in the economic analysis for the feasibility study are shown in Table 1.3.

Table 1.3
Avalon Forecast Prices for Rare Earth Oxides

Rare Earth Oxide	Feasibility Study Forecast 2016 Forecast (US$/kg, FOB China)
LREE	
La_2O_3	8.75
CeO_2	6.23
Pr_6O_{11}	75.20
Nd_2O_3	76.78
Sm_2O_3	6.75
HREE	
Eu_2O_3	1,392.57
Gd_2O_3	54.99
Tb_4O_7	1,055.70
Dy_2O_3	688.08
Ho_2O_3	66.35
Er_2O_3	48.92
Lu_2O_3	1,313.60
Y_2O_3	67.25

Prices for rare earths beyond 2016 have been held constant at 2016 levels for the purposes of the feasibility study. Demand is expected to continue to grow steadily and could be significantly higher if one or more of the potentially important end-use sectors are adopted widely, for example, electric/hybrid electric vehicles and wind turbines. Avalon believes that it is both reasonable and conservative to hold prices constant from 2016 onwards. Micon agrees with this approach.

The EZC produced at Pine Point will carry the niobium and tantalum values recovered from the Nechalacho project. Projected revenue from EZC in the feasibility study is based on the terms of a memorandum of understanding (MOU), net of the freight cost to China. Given the small number of players in this sector of the zirconium industry, Avalon considers pricing for EZC to be sensitive and proprietary.

Avalon has entered into several MOUs for the sale of its rare earth products. Negotiations to convert these MOUs into binding agreements are ongoing. Avalon entered into an MOU for the sale of EZC in January, 2013. In Avalon's opinion, these MOUs show that there is significant interest in its EZC product.

1.14 ENVIRONMENTAL STUDIES, PERMITTING AND SOCIAL OR COMMUNITY IMPACT

1.14.1 Northwest Territories

A complete description of the environmental setting and existing baseline conditions for the mine, concentrator and hydrometallurgical plant is provided in the Developers Assessment Report (DAR) as well as in various environmental baseline studies and recent fieldwork undertaken by Avalon's consultants.

1.14.1.1 Permitting

In 2010, Avalon filed a detailed Project Description Report (PDR) with the MVLWB as the first step in its application for a Type A Water Licence and a Type A Land Use Permit and was advised by the MVEIRB that the permit application would be referred for environmental impact assessment. In 2011, the DAR was submitted to the MVEIRB and the conformity review and information request stages were completed. The Technical Sessions were held during the period August 14-17, 2012 and the Public Hearings were held during the period February 18-26, 2013.

Preparation and issuance of the MVEIRB Report of Environmental Assessment is expected by the summer of 2013. This report will then be submitted to the Minister of Aboriginal Affairs and Northern Development. Following Ministerial approval, the process will revert back to the MVLWB to determine the conditions for the Type A Water Licence(s) and Type A Land Use Permit(s) that are anticipated to be required for the Nechalacho and Pine Point sites. The MVLWB process for the Type A Water Licence(s) is anticipated to take eight to 12 months to obtain and will include a public hearing process. The Type A Land Use Permit(s) will be processed concurrently.

1.14.1.2 Reclamation and Closure

Reclamation and closure of all site facilities in the Northwest Territories will be based on the Mine Site Reclamation Policy for the Northwest Territories and the Mine Site Reclamation Guidelines for the Northwest Territories.

1.14.1.3 First Nations Agreements

Avalon has entered into Negotiation Agreements with the LKDFN, YKDFN and the DKFN First Nations. As of March 2013, Avalon has finalized Accommodation Agreements with the DKFN. Negotiations are ongoing with the LKDFN and the YKDFN. In the Pine Point area, negotiation agreements have been signed with the NWTMN.

1.14.1.4 Tailings Disposal

Feasibility level designs have been completed for the tailings disposal facilities at both the Nechalacho and Pine Point sites in order to provide permanent and secure tailings storage, in addition to water storage and control to ensure protection of the environment during operations and in the long-term, post-closure.

1.14.2 Louisiana

Baseline environmental studies have been completed for the Geismar site.

The construction and operation of the Geismar site will require environmental permits issued by the Louisiana Department of Environmental Quality.

A “404 Permit” issued by the USACE is required for disturbance of wetlands at the site, if wetlands are disturbed. Avalon will obtain a 404/Section 10 Permit for the site, the access road and the LPDES discharge location to the Mississippi River.

The conceptual closure and reclamation plan for the Geismar site is based on a “design for closure” approach and will be updated throughout the life of the project.

1.15 CAPITAL AND OPERATING COSTS

1.15.1 Capital Costs

A summary of the feasibility study capital cost estimate for the project is presented in Table 1.4.

Table 1.4
Project Capital Cost Summary

Cost Category	NWT[1] ($ million)	LA[2] ($ million)	Total ($ million)
Mine development	81.58	-	81.58
Main process facilities	351.24	192.51	543.75
Infrastructure	150.68	78.82	229.50
EPCM	119.27	38.57	157.84
Indirect construction costs	175.56	27.25	202.81
Owner's costs	36.76	18.95	55.71
Contingency	120.91	44.90	165.81

Cost Category	NWT[1] ($ million)	LA[2] ($ million)	Total ($ million)
Closing costs / bond	13.00	3.16	16.16
Pre-production capital cost	**1,049.00**	**404.16**	**1,453.16**
Sustaining capital	102.72	19.12	121.84
Total capital cost	**1,151.72**	**423.28**	**1,575.00**

[1] NWT – Costs applicable to the Nechalacho and Pine Point sites in the Northwest Territories.
[2] LA – Costs applicable to Geismar, Louisiana.

The capital cost estimate was prepared by SLI. The scope of the estimate encompasses the engineering, administration, procurement services, construction, pre-commissioning and commissioning of the Project. The estimate was completed to a level consistent with an AACEI (Association of Advanced Cost Engineering International) Class 3 estimate with target accuracy level of ±15%, based on second quarter 2012 prices, excluding escalation.

The total estimated pre-production capital cost is $1,453 million. The life-of-mine sustaining capital is estimated at $122 million.

1.15.2 Operating Costs

A summary showing the average annual and life-of-mine unit operating costs by project cost area is presented in Table 1.5.

Table 1.5
LOM Average Operating Costs per Project Cost Area

Project Section	Average Annual Costs ($ million)	LOM Average Unit Costs ($/t milled)
Reagents & Consumables	97.2	133.20
Fuel & power	50.7	69.48
Labour	36.7	50.26
Freight	29.4	40.31
G&A	26.8	36.74
Other	23.7	32.29
Project total	**264.5**	**362.28**

The feasibility study operating cost estimates have been prepared on an annual basis for the life of the mine by SLI. The operating cost estimate has been prepared with an estimated level of accuracy of ±15% based on the design of the plant and facilities as described in detail in the feasibility study.

1.16 ECONOMIC ANALYSIS

Micon has prepared its assessment of the project on the basis of a discounted cash flow model, from which net present value (NPV), internal rate of return (IRR), payback and other measures of project viability can be determined. Assessments of NPV are generally accepted within the mining industry as representing the economic value of a project after allowing for the cost of capital invested. The sensitivity of this NPV to changes in the base case assumptions is then examined.

1.16.1 Base Case Evaluation

Figure 1.3 presents a summary of the LOM cash flows, and the cumulative discounted and undiscounted cash flows.

Figure 1.3
LOM Cash Flow

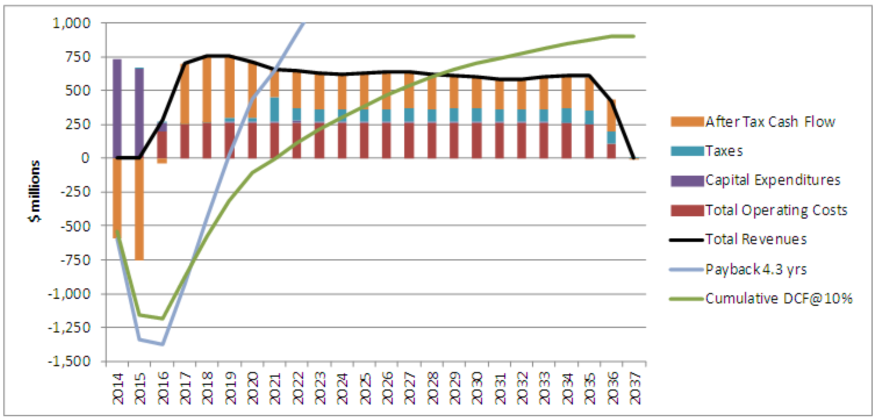


Table 1.6 shows the results of the economic evaluation of the base case cash flows.

Table 1.6
Economic Evaluation of Base Case

	Discount rate (%/y)	Pre-Tax $ million	After Tax $ million	Payback (y)
Undiscounted Cash Flow		6,052	4,381	4.3
Net Present Value	8	1,833	1,262	5.5
	10	1,351	900	6.3
	12	981	620	7.2
Internal Rate of Return		22.5%	19.6%	

The project provides a payback of 4.3 years on the undiscounted cash flow, or 6.3 years on the cash flow discounted at 10%/y, leaving a considerable reserve "tail". The cash operating margin is seen to remain positive over the whole LOM period, and is particularly strong in the first four years of full production.

1.16.2 Sensitivity Analysis

Revenue factors and exchange rate have the greatest impact on project returns, but an adverse change of more than 25% in project revenues would be required to reduce NPV to near zero. The project is less sensitive to capital and operating costs, with the project being only slightly more sensitive to operating costs than to capital expenditure.

The project is seen to be more sensitive to revenues from TREO than EZC, and the project is more sensitive to HREO prices than to those for LREO.

1.16.3 Conclusion

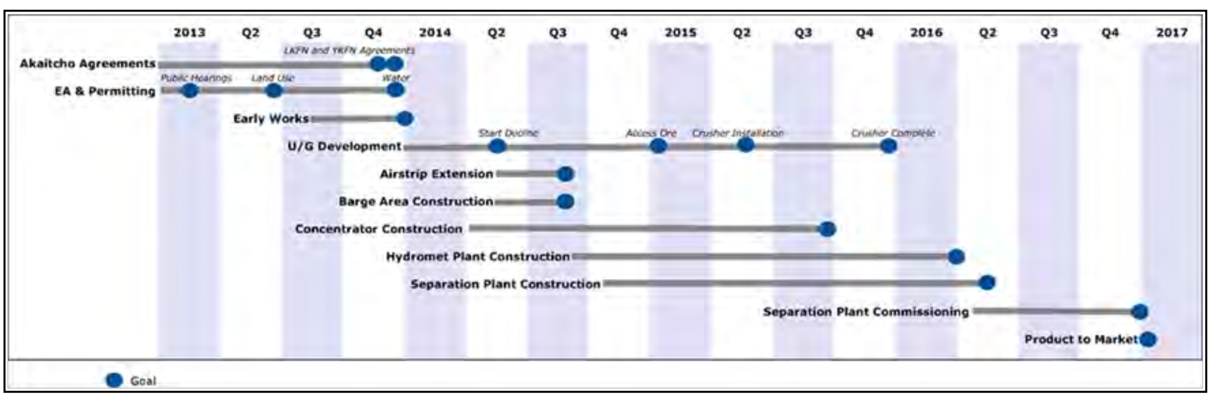
Micon concludes that the base case cash flow and sensitivity studies demonstrate robust economics that support the declaration of mineral reserves and warrant continued development of the Nechalacho REE project as described in this study.

1.17 OTHER RELEVANT DATA AND INFORMATION

1.17.1 Project Implementation

A project master schedule provided in Figure 1.4 shows the overall major activities and milestone dates. The schedule was developed based on Avalon's requirement to have the refinery at Geismar mechanically complete at the beginning May, 2016, and it takes into consideration equipment delivery and limitations related to Nechalacho site access.

Figure 1.4
Project Master Schedule



The Nechalacho site is considered to be the most critical of the three sites from a schedule perspective, due primarily to the limited access. A detailed critical path schedule has been developed for each of the project sites.

Successful implementation of Avalon's human resources policies and procedures will enable Avalon to achieve its goals and objectives for its employees and contractors. Avalon's goal of increasing local participation rates will be facilitated through its training and development initiatives. Avalon is committed to worker training throughout the life of the project.

Avalon's program will initially be designed to fill apprenticeship and technological occupations. In addition, all project contractors and sub-contractors will be required to adhere to Avalon's goal of maximizing Northern and Aboriginal employment.

1.17.2 Risk Analysis

Risk management for the project follows the principle of creating an environment and a context for dealing pro-actively with threats and opportunities. The risk management process is a continuous process to deal with potential threats and opportunities that could impact the expected outcome of the project.

During the preparation of the feasibility study, two joint Owner and Contractor risk identification workshops were conducted to systematically explore all critical processes and project areas in identifying potential risks and establishing the initial risk register. The areas of project management, project controls, procurement, engineering, construction, mining, geotechnical, environmental, and health and safety were represented. Following the workshops, mitigation plans and actions were developed and carried out to the extent possible during the feasibility study phase. Going forward, a plan will be developed for the EPCM and execution phases of the project that would start with the risk register developed in the feasibility study and build on the basic process described below and followed through into operations.

1.17.3 Health and Safety

Avalon has confirmed that safety is a core value above all others in its operations and that it will implement a philosophy of zero accidents. Health and safety management plans will be developed for construction, commissioning and operations. Appropriate training will be completed as required by all employees, contractors and subcontractors, and the training will be documented. While the focus will be on prevention of accidents, emergency and crisis management plans will be developed.

1.18 INTERPRETATION AND CONCLUSIONS

SLI has completed a feasibility study for the Nechalacho rare earth elements project on behalf of Avalon. The results of the feasibility study demonstrate that the project is technically feasible and economically robust. Table 1.7 presents the key project parameters, based on 100% equity financing.

Table 1.7
Key Project Parameters

Metric	Units	Quantity
Pre-tax economics		
IRR	%	22.5
NPV at 8%/y	$ million	1,833
NPV at 10%/y	$ million	1,351
NPV at 12%/y	$ million	981
After tax economics		
IRR	%	19.6
NPV at 8%/y	$ million	1,262
NPV at 10%/y	$ million	900
NPV at 12%/y	$ million	620
Payback period	y	4.3
Mining		
Mineral reserves	t	14,600,000
Production rate	t/y	730,000
Initial mine life	y	20
Total revenue	**$ million/y**	**645.8**
Rare earth oxides	$ million/y	456.5
EZC	$ million/y	189.3
Unit basis	$/t mined	884.65
Operating costs	**$ million/y**	**264.5**
Reagents	$ million/y	97.2
Fuel and power	$ million/y	50.7
Labour	$ million/y	36.7
Freight	$ million/y	29.4
General and Administration	$ million/y	26.8
Other	$ million/y	23.7
Unit basis	$/t mined	362.28

Revenues from HREE are in excess of 50% of total project revenue and revenue from neodymium, europium, terbium, dysprosium and yttrium account for over 82% of revenue for separated REO. Sales of lanthanum and cerium represent less than 4.5% of total revenue.

Sensitivity analysis demonstrates that revenue factors and exchange rate have the greatest impact on project returns, but that an adverse change of more than 25% in project revenues is required to reduce NPV to near zero. The project is less sensitive to capital and operating costs, with the project being only slightly more sensitive to operating costs than to capital expenditure.

The project is seen to be more sensitive to revenues from TREO than EZC, and the project is more sensitive to HREO prices than to those for LREO.

Micon concludes that the base case cash flow and sensitivity studies demonstrate robust economics that support the declaration of mineral reserves and warrant continued development of the Nechalacho rare earth elements project as described in the feasibility study and in this technical report.

1.19 RECOMMENDATIONS

Given that the base case cash flow and sensitivity studies warrant continued development of the Nechalacho project as presented in the feasibility study, a number of recommendations are made for further work during project development.

1.19.1 Geology and Exploration

The Nechalacho deposit is not fully delineated in terms of the definition of the limits of the mineralization. The greatest potential for increase of tonnage of mineralization similar to the known mineralization is to the north of the present inferred resources. It is recommended that additional drilling be undertaken in order to define the northern limit of the Nechalacho deposit and to demonstrate that mineralization may exist under apparently un-mineralized Thor Lake Syenite.

1.19.2 Mining

It is recommended that:

- Optimization of the stoping sequence and stoping plans be undertaken, using specialized software, in order to determine whether further increases in the feed grades in the early years of the mine life can be achieved.
- The opportunity for higher mine production rates be assessed.
- The opportunity to increase mining of direct shipping ore be assessed.
- Additional geotechnical analysis and optimization analysis be undertaken relating to the proposed stope excavation spans and the extraction ratio in order to minimize and mitigate the associated rock rock mechanics risks and challenges of mining under large excavations and high extraction ratio. A numerical analysis is recommended to assess whether the location of pillars in ore or low grade/waste can be incorporated into the mine plan with a view to recovery at the end of mine life.
- Further paste backfill studies be undertaken in order to increase the effectiveness of paste fill contribution to mine stability. Cement or binder screening and optimizations are recommended at an early stage to reduce the overall operating cost for paste fill.
- A laboratory or on-site flow loop be completed to confirm the flow characteristics of the backfill paste to confirm that the positive displacement pumps located at the concentrator can service the entire mine without the need of a booster station.

Further mine design optimization analysis can be performed in the following areas to reduce the mine operating costs:

- Stope optimization with mine scheduling and optimization software to increase the feed grade to the concentrator in early years of project life.

- Review of sill excavation from 15 m to 5 m wide and perform fan drilling especially in the long hole stopes to parallel holes drilling from a 15 m slashed sill.

- Investigation into the use of friction bolts on the side walls to reduce operating costs and increase efficiencies for the development of non-permanent excavations such as backfill drifts, headings in ore.

1.19.3 Mineral Processing

1.19.3.1 Concentrator

It is recommended that:

- Avalon continues with mineralogical studies to improve the understanding of compositional and textural variation of minerals laterally and vertically within the deposit, the chemical nature of the complex pseudomorphs within the lower parts of the BZ, and the relationship of mineral texture with metallurgical performance.

- Avalon continues to investigate means to improve process efficiency and reduce operating costs, including optimization of the comminution circuit, alternative flotation reagents and optimization of the cleaner flotation circuit.

1.19.3.2 Hydrometallurgical Plant

It is recommended that:

- Avalon continues with testwork to further optimize the hydrometallurgical flowsheet and identify potential improvements to the process that increase metallurgical efficiencies and/or reduce operating and capital costs including:

  - Optimization of the acid bake process.

  - Improvement of the HREE recoveries into the rare earth precipitate product using caustic cracking.

  - Optimization of the HREE recoveries by acid leaching at elevated pressures.

  - Direct treatment of high grade mineralization.

1.19.3.3 Rare Earth Refinery

It is recommended that Avalon continues:

- Further testwork with the primary objective of optimizing the impurity removal process.

- To work with Mintek to investigate enhancements to the process flowsheet which offer the potential to reduce capital and operating costs in the rare earth solvent extraction separation plant.

- To assess opportunities for toll processing of rare earth feedstocks for third parties.

- To assess opportunities to have a third party process output from the hydrometallurgical plant under a tolling agreement.

1.19.4 Infrastructure and Transportation

It is recommended that Avalon carries out a detailed trade-off study to assess the merits of re-establishing the rail line between Hay River and Pine Point in order to reduce trucking costs.

1.19.5 Community Engagement

It is recommended that Avalon continues to engage with its Aboriginal partners and to utilize traditional knowledge with the objective of improving environmental performance.

1.19.6 Marketing

It is recommended that Avalon continues to move forward to convert MOUs into binding contracts and continues to develop marketing and strategic relationships that will establish the company as a credible participant in the rare earth industry.

1.19.7 Project Development

Micon recommends that Avalon proceeds with continued development of the Nechalacho rare earth element project as described in the feasibility study prepared by SLI.

1.19.7.1 Potential Opportunities

Avalon has identified a number of opportunities for project optimization that may improve project economics, reduce technical risk, enhance metallurgical recoveries and improve operational efficiencies. These include:

- Optimization of the crushing and grinding circuit, plant layouts and materials of construction.

- Metallurgical testwork to further improve reagent selection and flotation recoveries.

- Improvements to the hydrometallurgical plant processes.
- Alternative impurity removal scenarios.
- Potential to separate lanthanum and cerium at the hydrometallurgical plant and stockpile for future sales.
- Potential to reintroduce cracking of zircon at a later date to increase direct production of HREE and separate the by-products from the EZC.
- Potential sales of magnetite by-product from the concentrator.
- Potential to defer construction of the refinery and toll process mixed rare earth concentrate through a refinery or refineries built and operated by others.
- Potential to use excess capacity in the refinery to toll process third party production and reduce operating costs.

These opportunities are under consideration and will continue to be investigated as the project proceeds. Avalon has a budget of $3 million to carry out this work over the next six to 12 months.

Micon concurs with Avalon's proposed work to optimize the project and recommends that the work proceeds as planned.

2.0 INTRODUCTION

2.1 SCOPE OF WORK AND TERMS OF REFERENCE

Micon International Limited (Micon) has been retained by Avalon Rare Metals Inc. (Avalon) to compile the Technical Report under Canadian National Instrument 43-101 (NI 43-101) which discloses the results of the feasibility study prepared by SNC-Lavalin (SLI) on Avalon's Nechalacho rare earth elements project.

SLI was engaged by Avalon to complete the feasibility study for development of the Nechalacho project under an agreement dated 14 December, 2011. The study was completed in April, 2013 and the results summarized in a press release dated 17 April, 2013.

2.1.1 Scope of the Feasibility Study

The Nechalacho mine and the three processing plants comprise the following principal components:

- The Nechalacho site on the north shore of Great Slave Lake, Northwest Territories, which includes:
    - Underground mine and paste backfill plant.
    - Concentrator and tailings management facility (TMF).
    - Site utilities and infrastructure including power plant, accommodation, airstrip and dock facilities.
- The Pine Point Site on the south side of Great Slave Lake, Northwest Territories, which includes:
    - Hydrometallurgical plant, including sulphuric acid plant, hydrometallurgical tailings facility (HTF) and dock facility.

The Geismar site near Baton Rouge, Louisiana, United States, which includes:

- Rare earth refinery, including administration and maintenance facilities, and gypsum impoundment.

The locations of these facilities are shown in Figure 2.1.

The flotation concentrate produced at the Nechalacho site will be transported to the hydrometallurgical plant at Pine Point where a mixed rare earth precipitate and an enriched zirconium concentrate (EZC) will be produced. The EZC will be sold to one or more third parties. The mixed rare earth concentrate will be transported to the leaching and rare earth separation facility at Geismar where a range of separated rare earth products will be recovered for sale.

Figure 2.1
Project Site Locations

The Nechalacho project is 100% owned by Avalon.

In addition to SLI, the principal contributors and consultants providing input to the feasibility study, and their roles, are summarized below:

Avalon

- General scope items applicable to all project sites:
  - Overall project management.
  - Financing and financial analysis.
  - Process testwork and design criteria.
  - Land concessions, acquisition and rights-of-way.
  - Environmental data, studies, monitoring and reporting plans.
  - Environmental and operating permits.
  - Radiation pathways assessments.
  - Geotechnical, hydrogeological investigations and foundation recommendation.
  - Site topography and survey.
  - Community relations and training program development.
- Nechalacho site:
  - Geology and resource definition (together with RPA).

  - Underground mine design including infrastructure, utilities and equipment.
  - Ore reserves definition.
  - Site development for mine and concentrator including docking facilities, power plant, accommodation, warehousing, and access and internal roads.
  - Airstrip improvements (with input from local contractors).
  - Barging and off-shore docking facilities in conjunction with Northern Transportation Company Limited (NTCL) and Red Sky Enterprises Inc.
- Pine Point site:
  - Railway trans-load facilities at Hay River.
  - Barging and off-shore docking facilities in conjunction with NTCL and Red Sky Enterprises Inc.

Mine Ventilation Services Inc. (MVS)
- Nechalacho site:
  - Mine ventilation design.

Roscoe Postle Associates Inc. (RPA)
- Nechalacho site:
  - Geology and resource definition (together with Avalon).

EBA Engineering Consultants Ltd. (a Tetra Tech Company) (EBA)
- Nechalacho and Pine Point sites:
  - Environmental conditions.
  - Socio-economic conditions.

Knight Piésold Ltd. (KPL)
- Nechalacho site:
  - Tailings management facility (TMF) design.
  - TMF water balance.
  - Underground mine geomechanics review.
- Pine Point site:
  - TMF design.
  - TMF water balance.
  - TMF capital cost estimating.

FLSmidth (FLS)
- Nechalacho site:
  - Concentrator process design.
  - Equipment budgetary costs (initial).

Golder Associates Ltd. (Golder)
- Nechalacho site:
  - Development and application of paste backfill.
  - Paste backfill plant design.

- Equipment and material capital cost estimating of paste backfill facilities.

URS Corporation (URS)

- Geismar site:
  - Environmental assessment.
  - Permitting.

Hwozdyk Inc. (mining consulting services)

- Nechalacho site:
  - Underground mine cost estimation.

Other technical consultants retained by Avalon include:

- SGS Canada Inc. (metallurgical testwork).
- Xstrata Process Support (XPS).
- J.R. Goode and Associates (metallurgical consultant).
- Jerald Tabor (independent metallurgical consultant).
- PRS & Associates (independent project management consultant).

The feasibility study was completed over a period of two years at a cost of approximately $60 million, as detailed in Table 2.1.

Table 2.1
Feasibility Study Costs

Category	$ million
Resource definition	23.0
Metallurgy	17.0
Feasibility studies	15.0
Environmental and permitting	3.0
Community engagement	2.0
Total	**60.0**

Capital cost estimates were completed to an intended level of accuracy of +15%, consistent with an AACEI Class 3 estimate, based on prices as of second quarter 2012 and excluding escalation. Operating cost estimates were completed to an intended level of accuracy of +15%, based on prices as of fourth quarter 2012.

2.2 QUALIFIED PERSONS AND SITE VISITS

This report is based on the feasibility study prepared by SLI and supporting documentation. Micon has held discussions with the responsible individuals at SLI and Avalon in reviewing those sections of the report for which it is taking responsibility.

As noted above, RPA and KPL contributed to specific sections of the feasibility study.

The Qualified Persons (QPs) for this Technical Report are the following:

- Tudorel Ciuculescu, M.Sc., P.Geo., RPA (geology, mineral resource estimate and all aspects of the resource database).
- Barnard Foo, P.Eng., Micon (mine design, mineral reserve estimate, capital and operating costs).
- Richard Gowans, P.Eng., Micon (metallurgical testwork and processing, capital and operating costs).
- Kevin Hawton, P.Eng., KPL (tailings management design, Nechalacho and Pine Point).
- Christopher Jacobs, CEng, MIMMM, Micon (economic analysis).
- Jane Spooner, P.Geo., Micon (market studies and overall management of the Technical Report preparation).

Site visits have been carried out on the following dates:

Richard Gowans and Barnard Foo	13 March, 2013
Tudorel Ciuculescu	25-27 April, 2011, 9-11 October, 2012
Kevin Hawton	August, 2011

2.3 USE OF REPORT

This report is intended to be used by Avalon subject to the terms and conditions of its agreement with Micon. Subject to the authors' consent, that agreement permits Avalon to file this report as a Canadian National Instrument 43-101 (NI 43-101) Technical Report on SEDAR (www.sedar.com) pursuant to Canadian provincial securities legislation. Except for the purposes legislated under provincial securities laws, any other use of this report, by any third party, is at that party's sole risk.

The requirements of electronic document filing on SEDAR necessitate the submission of this report as an unlocked, editable pdf (portable document format) file. Micon accepts no responsibility for any changes made to the file after it leaves its control.

The conclusions and recommendations in this report reflect the authors' best judgment in light of the information available to them at the time of writing. The authors and Micon reserve the right, but will not be obliged, to revise this report and conclusions if additional information becomes known to them subsequent to the date of this report. Use of this report acknowledges acceptance of the foregoing conditions.

2.4 UNITS

Cost estimates and other inputs to the cash flow model for the project have been prepared using constant money terms, i.e., without provision for escalation or inflation. All costs are presented in Canadian dollars ($, CAD), unless otherwise noted. Prices for rare earth products are given in United States dollars (US$).

Units in the project feasibility study and this report are in the Système international d'unités (SI), unless otherwise noted.

The CAD:US$ exchange rate assumption used in the financial analysis was based on a major bank's forecast prepared during January, 2013, shown as follows:

To end of 2015	CAD1.00 = US$0.97
2016	CAD1.00 = US$0.96
2017 and thereafter	CAD1.00 = US$0.95

It should be noted that rare earth and yttrium oxides have been calculated and expressed herein as RE_2O_3 although oxides of cerium, praseodymium and terbium are often given using the formulae CeO_2, Pr_6O_{11} and Tb_4O_7, respectively. The latter convention is used for oxide pricing.

2.5 LIST OF ABBREVIATIONS

Table 2.2 provides a list of the abbreviations used in this report.

Table 2.2
List of Abbreviations

Abbreviation	Term
°	degree(s)
°C	degree(s) Centigrade
°F	degree(s) Fahrenheit
<	Less than
>	Greater than
µm	Micrometre (micron = 0.001 mm)
%	percent, percentage
A	Ampere
ABA	Acid Base Accounting
Acme	Acme Analytical Laboratories (Vancouver) Limited
Actlabs	Activation Laboratories Limited
Ag	silver
Al	aluminum
ALS	ALS Limited
ANFO	Ammonium Nitrate and Fuel Oil
As	arsenic
Avalon	Avalon Rare Metals Inc.

Abbreviation	Term
B	billion
Ba	barium
BCC	BCC Research
bi	bismuth
BLC	Blatchford Lake Complex
BOO	Build-Own-Operate
Btu	British thermal units
BZ	Basal Zone
BZ-B	BZ-Bottom
BZ-T	BZ-Top
C	carbon
Ca	calcium
$, CAD	Canadian dollar(s)
CATV	cable television systems
CCR	central control room
CDA	Canadian Dam Association
CDN	CDN Resource Laboratories Limited
Ce	cerium
cfm	cubic feet per minute
CIF	cost insurance freight
cm	centimetre
CN	Canadian National Railway
CNF	cost and freight
Co	cobalt
Cu	copper
CV	coefficient of variation
d	day
db(a)	decibel (a-weighted)
DFKN	Deninu Kue First Nation
dmt	dry metric tonne(s)
dtpd	dry metric tonnes per day
dwt	dead weight tonnes
Dy	dysprosium
EBA	EBA Engineering Consultants Ltd.
edxrf	energy dispersive x-ray fluorescence
Entergy	Entergy Louisiana, LLC
EPCM	Engineering, procurement and construction management
Er	erbium
Eu	europium
EZC	Enriched Zirconium Concentrate
F	fluorine
Fe	iron
FOB	Free On Board
ft	foot, feet
FW	footwall
g	gram(s)
g	acceleration due to gravity
g/L	gram per litre
g/t	gram per tonne
Ga	gallium
GA	General Arrangement

Abbreviation	Term
gal	gallon
Gd	gadolinium
GHG	Green House Gas (emissions)
Golder	Golder Associates Ltd.
gpm	gallons per minute
GPS	global positioning system
GWh	gigawatt-hour
H	hydrogen
h	hour
h/d	hours per day
ha	hectare
HDPE	high density polyethylene
Ho	holmium
HP	horsepower
HQ	Diamond Drill Core Size 63.5 mm (inside diameter of core tube)
HREE	Heavy Rare Earth Elements
HREO	Heavy Rare Earth Oxides
Hz	Hertz
ICP	Inductively Coupled Plasma
ICP-MS	Inductively Coupled Plasma Mass Spectrometry
in	inch(es)
INAC	Indian and Northers Affairs Canada
IRR	Internal Rate of Return
J	joule
K	potassium
k	kilo (thousand)
kg	kilogram(s)
kg/h	kilograms per hour
kg/m^3	kilograms per cubic metre
km	kilometre(s)
km/h	kilometres per hour
kPa	kilopascal
KPL	Knight Piésold Ltd.
kV	kilovolt(s)
kVA	kilovolt-ampere
kW	kilowatt(s)
kWh	kilowatt hour
kWh/t	kilowatt hours per tonne
L	litre
La	lanthanum
L/s	litre per second
LAN	local area network
lb	pound
LCT	locked cycle test
LHD	load-haul-dump
LIMS	low intensity magnetic separation
LKDFN	Lutsel K’e Dene First Nations
LOI	loss on ignition
LOM	life of mine
LPDES	Louisiana Pollutant Discharge Elimination System
LREE	light rare earth element(s)

Abbreviation	Term
LREO	light rare earth oxide(s)
Lu	lutetium
M	mega (million)
m	metre(s)
m/min	metres per minute
m/s	metres per second
mA	milliampere
masl	metres above sea level
MCC	Motor Control Centre
min	minute
ML	million litres
mm	millimetre(s)
mg/L	milligrams per litre
Mg	magnesium
Mo	molybdenum
MOU	Memorandum of Understanding
mPa.s	millipascal second
MPa	megapascal
MPP	mini-pilot plant
MW	megawatt(s)
MVEIRB	Mackenzie Valley Environmental Impact Review Board
MVLWB	Mackenzie Valley Land and Water Board
MVRMA	Mackenzie Valley Resource Management Act
MVS	Mine Ventilation Services Inc.
MW	megawatt(s)
MWh	megawatt hour
Na	sodium
NAG	net acid generating
Nb	niobium
Nd	neodymium
Ni	nickel
NI 43-101	Canadian National Instrument 43-101
NMR	net metal return
NPV	net present value
NSMA	North Slave Métis Alliance
NSR	net smelter return
NWTMN	Northwest Territories Métis Nation
oz	ounce
P	phosphorus
P&ID	process and instrument diagram
Pa	pascal
Pa.s	pascal-second
Pb	lead
ppb	parts per billion
ppm	parts per million
Pr	praseodymium
QA	quality assurance
QA/QC	quality assurance/quality control
QC	quality control
RBC	rotating biological contactor
REE	rare earth element(s)

Abbreviation	Term
REO	rare earth oxide(s)
RMB	Chinese Renminbi
RMR	rock mass rating
ROM	run-of-mine
RPA	Roscoe Postle Associates Inc.
RQD	rock quality designation
s	second
S	sulphur
SAG	semi-autogenous grind
Sb	antimony
SD	standard deviation
SEM	scanning electron microscope
SG	specific gravity
SI	international system of units
Si	silicon
SLI	SNC-Lavalin Inc.
Sm	samarium
Sn	tin
t	tonne(s) (metric = 1,000 kg)
t/h	tonnes per hour
t/y	tonnes per year
Ta	tantalum
Tb	terbium
Th	thorium
Tn	thulium
TMF	Tailings Management Facility
tpd	tonnes per day
TREE	total rare earth element(s)
TREO	total rare earth oxide(s)
TSS	total suspended solids
U	uranium
UCS	unconfined compressive strength
USACE	United States Army Corps of Engineers
US$	United States dollar(s)
UZ	Upper Zone
V	volt(s)
W	tungsten
wmt	wet metric ton
Wardrop	Wardrop Engineering Inc.
XPS	Xstrata Process Support
XRF	energy-dispersive x-ray fluorescence
Y	yttrium
y	year
Yb	ytterbium
YKDFN	Yellowknives Dene First Nation
Zn	zinc
Zr	zirconium

3.0 RELIANCE ON OTHER EXPERTS

Micon has reviewed and analyzed data provided by Avalon and SLI relating to the feasibility study for the Nechalacho rare earth elements project.

While exercising all reasonable diligence in checking, confirming and testing it, Micon has relied upon Avalon's presentation of the feasibility study for the Nechalacho rare earth elements project and the data contained therein.

Micon has not carried out any independent exploration work, drilled any holes or carried out any sampling and assaying on the Nechalacho property.

Micon has not reviewed any of the documents or agreements under which Avalon holds title to the Nechalacho, Pine Point and Geismar sites, or the underlying mineral concessions and Micon offers no opinion as to the validity of the mineral titles claimed. A description of the properties, and ownership thereof, presented in Section 4.0, based on material prepared by Avalon, is provided for general information purposes only.

Micon has relied upon the expertise of Rick Hoos, M.Sc., R.P.Bio., Principal Consultant with EBA, who has reviewed Section 20 of this report, dealing with Environmental Studies, Permitting and Social or Community Impact, with the exception of Sections 20.2.3.1 through 20.2.3.4, 20.2.3.6 through 20.2.3.10, 20.2.4 and 20.3.3 all of which deal with design of tailings disposal facilities at the Nechalacho and Pine Point sites and for which Kevin Hawton, P.Eng., is the Qualified Person.

Mr. Hoos directed the completion of environmental, archeological and socioeconomic baseline studies at both the Nechalacho and Pine Point sites, the completion of the Project Description Report (PDR) required by the MVLWB and the Developer's Assessment Report (DAR) required by the MVEIRB. Environmental and permitting aspects of the feasibility study for the Geismar site were prepared by URS Corporation and this work is summarized in Section 20.4 of this Technical Report. Mr. Hoos has reviewed the work completed by URS.

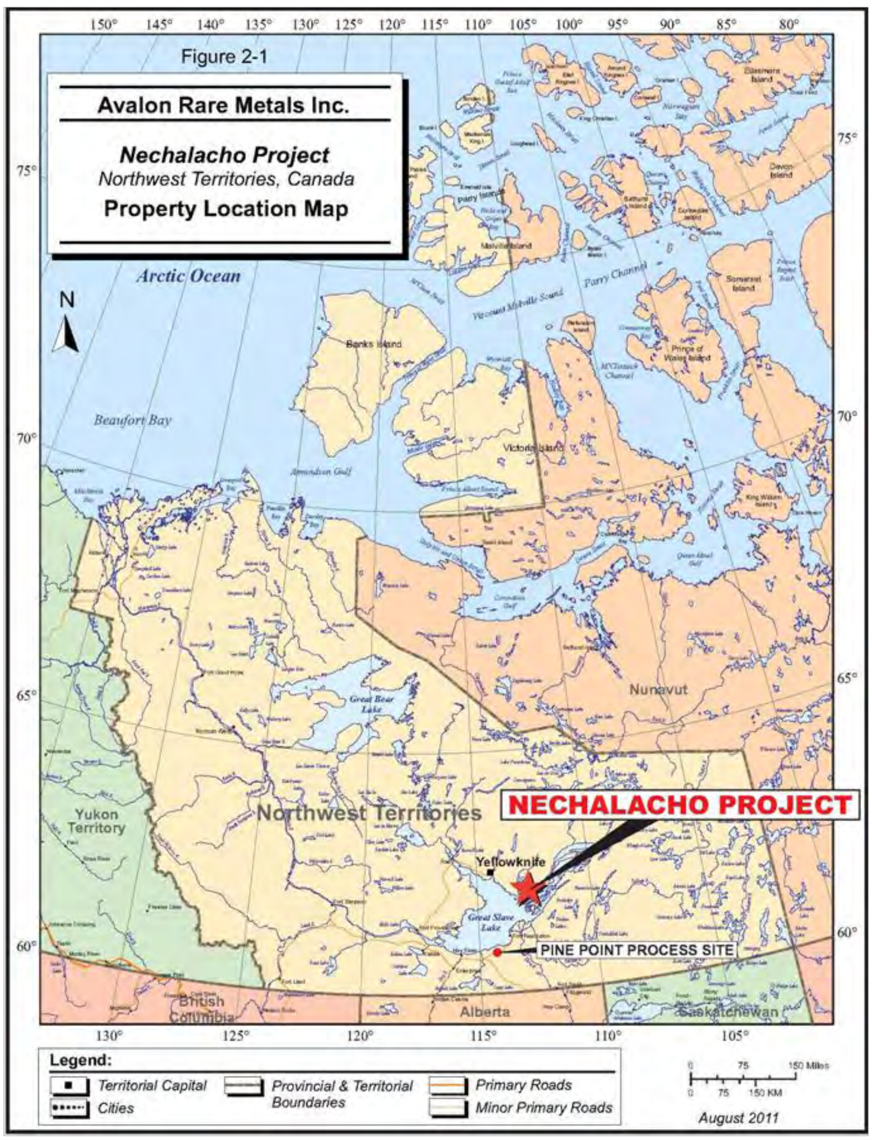
4.0 PROPERTY DESCRIPTION AND LOCATION

The Nechalacho project consists of a mine and concentrator at Nechalacho and two further processing sites located at Pine Point (hydrometallurgical plant) in the Northwest Territories, and at Geismar (rare earth refinery) near Baton Rouge, Louisiana, United States.

4.1 NECHALACHO SITE

The Thor Lake property, part of the Nechalacho project, is located on the north shore of the Great Slave Lake in the Northwest Territories of Canada, approximately 100 km east of the territorial capital of Yellowknife and 5 km north of the Hearne Channel. See Figure 4.1.

Figure 4.1
Nechalacho Site, General Location Map



The property lies within the Mackenzie Mining District of the Northwest Territories and Thor Lake is shown on the National Topographic System (NTS) map sheets 851/1 and 851/2. It is located at approximately 62°06'30" N and 112°35'30" W (6,886,500mN and 417,000mE).

4.1.1 Land Tenure

4.1.1.1 Legal Survey

The Thor Lake mineral leases have been legally surveyed and are recorded on a Plan of Survey, Number 69408 M.C. in the Legal Surveys Division of the Federal Department of Energy, Mines and Resources, Ottawa, now Natural Resources, Canada. The perimeter boundaries of the lease lots were surveyed as part of the leasing requirements.

4.1.1.2 Land Tenure

The Thor Lake property consists of five contiguous mineral leases (totalling 4,249 ha, or 10,449 acres) and three claims (totalling 1,869 ha, or 4,597 acres). The claims were staked in 2009 to cover favourable geology to the west of the mining leases.

Pertinent data for the mining leases are shown in Table 4.1 while the mineral claims data are shown in Table 4.2.

Table 4.1
Mineral Lease Summary

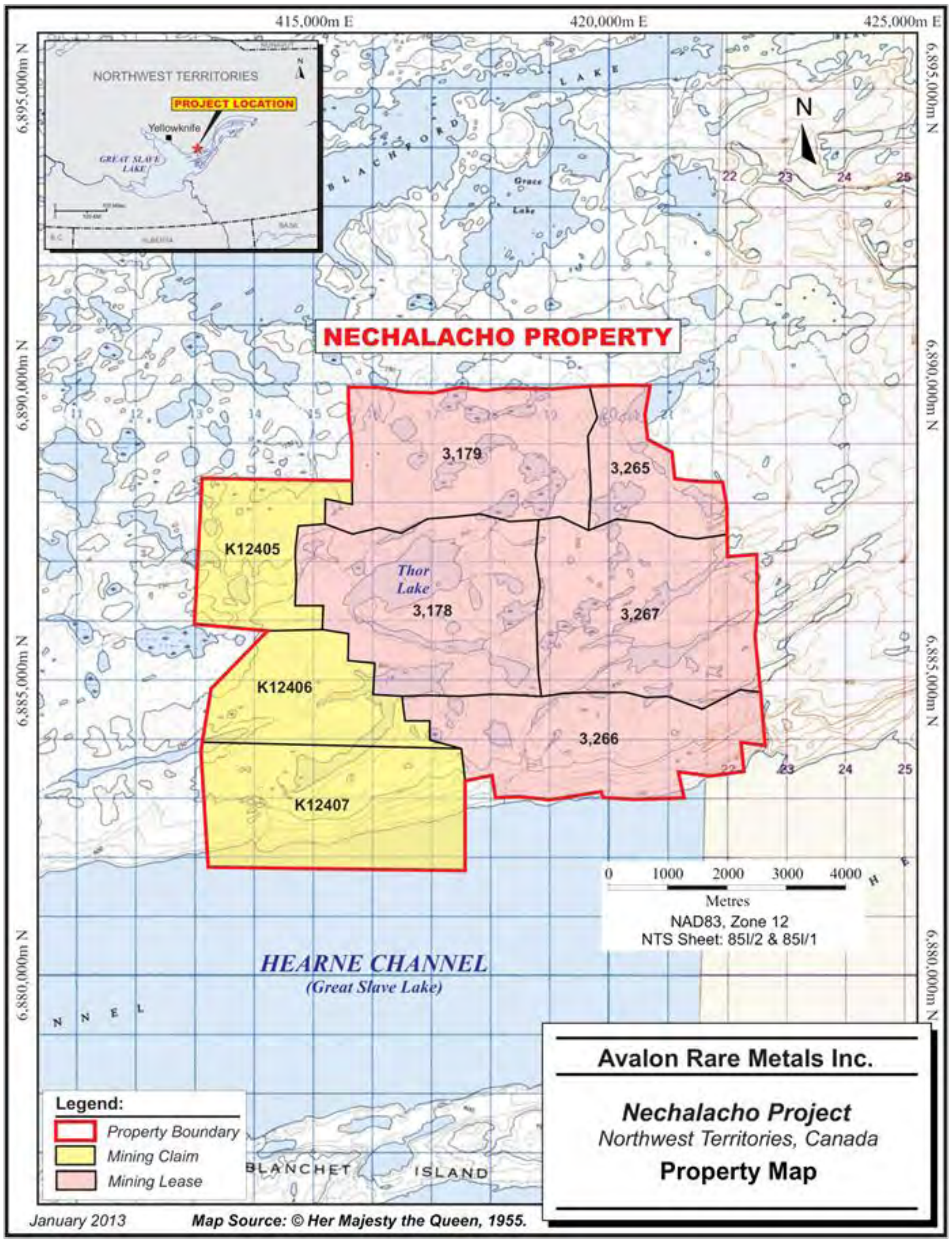
Lease Number	Area (ha)	Legal Description	Effective Date	Expiration Date
3178	1,053	Lot 1001, 85 I/2	05/22/1985	05/22/2027
3179	939	Lot 1000, 85 I/2	05/22/1985	05/22/2027
3265	367	Lot 1005, 85 I/2	03/02/1987	03/02/2029
3266	850	Lot 1007, 85 I/2	03/02/1987	03/02/2029
3267	1,040	Lot 1006, 85 I/2	03/02/1987	03/02/2029
Total	**4,249**			

Table 4.2
Mineral Claims Summary

Mineral Claim Number	Mineral Claim Name	Claim Sheet Number	Mining District
K12405	Angela 1	85I2	Mackenzie
K12406	Angela 2	85I2	Mackenzie
K12407	Angela 3	85I2	Mackenzie

The property map is shown in Figure 4.2.

Figure 4.2
Property Map, Nechalacho Deposit, Thor Lake Project



The mining leases have a 21-year life and each lease is renewable in 21-year increments. Annual payments of $2.47/ha ($1.00 per acre) are required to keep the leases in good standing. Avalon owns 100% of all of the leases subject to various legal agreements described below.

The mineral claims are in good standing with assessment work filed.

4.1.2 Royalty Interests

One underlying royalty exists on the Thor Lake property and originated with Highwood Resources Ltd. (Highwood), the initial developer of the property. Signed in 1977, the Murphy Royalty Agreement entitles J. Daniel Murphy to a 2.5% NSR royalty which applies to the entire Thor Lake property and is capped at an escalating amount indexed to the rate of inflation. Under the agreement, Avalon is permitted to purchase the royalty at the commencement of production at a fixed, inflation-adjusted price.

In July, 2012 Avalon bought out the 3% Calabras/Lutoda NSR royalty on the Thor Lake property which entitled Calabras (Canada) Ltd. and Lutoda Holding Ltd. to NSR royalties of 2% and 1%, respectively.

4.1.3 First Nation Communities

The Nechalacho site is located within the Akaitcho Dene asserted traditional territory and is subject to the comprehensive land claim negotiation between the Akaitcho Dene First Nations (Yellowknives Dene First Nation (Dettah), Yellowknives Dene First Nation (Ndilo), Lutsel K'e Dene First Nation and Deninu Kue First Nation) and the Federal government.

Further details on Avalon's engagement with First Nation communities are provided in Section 20.

4.1.4 Permits

A two-year exploration permit was granted to Avalon effective July, 2007 by the MVLWB which administers land use permits.

The permit was renewed in July, 2009 for a further two years and an amendment granted to include the operation of two diamond drills. On 23 June, 2011, Avalon received a new land use permit from the MVLWB to cover its activities at site and to replace the existing permit scheduled to expire on 4 July, 2011. The present land use permit extends for five years from 5 July, 2011.

In addition, Avalon was granted a five-year permit effective 29 May, 2012 to extend the airstrip to 1,000 m and, subsequently, a quarry permit for material for fill and the surface of the airstrip.

Permitting for the Nechalacho and Pine Point sites is discussed in more detail in Section 20 of this report.

4.1.5 Environmental Liabilities

Avalon has carried out reclamation of areas on the property disturbed by previous exploration programs. Among the activities completed are the use of historic waste piles for construction of the airstrip, refurbishment of part of the trailer camp for camp and office facilities and consumption of diesel fuel remaining in the tank farm for exploration activities.

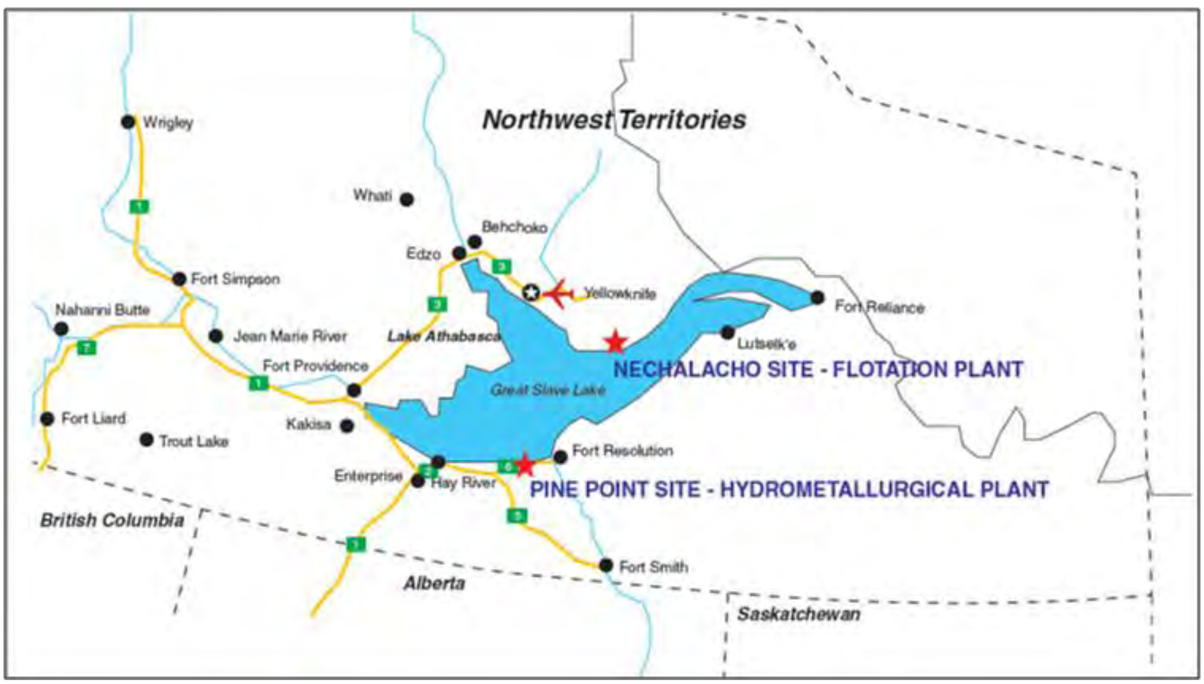
The environmental status of the Nechalacho site and studies undertaken are discussed in Section 20.0 of this report.

4.2 PINE POINT SITE

The Pine Point site is located in the Northwest Territories about 12 km south of the south shore of Great Slave Lake at Latitude 60°52’10.33”N Longitude 114°25’10.42”W. It will host the hydrometallurgical plant and related facilities.

The location of the historic Pine Point mine operated by Pine Point Mines Ltd., a subsidiary of Cominco Ltd., the Pine Point site lies approximately 90 km east of Hay River and 65 km west of Fort Resolution. See Figure 4.3.

Figure 4.3
Location of Pine Point Site



4.2.1 First Nation Communities

The Pine Point site is located within the Akaitcho Dene asserted traditional territory and is subject to the comprehensive land claim negotiation between the Akaitcho Dene First Nations and the Federal government. The Northwest Territory Métis Nation (NWTMN) is also negotiating a land claim in the South Slave region which encompasses a membership in Fort Resolution, Fort Smith and Hay River.

Further details on Avalon's engagement with First Nation communities are provided in Section 20.0.

4.2.2 Permits

Avalon has applied to the Northwest Territories government, through a Commissioner's Land Application, for a lease on the site of the proposed hydrometallurgical plant at Pine Point. The Northwest Territories Department of Municipal and Community Affairs (MACA) has deemed the application complete. However, MACA has stated that the application will not be processed until the environmental assessment process conducted by the Mackenzie Valley Environmental Impact Review Board (MVEIRB) is complete. Avalon has written confirmation from MACA that, in the meantime, the application has been recorded in the land register and the parcel of interest will be held for Avalon's use. Similarly, Avalon has applied to MACA for a lease on land covering the access route from the proposed barge docking facility to the proposed Pine Point hydrometallurgical plant site. As a result, Avalon does not believe there is any reason that it will not be issued the leases once the MVEIRB environmental assessment process is completed. (See Figure 4.4 and Figure 4.5) The total area applied for is 48.7 ha.

The remaining areas to be utilized for the hydrometallurgical facility, including the fresh water supply, dolomite waste piles, tailings waste water disposal (infiltration) areas and the barge docking location, are on federally owned Crown land. Applications have been made to Aboriginal and Northern Development Canada (AANDC) for surface leases applicable to these facilities. AANDC has processed the lease applications. However an exploration company has mineral claims in the same area which grant subsurface rights overlapping the location of the proposed hydrometallurgical plant, hydrometallurgical plant TMF, Pit J-44 (the proposed process water supply source) and Pit N-42 (proposed excess water infiltration pit). At the time of this report, Avalon is seeking to negotiate with the subsurface rights owner and also seeking clarification from the appropriate government authorities. Should the conflict of rights relating to the proposed location of the hydrometallurgical plant TMF not be resolved within a reasonable time, Avalon has identified alternative locations for tailings disposal. Process water can be acquired from wells and ground water recharge wells can also be installed in place of the excess water infiltration pit. The total area applied for is 215.9 ha.

Permitting for the Pine Point site is described in more detail in Section 20.0.

Figure 4.4
Pine Point Lease Application Area for Access Road

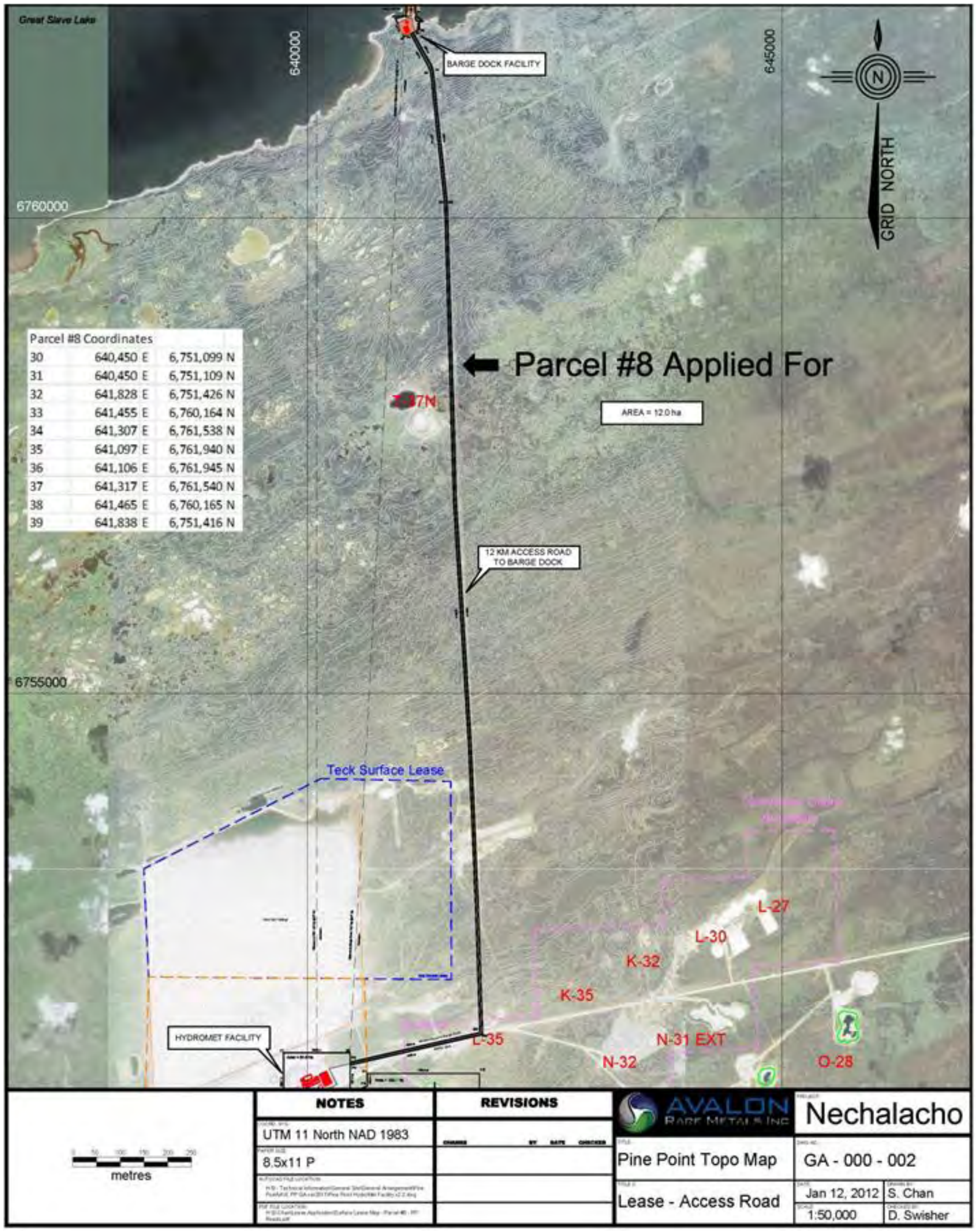
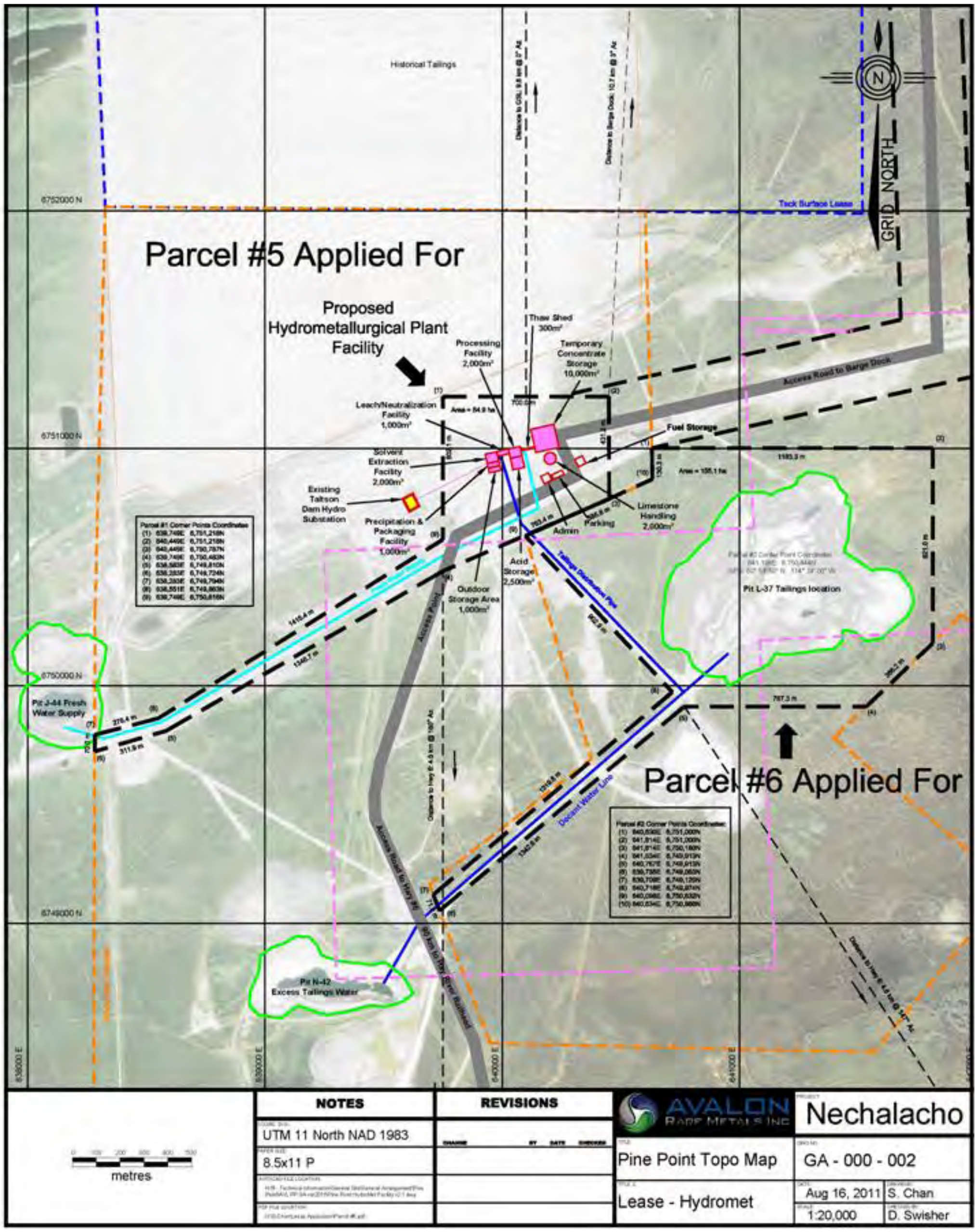


Figure 4.5
Pine Point Lease Application Area for Hydrometallurgical Plant



Should Avalon require permits for additional small scale work such as geotechnical drilling or reclamation experimentation prior to receipt of the MVLWB Land Use Permit, permits of this nature can normally be obtained within 60 days of application.

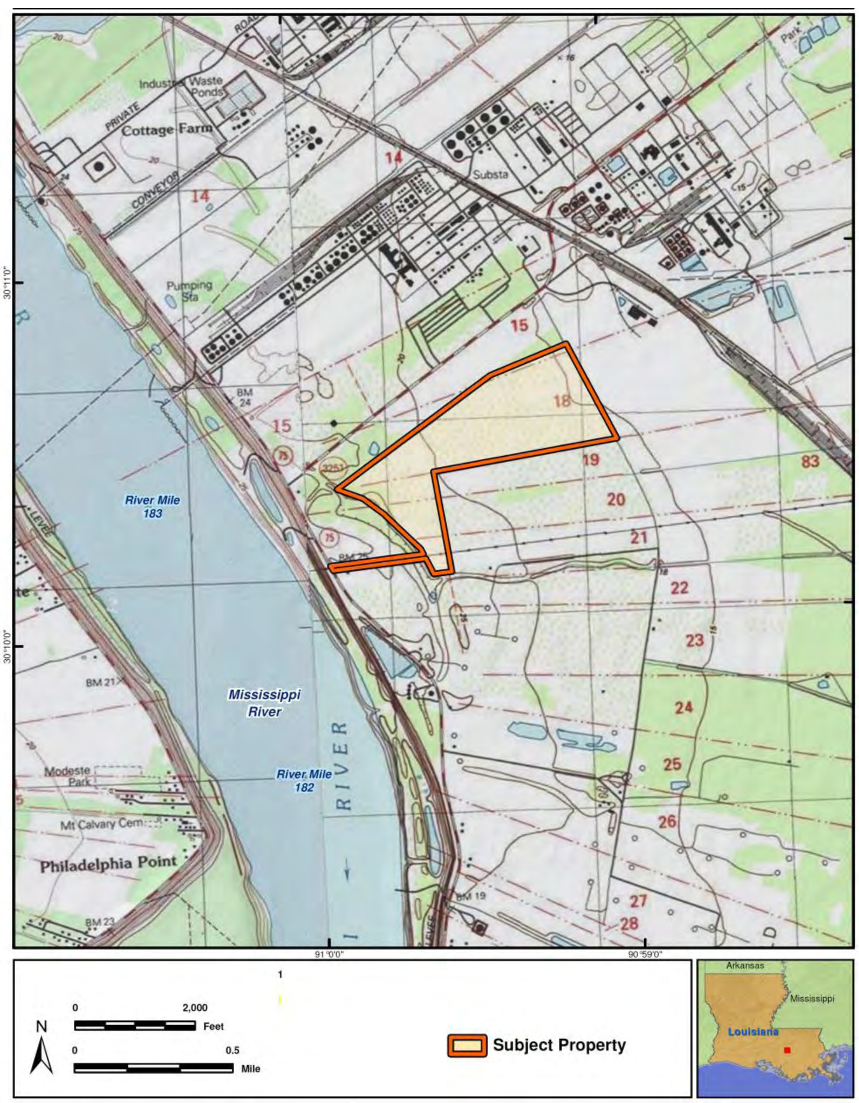
4.2.3 Environmental Liabilities

The environmental status of the Pine Point site and studies undertaken are discussed in Section 20.0.

4.3 GEISMAR SITE

The Geismar site is located in the industrial zone of Baton Rouge, approximately 27 km (17 miles) south of Prairieville, in the Ascension Parish of Louisiana, United States. Its coordinates are Latitude 30°10'36.7"N Longitude 90°59' 06.6"W. See Figure 4.6.

Figure 4.6
Location of Geismar Site



The site has an area of 60.6 ha (150 acres).

Avalon has purchased an exclusive option for US$150,000 to buy the land. The price of the land is approximately US$4 million.

Avalon, through its wholly owned subsidiary Avalon Rare Metals Processing Inc., entered into the option agreement on 31 July, 2012 with the vendor. Avalon has made all interim payments due under the option to maintain its rights and a final payment is due on 31 May, 2013 in order to complete the purchase.

4.3.1 Permits

The construction and operation of the Geismar site will require environmental permits issued by the Louisiana Department of Environmental Quality covering emissions to air, storage of solid waste (gypsum) and discharge of water. A "404 Permit" issued by the United States Army Corps of Engineers (USACE) is required for disturbance of wetlands at the site, should this be determined to be necessary.

Authorization from the Pontchartrain Levee District is required for subsurface work (such as drilling or excavation) conducted within 1,500 ft (457 m) of the Mississippi River levee and a Phase I Cultural Resources Survey under the Historical Preservation Act will be required prior to construction.

Permits are proceeding in a timely fashion with a contractor from Louisiana experienced in this work and all required permits are anticipated to be in place well in advance of construction. Further detail is provided in Section 20.0.

4.3.2 Environmental Liabilities

The environmental status of the Pine Point site and studies undertaken are discussed in Section 20.0 of this report.

5.0 ACCESSIBILITY, CLIMATE, LOCAL RESOURCES, INFRASTRUCTURE AND PHYSIOGRAPHY

Project facilities within the Northwest Territories comprise the Nechalacho mine and concentrator site and the hydrometallurgical processing site at Pine Point. See Figure 4.1, above.

The proposed site for the rare earth refinery is located at Geismar, Louisiana in the United States.

5.1 NECHALACHO SITE AREA

5.1.1 Location and Access

The Nechalacho site is located approximately 100 km east of Yellowknife and 5 km north of the Hearne Channel on the East Arm of Great Slave Lake. The property is within the Mackenzie Mining District of the Northwest Territories.

There is no road access into the Nechalacho site. It is accessible by barge during the summer months and year-round by aircraft. The site has been serviced using float planes and ski planes, with helicopter support over freeze-up and break-up.

A 30.4 m wide by 305 m long airstrip provides all-year fixed-wing service to the site, if necessary. In addition, ice roads on Great Slave Lake have been utilized to mobilize equipment to the site in winter as and when required.

5.1.2 Climate

Air temperature at the Nechalacho site recorded from June, 2008 to October, 2010 displayed typical seasonal fluctuations, with warm temperatures occurring from late May to August, with the coldest period occurring from December to February. The monthly average temperatures expected at site range from -26°C in January to 16°C in July. Monthly average temperatures rise above 0°C for significant periods of time in May and fall below 0°C for significant periods in October.

Average annual total precipitation at Thor Lake is approximately 275 mm. Rainfall predominates during May to October, and snowfall predominates during October to April.

Six snow courses were established throughout the Nechalacho site in March, 2009. Mean snow depths varied from 31.3 cm to 66.6 cm in the vicinity of Thor Lake. Forested areas that were generally less exposed to wind had a tendency to accumulate the thickest snowpacks.

Relative humidity is generally highest during the winter months, while summers are generally drier.

The dominant wind direction at the site is from the east-northeast during November to June. Wind directions had a tendency to be more dispersed from July to October; however, an east-northeast trend was still evident. The average hourly wind speed at 20 m above ground level is 4.54 m/s. Wind speeds at 20 m above ground are generally in the range of 2 to 6 m/s, with occasional wind speeds exceeding 10 m/s.

5.1.3 Topography and Vegetation

The Nechalacho landscape consists of a gently undulating relief that gradually decreases in elevation towards Great Slave Lake. Elevation ranges from 235 m along the shore of Thor Lake, to about 265 m on top of the highest bedrock knobs. Elevations progressively drop to approximately 160 m along the north shore of Great Slave Lake.

The landscape is dominated by subdued topography and fractured bedrock plains. Black spruce, jack pine, paper birch, and trembling aspen form discontinuous forested patches that are interspersed with exposed rock. Wetlands and peat plateaus commonly form around the margins of shallow lakes, as well as in wetter depressions and lowlands.

Existing dock facilities on the Hearne Channel on Great Slave Lake will be upgraded, as will the existing access road to the Nechalacho site, a distance of approximately 8 km. During the winter, heavy equipment and bulk materials can access the site using winter roads on the ice cover of Great Slave Lake.

Northern Transportation Avalon Ltd. (NTCL) operates a commercial barging service on Great Slave Lake and serves the communities of Lutsel K'e and Fort Reliance from the Port of Hay River. Hay River is a major transportation hub for the Northwest Territories and is one of Canada's largest inland ports. Hay River also has the only rail connection in the Northwest Territories.

Most lakes in the area do not freeze to the bottom and process water will be available year-round. Freeze-up commences in late October and break-up of the majority of the lakes in the area is generally complete by late May. Great Slave Lake freezes later and remains ice-free longer than the smaller lakes.

5.1.4 Local Resources and Infrastructure

Yellowknife (population 20,000) and Hay River (population 3,500) are two key transportation hubs in the Northwest Territories. Both communities have very good supporting infrastructure and are located in relatively close proximity to the Nechalacho site. The local economy is generally dependent upon government services, although both communities act as transit sites for mining and mineral exploration activities throughout the Northwest Territories and Nunavut.

The North Slave region has experienced significant economic growth over the last decade. A considerable number of residents work directly for the major diamond mines, or with companies supplying mining goods and services. Future development of new mines will further expand employment and business opportunities. Opportunities also exist to expand services in retail trade and tourism. The traditional economy also remains strong in the Northwest Territories through hunting, harvesting, and arts and crafts.

Development plans are also being considered to expand the region's hydro capacity. Other industries include tourism, transportation, manufacturing, commercial fishing, forestry, trapping and arts and crafts.

The Thor Lake project is situated in the Akaitcho Territory, an area that is subject to a comprehensive land claim negotiation involving communities belonging to the Yellowknives Dene, Lutsel K'e Dene and the Deninu Kue First Nations. The Yellowknives Dene consists of two communities, known as Ndilo and Dettah, each having over 250 residents. Ndilo is located on Latham Island in the northern part of the City of Yellowknife. Dettah, accessible by road, is located southeast of Yellowknife, across Yellowknife Bay. The Yellowknives Dene assert that the project lies within their traditional territory known as the Chief Drygeese Territory.

The community of Lutsel K'e is located on Christie Bay on the East Arm of Great Slave Lake and is accessible by air or boat. It has a population of over 250. Fort Resolution is located on the southeast coast of the main body of Great Slave Lake in Resolution Bay. The Deninu Kue First Nation is based in Fort Resolution and has a population of over 500. The community is serviced by road from Hay River and by air.

The town of Hay River, located on the south shore of Great Slave Lake where the Hay River enters the lake, extends south from the lake along the west bank of the river. The largest Aboriginal community in the Hay River area is the Katlodeeche First Nation, often referred to as the Hay River Reserve, which is located on the east bank of the Hay River across from the town. Hay River is accessible by air, rail and by using Highway 3 from Edmonton, Alberta.

Both the north and south sides of Great Slave Lake are occupied by two groups of Métis. The North Slave Métis Alliance (NSMA) is located in Yellowknife, while the Northwest Territory Métis Nation is located in Fort Smith (and represents the communities of Fort Smith Métis, Fort Resolution Métis and Hay River Métis).

Yellowknife uses diesel and hydroelectric facilities to generate its power and this is the closest source of power to Thor Lake. However, there is no transmission line and the generating capacity is limited.

Water is available at Thor Lake from any one of the surrounding lakes.

Reliable phone and e-mail communications currently exist at Nechalacho and will be upgraded to serve the larger crews for future construction and operations activities. Similar communication facilities will be installed for the hydrometallurgical plant facilities

5.2 PINE POINT SITE AREA

5.2.1 Location and Access

The Pine Point Site is located in the Northwest Territories about 12 km south of the south shore of Great Slave Lake, and is accessible year-round from Hay River via Highways 5 and 6 and an access road from Highway 6 to the site. All access roads are in good condition. There is also a network of local gravel roads, developed for past operations, that provides alternative access to the site.

Nearby communities include Hay River and Fort Resolution. Hay River is located approximately 90 km to the west along Highways 6 and 5 while Fort Resolution is located 65 km east along Highway 6.

As part of the Pine Point Mine development, the Mackenzie Northern Railway was completed by 1964 and resulted in Hay River becoming the northernmost point in Canada which is connected to the continental railway system.

Hay River is served by Buffalo Airways which provides scheduled flights to Yellowknife, as well as charter services and a courier service throughout the north, First Air which provides scheduled services to Yellowknife and Northwestern Air which provides scheduled flights to Edmonton. Other companies offering charter services in Hay River include Landa Air, Carter Air Services (fixed-wing aircraft), Denendeh Helicopters and Remote Helicopters.

5.2.2 Climate

Thirty-year climate normals for temperature and precipitation were calculated for the period between June, 1980 and May, 2010 from daily data at Hay River Airport published by Environment Canada. The mean annual temperature at Hay River is -2.6°C. July is the warmest month on average (16.1°C) while January is the coldest (-21.7°C). Temperatures are typically below zero from November to March and above zero between May and September.

Mean annual precipitation at Hay River is 335 mm, of which 65% (217 mm annually, on average) falls as rain and 35% as snow (137 cm annually, on average). Precipitation is typically higher during the summer, with 53% of the annual total occurring between June and September. Based on the 30-year period, rainfall has occurred in every month but snowfall has not been recorded in July and August.

Mean annual and monthly wind speeds and predominant wind direction were determined from hourly wind observations over the five-year period between June, 2006 and May, 2010. Winds at Hay River are most frequent from the east-northeast, south-southeast and west-northwest. East winds occur predominantly during the summer months and southeast and northwest winds occur more typically during the winter. Mean annual wind speed at Hay

River is 3.2 m/s. Higher wind speeds tend to occur between March and May and between September and November.

5.2.3 Topography and Vegetation

The topography of the Pine Point site is flat to gently sloping, generally to the north. Topography around the proposed Pine Point Site typically ranges between 190 and 220 masl and decreases to approximately 160 masl along the south shore of Great Slave Lake.

The area supports extensive wetlands characterized as tree, shrub, and sedge-dominated fens. On drier, coarse-textured soils, forests of jack pine and mixed jack pine/trembling aspen develop, while other upland areas are characterized by pure or mixed forests of black and white spruce, trembling aspen, paper birch, and balsam poplar.

5.2.4 Local Resources and Infrastructure

Fresh water will be supplied to Pine Point site facilities from an historic open pit designated as J-44 and located 2.2 km from the hydrometallurgical plant. Fresh water will be treated in a potable water treatment plant located at the Pine Point site.

A hydroelectric generating facility is located on the Taltson River approximately 200 km south of Fort Smith. The power line from the Taltson facility passes through Pine Point and the plant will be supplied with power from an existing substation.

The site will be interconnected to the Internet and public switched telephone network (PSTN) via Northwestel's terrestrial microwave network.

5.3 GEISMAR, LOUISIANA SITE AREA

5.3.1 Location and Access

Geismar is located in Louisiana, United States, in an unincorporated area in Ascension Parish, which is part of the Baton Rouge Metropolitan Statistical Area. Geismar is on State Highway 75 approximately 27 km south of Prairieville. The Mississippi River runs through the town.

Geismar is accessible year-round by road, air, river and railroad. By road, Geismar is located approximately 40 minutes from Baton Rouge Airport and approximately one hour from New Orleans International Airport. Avalon reports that an existing river dock, owned by Kinder Morgan, in the vicinity of the Geismar site will be available for use by Avalon.

5.3.2 Climate

The Geismar area has a humid subtropical climate, with mild winters, hot and humid summers, moderate to heavy rainfall, and the possibility of damaging winds and tornadoes yearlong. The proximity to the coastline exposes the metropolitan region to hurricanes. Snow is very rare, although it occurred for three consecutive winters in 2008-2010.

Geismar experiences an average winter temperature of 10.6°C, and an average summer temperature of 26.7 °C.

Rainfall tends to increase in spring and peak in July. Total annual precipitation averaged 1,539 mm in the period 1981-2010.

The dominant wind direction at the site is from the east-southeast between September and March. From April to August, wind directions are more dispersed but are primarily from the north.

5.3.3 Topography and Vegetation

Louisiana lies wholly within the Gulf Coastal Plain. Alluvial lands, chiefly of the Red and Mississippi rivers, occupy the north-central third of the state. East and west of this alluvial plain are the upland districts, characterized by rolling hills sloping gently toward the coast. The coastal-delta section, in the southernmost portion of the state, consists of the Mississippi Delta and the coastal lowlands. The highest elevation in the state is Driskill Mountain at 163 masl, Baton Rouge lies at 16 masl and Geismar at 7 masl. The lowest elevation, at 2 m below sea level, is in New Orleans. The Geismar area is vulnerable to hurricanes and tropical systems due to its low elevation and its proximity to the coast of southeast Louisiana.

The rare earth refinery site is located within the Mississippi River Delta Plain Region of the Coastal Plain Province of North America. The major ecological regions in Coastal Louisiana are palustrine forests, and coastal marshes. The physiography of the Geismar site is characterized by terrace uplands, natural levees of the Mississippi River, and back swamps which formed in depressional areas between natural levees and terrace uplands. Major waterbodies near the area include the Mississippi River, the New River and Bayou Conway.

The wide variety of flora and fauna in the area are typical of humid subtropical regions nourished by abundant rainfall and normally mild winters. The back swamp and hardwood forest is an active ecosystem providing cover and water for birds, mammals, reptiles and amphibians.

The Geismar site is located on an old sugar plantation. The terrain is wooded with mature cypress trees, and is elevated above the swamp area.

5.3.4 Local Resources and Infrastructure

Power will be provided by the local power company, Entergy Louisiana, LLC (Entergy).

The adjacent chlor-alkali plant, located approximately 1.6 km to the northeast will provide both fresh and fire water, and key reagents (sodium hydroxide and hydrochloric acid). Waste brine from Avalon's operation will be returned to the chlor-alkali plant for disposal.

Potable water will be supplied on site, using water wells and a potable water treatment plant.

6.0 HISTORY

The history of the project site has been described in the Technical Reports completed by Scott Wilson RPA in 2010 (Scott Wilson RPA, 2010) and RPA in 2011 (RPA, 2011) and the information is summarized here.

The Thor Lake site area was first mapped by J.F. Henderson and A.W. Joliffe of the Geological Survey of Canada (GSC) in 1937 and 1938. The first staking activity at Thor Lake dates from July, 1970 when the Odin 1-4 claims were staked for uranium by K.D. Hannigan. Shortly after, the Odin claims were optioned to Giant Yellowknife Mines Ltd. and, in 1970, were acquired by Bluemount Minerals Ltd.

In 1971, the GSC outlined a radioactive anomaly over the Thor Lake area. A. Davidson of the GSC initiated mapping of the Blatchford Lake intrusive complex. It subsequently became clear that this radiometric anomaly was largely due to elevated thorium levels in the T-Zone within the Thor Lake project area.

In 1976, Highwood Resources Ltd. (Highwood) discovered niobium and tantalum on the Thor Lake property. From 1976 to 1979, exploration programs included drilling, geological mapping, sampling and trenching on the Lake, Fluorite, R-, S- and T-zones. Drilling of 22 holes was completed, seven on the Lake Zone. This work resulted in the discovery of significant concentrations of niobium, tantalum, yttrium and REEs. Recognizing a large potential resource at Thor Lake, Placer Development Ltd. (Placer) optioned the property from Highwood in March, 1980 to further investigate the tantalum and related mineralization. The exploration included drilling 21 holes in 1980-1981. Preliminary metallurgical scoping work was also conducted, but Placer relinquished its option in April, 1982.

From 1983 to 1985, the majority of the work on the property was concentrated on the T-Zone and included geochemical surveys, berylometer surveys, surface mapping, significant drilling, surface and underground bulk sampling, metallurgical testing and a detailed evaluation of the property by Unocal Canada. However, seven holes were also drilled in the Lake Zone (later renamed the Nechalacho deposit) to test for high grade tantalum-niobium mineralization. In August, 1986, the property was joint-ventured with Hecla Mining Company of Canada Ltd. (Hecla). However, in 1990, Hecla withdrew from the project. In 1990, control of Highwood passed to Conwest Exploration Company Ltd. (Conwest) and the Thor Lake project remained dormant until 1996. Through subsequent ownership changes, Highwood became controlled by Royal Oak Mines Ltd. (Royal Oak).

Royal Oak's subsequent bankruptcy in 1999 resulted in the acquisition of the control block of Highwood shares by Dynatec Corporation (Dynatec). In 2000, Highwood initiated metallurgical, marketing and environmental reviews by Dynatec. Through 2001 to 2004, Navigator Exploration Corp. optioned the property and conducted additional metallurgical research on the possibility of producing a tantalum concentrate, but dropped the property when tantalum prices fell. Meanwhile, Beta Minerals Inc. (Beta) acquired Highwood's interest in the property in November, 2002 under a plan of arrangement with Dynatec. No work was conducted by Beta and, in May, 2005, Avalon purchased from Beta a 100% interest and full title, subject to royalties, in the Thor Lake property.

In 2005, Avalon conducted extensive re-sampling of archived Lake Zone (Nechalacho deposit) drill core to further assess the yttrium and HREE resources on the property. In 2006, Wardrop was retained to conduct a preliminary economic evaluation of the Thor Lake deposits (Wardrop, 2006). In 2007 and 2008, Avalon commenced further drilling of the Lake Zone. This led to a second technical report on the property by Wardrop (Wardrop, 2009). In 2010, Scott Wilson RPA completed a prefeasibility study on the project (Scott Wilson RPA, 2010). Avalon has continued to drill the Nechalacho deposit in campaigns in 2010 to 2012 and, in 2011 published a technical report on the deposit (Avalon, 2011).

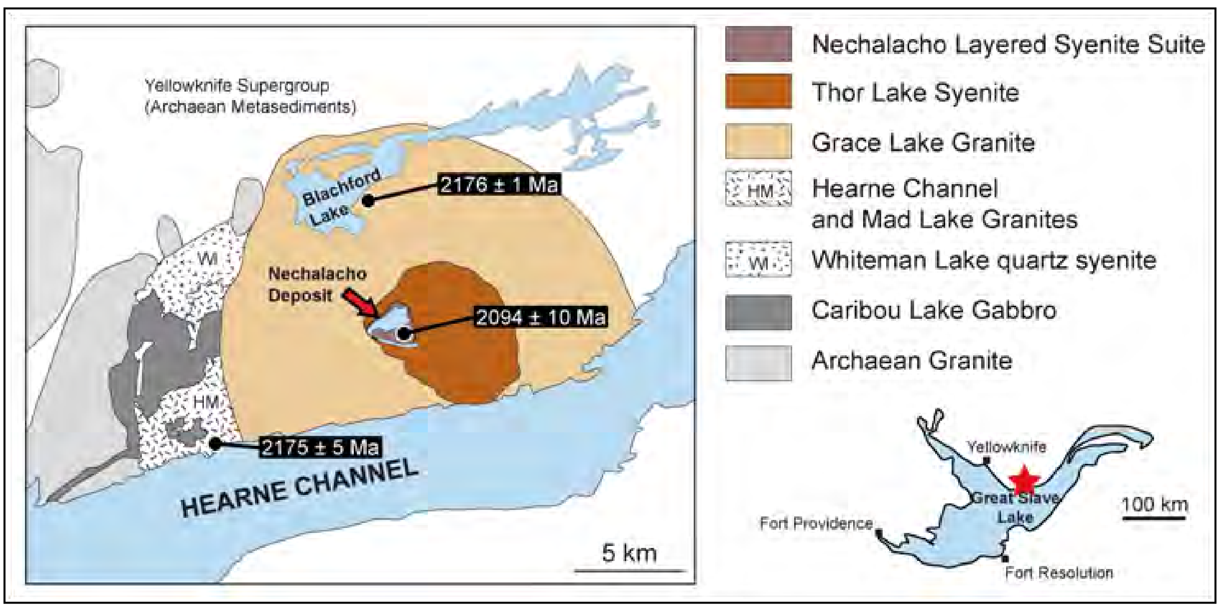
7.0 GEOLOGICAL SETTING AND MINERALIZATION

7.1 REGIONAL GEOLOGY

The following section is summarized from Trueman et al. (1988), LeCouteur (2002), Pedersen et al. (2007), and supplemented with observations made by Avalon geologists during the drill programs of 2007 to 2010. In addition, research at McGill University (Emma Sheard, M.Sc. thesis, 2011; personal communications: Volker Muller, Ph.D. candidate; Kent McWilliams, M.Sc. candidate) and University of Windsor (personal communications: Yonggang Feng, Ph.D. candidate) has advanced the understanding of the local geology of the Nechalacho deposit.

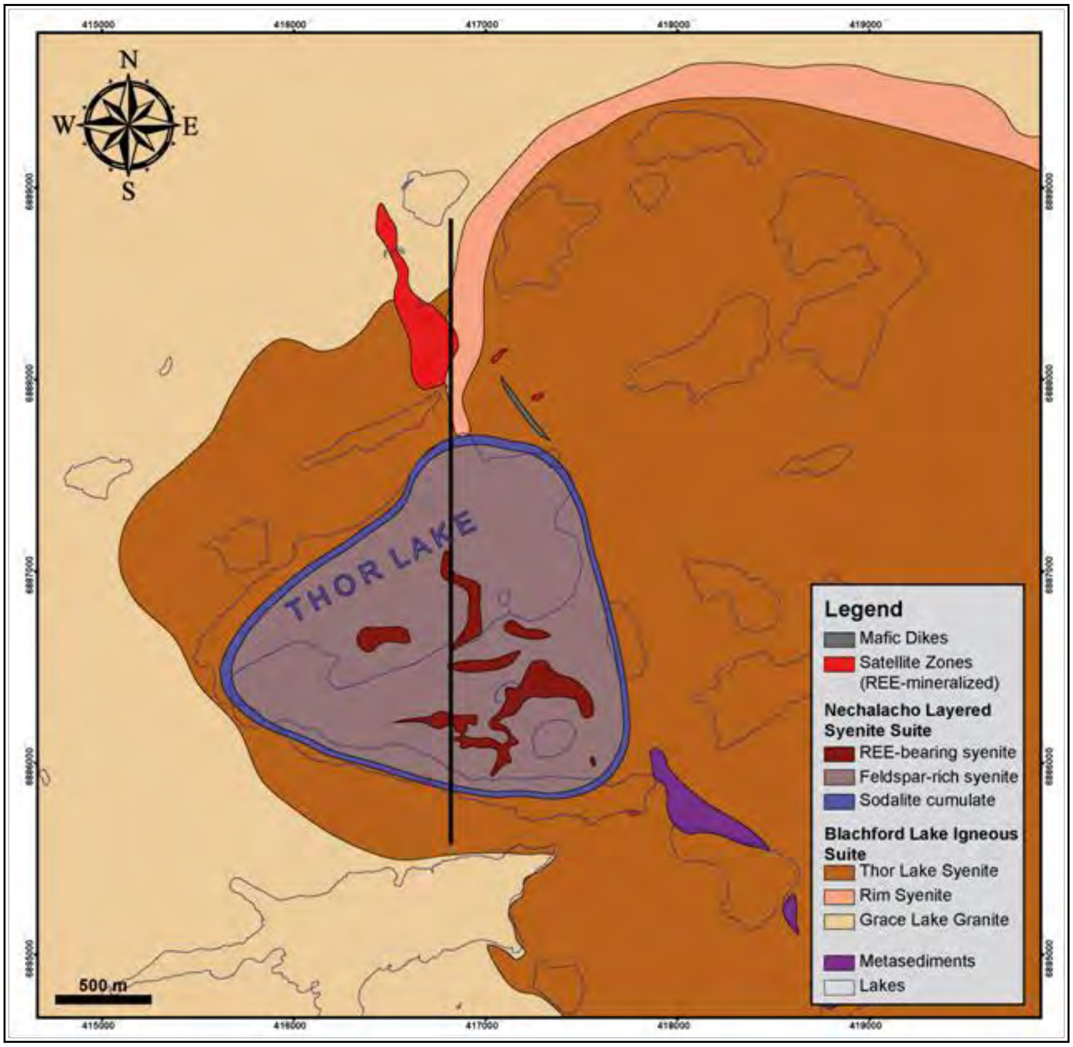
A map showing the regional geology is shown in Figure 7.1 and a Nechalacho deposit geological map is provided in Figure 7.2.

Figure 7.1
Regional Geology, Blachford Lake Complex



Modified after Davidson (1981), U-Pb zircon dates from Bowring et al. (1984) and Sinclair et al. (1994).

Figure 7.2
Nechalacho Deposit Geological Map



Volker Möller, 2012, McGill University, unpublished map.

The various Thor Lake mineral deposits occur within the Aphebian Blatchford Lake Complex (BLC), which includes Archaean Yellowknife Supergroup metasedimentary rocks of the southern Slave geologic province. The BLC has an alkaline character and intrusive phases vary successively from early pyroxenite and gabbro through to leuco-ferrodiorite, quartz syenite and granite, to peralkaline granite and a late syenite (Davidson, 1982). There appear to be successive intrusive centres with an earlier western centre truncated by a larger centre that consists of the Grace Lake Granite and the Thor Lake Syenite. Nepheline syenite underlies the Thor Lake Syenite of the Nechalacho deposit. This unit was recognized in drilling by Avalon during 2007 to 2010. Outcrops of the nepheline syenite within the area of the Nechalacho deposit display strong hydrothermal alteration and, consequently, the unit was not originally mapped as distinct from the Thor Lake Syenite.

Davidson (1978) subdivided the BLC into six texturally and compositionally distinct plutonic units known as the Caribou Lake Gabbro, the Whiteman Lake Quartz Syenite, the Hearne Channel Granite, the Mad Lake Granite, the Grace Lake Granite and the Thor Lake Syenite. Based on exposed cross-cutting relationships of dykes and the main contacts, Davidson recognized a sequence of five intrusive events. The rocks of the last intrusive event, being compositionally and spatially distinct, are sub-divided by Davidson into the Grace Lake Granite and the Thor Lake Syenite. Although these two units are defined as separate entities, there are no known cross-cutting relationships and they are in fact believed by Avalon's geologists to be time-equivalent. It is now believed that the only real differences between the Thor Lake Syenite and Grace Lake Granite are their varying quartz contents and the degree of silica saturation. In fact, the two sub-units likely reflect a single early intrusive magma pulse which preceded a second related pulse of nepheline sodalite-bearing peralkaline magma. Until 2010, the hydrothermally altered apical portion of this nepheline syenite was believed to be exposed only under and between Thor and Long Lakes. Previously, it was described as altered Thor Lake Syenite. Now, the nepheline syenite unit has been encountered in drilling north of Thor Lake, and under Cressy Lake, thus establishing that it is more extensive than originally believed. Drilling of the Nechalacho deposit has also shown that the same nepheline-sodalite peralkaline syenite that underlies the Thor Lake Syenite is, in fact, a distinct intrusion. The nepheline syenite is now informally referred to as the "Nechalacho Nepheline Syenite".

Recent age-dating of the BLC supports the view that all of the intrusions are related since the main eastern and western intrusive centres have comparable ages. The Hearne Channel Granite has been dated at 2,175 + 5 M years while the Whiteman Lake Syenite is dated at 2,185 + 5 M years (Bowring et al., 1984) and the Grace Lake Granite is dated at 2,176 + 1.3 M years (Sinclair and Richardson, 1994).

Henderson (1985) reports that small dioritic plugs, which have been assigned to the Compton Lake Intrusive Suite, cross-cut the Grace Lake Granite. As well, diabase dykes of the 1,200 M year old Mackenzie swarm and the 2,000 M year old Hearne dyke swarm cut most of the members of the BLC.

Most of the Thor Lake Property is underlain by the Thor Lake Syenite and Nechalacho Nepheline Syenite within the central part of the Grace Lake Granite. The T-Zone deposits cross-cut both rock types whereas the Nechalacho deposit is confined to the area of the hosted in the underlying Nechalacho Nepheline Syenite.

The Grace Lake Granite is a coarse-grained, massive, equigranular, riebeckite-perthite granite with about 25% interstitial quartz. Near the contact between the Grace Lake Granite with the Thor Lake Syenite the two units are texturally similar and the contact appears to be gradational over a few metres. Because of their textural similarity and gradational contact relations, Davidson suggested that both rock types are derived from the same magma.

The Thor Lake Syenite is completely enclosed by the Grace Lake Granite. The most distinctive sub-unit is a fayalite-pyroxene mafic syenite which locally has a steep dip and is located close to the margin of the main amphibole syenite and the Grace Lake Granite. It forms a distinct semi-circular ridge, locally termed the rim syenite, which can be traced for a distance of about 8 km and has the appearance of a ring dyke, most prominent on the east side of the Thor Lake body. The rim syenite is clearly identifiable on the airborne magnetic map.

The nepheline-sodalite syenite hosting the Nechalacho deposit, here termed the Nechalacho Nepheline Syenite, has the following key distinctive features which contrast it to the Thor Lake Syenite and Grace Lake Granite:

- It has a distinct chemical composition with under-saturation in quartz as shown by the presence of nepheline and sodalite as primary rock-forming minerals.

- It displays cumulate layering.

- It contains agpaitic zircon-silicates (including trace eudialyte and pseudomorphs after eudyalite).

- It is the host to the Nechalacho zirconium-niobium-tantalum-rare earth element mineralization.

The Nechalacho Nepheline Syenite, both mineralized and unmineralized, is only exposed at surface in a small portion of the Thor Lake Syenite between Long and Thor Lakes. It is believed that the Nechalacho Nepheline Syenite dips underneath the Thor Lake syenite in all directions. This is supported by drilling north of Thor Lake, within and close to Cressy Lake and also by drilling south of Long Lake. Also, the Nechalacho mineralization occurs in the top, or apex, of the Nechalacho Nepheline Syenite.

The Nechalacho Nepheline Syenite consists of a layered series of rocks with increasing peralkaline characteristics at depth. A consistent, downward progression is noted from the hanging wall sodalite cumulates, through the coarse-grained or pegmatitic nepheline aegirine syenites (which are locally enriched in zircon-silicates), to foyaitic syenite within a broad zone of altered “pseudomorphs-after-eudialyte” cumulates (referred to as the Basal Zone). This upper sequence is also intensely altered by various sodium and iron hydrothermal fluids. Pre-existing zircon-silicates are completely replaced by zircon, allanite, bastnaesite, fergusonite and other minerals. Beneath the Basal Zone cumulates, mineralization decreases rapidly, but alteration decreases more gradually, with relict primary mineralogy and textures increasingly preserved. Aegirine- and nepheline-bearing syenites and foyaitic syenites progress downward to sodalite foyaites and naujaite. Drilling has not extended beyond this sodalite lithology to date. Minerals related to agpaitic magmatism identified from this lower unaltered sequence include eudialyte, catapleite, analcime and, possibly, mosandrite.

7.1.1 Regional Structures

The BLC was emplaced in a setting that has been postulated to be initially extensional with a triple junction rift consisting of structures oriented at azimuths of 060° to 070°, 040°, and 330°. These structures are readily seen on large-scale topographic and magnetic maps but their presence can be detected at the outcrop scale and within the distribution of the structurally influenced mineralized zones (R-, S-, and T-zones). The 060° to 070° and the 040° structures represent orientations of the failed "East Arm Aulacogen" now occupied by the Hearne Channel in the vicinity of the Nechalacho deposit (Hoffman, 1980). The presence of younger, Aphebian-age, metasedimentary and metavolcanic rocks of the Great Slave Supergroup to the south of Hearne Channel demonstrates that the two structures represent extensional fractures bordering a basin that was subsequently filled with sedimentary and volcanic rocks.

Later phases of tectonic movement were principally compressional and relate to closure of the rift, over-thrusting, nappe emplacement and recumbent folding in the East Arm, and collision of the Great Bear Magmatic terrain. Younger (Proterozoic) metasedimentary and metavolcanic rocks south of Thor Lake were deposited in the failed arm of the triple junction rift, and their position now represents the location of this feature.

7.1.2 Diabase Dykes

Two ages of diabase dyke swarms are present, known as the Mackenzie and the Hearne. The Mackenzie dykes are dated at 1.27 B years, have a north-northeast strike orientation and are part of the largest dyke swarm on earth. Although there are Mackenzie dykes in the general vicinity of Thor Lake none are known to cut the Nechalacho deposit.

The Hearne dykes are dated at 1.902 B years and trend east-northeast. Diabase dykes locally cut the Nechalacho deposit and these are interpreted as Hearne-age dykes. In previous resources models, the dykes were not well constrained as drill intercepts were infrequent due to the near vertical nature of the dykes and the mainly vertical drilling. However, specific drilling has recently focused on improving dyke definition and, between vertical drilling that has gone down the interior of dykes and angle holes specifically targeting the dykes, has resulted in greatly improved knowledge of their geometry. The steep nature of these intrusives is clear with steep dips to both south and north. Diabase dyke emplacement appears to have been quite passive, exhibiting undulating and anastomosing lateral morphology. Thicknesses range from less than 1 m up to approximately 15 m. At present, there is no evidence that the Hearne dykes are emplaced in fault structures that have experienced significant displacement.

The Hearne dykes appear to be somewhat clustered within the Nechalacho deposit, with a band of dykes through the centre of the deposit, but only intersecting the mine plan areas in the northwest of the mineral reserve zone.

Two principal diabase dykes occur in the West Long Lake mine plan, with several subsidiary or off-shoot dykes occurring erratically. These subsidiary dykes often appear to be anastomosing, and can begin or terminate without warning, both laterally and vertically. It is possible that at least some dykes will be blind, in that drilling may have overshot the apical portion of a deeper dyke.

7.1.3 Structure and Tectonics

It is interpreted that the Nechalacho deposit is a virtually undeformed deposit where most of the features observed were generated by magmatic and hydrothermal processes. In the least altered portions of the deposit, delicate primary textures are well preserved and no penetrative tectonic or metamorphic fabric is observed.

The distribution of most of the rock units and the mineralization generally follow a sub-horizontal pattern that can be traced for several hundreds of metres. However, the sub-horizontal pattern is interpreted to be locally disturbed by changes in elevation of up to 40 m. These changes in elevation may occur erratically or along linear trends. As noted above, the deposit is also cut by late diabase dykes, which are part of the east-northeast-trending Hearne dyke swarm. It is believed by Avalon's geologists that the changes in elevation of the mineralization are largely due to events during the magmatic stages of the nepheline syenite formation.

7.1.4 Late Tectonic Faults

Faults are present, are generally less than a metre in thickness and are characterized by fault gouge, breccia, frequent red hematite, and variable amounts of carbonate-quartz veining. However, it is believed that these are minor local features relating to late release of pressure in the solidifying magma chamber. There is little evidence of significant displacement along late faults.

7.2 MINERALIZATION

Mineralization in the Nechalacho deposit includes LREE found principally in allanite, monazite, bastnaesite and synchysite; yttrium, HREE, niobium and tantalum found in fergusonite; niobium in ferrocolumbite; HREE, niobium, tantalum and zirconium in zircon; and gallium in biotite, chlorite and feldspar in albitized feldspathic rocks.

There are some trends in chemical and mineralogical characteristics in the Nechalacho deposit that have both geological and metallurgical significance. These trends are most apparent in vertical zoning of the deposit as follows:

- The proportion of HREE to LREE gradually increases from top to bottom of the deposit. The proportion of HREE within TREE at the top of the deposit may be 6% and at the bottom of the deposit it may exceed 30%.

- The amount of certain minerals increases towards the bottom – specifically as HREE increases, the amount of fergusonite increases.

- Bastnaesite shows a bipolar distribution, where it correlates with HREE in being present at the lowest parts of the deposit, but also occurs locally at surface in exceptionally high grades.

- Mineralogical studies (Sheard, M.Sc. thesis, McGill University; Muller, unpublished McGill thesis studies) have shown that the textures and chemistry of zircon minerals, in general, increases in complexity from top to bottom.

- Some five types of zircon textures have been identified, which show imperfect vertical zoning, with "cumulate" textures being more common towards the bottom of the Basal Zone and "wavy banded zircon" textures more common in the upper parts of the Basal Zone and in Upper Zones.

- Liberation data from QEMSCAN® studies (SGS Mineral Services, SGS-M, 2012, unpublished data) have shown that liberation decreases downwards in the deposit – that is at specific grain sizes, the proportion of free and liberated zircon decreases downwards.

These trends are discussed in more detail below.

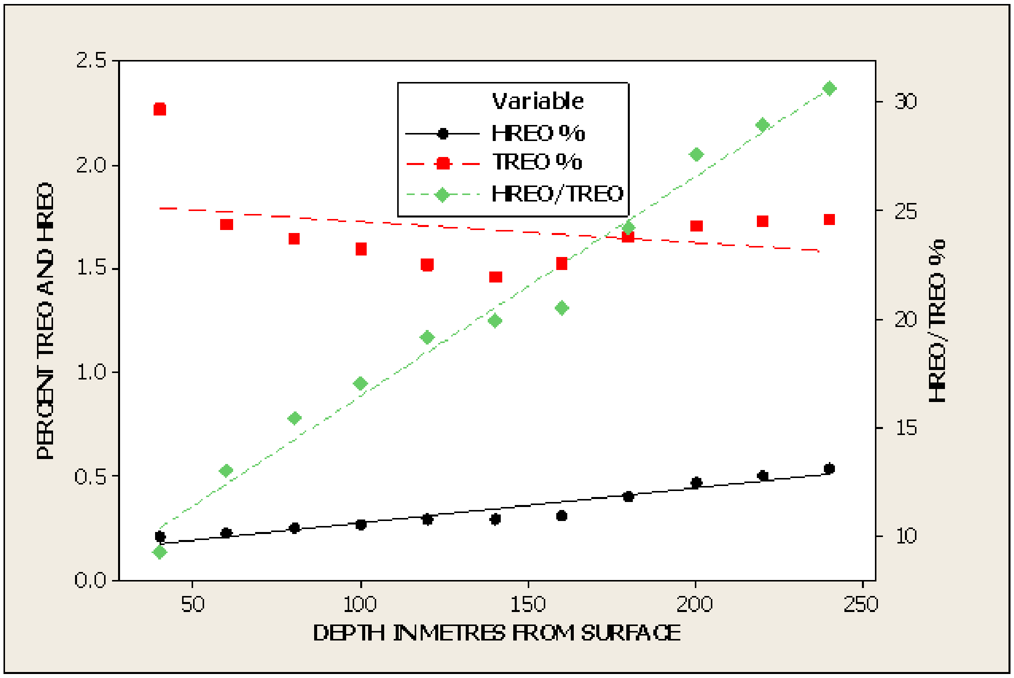
7.2.1 Trends of REE with Depth

The Nechalacho deposit alteration system varies between 80 m (L08-65) and 190 m (L08 127) in vertical thickness, with the alteration typically starting at the surface. The complete alteration system is enriched to varying degrees in REE, zircon, niobium and tantalum, relative to unaltered syenite, with average values over the whole alteration package of approximately 0.75% to 1.0% TREO. Within this alteration envelope, there are sub-horizontal zones of increased alteration accompanied by increased REE enrichment alternating with less enriched REE zones. Within the more intensely altered zones, the original textures and mineralogy of the host rock are no longer apparent.

These zones of increased alteration, which can vary in thickness from a few metres to tens of metres, can frequently contain TREO grades in the range of 2% and higher. The lowermost band, referred to here as the Basal Zone, contains the highest proportion of HREO. Overall, the HREO proportion of the TREO within the 80 m to 190 m thick alteration system is typically between 7% and 15%. However within the Basal Zone this proportion can exceed 30%.

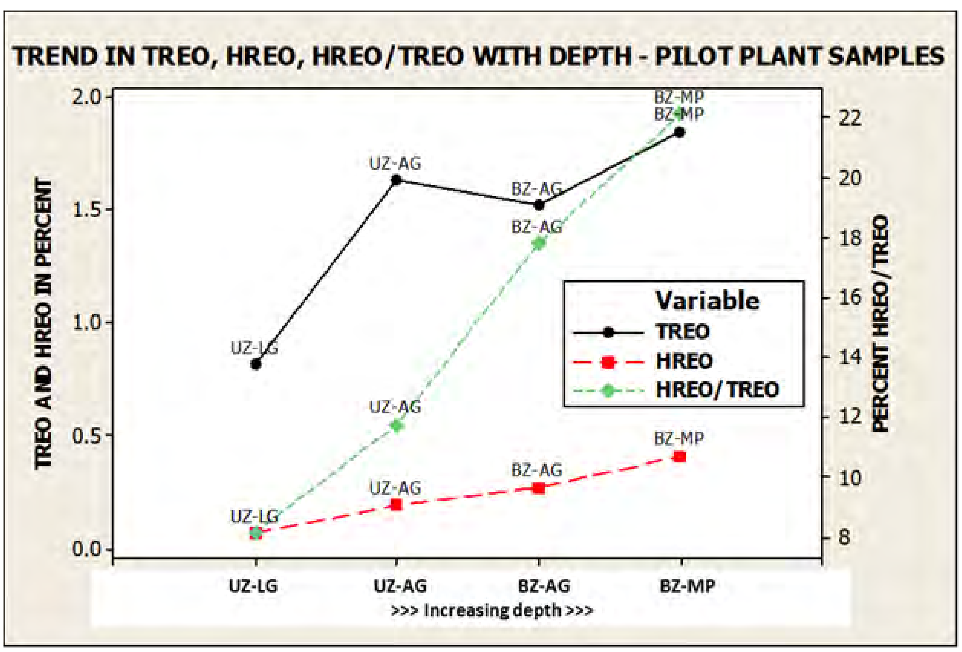
In general, the HREE relative to the LREE show a distinct vertical zonation with increasing HREE to depth. This is not always consistent in individual drill holes, but when averaged over a number of holes, the pattern becomes clear as illustrated in Figure 7.3 and Figure 7.4.

Figure 7.3
TREO, HREO, HREO/TREO Against Depth in 20-m Depth Slices



From resource model.

Figure 7.4
Trends in TREO, HREO, HREO/TREO Against Depth



From resource model.

This pattern of increasing HREE to depth is clearly important to the economics of any potential mine, as the HREE have higher average prices than the LREE.

Although gallium is anomalous in the intrusive relative to typical granites, it is not contained in the same minerals as the REE, and is in fact mainly in silicates such as chlorite, biotite and feldspar. As a result, the gallium actually varies inversely to the REE and is lower in REE- and zircon-enriched rocks than in the less mineralized rocks.

7.2.2 Mineral Textures in the Mineralization

The mineralogy of the Nechalacho deposit has been studied by SGS-M, XPS and McGill University utilizing optical microscopes, scanning electron microprobe analysis and QEMSCAN® equipment.

The formulae of the key ore minerals in the Nechalacho deposit are summarized in Table 7.1.

Table 7.1
Ore Minerals Present in Nechalacho Deposit

Light Rare Earth-bearing Minerals		
Monazite-(Ce)	$(REE,Th)PO_4$	
Allanite-(Ce)	$Ca(REE,Ca)Al_2(Fe^{2+},Fe^{3+})(SiO_4)(SiO_2O_7)O(OH)$	
Bastnaesite-(Ce)	$REE(CO_3)F$	
Synchisite/Parisite-(Ce)	$Ca(REE)_2(CO_3)_3F_2$	
Heavy Rare Earth-bearing Minerals		
Zircon	$Zr(HREE)SiO_4$	Also contains Nb, Ta
Fergusonite-(Y)	$Y(HREE)(Nb,Ta)O_4$	
Other Potential Ore Minerals		
Columbite	$(Nb,Ta)_2O_6$	

It has become apparent that there are a number of mineral texture types within the Nechalacho deposit. These texture types are manifest both at the macroscopic scale when logging drill core, and at the microscopic scale through optical petrographic microscope, QEMSCAN® and electron microprobe. Initially, the various textures were not logged as individual units due to the rapid changes in drill core. However, it has subsequently been recognized that these textures may be critical in understanding metallurgical performance in mineral processing. In addition, it is possible that these textures may play a role in recoveries during subsequent hydrometallurgical processing of mineral concentrates.

When logging the drill core in the field, zircon textures are clearly visible, but other REE-bearing minerals are not often apparent due to the fine grained nature. The most commonly identified mineral in core is bastnaesite, due to its usual brick red colour in the Nechalacho deposit, though QEMSCAN® indicates that it is frequently not the dominant LREE-bearing mineral. As a result, logging in the field tends to focus on the textures of zircon.

Zircon ore textures currently logged since the winter 2012 drill program include (1) fine-grained, disseminated pseudomorphs; (2) medium to coarse-grained pseudomorphs; (3) poikilitic pseudomorphs; (4) cumulate textures; and (5) wavy laminations.

These textures reflect the magmatic and hydrothermal crystallization history of the mineralization. Thus, cumulate textures suggest a disturbance of the cotectic that caused the mono-mineralic precipitation of magmatic zircon-osilicates, interpreted to be primarily eudialyte. These crystals were subsequently concentrated in cumulate horizons by processes such as crystal settling.

Poikilitic intergrowths suggest in-situ crystallization of possible eudialyte, where eudialyte crystallized at the expense of feldspar or feldspathoid.

Pseudomorph textures suggest co-magmatic crystallization of rock-forming minerals and zircono-silicates.

The significance and origin of wavy laminations is still in question. Proposed theories include the precipitation of magmatic zircon from a miaskitic magma and the hydrothermal mobilization of pre-existing eudialyte.

Basal zones logged during the 2012 winter/spring field season display high variability of ore textures, commonly on the sub-metre scale. A general trend of zircono-silicate distribution could be summarized as fine-grained pseudomorphs and wavy laminations being most common near the top of the basal zone and poikilitic intergrowths and cumulate textures occurring primarily in the heart of the basal zone. The bottom contact of the basal zone is commonly abrupt and is underlain by unmineralized alkaline syenites. However, gradational lower contacts occur locally, characterized by intermittent bands of fine- to coarse-grained pseudomorphs interlayered with weakly or unmineralized syenite.

7.2.3 Uranium and Thorium Distribution

Uranium (U_3O_8) averages 28 ppm and thorium (ThO_2) averages 139 ppm in the current Basal Zone resource estimate for measured plus indicated material above the cut-off grade of US$320/t Net Metal Return (NMR). Thorium increases slightly with REE grade and averages 179 ppm in material estimated above US$1,000/t NMR, while uranium reaches 33 ppm.

Uranium content varies throughout the Nechalacho deposit with no significant correlation with REEs (r = 0.25). Higher-grade zones occur in the Basal and Upper Zones and are most likely lens-shaped but may be modified by post-mineralization alteration. Cross-sections show little connection between high-uranium assays. Thorium correlates modestly with phosphate (r = 0.85) and the light rare earths (r = 0.71) with lens-shaped distribution in both the Upper and Basal Zones.

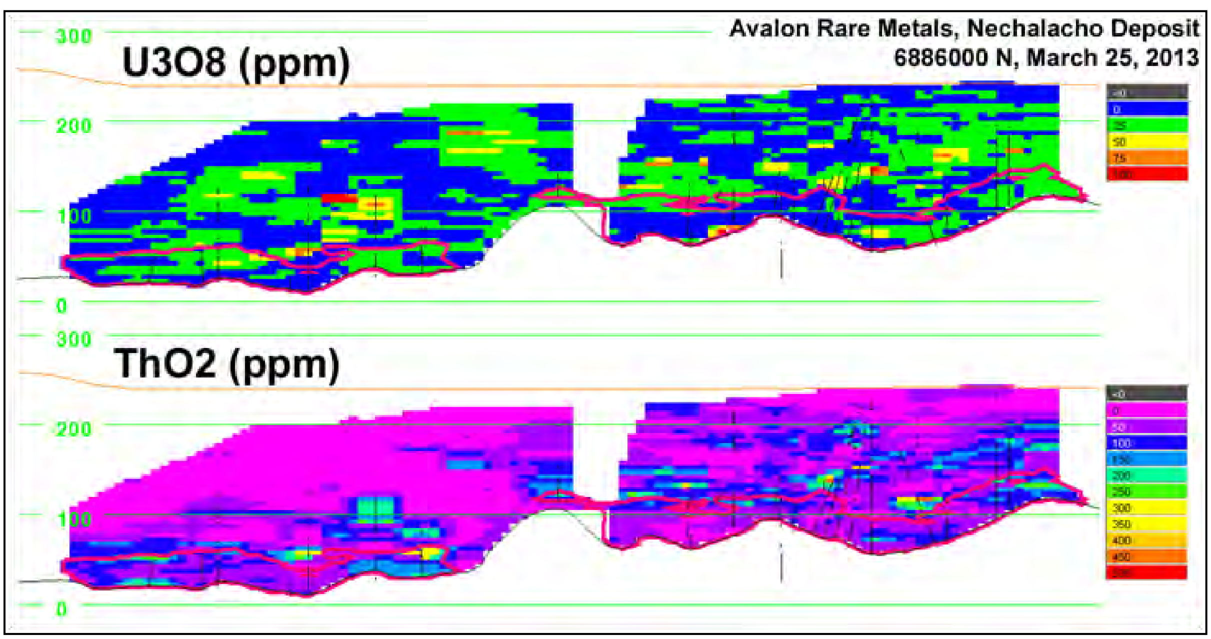
The uranium and thorium contents at various cut-off grades are given in Table 7.2.

Table 7.2
Basal Zone Thorium and Uranium Estimates at Various NMR Cut-off Grades

Basal Zone	NMR Cut-off (US$/t)	ThO_2 (ppm)	U_3O_8 (ppm)
Measured			
	≥320	105	26
	≥600	109	27
	≥800	112	27
	≥1000	119	28
Indicated			
	≥320	146	28
	≥600	158	29
	≥800	174	31
	≥1000	200	35
Measured + Indicated			
	≥320	139	28
	≥600	149	29
	≥800	161	30
	≥1000	179	33

Uranium and thorium distribution is illustrated in Figure 7.5.

Figure 7.5
Nechalacho Deposit East-West Cross-section at 688600N Showing Uranium (U_3O_8) and Thorium (ThO_2) Distribution



7.2.4 Mineralogical Trends with Depth

The abundance of the rare earth minerals can be summarized using the composites processed in the 40 t pilot plant testwork in 2012 as in Table 7.3. The samples in the table are as follows:

The UZ-LG is an Upper Zone composite of low grade – about 1% TREO – that illustrates the mineral composition of low HREO bearing mineralization as seen by the low fergusonite content and the relatively low zircon content.

The UZ-AG is an Upper Zone composite with average TREO grade but low HREO, illustrated by the typical zircon content of 4.1%, LREO minerals totalling 7.7% of the rock, but the low content of fergusonite, the HREO bearing mineral.

The BZ-AG is a composite of the upper half the Basal Zone, showing similar zircon and overall REE mineral content to the UZ-AG but with higher fergusonite content, reflecting the higher HREO in the Basal Zone.

The BZ-MP is a composite of the bottom half of the Basal Zone, showing overall higher zircon, LREO mineral and fergusonite content, reflecting the higher grade and higher HREO/TREO of the lower parts of the Basal Zone.

Table 7.3
Ore and Gangue Mineral Content in Composites from 40-t Pilot Plant Testwork
(Mass Distribution, %)

Ore Minerals	UZ-LG Comp	UZ-AG Comp	BZ-AG Comp	BZ-MP Comp
Columbite (Fe)	0.3	0.3	0.2	0.3
Fergusonite	0.1	0.1	0.3	0.7
Bastnaesite	0.1	0.3	0.4	0.5
Synchysite	0.2	0.7	0.7	0.7
Allanite	1.1	1.5	1.0	1.0
Monazite	0.3	0.7	0.3	0.3
Other REE	0.0	0.0	0.0	0.0
Zircon	2.5	4.1	4.1	5.6
Ore Minerals, %	4.6	7.7	7.1	9.1
Gangue Minerals	**UZ-LG Comp**	**UZ-AG Comp**	**BZ-AG Comp**	**BZ-MP Comp**
Apatite	0.0	0.2	0.0	0.1
Quartz	17.9	19.4	19.2	21.3
Plagioclase	26.6	17.7	16.3	15.7
K-Feldspar	23.5	20.1	21.0	15.6
Microcline	4.6	3.1	3.1	2.5
Biotite	16.6	22.7	22.9	23.0
Muscovite/clays	0.2	0.1	0.1	0.1
Chlorite	0.6	0.6	0.4	0.2
Other silicates	0.2	0.2	0.5	0.5
Calcite	0.3	0.3	0.6	1.3
Dolomite	0.5	0.4	0.6	0.9

Ankerite	0.9	1.3	1.8	2.8
Other carbonates	0.0	0.0	0.0	0.0
Fluorite	0.2	0.5	0.4	0.6
Fe-oxides	3.0	5.5	5.9	6.3
Other	0.2	0.2	0.1	0.1
Gangue Minerals, %	**95.4**	**92.3**	**92.9**	**90.9**
Total	**100.0**	**100.0**	**100.0**	**100.0**

Among the LREO bearing minerals, there is a clear trend from 70% allanite+monazite and 30% synchisite+bastnaesite at the upper parts of the deposit to 50% allanite+monazite with 50% synchisite+bastnaesite in the lower parts of the deposit.

For the non-ore minerals, the QEMSCAN® demonstrates that the feldspar content decreases with depth, but iron oxides, biotite and carbonates increase with depth.

8.0 DEPOSIT TYPES

The mineral deposits at Thor Lake site include the Nechalacho deposit, and the separate and distinct R, S and T Zones, with the Nechalacho deposit bearing many of the attributes of an apogranite (Beus et al., 1962) originating as an apical or domal facies of the parental syenite and granite. The deposits are extensively metasomatized with pronounced magmatic layering and cyclic ore mineral deposition. The Nechalacho deposit essentially forms part of a layered, igneous, peralkaline intrusion.

According to Richardson and Birkett (1995) other comparable rare metal deposits associated with peralkaline rocks include:

- Strange Lake, Canada (zircon, yttrium, beryllium, niobium, rare earth elements).
- Mann, Canada (beryllium, niobium).
- Illimausaq, Greenland (zircon, yttrium, REE, niobium, uranium, beryllium).
- Motzfeldt, Greenland (niobium, tantalum, zircon).
- Lovozero, Russia (niobium, zircon, tantalum, REE).
- Brockman, Australia (zircon, yttrium, niobium, tantalum).

Richardson and Birkett (1995) further comment that some of the characteristics of this type of deposit are:

- Mineralizing processes are associated with peralkaline intrusions and the latter are generally specific phases of multiple-intrusion complexes.

- Elements of economic interest include tantalum, zircon, niobium, beryllium, uranium, thorium, REE, and yttrium, commonly with more than one of these elements in a deposit. Volatiles such as fluorine and carbon dioxide are typically elevated.

- End members may be magmatic or metasomatic although deposits may show the influence of both processes. Alteration in magmatic types is often deuteric and local while alteration in metasomatic types is generally more extensive.

- This type of deposit is typically large but low grade. Grades for niobium, tantalum, beryllium, yttrium and REE are generally less than 1%, while the grade for zircon is typically between 1% and 5%.

- These deposits display a variety of rare metal minerals including oxides, silicates, calcium phosphates and calcium fluoro-carbonates. Niobium and tantalum mineralization is typically carried in pyrochlore and less commonly in columbite.

The main chemical features of the Nechalacho deposit that contrast to those overall features are that the rare earth levels are higher than typical, uranium is not particularly high, with anomalous but modest levels of thorium, and the lack of beryllium mineralization. Beryllium is present in the North T deposit, a separate smaller deposit to the north with dissimilar geology. The R, S and T Zones are small separate pegmatitic bodies to the north and northeast of the Nechalacho deposit.

The preferred genetic model is that of igneous differentiation within a closed-system with rare earth element concentration within a residual magma, aided by depression of the freezing temperature of the magma by fluorine and possibly carbon dioxide.

9.0 EXPLORATION

The following description of exploration on the Thor Lake property has been extracted from the March, 2011 Technical Report prepared by Avalon (Avalon, 2011).

The Thor Lake property has been systematically explored for several different metals at different deposits within the property over a period of 30 years (see Section 6.0). The exploration focus has shifted as new discoveries were made, such as beryllium at the T Zone deposit, were made, or in response to price increases for tantalum, yttrium and HREE, or for example, because of improved methods of recovery for tantalum.

Since taking over the property in 2005, Avalon has sampled archived drill cores from the Nechalacho deposit to extend the area of known yttrium and REE mineralization. This led to the completion of a technical report by Wardrop in 2007 (Wardrop, 2007) which included a mineral resource estimate and recommendations for further work, including diamond drilling.

Starting in August, 2007, Avalon has conducted continuous drilling campaigns, except during freeze- and break-up periods. The details of these drilling campaigns are given in Section 10.0.

An airborne magnetic survey was completed in winter 2009 to aid in mapping the local geology and structures.

In addition to drilling, mapping and geophysical work, Avalon has supported four Masters' theses (two from McGill University and two in Switzerland) and two Doctoral theses (McGill University and the University of Windsor). These have contributed to the understanding of the regional and local geology and detailed mineralogy of the Nechalacho deposit. In addition, the company has supplied logistical support to a Doctoral thesis (Carlton University) on the entire Blatchford Lake Complex.

The exploration work completed on the Thor Lake property has led to the interpretation of the geology and mineralization as provided in Sections 7.0 and 8.0 of this report.

9.1 FUTURE EXPLORATION POTENTIAL

The Nechalacho deposit is not fully delineated in terms of the definition of the limits of the mineralization. The southern and eastern limits of the deposit are reasonably well defined, and although increases in tonnage may be possible in those directions, the increase is unlikely to be significant.

The western boundary of the deposit is not well defined, and some potential for additional tonnage likely exists in that direction.

The greatest potential for increase of tonnage of mineralization similar to the known mineralization is to the north of the present inferred resources. Three drill holes at Cressy

Lake intersected Basal Zone type mineralization and are about 450 m beyond the next nearest drill holes included within the present limit of inferred resources. These holes do not define a northern limit to the deposit and more importantly they prove the concept that mineralization may exist under apparently unmineralized Thor Lake Syenite.

The inferred resource is open to the north of a line through drill hole L08-67 through L08-87 to L10-249, a distance of about 1.5 km, for at least 400 m width.

It is apparent that the grade of TREO and the proportion of HREO within the deposit vary from place to place, with higher HREO tending to occur within basins or depressions in the Basal Zone.

10.0 DRILLING

10.1 DRILL PROGRAMS

The following description of drilling programs prior to 2011 at the Thor Lake site has been extracted from RPA, 2011, to which the reader is referred for more detail.

Since 1977, diamond drilling has been carried out intermittently by various operators over five separate mineralized zones at Thor Lake. A total of 51 holes (5,648 m) had been completed on the Nechalacho deposit through 1988 (see Section 6.0, History). As the geology was poorly understood, the drilling frequently did not penetrate the Basal Zone (BZ). Modern QA/QC practices were not followed and samples were analyzed for only four to six elements. Modern, inexpensive and reliable multi-element analytical methods were not available. Consequently, the historic drilling, in general, is not useful for the recent mineral resource estimates.

Avalon commenced diamond drilling on the Nechalacho deposit in July, 2007 the goal of which was to further delineate zones of REE and tantalum mineralization. The initial drilling (2007-2008) was completed largely at a spacing of approximately 150 m by 150 m. Eleven tightly-spaced inclined holes (L08-099 to L08-109) were drilled to obtain a mini-bulk sample for continued metallurgical work on the REE-enriched zones. Six of the earlier holes were reassayed to test for the full suite of REE, as was done on the recent drilling.

From February, 2009, the drill spacing was reduced and this resulted in intercepts at approximately 50-m centres. The drilling focused on the southeastern part of the deposit where the BZ has higher TREE grades but, also, higher HREE grades, along with thicker intercepts.

As the mineralized zone is subhorizontal, and many of the drill holes are vertical, the drilled widths approximate true widths for vertical holes. For angled holes, true width varies with the angle of the hole.

In January, 2010, one drill was converted to larger PQ core in order to obtain larger weights of core for metallurgical testwork purposes.

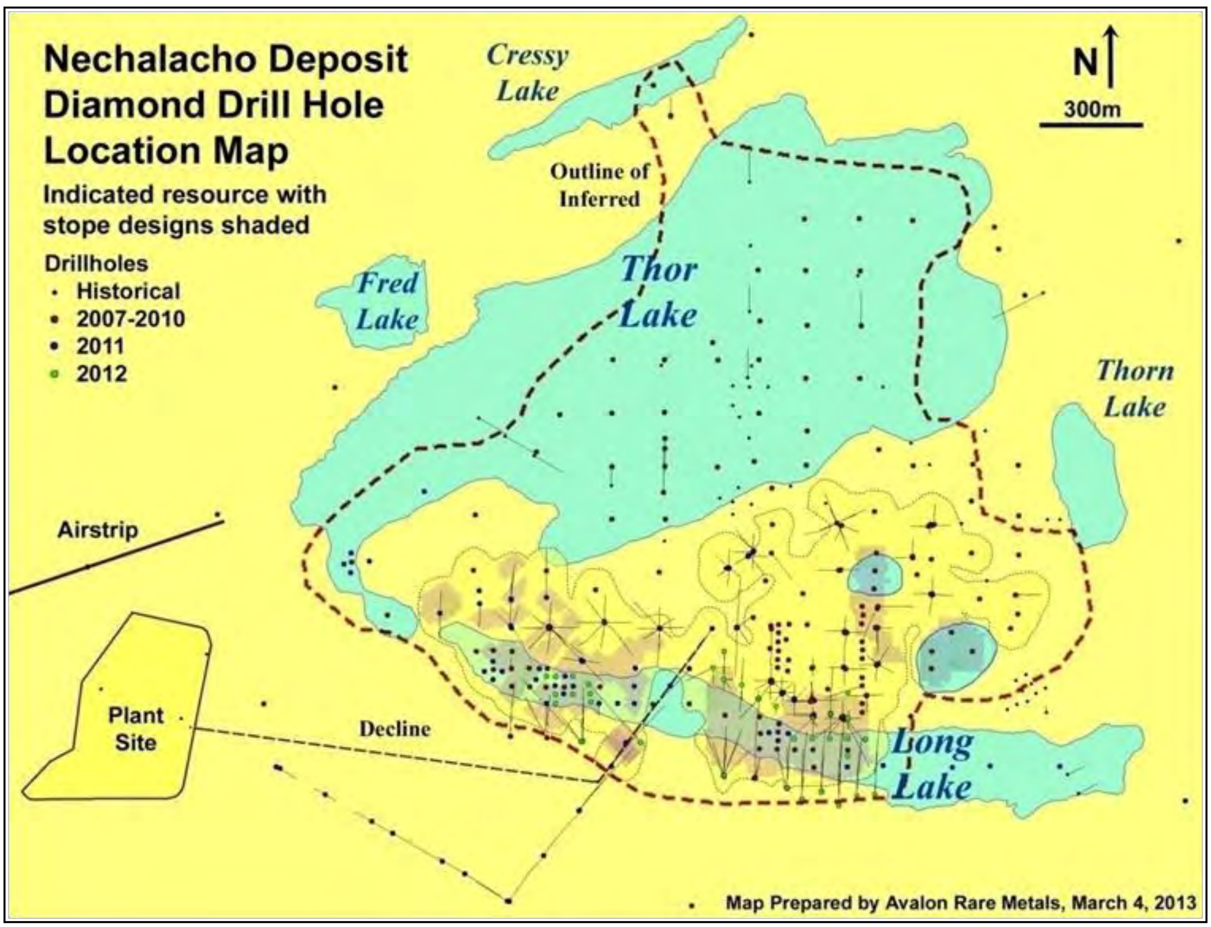
Core from both the historic drilling and the Avalon programs is stored at the Thor Lake site. Archived core has been re-boxed as necessary. Core pulps and rejects are stored in a secure warehouse in Yellowknife and at site.

Starting with 2007, and continuing throughout the Avalon drilling programs at the Nechalacho deposit, the core material collected was sampled using standard core drilling methods. Core samples and quality control materials have been submitted to independent commercial laboratories for assay analysis.

Avalon's drilling operations were supervised by J.C. Pedersen, P.Geo., and Avalon's Vice-President, Exploration, Bill Mercer, Ph.D., P.Geo. (Ontario), P.Geol. (Northwest Territories), monitored the QA/QC and provided overall direction on the project.

Figure 10.1 gives the location of all drill holes in relation to the outline of the Nechalacho deposit, the Indicated Mineral Resources and the planned location of mine stopes.

Figure 10.1
Diamond Drill Hole Location Map



Dashed red line = present limit of mineralization.
Yellow outlined area = Indicated Mineral Resources.
Pink outlines area = Mining stopes.

Core sizes range from BTW diameter for the initial 2007 drill program to NQ2 in the winter/summer 2008 program, NQ2 or HQ in the 2009-2010 programs, and HQ and PQ in the 2010 and subsequent programs (see Table 10.1).

Table 10.1
Core Diameter for 2007-2012 Drilling Programs

Year	Hole Diameter	Drill Holes[1]	Metres
pre-2007	Not specified	51	5,588
2007	BTW	13	2,440
	Total	13	2,440
2008	NQ2	72	14,259
	Total	72	14,259
2009	HQ	43	8,794
	NQ	26	5,477
	Total	69	14,271
2010	HQ	86	24,147
	PQ	20	3,754
	Total	106	27,901
2011	HQ	44	11,057
	NQ	21	3,924
	PQ	46	10,865
	Total	111	25,846
2012	HQ	73	18,076
	PQ	14	3,277
	Total	87	21,353
Avalon		**458**	**106,070**
Grand Total		**509**	**111,659**

[1] Not including abandoned holes.

10.2 CORE LOGGING AND CORE RECOVERY

The geological, geophysical, geotechnical, and assay data are entered into a database. Until the end of 2011, a custom designed database and managed by a consultant, with regular back-ups off-site. Since the beginning of 2012, the drill hole database has been rebuilt into DataShed of Maxwell Geoservices and is maintained by Avalon geologists. The geologists at site can access the drill database to review previous drill results.

Due to the strong hydrothermal alteration of all lithologies, identifying specific precursor lithologies has proven quite difficult, particularly in the early drill programs. Early lithological coding tended to incorporate hydrothermal alteration, commonly making it difficult to correlate units between drill holes. As more information became available from deeper drilling and specific textures and lithologies were compared to other unaltered, alkaline deposits elsewhere, such as Illimausaq in Greenland, a new lithological code was produced using, as a basis, the recognizable precursor lithologies. This has greatly advanced the understanding of the lithology, mineralogy, and to a lesser degree the petro-genesis of the deposit.

The REE-bearing minerals in the Nechalacho deposit are generally not visible with the naked eye due to their disseminated and fine-grained nature. The dominant minerals identified easily are zircon and (infrequently) traces of bastnaesite although visual grade estimates of bastnaesite are rarely possible. To map the relative grades in the core, Avalon utilizes a Thermo-Scientific Niton® XLP-522K handheld analyzer for assistance to the geologist during core logging. The Niton® energy-dispersive X-ray fluorescence (EDXRF) analyzers, commonly known as XRF analyzers, are able to quickly and non-destructively determine the elemental composition of the drill core.

A number of elements may be analyzed simultaneously by measuring the characteristic fluorescence X-rays emitted by a sample. EDXRF analyzers determine the content of a sample by measuring the spectrum of the characteristic X-rays emitted by the different elements in the sample when it is illuminated by X-rays, in the case of the XLP-522K, from a small, sealed capsule of radioactive material.

Due to variations in analysis conditions – the physical surface of the sample, the dampness of the sample, and the small window for analysis – the readings for individual elements cannot be considered as quantitative and representative of long core lengths. However, the readings can assist the geologist to identify mineralized sections, and determine whether these sections are relatively higher or lower in heavy rare earth elements.

Core recoveries are generally high at the Nechalacho deposit, due to the exceptionally competent nature of the rock, with an average of 97% in the mineralization, with no bias in the sampling due to incomplete sample recovery. Also, there is no apparent bias in results between various core sizes suggesting that there is no issue with respect to a nugget effect on sampling. As a result, the authors believe that there are no drilling, sampling, or recovery factors that could materially affect the accuracy or reliability of the results.

A comprehensive core logging and sampling protocol was established at the beginning of the July 2007 drilling program. This protocol has been strictly applied for all of the subsequent drilling programs. A comprehensive geotechnical logging protocol was introduced at the start of the summer 2009 drill program.

Core was placed in standard wooden core boxes at the drill by the driller helper, with a wooden marker placed at the end of each core run marking the metreage from the surface. Throughout the BTW-NQ programs drill rods were imperial lengths of 10 ft (3.048 m), and core markers were written in feet on one side of the wooden block, and using a metric conversion chart, written in metres on the opposite side of the block. The HQ drilling initially used both imperial and metric rods, so markers were in both feet and metres to ensure proper measurement. Imperial rods were used exclusively in the latter part of the 2009 drill program.

After inspection by the geologist at the drill, the boxes were closed with wooden lids and taken to the core logging facility at the camp by snowmobile in the winter and by boat and all-terrain vehicle in the summer. At the camp, the boxes were opened by the geologist on outdoor racks. In good weather, logging and other geotechnical measurements were done outside; in poor weather and in winter, core was processed in a heated core shack.

Core was initially measured to determine recoveries, and marked incrementally every metre. This marking served as a guide for magnetic susceptibility, rock quality designation (RQD), and density measurements. Magnetic susceptibility was measured every metre with a hand-held “KT-10 magnetic susceptibility meter”. Density was measured every 5 m by weighing a section of drill core in air and then weighing by submersing the sample in water and comparing the difference between dry and submersed weight. A typical core sample for density measurement averages 10 cm in length.

Core was generally very clean when brought to camp, and required no washing except for occasional sprays of water when mud is present. The geologist marked out major rock units and completed a written description for the entire core sequence. Frequent readings, using a Thermo-Scientific Niton® XLP-522K handheld analyzer, act as a guide to areas of mineralization and general chemistry of a specific interval. The final task was to mark out with a china marker specific sample intervals for the length of the entire drill hole.

After all tests and core observations were completed, and prior to splitting, the core was photographed outdoors using a handheld digital camera. Metreages and hole number were marked so as to be visible in all photos. Core was generally photographed in groups of six boxes. Core photographs were stored on the camp computer, in addition to an external hard drive.

As indicated above, starting in the 2009 summer drill program, drill core was also logged for geotechnical characteristics. Some of the holes were logged from top to bottom, while others were logged above, below, and within the Basal Zone, to determine rock quality characteristics of both the mineralized zones and country rocks. Efforts were made to select holes with varying orientations to provide comprehensive orientation characteristics of planar structural features.

Geotechnical logging, comprising RQD, was performed for each run. The geotechnical logging was done on core logging sheets and entered electronically into a custom-designed Excel spreadsheet provided by KPL. A total of 385 holes were logged in whole or in part. Holes which were partially logged included the Basal Zone and a minimum 10 m interval above and below.

When the core had been logged and photographed, it was stored in core racks outside the core splitting tent, from which they were then brought in to the core shack to be split and sampled.

10.3 SAMPLING METHOD AND APPROACH

On average, and except for PQ drilling, assay samples were 2 m long except where, in the geologist's opinion, it was advisable to follow lithological boundaries. In 2010, when PQ drilling commenced at the project, the sample interval was reduced to typically 1 m due to health and safety concerns with the weight of PQ core. A 1-m sample of PQ core weighs about 18 kg.

Due to the long widths of mineralization within the Basal Zone, averaging over 20 m thick, even spaced sampling was not considered a significant factor in resource estimation. Consequently, individual samples can vary in length when encountering lithological changes, as efforts were made not to split across well-defined lithological boundaries. A list was made of all sample intervals as a record and also a guide to the core splitting technicians.

The exception to the above procedures was when PQ core was being handled. For mineralized zones, PQ core was submitted to the independent laboratory for crushing of whole core to 6 mesh (approximately 3.3 mm, a size suitable for metallurgical testing). The details are given below. Unmineralized intervals of PQ core were sawn with about one third of the PQ core sent for assay (typically about 5 kg) and the remainder boxed on site.

11.0 SAMPLE PREPARATION, ANALYSES AND SECURITY

All sample preparation, from identification of sample intervals to bagging of drill core, was completed by employees of Avalon. Subsequent preparation, such as crushing, grinding and further steps, were all completed by commercial laboratories, ALS Limited (ALS), Acme Analytical Laboratories (Vancouver) Limited (Acme) and Activation Laboratories Limited (Actlabs). For the drill programs of 2007-2008, the core splitter broke the core into smaller lengths to fit into the mechanical core splitter, split the core in half, and placed one half in a plastic sample bag with the other half placed back into its sequence in the core box to serve as a permanent record. Starting in 2009, it became standard practice when handling HQ core to initially split the core in half, then one half in quarters, with one quarter for assay, one quarter as library core and half core retained for metallurgical purposes. For core prior to 2009, limited metallurgical sampling was completed.

While the majority of the sample splitting has been with mechanical core splitter to produce a half core for a sample, some core has also been sawn and quartered when required for metallurgical testing or standard preparation; however, this method was abandoned due to slow production.

As noted above, an exception to the above procedures was initiated when drilling with PQ core. Due to the large size and weight, PQ core creates special problems for handling and sampling. Due to weight handling, and safety concerns, it was decided to sample PQ core at 1 m intervals, not 2 m, as is used for BTQ, HQ and NQ2 core.

Due to the inability to split PQ core with a mechanical splitter, it was decided at first to saw the core using an electric core saw. As a result, a core saw was custom prepared for PQ core. However, it was quickly discovered that the productivity was too slow and could not keep up with core production. As a result, a change in procedures occurred with the decision to crush whole core from mineralized zones. It was considered, as sufficient library drill core from 2007 to 2012 was on hand, that preserving PQ core for due diligence purposes was not necessary. The procedure followed for mineralized PQ core was as follows:

- Bag 1 m intervals of whole core in camp and ship to ALS in Yellowknife. Each 1 m weighed about 18 kg.
- At ALS the complete core was crushed to 6 mesh (about 3.3 mm) and about 2 kg was split off using a rotary splitter. The larger portion, about 16 kg, was bagged as a metallurgical sample and given the same sample number as the original sample.
- The smaller portion, 2 kg, was then crushed to 10 mesh and, from that point on, followed the normal assay procedure.

The rationale behind the above approach is that 6 mesh samples are suited to metallurgical processing whereas finer grain sizes (10 mesh) would not be suited to metallurgical processing. Also, the split-off section of 2 kg is very similar to quarter HQ core, or half NQ core, and so is just as representative as these samples. For unmineralized PQ core, sawing was utilized in order to reduce sample size and lower cost. In this case, about one-third of the core was sawn off for a representative sample and two-thirds retained as library samples.

The sample interval was marked on a sample tag in a three-part sample book and a tag with the corresponding sample number was placed in the sample bag. The sample bag was also marked with the corresponding sample number using a felt marker. The bag was then either stapled or zip-tied closed, and placed in a rice bag with two other samples. Most rice bags contained three samples to keep weight to a manageable level. The rice bag was then marked on the outside with corresponding sample numbers contained within, and a second number identified the rice bag itself. A sample shipment form was then completed, generally in increments of 50 rice bags, which constituted a single shipment.

The sample form was enclosed in an appropriately marked rice bag, with a duplicate paper copy kept in camp, and also kept on electronic file.

Internally developed standards were inserted routinely. Blanks, composed of split drill core of unaltered and unveined diabase dyke intersected in drilling beneath Thor Lake, were inserted every 40th sample.

Avalon has developed a number of standards for the Thor Lake project to date. These have been as follows:

- In 2007, three standards were developed utilizing historic drill core, with three grade ranges designated “Low”, “Medium” and “High”. The standards were subject to a Round Robin analysis in a number of laboratories where certified values were established.

- In 2009 a second standard, designated STD-04_09, was prepared, this time from current drill core, and subject to Round Robin analysis for certified values.

- In 2010, two further standards were developed, being STD-229 and STD-236, and again subject to Round Robin analysis for certified values.

Drill core samples were shipped generally by air from Thor Lake to Yellowknife except in unusual cases when high weight metallurgical batches were shipped by ice road or barge on Great Slave Lake. The standard shipment was 50 rice bags, or a total of 150 samples per shipment. The rice bags were zip-tied for security, and were met and unloaded in Yellowknife by a representative of Discovery Mining Services (Discovery). In 2010-2011, for a limited period, Deton Cho Logistics, Yellowknife, completed the role taken at other times by Discovery. Discovery took the samples to its warehouse, inventoried all samples, and produced a manifest which was sent electronically to Thor Lake camp and accompanied the shipment. The samples were then taken by Discovery to the core processing facilities of initially Acme and subsequently ALS, both in Yellowknife. At this point, the laboratories took custody of the samples.

Core was sent to the preparation laboratory with specification that all core should be crushed to 90% passing 10 mesh with a supplementary charge if necessary (see note concerning PQ core above). In the first program in 2007, two 250 g pulps were prepared from each sample, one for the primary laboratory, and one to be shipped to Avalon and used for the check analysis. As noted, for samples from drill holes completed in 2007, every sample was duplicated and sent to a secondary laboratory for check analyses. Subsequent to this, approximately every tenth pulp was sent for duplicate analysis in the secondary laboratory. Standards were inserted in the duplicate sample stream by Avalon employees prior to shipping to the secondary laboratory.

All remaining drill core is stored on site at Thor Lake. Core is racked at the exploration camp, and additional storage facilities have been utilized at the former Highwood mine site buildings at the T-Zone. Historic core, particularly T-Zone core, is stored at the mine site, while Nechalacho deposit core is stored at both the T-Zone mine site and the camp for the most current core.

Since December, 2009, Avalon has rented a storage facility at the Yellowknife airport, and laboratories are requested to return all pulps and rejects to Avalon. The material is stored in the location and a computer database holds the sample numbers and type. In addition, samples destined for metallurgical testing, including pilot plant testing, are stored in the Yellowknife facility.

11.1 ANALYTICAL PROCEDURES

Any assay results obtained prior to 2007 (holes 1 to 51) are referred to as the “older holes”. These did not have internal QA/QC and were analyzed for a limited set of elements; however, six of the old holes were re-assayed in 2008 for the complete suite of elements.

Avalon has changed the laboratories used for analysis over time. Table 11.1 summarizes the laboratory usage.

Table 11.1
Laboratory Usage

Program	Preparation Laboratory	Prime Laboratory	Secondary Laboratory	Notes
2007	Actlabs	Actlabs	Acme	All samples double analysed
2008	Acme	Acme	ALS	Every 10th sample to secondary
2009 – 2012	ALS	ALS	Acme	Every 10th sample to secondary

For the first year of drilling by Avalon (2007), the primary laboratory was Actlabs of Ancaster, Ontario, and the secondary laboratory was Acme in Vancouver, British Columbia. Samples were shipped to the Actlabs facility in Ancaster for preparation, and a duplicate pulp was submitted to Acme in Vancouver for complete check analysis. The Actlabs procedures used are Codes 4B, 4B2-STD, 4B2-RESEARCH, 4LITHO, and 4LITHORESEARCH.

The Actlabs method involved lithium metaborate/tetraborate fusion inductively coupled plasma (ICP) Whole Rock package Code 4B and a trace element inductively coupled plasma mass spectrometry (ICP-MS) package Code 4B2. The two packages are combined for Code 4Litho. The fusion process ensures total metals particularly for elements like REE in resistate phases (this may not be the case for acid digestions, particularly for heavy rare earths and other elements contained in refractory minerals like zircon, sphene, monazite, chromite, gahnite and several other phases). If refractory minerals are not digested, a bias may occur for certain REE and high field strength elements with standard acid digestions. The trace element package using ICP-MS (Codes 4B2-STD or 4B2-RESEARCH) on the fusion solution provides research quality data whether using standard or research detection limits. Note that europium determinations are semi-quantitative in samples having extremely high barium concentrations (greater than 1%). This package is intended primarily for unmineralized samples. Mineralized samples can be analyzed, but the results will be semi-quantitative for the chalcophile elements (Ag, As, Bi, Co, Cu, Mo, Ni, Pb, Sb, Sn, W, and Zn).

For the 2008 winter and summer programs, the preparation laboratory was the Acme facility in Yellowknife and the primary analytical laboratory was the Acme facility in Vancouver. A split of every tenth sample reject was sent to ALS in Vancouver for check analyses. All core was analyzed by Acme using two analytical packages: Group 4A and Group 4B. ALS analyzed the samples with the MS81 method.

Acme's Group 4A is a whole rock characterization package comprising four separate analytical tests. Total abundances of the major oxides and several minor elements are reported using a 0.1 g sample analyzed by ICP-emission spectrometry following a lithium metaborate/tetraborate fusion and dilute nitric digestion. Loss on ignition is by weight difference after ignition at 1,000°C.

Acme's Group 4B is a Total Trace Elements by ICP-MS. This package comprises two separate analyses. Rare earth and refractory elements are determined by ICP-MS following a lithium metaborate/tetraborate fusion and nitric acid digestion of a 0.1 g sample (same decomposition as Group 4A). In addition, a separate 0.5 g split is digested in aqua regia and analyzed by ICP-MS to report the precious and base metals.

For 2008, secondary samples, comprising roughly every tenth reject sample supplied by Acme, were shipped to ALS, where the samples were analyzed by the package MS81. This is a combination of lithium metaborate/ICP atomic emission spectrometry (ICP-AES) for whole rock values, lithium borate/ICP-MS for refractory mineral values and other elements, and aqua regia/ICP-MS for volatile elements.

Starting with the winter 2009 drilling campaign, all samples were prepared at the ALS preparation facility in Yellowknife, and a subsample shipped and analyzed at the ALS laboratory in Vancouver by lithium metaborate/tetraborate fusion and dilute nitric acid digestion, followed by whole rock and 45 element multi-element ICP analysis (ALS sample method ME-MS81d). All samples contained within intercepts above the 1.6% cut-off criteria and any additional samples exceeding analytical limits or of geological significance were re-run using similar ALS method ME-MS81h for higher concentration levels. ME-MS81h is a similar method but with greater dilution in the analytical procedure. Every tenth sample had a duplicate pulp prepared which, with inserted standards and blanks, was sent to Acme in Vancouver for check analyses. Results were monitored for key elements, and in cases of QA/QC issues, re-analysis was requested. The key elements that were monitored were those that have greatest financial impact on the project and typically would be neodymium, europium, terbium, dysprosium and niobium.

High values that were over limits for niobium and zirconium were routinely re-run with ALS XRF method "XRF10" which gives upper limits of 10% for Nb and 50% for Zr.

Statistics of samples submitted to ALS and Acme during the period 1 January, 2011 to 31 August, 2012 are shown in Table 11.2. The ratio of samples at ALS to Acme does not follow the protocol of 10:1 due to the delay in submission of the duplicates to Acme during the specific period.

Table 11.2
Statistics of Sample Submission, January, 2011-August, 2012

Laboratories	ACME_VAN	ALS_VAN
No. of Batches	8	215
No. of DH Samples	nil	16,908
No. of QC Samples	1,501	3,689
No. of Standard Samples	43	1,165

The above data gives a ratio of a standard per 15 samples with ALS and 35 samples with Acme. Blanks are considered as "standards" in these statistics.

Values were reported by the laboratories in element parts per million, transformed to oxide parts per million by Avalon geologists and then both element and oxide parts per million values were imported into the database.

As assay results became available from laboratories, Avalon consultants incorporated them into a database. Assay results of drill core, certified reference materials, and blank samples were merged into a master sheet according to a set of rules that establish the precedence of various types of assay methods. The master sheet is an Access database and various macros were used for database update and maintenance.

Starting with 2008 and current at the time of writing this report, Avalon used the following precedence algorithm:

- MS81h values for REE, Nb, Zr and Ta take precedence over MS81 values.
- MS81h values for Ta take precedence over XRF values.
- ALS XRF values for Nb and Zr take precedence over MS81h values.
- ALS XRF results take precedence over Actlabs XRF results.

Repeat pulp analysis by the same laboratory and method, or field sample duplicates, were incorporated into the Maxwell database and which datapoint is utilized depends on a priority ranking system.

In the opinion of RPA, the sample analyses are sufficiently reliable for resource estimation purposes.

11.2 SPECIFIC GRAVITY MEASUREMENT

Specific gravity (SG) is measured on core samples taken at 5 m intervals within the hole. Each sample is approximately 10 cm long. Breaking the drill core (if necessary) only occurs after other tests that require undisturbed core (such as photography and geotechnical analysis) have been completed.

The density method is as follows:

- Weigh the sample in air.
- Weigh the sample suspended in water.

A Mettler Toledo PL3001-S electronic scale was used for weighing in air. This scale has an accuracy of one decimal place (±0.1 g).

A small metal can suspended beneath the balance, set up on a table with a hole for the suspension of the basket, was used to weigh the sample in water (the Mettler balance has a hook underneath for SG measurement purposes). The balance was zeroed with the can hanging in a large container of water.

The calculation of the SG is as follows:

SG of sample = weight of sample in air/weight of sample in water.

SG measurements made at the Nechalacho site on the drill core, according to lithology, are summarized in Table 11.3.

Table 11.3
Statistics of Specific Gravity by Lithology

Lithology	Description	Count	Average SG	SD	Minimum	Maximum
43	Sodalite aegerine amphibole pegmatite	1	2.66	-	2.66	2.66
45	Foyaite III	38	2.7	0.05	2.59	2.84
47	Layered biotite sodalite albite syenite	27	2.72	0.14	2.59	3
48	Biotite sodalite aegerine syenite	36	2.66	0.11	2.43	2.98
49	Spotted aegerine amphibole syenite	12	2.66	0.09	2.48	2.82
60	Layered sodic syenites	1,272	2.78	0.12	1.07	3.52
61	K-feldspar pegmatite	73	2.7	0.12	2.48	3.23

Lithology	Description	Count	Average SG	SD	Minimum	Maximum
63	Foyaite II	1,198	2.79	0.1	2.44	3.41
65	Heterogenous syenite	1,402	2.79	0.11	2.3	3.4
67	Zircono-silicate cumulate	1,175	2.9	0.16	2.26	3.8
69	Heterogenous zircono-silicate syenite	2,881	2.87	0.16	1.91	4.07
70	Trachoidal microsyenite (altered)	107	2.79	0.13	2.58	3.71
75	Foyaite I	937	2.76	0.11	2.09	3.47
78	K-feldspar nepheline syenite (altered)	7,159	2.7	0.12	2.13	3.95
79	Sodalite cumulate (altered)	957	2.75	0.1	2.21	3.64
84	Thor Lake syenite	1,056	2.71	0.08	2.42	3.58
85	Grace Lake granite	966	2.68	0.06	2.11	3.22
90	Diabase	190	2.9	0.07	2.67	3.22
97	Breccia	54	2.72	0.07	2.54	2.91
99	Highly altered rock, precursor unknown	140	2.71	0.15	2.21	3.47
Total		**19,681**	**2.76**	**0.14**	**1.07**	**4.07**

In 2010, 32 samples of drill core were submitted to ALS for an independent check of the SG values. In 2012, a further 10 samples of drill core were submitted for independent check of SG. The same samples were measured at the Thor Lake camp site before shipment to ALS. ALS completed both water-only and wax-coated measurements on the core.

The statistics are summarized in Table 11.4. Figure 11.1 displays scatter plots of SG measurements determined by ALS and Avalon.

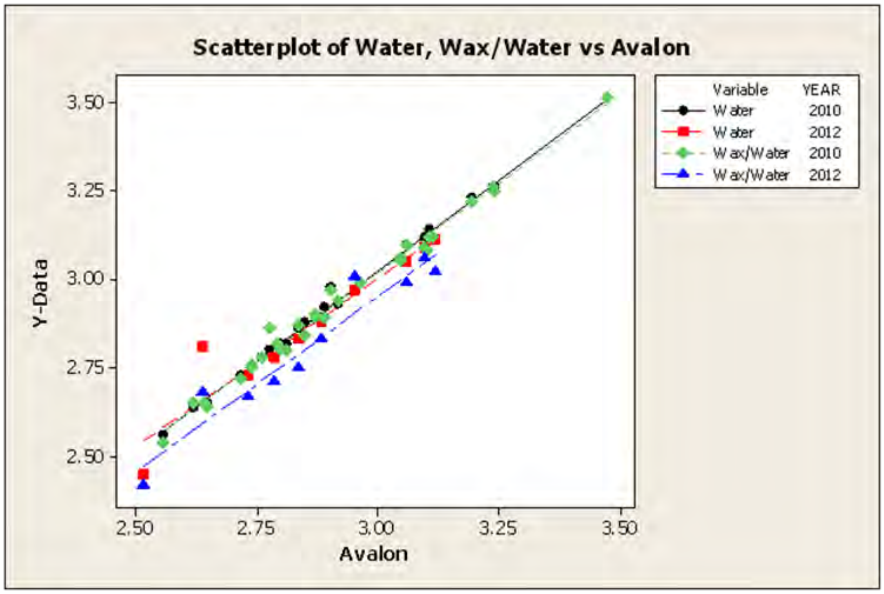
Table 11.4
Comparative Statistics of Specific Gravity Measurements Performed by ALS and Avalon

Variable	Mean	Standard Deviation	Median
ALS Water	2.9264	0.2132	2.8900
ALS Wax/Water	2.9090	0.2175	2.8900
Avalon	2.9076	0.2089	2.8775

Note: Standard Deviation in SG Units.

A two-sample t-test of the differences between the measurements (firstly ALS water-only against Avalon and, secondly, ALS wax against Avalon) gave a p-values of 0.69 and 0.98 respectively, indicating no significant difference at the 99% confidence level. As can be seen, the difference between Avalon and ALS mean measurements are in the third decimal place and for the median in the second decimal place, which is considered insignificant as the difference is of the order of 1%. Thus the field density measurement system is concluded to be reliable by RPA and Avalon.

Figure 11.1
Scatter Plots of ALS and Avalon SG Measurement



The general solid nature of the Nechalacho Nepheline Syenite rock is illustrated by the fact that the difference between SG measured with and without wax coating averages only 1%. Vuggy permeable rock would give a larger difference between the two methods.

RPA considers that the data are reliable and suitable for use in mineral resource estimation.

12.0 DATA VERIFICATION

The database used by Avalon for the resource estimate contained drill hole collar location, downhole deviation surveys, lithology, density, magnetic susceptibility, RQD, sample numbers, sample intervals, and analytical data. The database used for the November 21, 2012 resource estimate by Avalon and provided to RPA for verification and validation is based on DataShed of Maxwell Geosciences. Between 2008 and 2011, a custom database was maintained by consultants to Avalon.

The structure and content of the assay table are discussed in Section 11.0. Over the years, a number of steps have been taken to ensure the accuracy and validity of the database. Cross-referencing was routinely done over time to help validate the values of economic interest in the assay table by comparison with MS81, MS81h and XRF input tables, as well as various comparisons between sample duplicates, check samples, and comparison between laboratory results. Comparisons with snapshots of the database from October, 2008 and January, 2012 were also performed. At the beginning of 2012, the database solution was changed to DataShed and the database has been rebuilt by Avalon geologists.

RPA verification and validation procedures included checks between database and assay certificates, confirmation of conversion factors for element-to-oxide transformations, and comparisons with historic versions of the database. Also, collar locations have been reviewed, drill hole paths were inspected and GEMS project database validation routines were performed.

A total of 248 electronic assay certificates from 2007 to 2012, provided by Avalon, representing approximately a quarter of the assay database, were used for validation of rare earth and rare metal elements, as well as yttrium, uranium and thorium. Approximately 10% of the assays in the database were verified.

Additional assay spot checks were done for samples from five historical drill holes, by comparison with scans of old paper records, either analysis certificates or drill hole logs containing a variable mix of cerium, yttrium, gallium, zirconium and niobium values, in element parts per million or percent oxide.

Scatter plots were performed for pairs of rare metal, light REE, and heavy REE pairs of elements.

Collar locations of drill holes were compared with older printed records and GPS survey files. Downhole deviation surveys were visually inspected and spot checks were done for drill hole azimuth and dip.

The RPA database validation process resulted in the identification of seven samples missed during the transition to DataShed in 2012, and 14 cases where samples of inaccurate assay check analyses overwrote the original sample results, out of 41,567 samples in the database. The impact of the inaccuracies identified in the database had negligible influence over the resource estimate. RPA considers that the database is acceptable to use for resource estimation purposes.

12.1 INDEPENDENT SAMPLING BY RPA

During the April, 2011 site visit, RPA personnel collected eight core samples from six diamond drill holes from the 2010 drill program. Six of the samples were from the Basal Zone, and two from the Upper Zone. The core samples consisted of the second half core or quartered core retained by Avalon.

The samples were processed at SGS Canada Laboratories, Don Mills, Toronto, using the IMS95A and the IMS91B analytical packages. The presence of mineralization was confirmed and the assay results were similar to the original samples.

It is the opinion of RPA that Avalon follows the current industry practice, and that the analysis of standards, blanks, and duplicate reject samples show acceptable results.

12.2 QUALITY ASSURANCE/QUALITY CONTROL

In 2007, Avalon commissioned CDN Resource Laboratories Ltd. (CDN) in British Columbia to generate three standards called AVL-H, AVL-M, or AVL-L. These standards would be inserted into the assay stream. Avalon then commissioned Dr. Barry Smee to review the Round Robin procedure and assess the quality of the data.

In 2010, Avalon commissioned CDN to generate a further standard called STD-04_09. This standard would be inserted into the assay stream, alternating with the original three standards. Avalon then commissioned Dr. Smee to review the Round Robin and assess the quality of the data. The Round Robin on the new standard included samples of the original three standards, rare earth certified standards, all randomized for the Round Robin. When inserted into the sample database, this standard was referred to as STD-H2.

A further two standards were developed in 2010 referred to as STD-229 and STD-236. These too were subject to Round Robin analysis and certification by Charles Beaudry.

The statistics of 2007-2010 QA/QC samples were presented in the NI 43-101 report of 2010 (Scott Wilson RPA, 2010), and reviewed each time a new resource estimate was released.

12.2.1 January 2011 – August 2012 QA/QC

The control samples inserted into the sample stream from drill holes completed during 2011-2012 were presented in Table 11.2, above. Quality control procedures used during 2011 were continued through 2012 with the addition of core duplicate sampling for the summer 2012 drilling campaign. Overall results were satisfactory. All samples for batch VA11200410 (July, 2012) were rerun due to poor standards results. Also, 12 samples in batch YW11126454 (August, 2012) were rerun due to poor blank performance. New rare earth element standard samples at various grades have been certified for use starting in 2013. This report is for all samples assayed as ore-grade material (ALS method MS81h) which significantly affect the resource estimate. Quality control is performed on samples run using ALS method MS81 for non-ore grade material, but is not reported here since it does not significantly affect the resource estimate. The ratios of the various types of QA/QC samples utilized in the program from 1 January, 2011 to 31 August, 2012 are given in Table 12.1.

Table 12.1
QA/QC Samples Submitted From January, 2011 to August, 2012

QC Category	DH Sample Count	QC Sample Count	Ratio of QC Samples to DH Samples
Company Standards	16,914	1,117	1:15
Company Blanks	16,914	453	1:37
Laboratory Duplicates	16,914	2,019	1:8
Field Duplicates	16,914	88	1:192

The rate of insertion of standards was varied according to whether the samples were from a mineralized zone or not with a standard every 10 samples in mineralization and every 40 samples outside mineralization. The results of the standard analyses were checked against the certified or provisional means and tolerances listed in the standard certificates, as well as against the laboratory (ALS) precision tolerance level of $\pm$10%. The three rare earth elements with the potential highest value (neodymium, terbium and dysprosium) were routinely monitored along with the overall values for the TREE and HREE.

Precision results of the QA/QC program for all laboratories, as measured by relative standard deviation (standard deviation/sample population mean), are measured for the individual payable elements in the deposit. Representative examples for key individual elements are given below.

12.2.1.1 Standards Analysis Results

Cerium

Cerium is representative of the light rare earths and accounts for only about 4% of the value of the NMR of the Basal Zone. The certified values of the standards used are given on the left of Table 12.2 and compared to the results in the exploration program (shown as Calculated Values). An overall positive bias of 3% to 6.6% is seen in the standards results. The precision results, with a relative standard deviation of about 4%, are acceptable.

Table 12.2
Cerium Standards Results

Ce Standard(s)			No. of Samples	Calculated Values			
Standard Code	Value (ppm)	SD (ppm)		Mean Ce (ppm)	SD (ppm)	CV	Mean Bias (%)
AVL-H	6,189	308.0	77	6,412	258	0.040	3.60
STD-04_09	10,680	562.5	171	11,384	462	0.041	6.59
STD-229	4,714	309.5	192	5,007	222	0.044	6.22
STD-236	5,077	335.0	198	5,365	221	0.041	5.68

Figure 12.1 through Figure 12.4 show the cerium standards control charts.

Figure 12.1
Cerium Results for Standard AVL-H

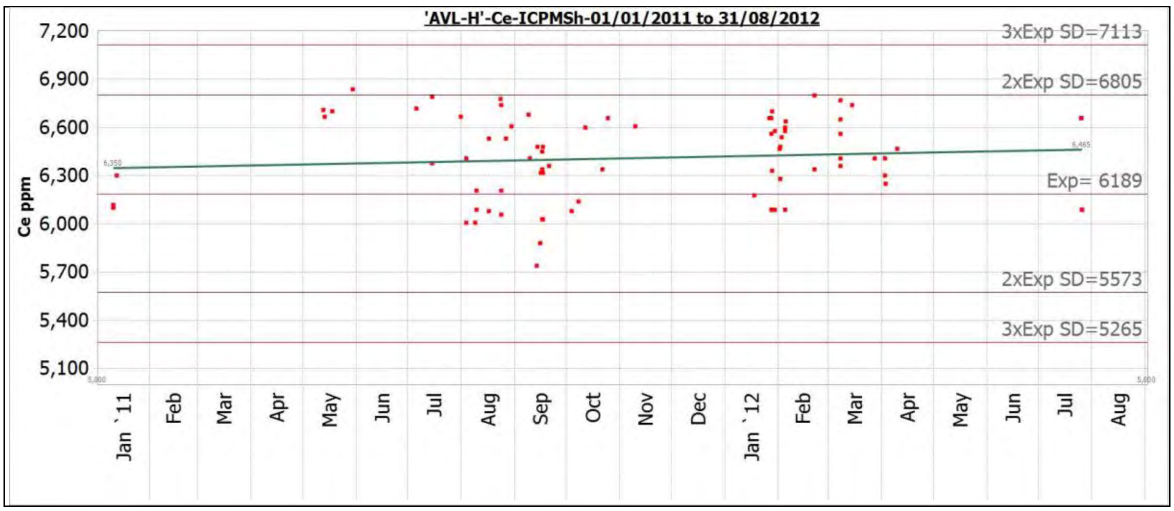


Figure 12.2
Cerium Results for Standard STD-04_09

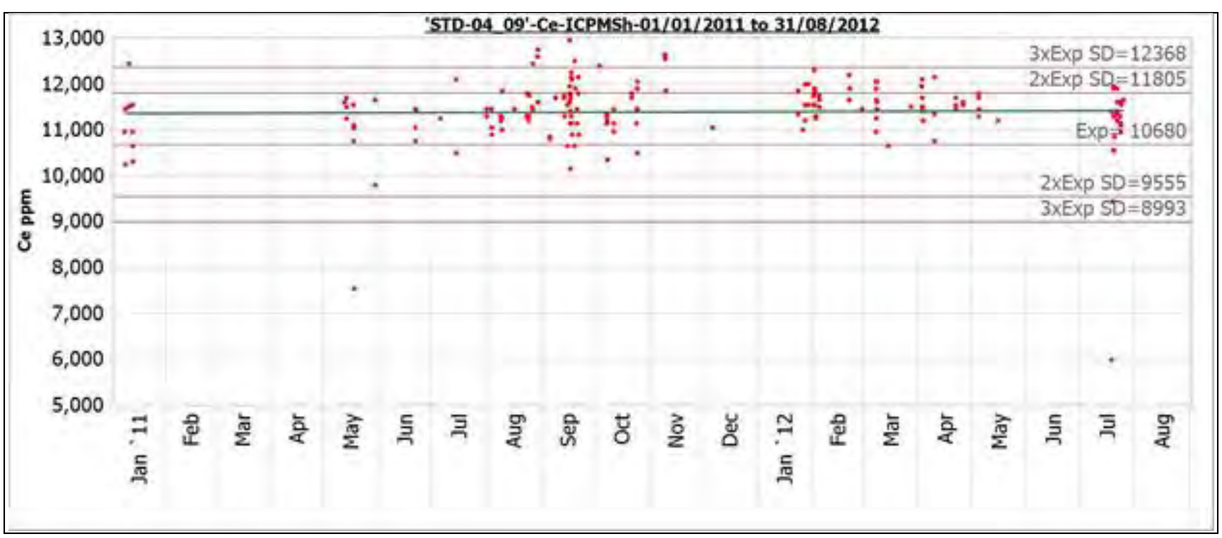


Figure 12.3
Cerium Results for Standard STD-229

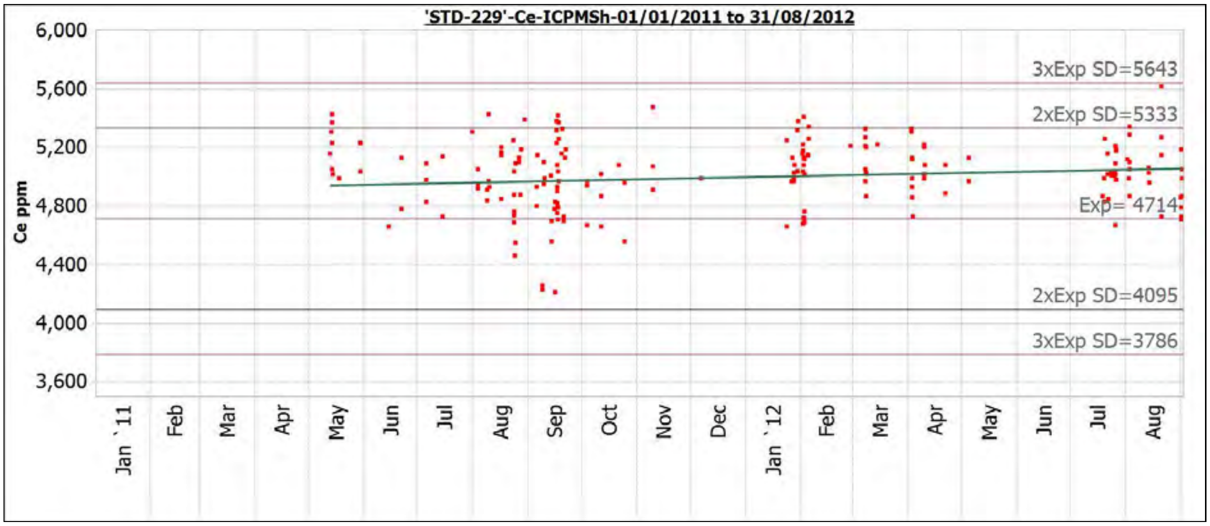
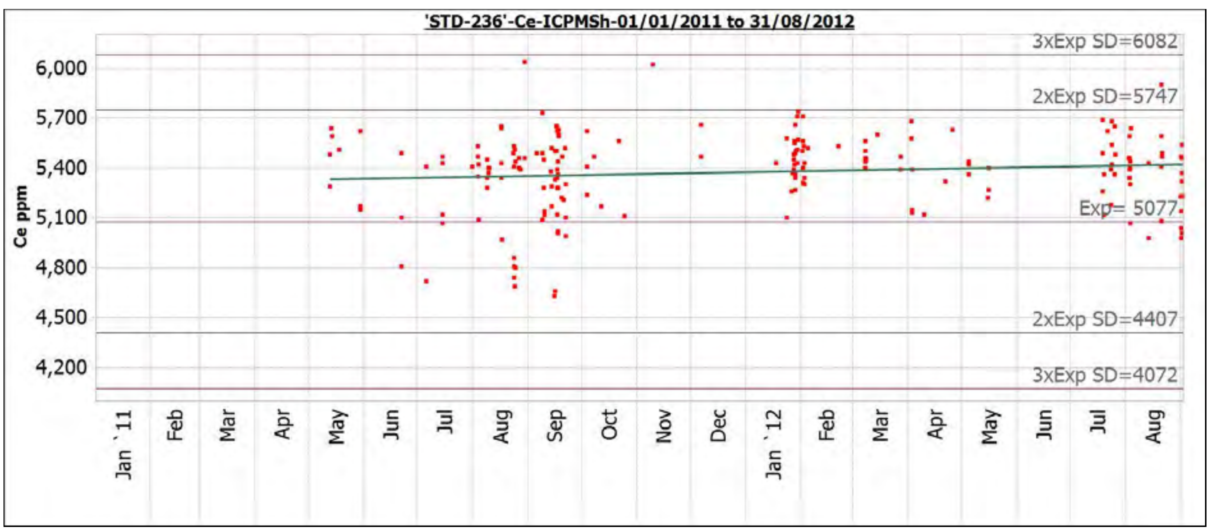


Figure 12.4
Cerium Results for Standard STD-236



Dysprosium

Dysprosium can be considered as representative of the heavy rare earths and comprises approximately 24% of the NMR of the Basal Zone. The mean bias for dysprosium analyses is between +0.1% and +1.9% for the three standards relative to the certified values. The precision results range from 4.3-5% CV and are acceptable. The dysprosium standards results are shown in Table 12.3 and the control charts shown in Figure 12.5 through Figure 12.8.

Table 12.3
Dysprosium Standards Results

Dy Standard(s)			No. of Samples	Calculated Values			
Standard Code	Value (ppm)	SD (ppm)		Mean Dy (ppm)	SD (ppm)	CV	Mean Bias (%)
AVL-H	569	19.0	77	572	24	0.043	0.61
STD-04_09	314	9.2	180	314	16	0.050	0.07
STD-229	252	10.0	193	255	12	0.049	1.05
STD-236	511	24.0	197	521	23	0.045	1.86

Figure 12.5
Dysprosium Results for Standard AVL-H

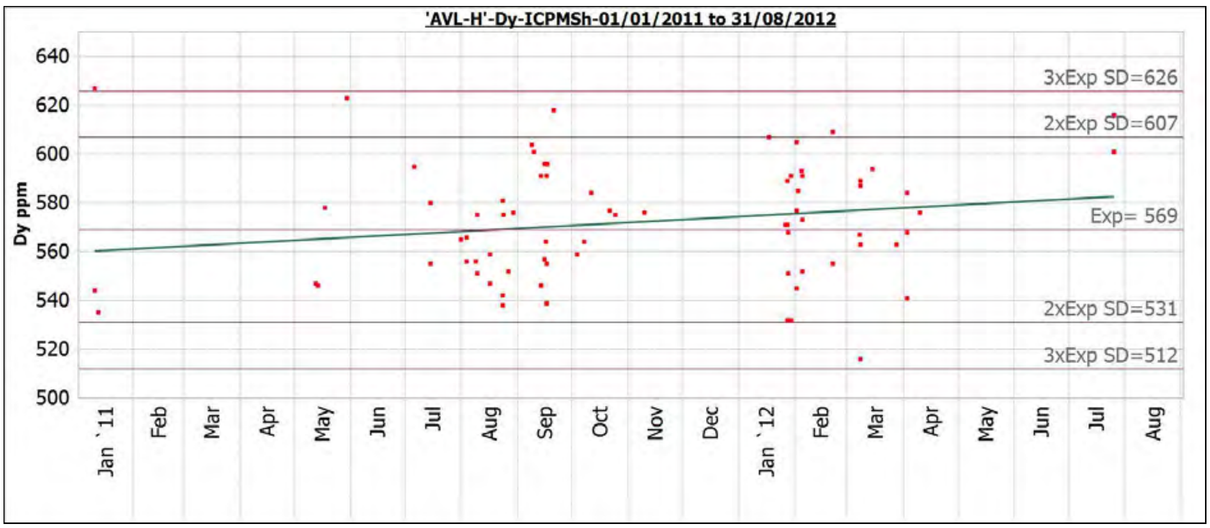


Figure 12.6
Dysprosium Results for Standard STD-04_09

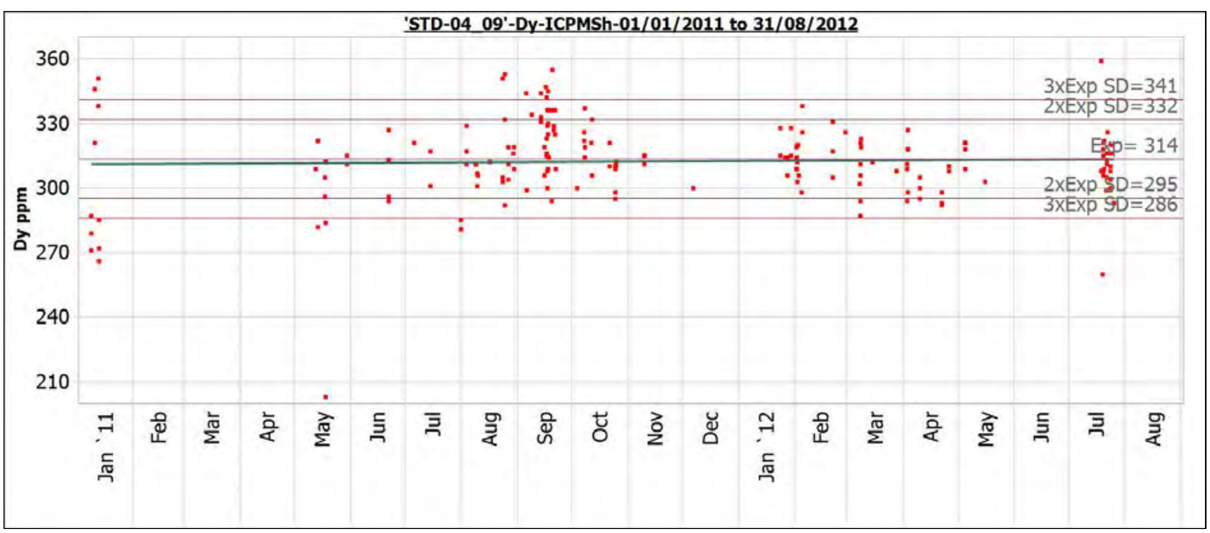


Figure 12.7
Dysprosium Results for Standard STD-229

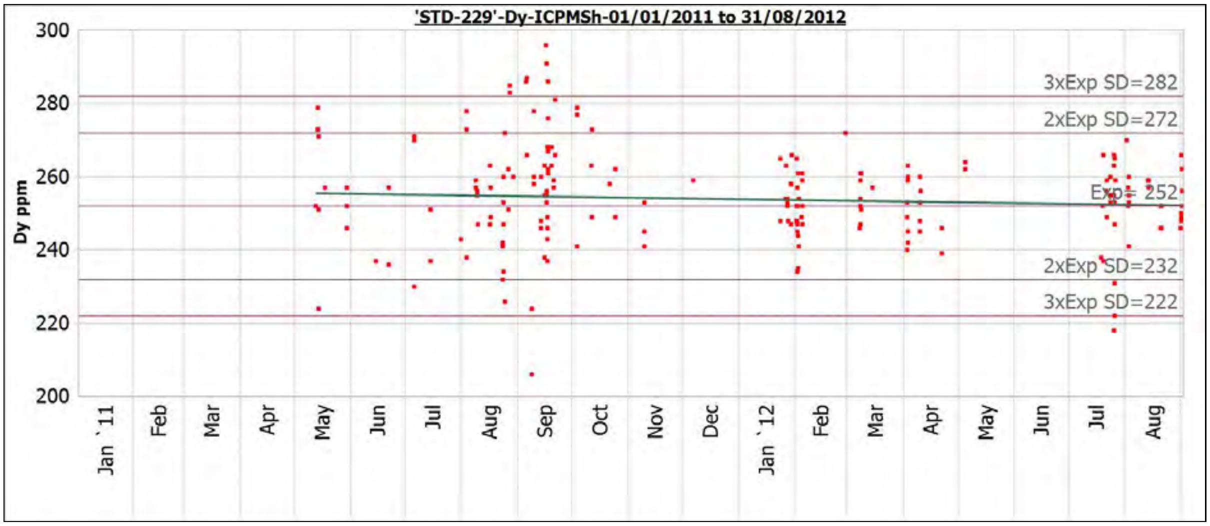
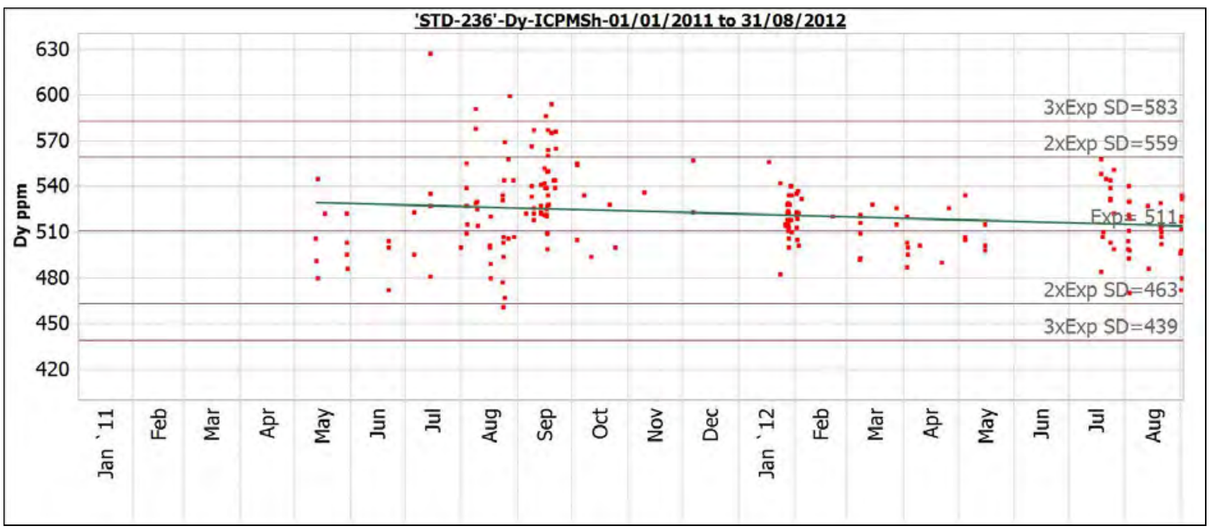


Figure 12.8
Dysprosium Results for Standard STD-236



Europium

Europium is representative of the middle rare earths and is a relatively high value element. The mean bias is between +0.02% and +3.0% for the four standards relative to the certified values. The precision results range from 4.5-4.9% CV and are acceptable. The europium standards results are shown in Table 12.4 and the control charts in Figure 12.9 through Figure 12.12.

Table 12.4
Europium Standards Results

Eu Standard(s)			No. of Samples	Calculated Values			
Standard Code	Value (ppm)	SD (ppm)		Mean Eu (ppm)	SD (ppm)	CV	Mean Bias (%)
AVL-H	73	3.1	77	75	3	0.045	2.82
STD-04_09	109	4.5	177	112	6	0.049	3.00
STD-229	64	3.5	195	65	3	0.045	1.10
STD-236	78	4.0	200	78	4	0.045	0.02

Figure 12.9
Europium Results for Standard AVL-H

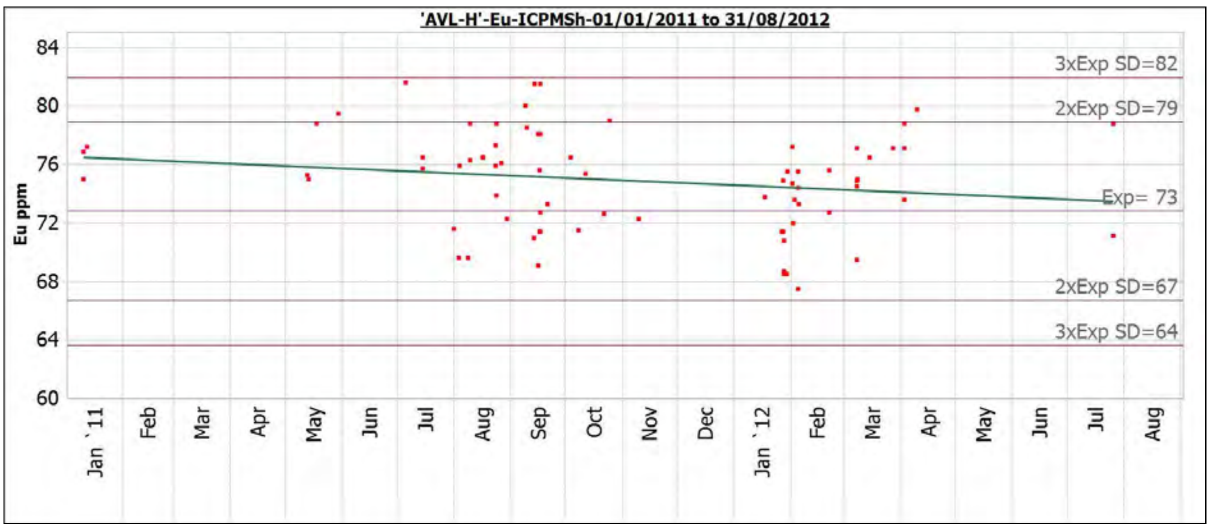


Figure 12.10
Europium Results for Standard STD-04_09

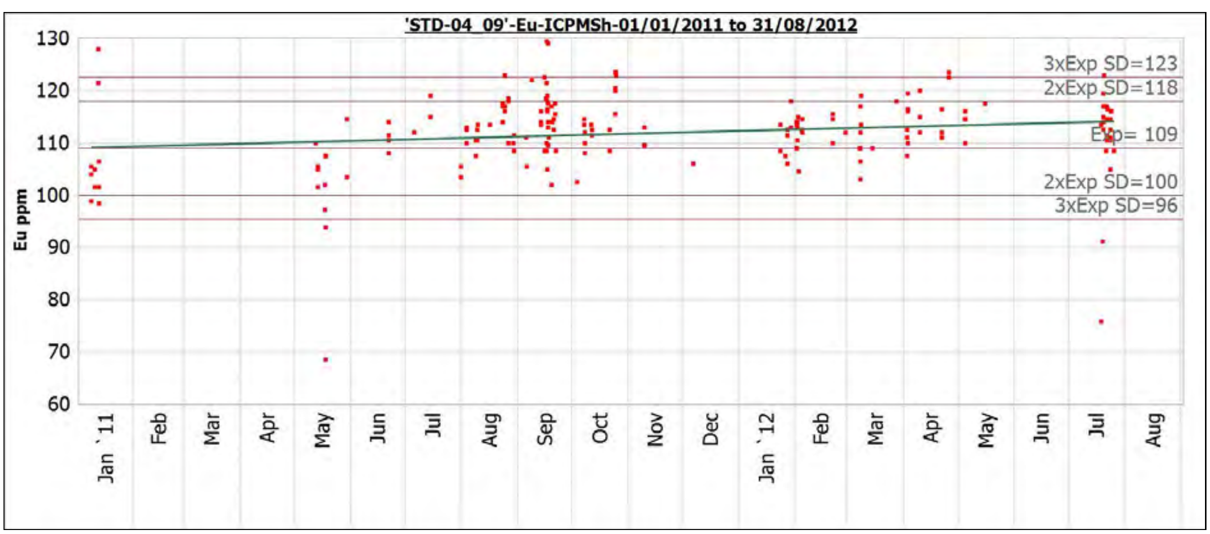


Figure 12.11
Europium Results for Standard STD-229

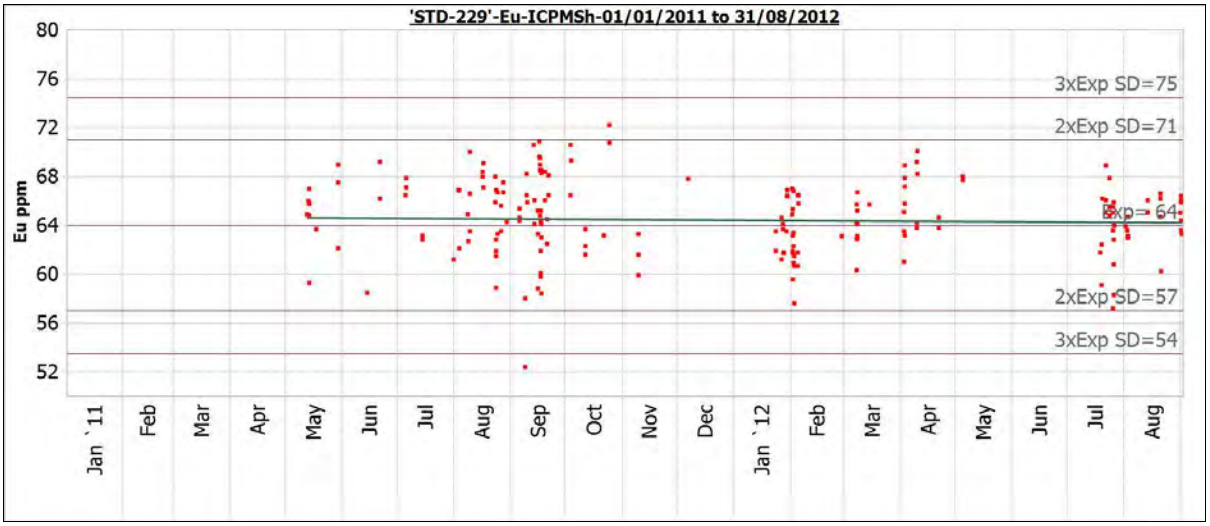
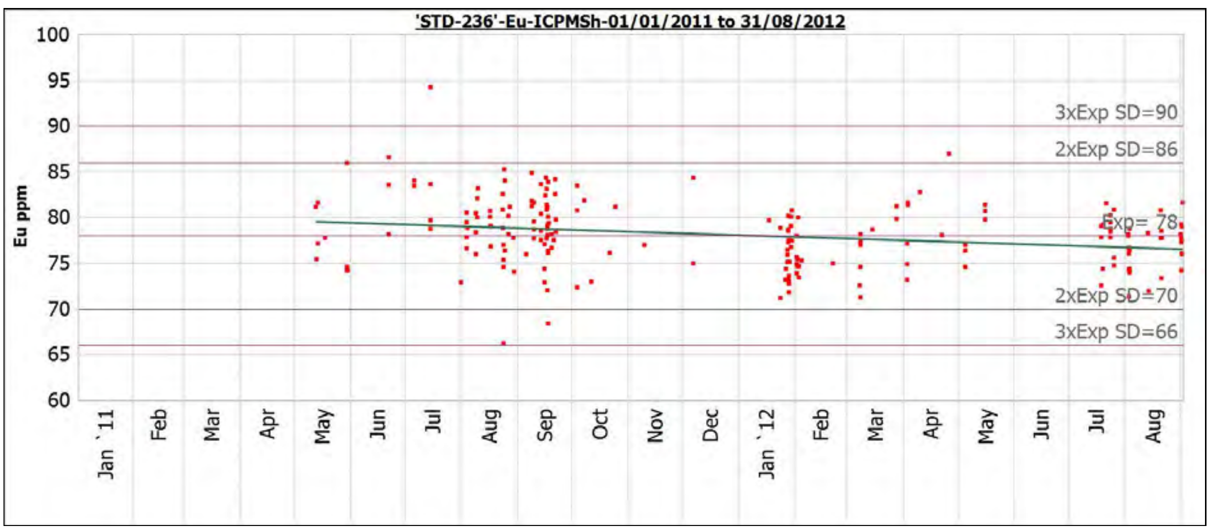


Figure 12.12
Europium Results for Standard STD-236



Niobium (ICP-MS)

Niobium is an economically important non-rare earth element present in the deposit and comprises 7% of the NMR value of the Basal Zone. The overall bias relative to the certified standard ranges from -8.7% to 4.5% for ICP-MS analysis and from 0 to -0.2% for XRF analysis. (XRF analysis is ordered on all over-limit ICP-MS samples.) The precision measured by the CV ranges from 5.5-7.2% for ICP-MS and 1.9-2.0% for XRF. The standards results for niobium ICP-MS are shown in Table 12.5 and the control charts shown in Figure 12.13 through Figure 12.16.

Table 12.5
Niobium ICP-MS Standards Results

Nb Standard(s)			No. of Samples	Calculated Values			
Standard Code	Value (ppm)	SD (ppm)		Mean Nb (ppm)	SD (ppm)	CV	Mean Bias (%)
AVL-H	2858	133.5	74	2738	150	0.055	-4.20
STD-04_09	4323	136.5	159	4156	271	0.065	-3.85
STD-229	2967	142.0	173	2708	196	0.072	-8.74
STD-236	2922	113.5	178	3053	209	0.069	4.49

Figure 12.13
Niobium ICP-MS Results for Standard AVL-H

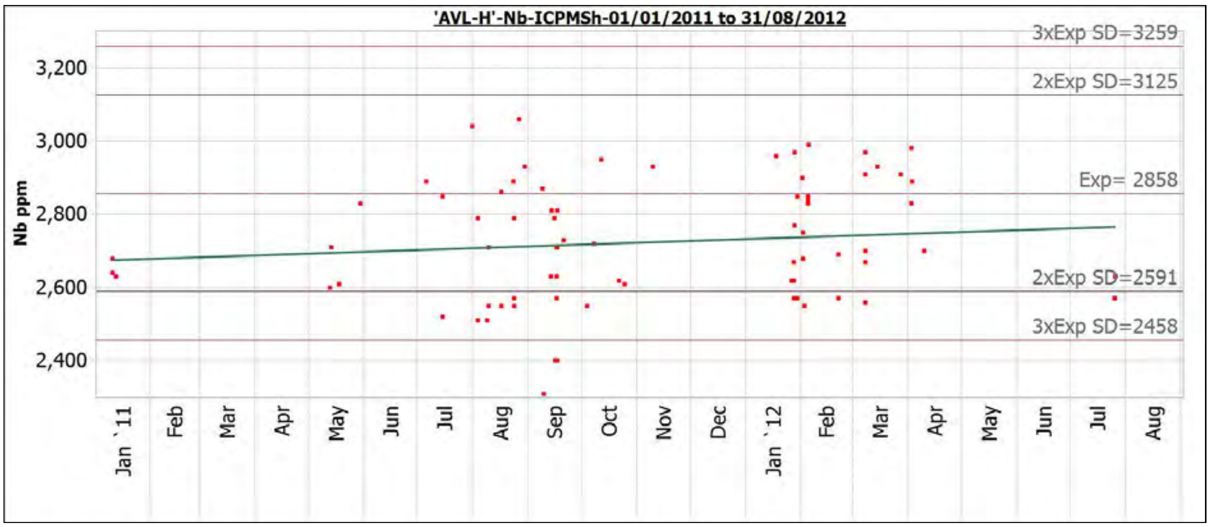


Figure 12.14
Niobium ICP-MS Results for Standard STD-04_09

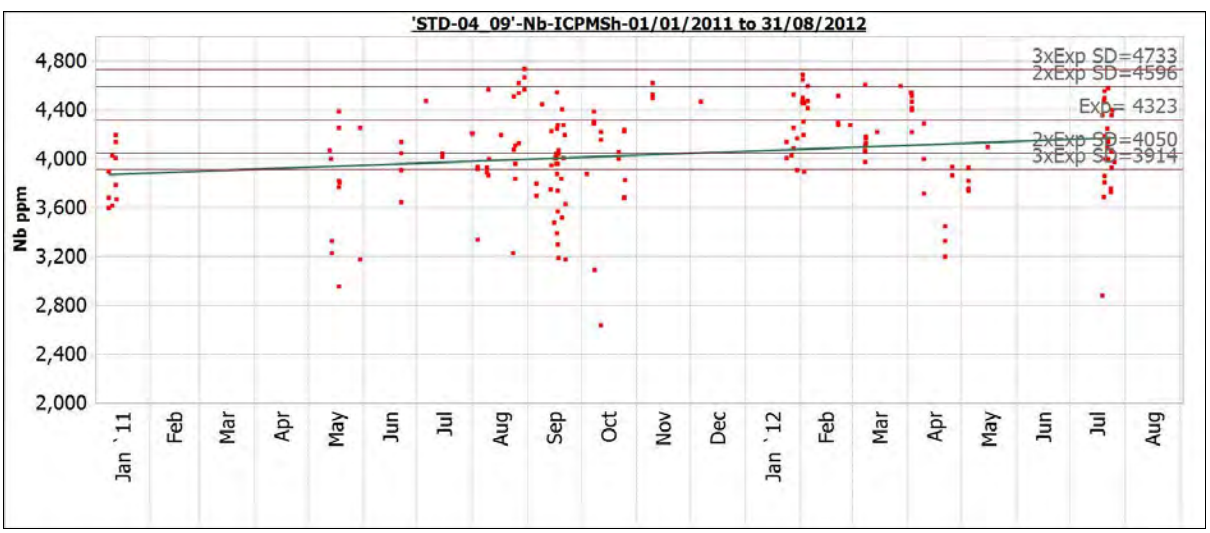


Figure 12.15
Niobium ICP-MS Results for Standard STD-229

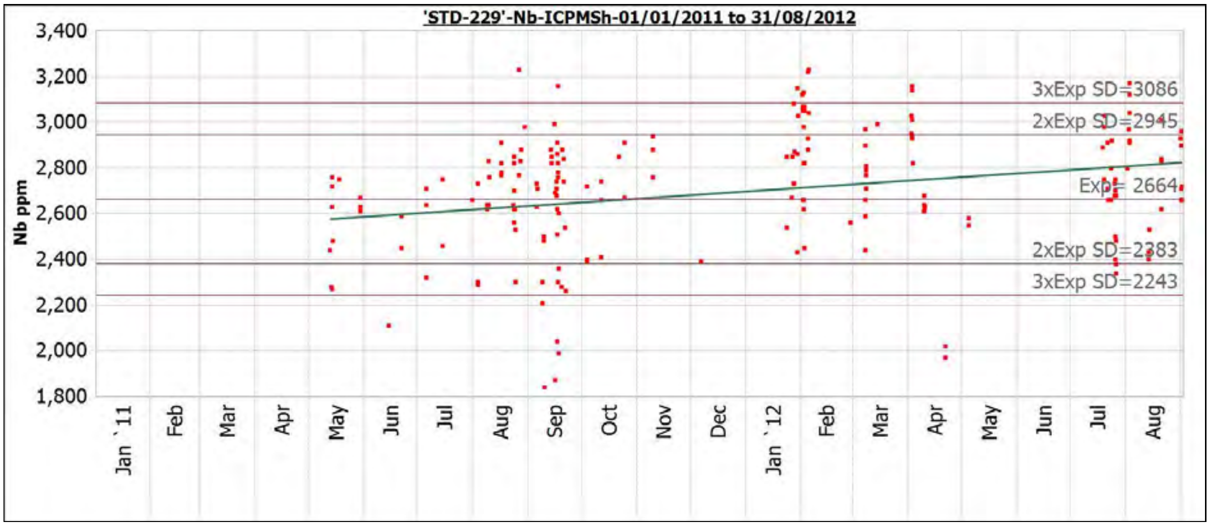
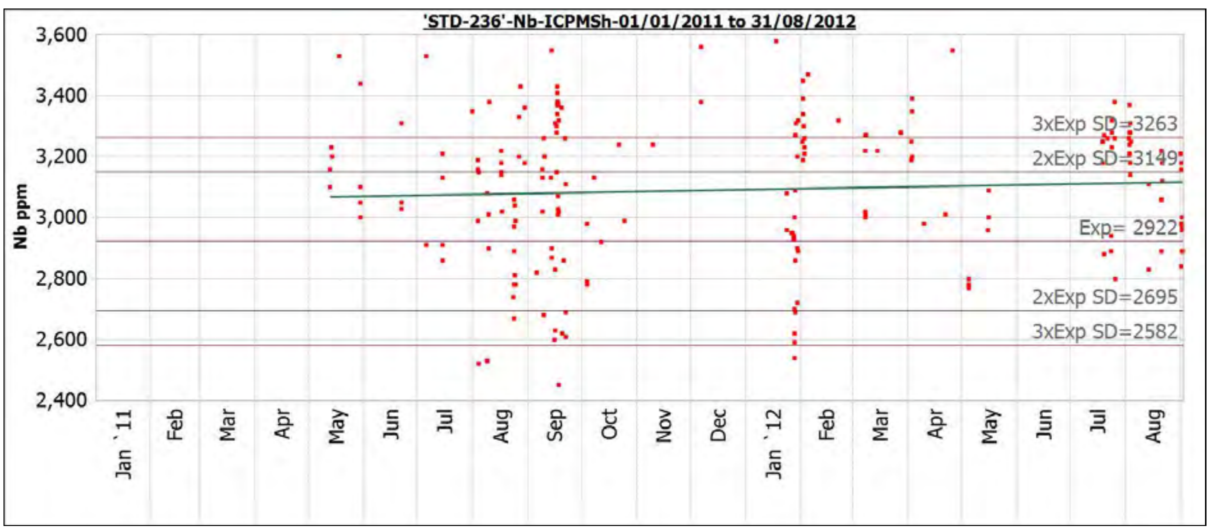


Figure 12.16
Niobium ICP-MS Results for Standard STD-236



Niobium (XRF)

The standards results for niobium XRF are shown in Table 12.6 and the control charts shown in Figure 12.17 and Figure 12.18.

Table 12.6
Niobium XRF Standards Results

Nb Standard(s)			No. of Samples	Calculated Values			
Standard Code	Value (ppm)	SD (ppm)		Mean Nb (ppm)	SD (ppm)	CV	Mean Bias (%)
STD-229	0	0.01	248	0	0	0.021	-0.15
STD-236	0	0.01	274	0	0	0.019	-0.01

Figure 12.17
Niobium XRF Results for Standard STD-229

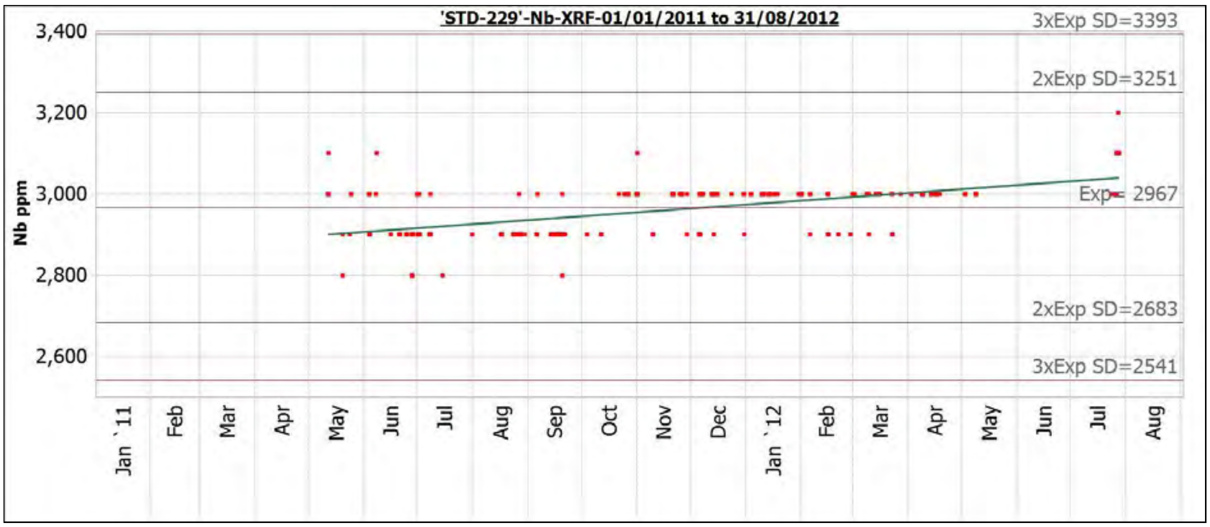
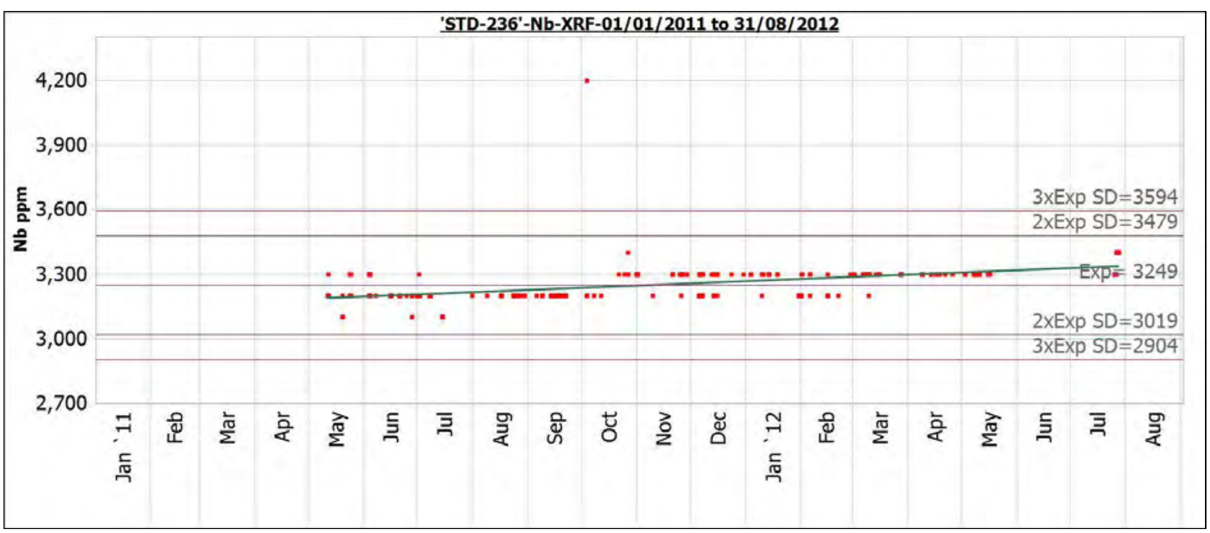


Figure 12.18
Niobium XRF Results for Standard STD-236



Neodymium

Neodymium standards results are shown in Table 12.7 and the control charts are shown in Figure 12.19 through Figure 12.22.

Table 12.7
Neodymium Standards Results

Nd Standard(s)			No. of Samples	Calculated Values			
Standard Code	Value (ppm)	SD (ppm)		Mean Nd (ppm)	SD (ppm)	CV	Mean Bias (%)
AVL-H	3038	104.0	77	3071	146	0.048	1.09
STD-04_09	5350	229.5	175	5510	234	0.043	2.99
STD-229	2340	117.5	194	2427	111	0.046	3.70
STD-236	2586	118.0	197	2697	113	0.042	4.29

Figure 12.19
Neodymium Results for Standard AVL-H

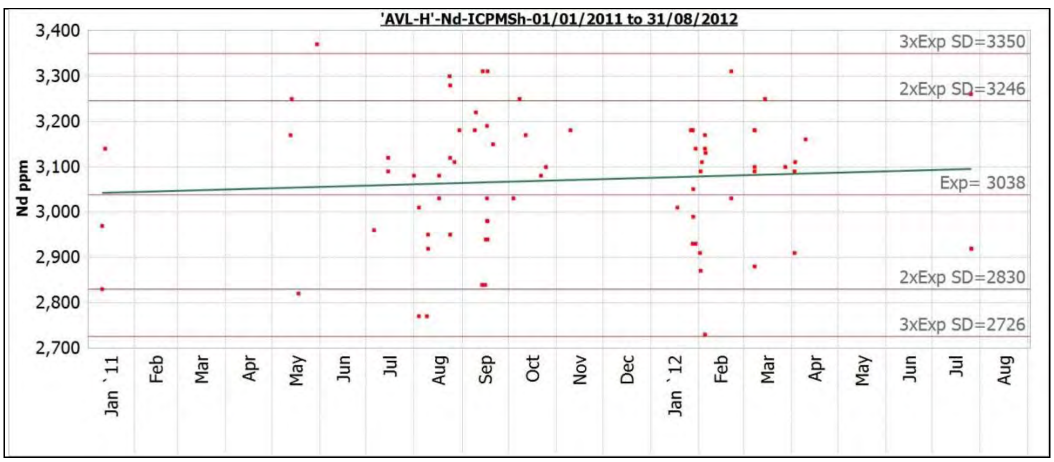


Figure 12.20
Neodymium Results for Standard STD-04_09

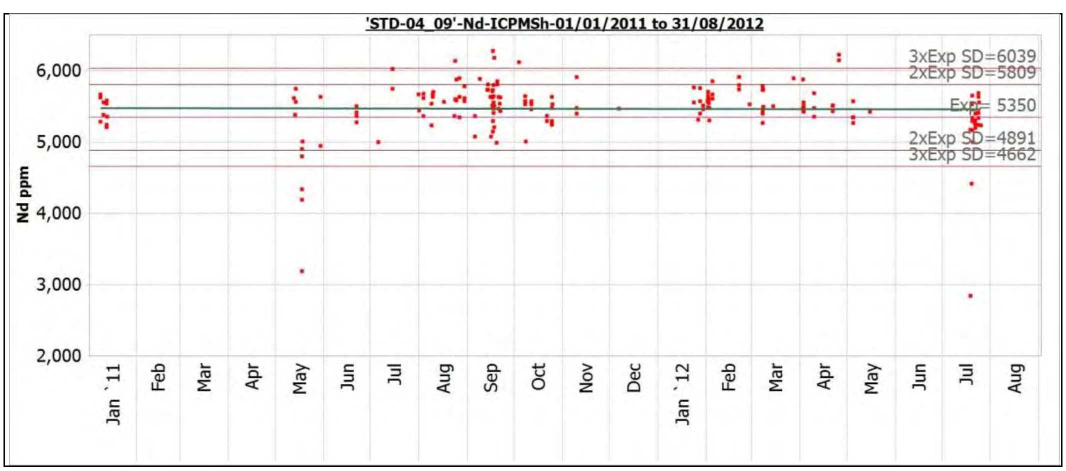


Figure 12.21
Neodymium Results for Standard STD-229

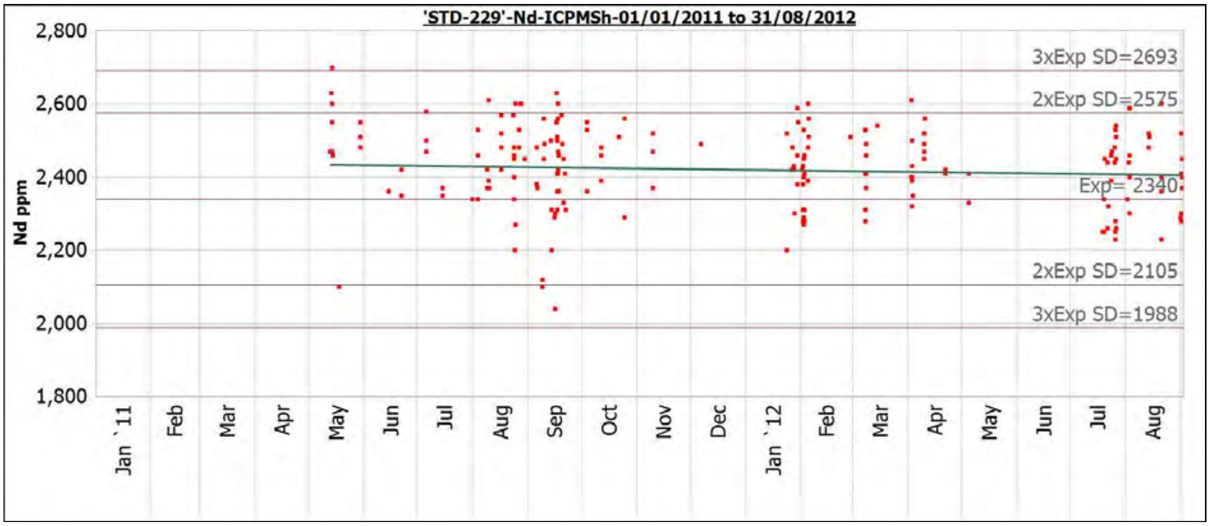
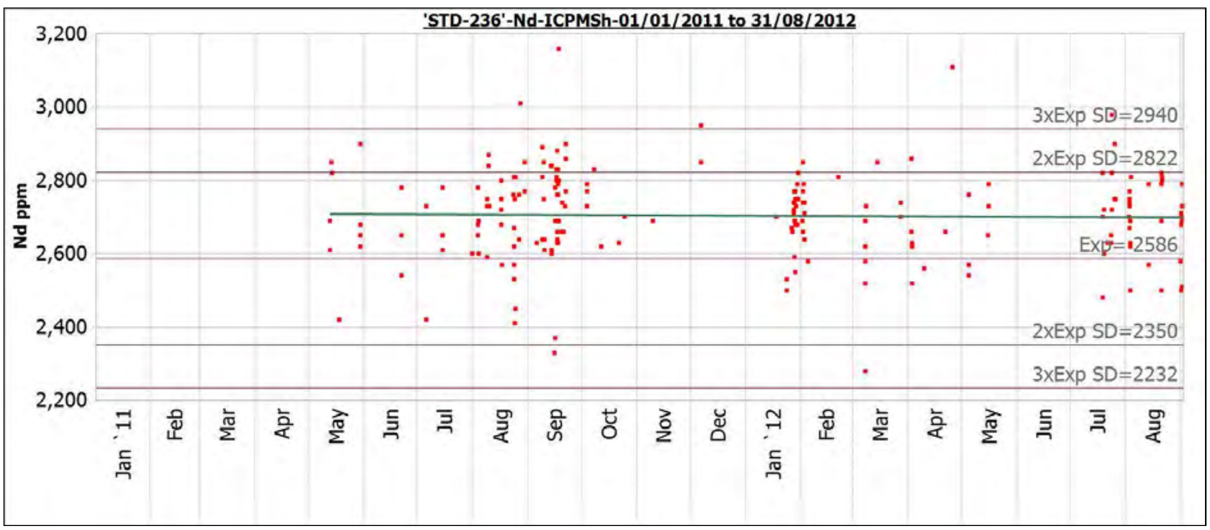


Figure 12.22
Neodymium Results for Standard STD-236



Terbium

Terbium, a high value heavy rare earth, comprising approximately 7% of the NMR of the Basal Zone, demonstrates an overall negative bias of -0.7% to -5.4% relative to the certified standards. The precision is acceptable as indicated by the CV ranging from 4.5% to 5.8%. Terbium standards results are shown in Table 12.8.

Table 12.8
Terbium Standards Results

Tb Standard(s)			No. of Samples	Calculated Values			
Standard Code	Value (ppm)	SD (ppm)		Mean Tb (ppm)	SD (ppm)	CV	Mean Bias (%)
AVL-H	106	5.2	77	100	4	0.045	-5.40
STD-04_09	75	2.9	176	73	4	0.058	-1.81
STD-229	58	2.5	186	57	3	0.052	-2.26
STD-236	94	4.0	196	93	5	0.053	-0.68

Zirconium (ICP-MS)

Zirconium is the most abundant economically significant element in the Nechalacho deposit comprising about 5% of the NMR value of the Basal Zone. The overall bias relative to the certified standard ranges from -2.6% to +8.5% for ICP-MS analyses and -0.1% to 0.3% for XRF analyses. Precision measured by the CV ranges from 4.7% to 5.7% for ICP-MS and 0.8% to 1.1% for XRF. These results highlight the reliability of the XRF method. The standards results for zirconium ICP-MS are shown in Table 12.9 and the control charts shown in Figure 12.23 through Figure 12.25.

Table 12.9
Zirconium ICP-MS Standards Results

Zr Standard(s)			No. of Samples	Calculated Values			
Standard Code	Value (ppm)	SD (ppm)		Mean Zr (ppm)	SD (ppm)	CV	Mean Bias (%)
AVL-H	18650	780.0	75	18157	852	0.047	-2.64
STD-229	20999	3934.5	91	22787	1080	0.047	8.51
STD-236	26014	895.0	190	26910	1520	0.057	3.44

Figure 12.23
Zirconium ICP-MS Results for Standard AVL-H

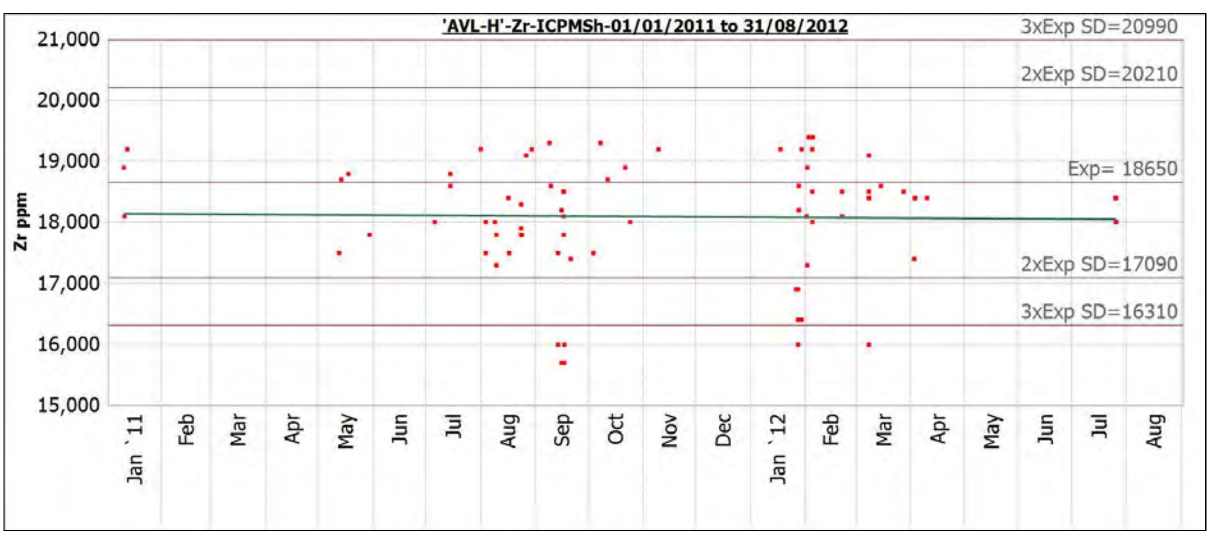


Figure 12.24
Zirconium ICP-MS Results for Standard STD-229

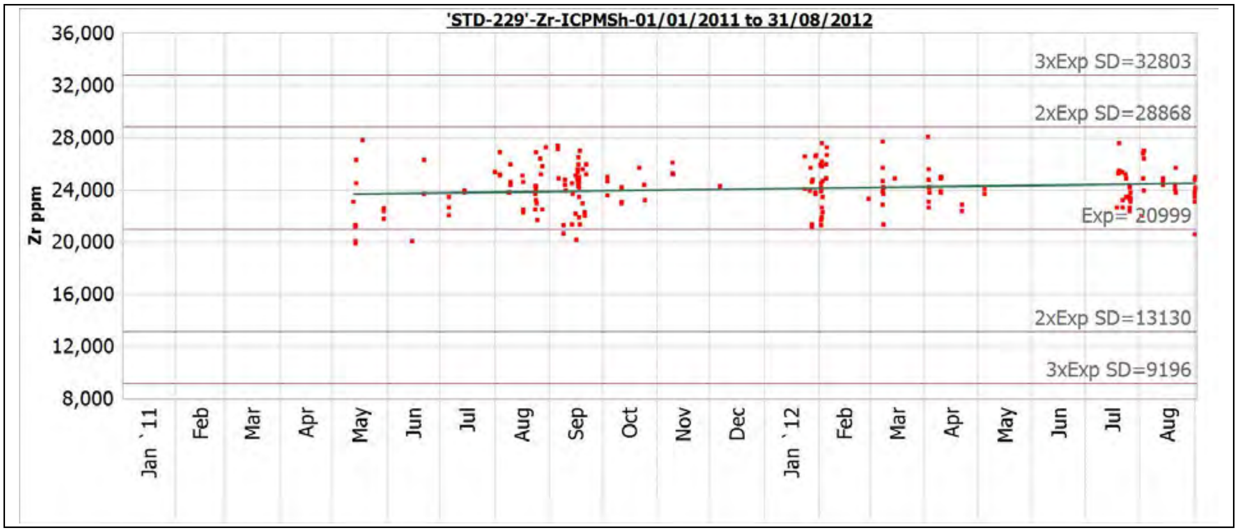
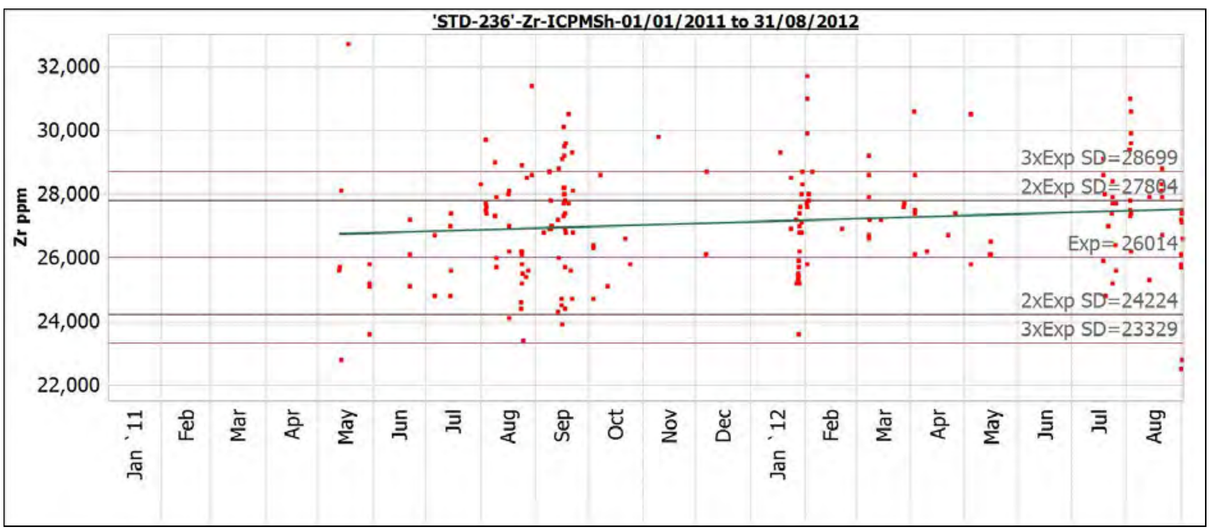


Figure 12.25
Zirconium ICP-MS Results for Standard STD-236



Zirconium (XRF)

Zirconium XRF standards results are shown in Table 12.10 and the control charts shown in Figure 12.26 and Figure 12.27.

Table 12.10
Zirconium XRF Standards Results

Zr Standard(s)			No. of Samples	Calculated Values			
Standard Code	Value (ppm)	SD (ppm)		Mean Zr (ppm)	SD (ppm)	CV	Mean Bias (%)
STD-229	3	0.107	288	3	0	0.009	0.26
STD-236	3	0.045	297	3	0	0.011	-0.14

Figure 12.26
Zirconium XRF Results for Standard STD-229

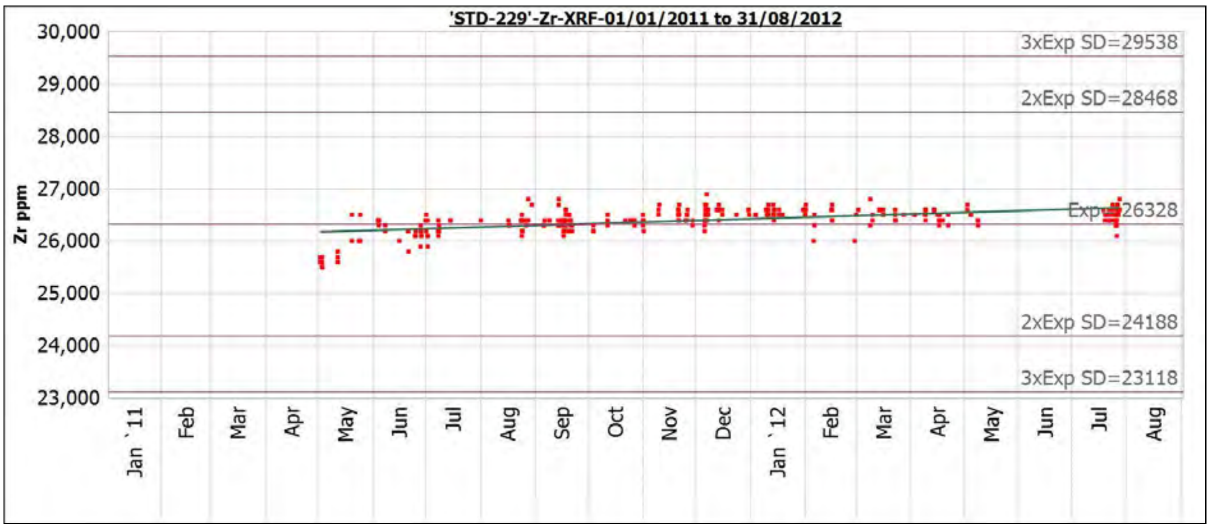
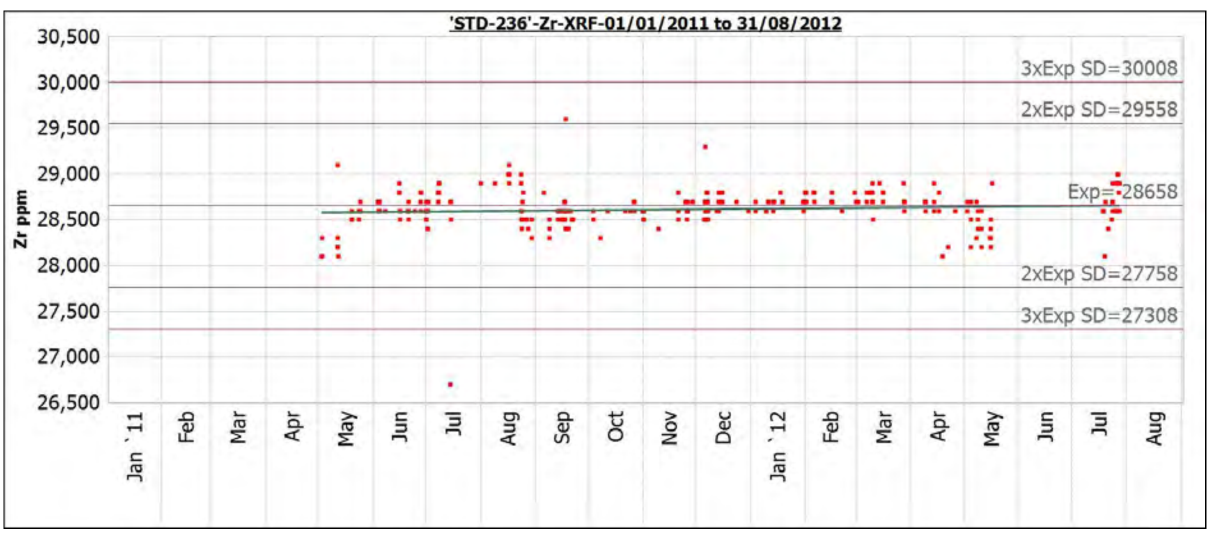


Figure 12.27
Zirconium XRF Results for Standard STD-236



Conclusions on Individual Element Analyses

The results of the analyses suggest an overall slightly high mean bias for most REE with the ALS method compared to the certified standard values. However, the bias is particularly small for neodymium, europium, terbium, dysprosium and lutetium, which are all high value elements. As a result, it is not considered that the net effect is material to the resource estimates. Similarly, the zirconium and niobium data indicate acceptable ranges of bias.

In all cases, except for perhaps standard STD-236 for niobium, the relative standard deviations indicate acceptable precision in the standard analyses.

12.2.1.2 Inter-laboratory Comparison

One sample in 10 was resubmitted to Acme Analytical Laboratories Ltd. (Acme) for an inter-laboratory comparison. The check sample was prepared by ALS Minerals (ALS) as a second pulp from the original coarse crush reject. ALS shipped the samples to Avalon, where Avalon geologists inserted standard samples into the stream and shipped the samples to Acme. The Acme analysis method used was AR_ICPMS.

In the past, it has been observed that Acme tended to give slightly lower results for REE than other laboratories (see comments in Wardrop, 2007). This effect is also observed in the laboratory comparison results for XRF methods.

In Table 12.11 the mean, standard deviation (SD), coefficient of variation (CV), and relative percent half difference (sRPHD) are presented for results from ALS and Acme, respectively.

Table 12.11
Laboratory Comparison Results for All Elements

Element	No. of Samples	Mean 1 (ALS)	Mean 2 (Acme)	SD 1 (ALS)	SD 2 (Acme)	CV 1 (ALS)	CV 2 (Acme)	sRPHD%
La	453	1996.72	1882.70	1153.64	1076.56	0.58	0.57	2.69
Ce	451	4398.52	4184.67	2535.14	2392.85	0.58	0.57	2.24
Pr	453	558.23	518.04	331.16	300.76	0.59	0.58	3.21
Nd	453	2166.97	2069.39	1297.18	1223.40	0.60	0.59	1.97
Sm	453	456.80	422.45	290.66	265.58	0.64	0.63	3.56
Eu	453	52.32	49.91	34.33	32.65	0.66	0.65	2.07
Gd	453	357.29	359.70	257.90	256.03	0.72	0.71	-0.80
Tb	453	48.63	48.37	44.61	43.68	0.92	0.90	0.01
Dy	452	240.93	235.89	258.54	252.82	1.07	1.07	1.05
Ho	453	41.09	38.98	50.96	50.06	1.24	1.28	5.24
Er	453	101.74	96.24	137.84	132.60	1.35	1.38	4.59
Tm	453	13.25	13.25	18.48	18.48	1.39	1.39	-0.44
Yb	453	80.59	81.97	112.45	112.15	1.40	1.37	-3.44
Lu	453	11.37	11.08	15.76	15.32	1.39	1.38	-0.39
Y	453	964.62	914.19	1144.75	1072.82	1.19	1.17	2.22
Zr-ICP-MSh	451	16794.83	16441.79	11635.23	11661.55	0.69	0.71	1.51
Zr-XRF	497	22748.89	20472.55	11023.60	9747.00	0.48	0.48	5.16

Nb-ICP-MSh	452	2045.91	1937.76	1173.36	1158.36	0.57	0.60	2.82
Nb-XRF	228	3645.18	3169.35	1189.18	994.35	0.33	0.31	7.00
Ta	453	217.29	207.36	169.17	157.83	0.78	0.76	1.45
Hf	453	380.31	369.85	274.54	268.91	0.72	0.73	1.47

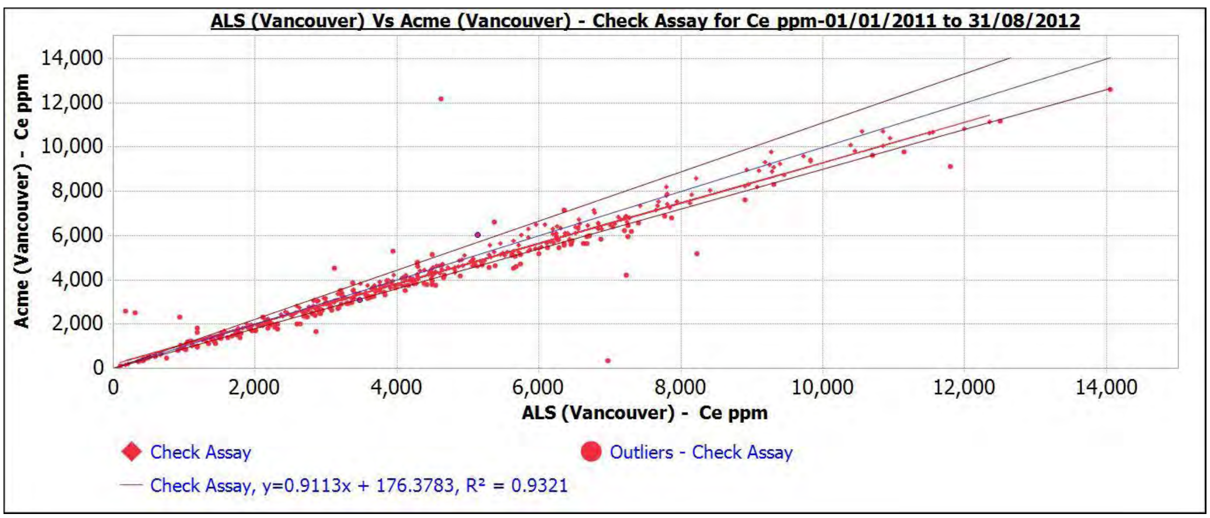
Figure 12.28 through Figure 12.35 display the X-Y plots of the laboratory comparison data with the X=Y trend line (blue), 10% error limits (brown) and a regression line (red). sRPHD is calculated by the formula:

$$\left(100 \times \left(\frac{\text{Original Value} - \text{Repeat Value}}{(\text{Original Value} + \text{Repeat Value}) \div 2}\right)\right) \div 2$$

Cerium

Figure 12.28 gives the laboratory comparison results of ALS method MS81h with Acme. It can be seen that ALS gives results about 2.6% higher than Acme on average. The comparison of ALS with the certified means of the Avalon standards indicates about 5% high bias suggesting that Acme has a 2% low bias compared to the certified reference standards.

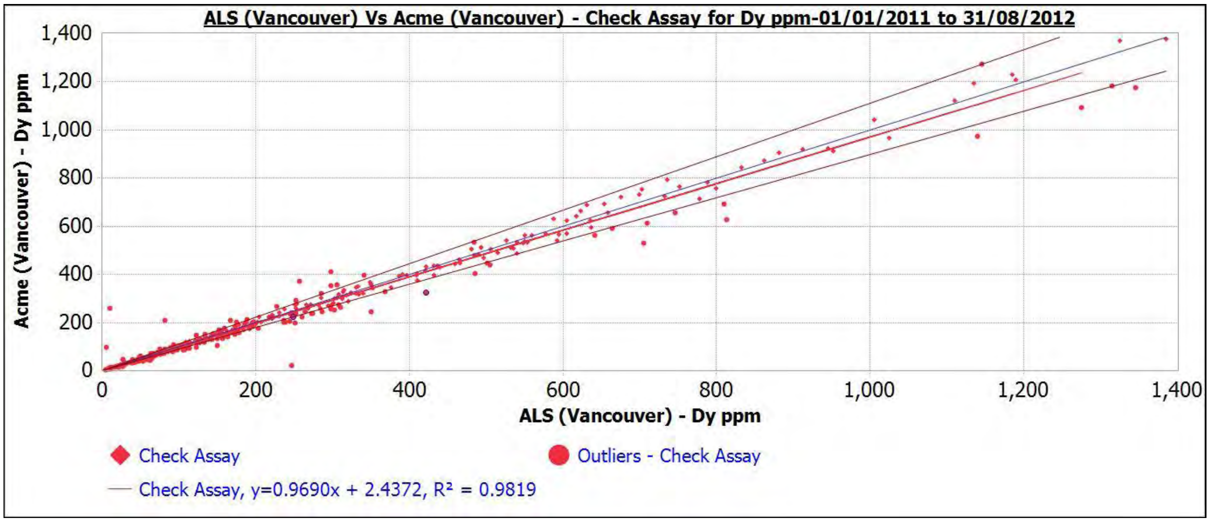
Figure 12.28
Cerium Laboratory Comparison



Dysprosium

Figure 12.29 gives the laboratory comparison results of ALS method MS81h with Acme. The results show near zero bias for dysprosium, an important high-value heavy rare earth. ALS comparison to the certified standards, given above, indicates a +1% positive bias compared to the certified reference standards.

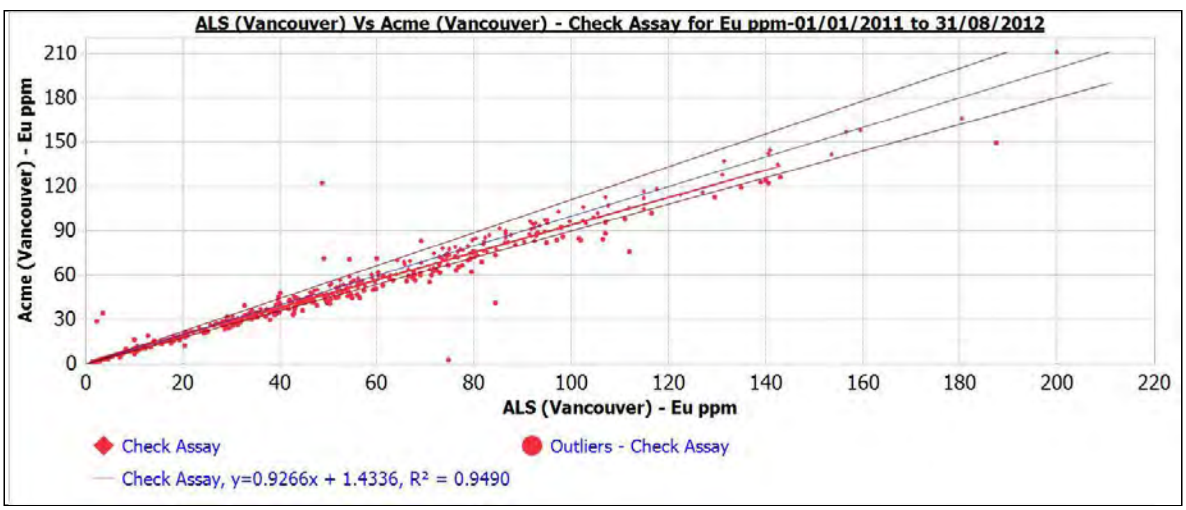
Figure 12.29
Dysprosium Laboratory Comparison



Europium

Figure 12.30 gives the laboratory comparison results of ALS method MS81h with Acme. The results show that ALS gives results about 2% higher than Acme for europium, an important high-value heavy rare earth. ALS comparison to the certified standards, given above, indicates a zero to 3% positive bias compared to the certified reference standards.

Figure 12.30
Europium Laboratory Comparison

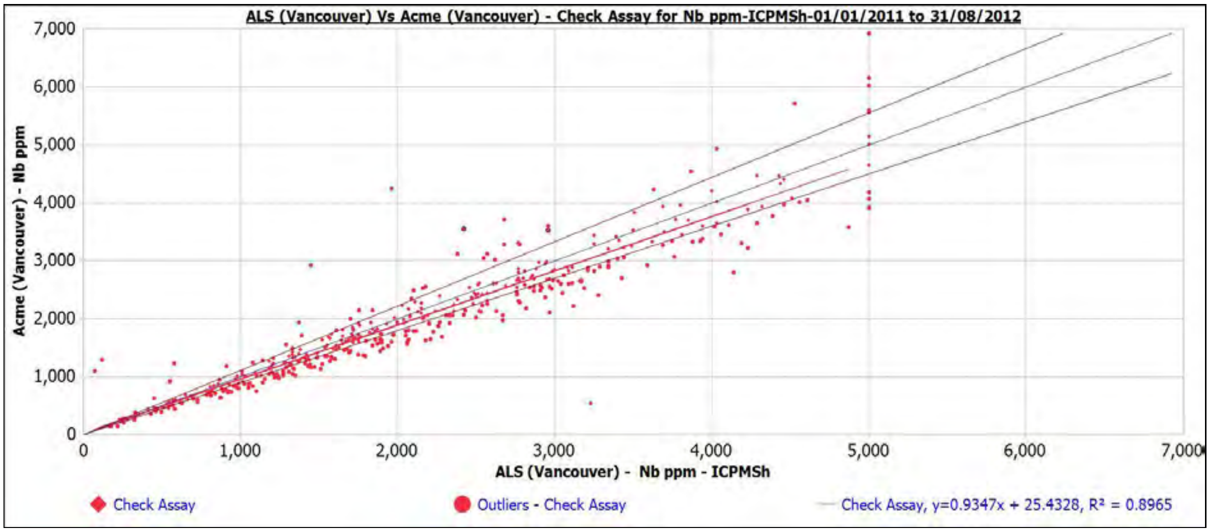


Niobium (ICP-MS)

Figure 12.31 gives the laboratory comparison results of ALS method MS81h with Acme. The results for niobium indicate that ALS results are, on average, about 2.8% higher than Acme.

However, the mean bias for niobium by ALS compared to the certified values for Avalon's standards ranges from -8.7% for standard STD-229 and +4.5% for STD-236, averaging -3% for all standards. Thus, it is judged that the ALS results are acceptable and that the Acme results do not indicate an unacceptable difference.

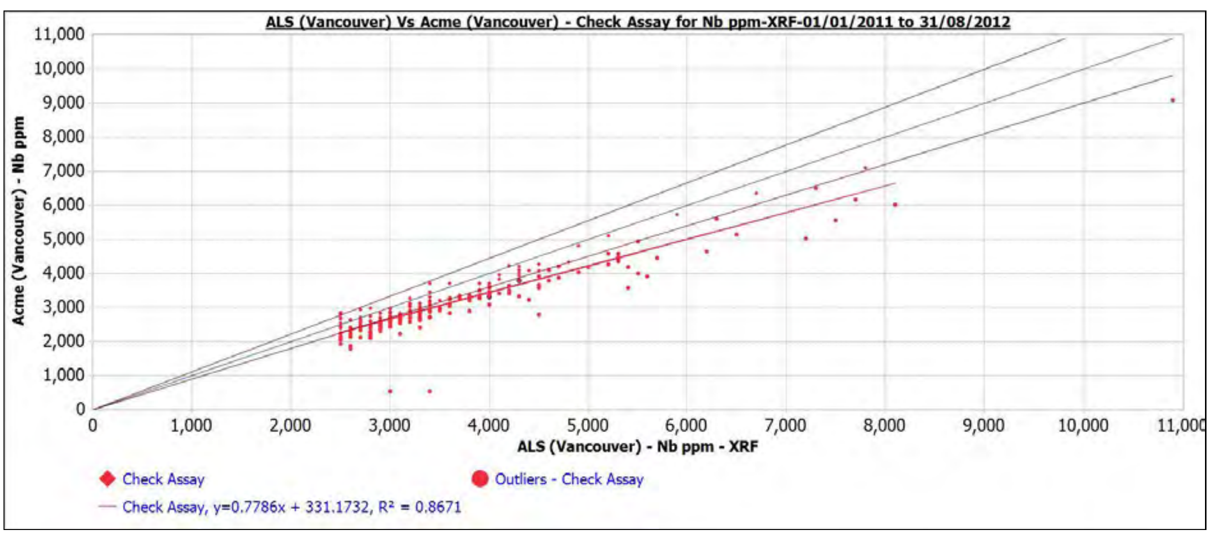
Figure 12.31
Niobium ICP-MS Laboratory Comparison



Niobium (XRF)

The niobium XRF laboratory comparison is shown in Figure 12.32.

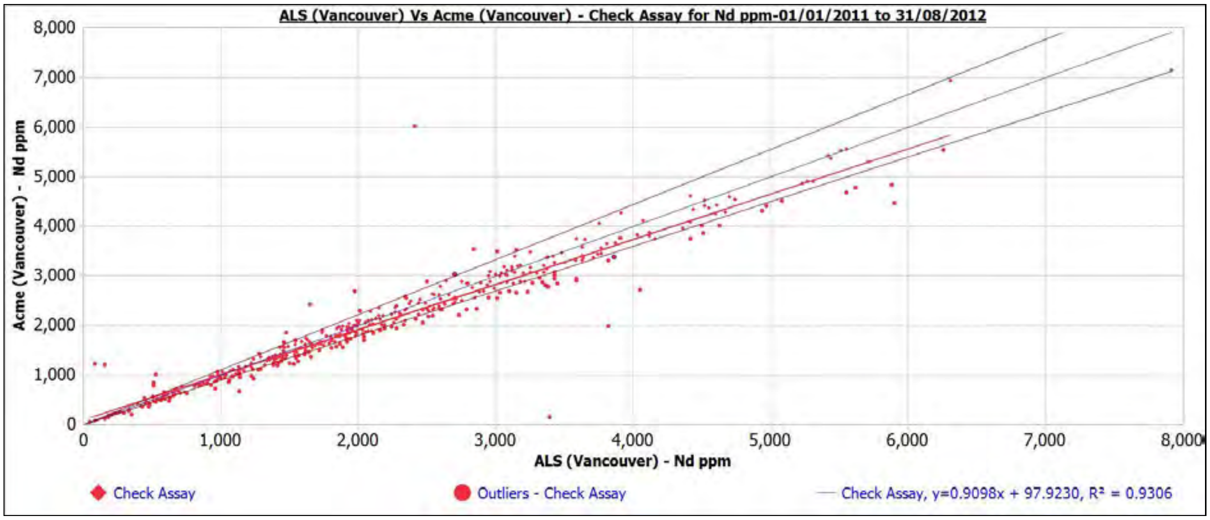
Figure 12.32
Niobium XRF Laboratory Comparison



Neodymium

The neodymium XRF laboratory comparison is shown in Figure 12.33.

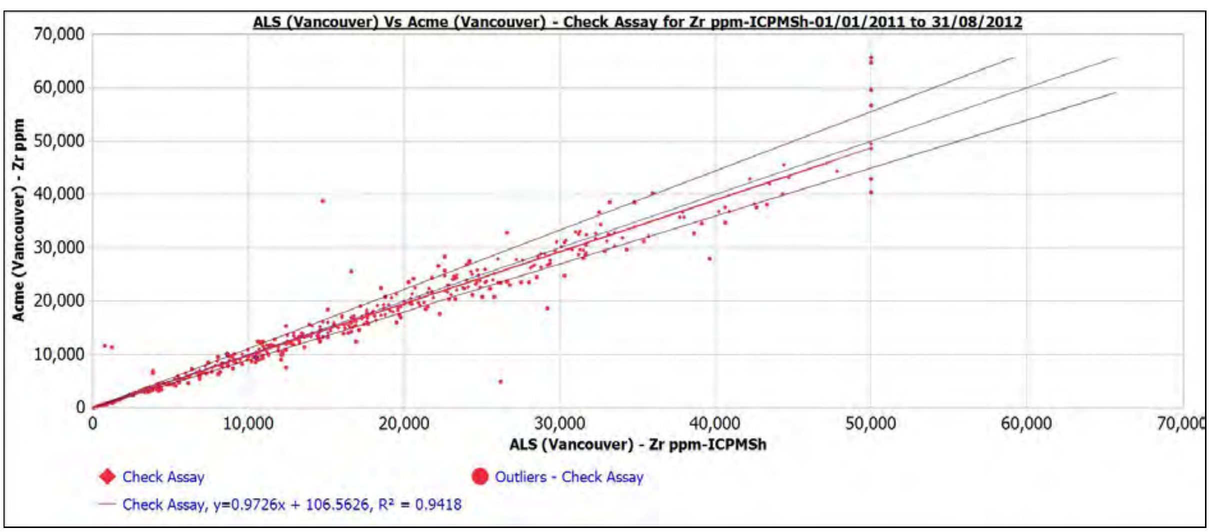
Figure 12.33
Neodymium XRF Laboratory Comparison



Zirconium (ICP-MS)

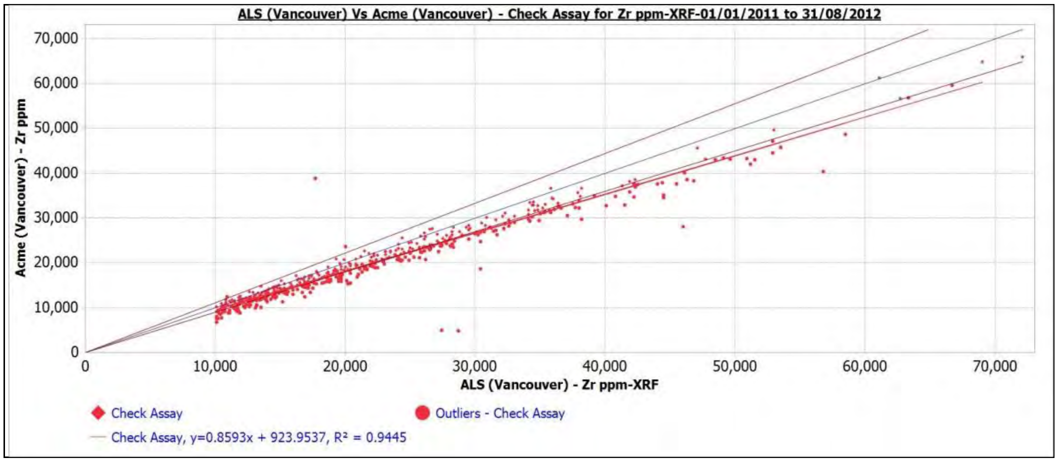
The comparison of Acme and ALS results for zirconium is similar to that for niobium, in this case with Acme XRF results about 5% lower values than ALS. See Figure 12.34 and Figure 12.35.

Figure 12.34
Zirconium ICP-MS Laboratory Comparison



Zirconium (XRF)

Figure 12.35
Zirconium XRF Laboratory Comparison



12.2.1.3 Field Duplicate Analysis

Two core duplicate samples were taken in the field for each hole drilled during the summer 2012 campaign for a total of 87 core duplicate pairs. The methodology comprised analysing a second quarter of core. Key elements cerium, neodymium, europium, dysprosium and zirconium are plotted though data is available for all elements.

In Table 12.12, the mean, standard deviation (SD), coefficient of variation (CV), and relative percent half difference (sRPHD) are presented for the field duplicate results. Figure 12.36 through Figure 12.43 display the X-Y plots of the laboratory comparison data with the X=Y trend line (blue), 10% error limits (brown) and a regression line (red).

The results are close to ideal and the overall results in the ore-grade range are acceptable. A consistent slight negative bias is observed in the data indicating the duplicate samples return slightly higher values. The cause is currently unknown, but sampling protocol will be reviewed before the next drilling campaign.

Table 12.12
Field Duplicate Results for All Elements

Element	No. of Samples	Mean 1 (Primary)	Mean 2 (Duplicate)	SD 1 (Primary)	SD 2 (Duplicate)	CV 1 (Primary)	CV 2 (Duplicate)	sRPHD%
La	87	2005.39	2050.37	1307.31	1258.53	0.65	0.61	-2.65
Ce	87	4235.55	4463.76	2903.34	2822.36	0.69	0.63	-4.53
Pr	87	534.66	547.99	367.72	359.49	0.69	0.66	-2.84
Nd	87	2023.44	2073.39	1485.74	1442.35	0.73	0.70	-3.22
Sm	87	418.89	423.20	345.79	340.34	0.83	0.80	-2.22
Eu	87	49.25	47.43	42.22	40.80	0.86	0.86	0.54
Gd	87	344.50	347.18	331.57	324.67	0.96	0.94	-2.44

Element	No. of Samples	Mean 1 (Primary)	Mean 2 (Duplicate)	SD 1 (Primary)	SD 2 (Duplicate)	CV 1 (Primary)	CV 2 (Duplicate)	sRPHD%
Tb	87	48.57	47.88	55.90	54.35	1.15	1.14	-1.23
Dy	87	250.09	247.18	326.57	321.22	1.31	1.30	-0.52
Ho	87	44.20	43.98	63.64	62.26	1.44	1.42	-1.34
Er	87	112.46	111.36	168.88	164.94	1.50	1.48	-0.82
Tm	87	15.37	15.10	22.54	22.10	1.47	1.46	0.66
Yb	87	89.70	88.20	128.44	125.67	1.43	1.42	1.06
Lu	87	12.35	12.25	16.98	16.53	1.38	1.35	-0.55
Y	87	1031.99	1020.86	1424.10	1400.70	1.38	1.37	0.41
Zr-ICPMS	87	19542.18	19505.06	14951.21	14636.86	0.77	0.75	-0.42
Zr-XRF	41	23304.88	24056.10	14259.24	13799.26	0.61	0.57	-2.35
Nb-ICPMS	87	2459.16	2560.48	1284.53	1259.70	0.52	0.49	-2.97
Nb-XRF	25	4208.00	4260.00	1556.13	1662.29	0.37	0.39	-0.30
Ta	87	257.27	259.46	211.57	208.92	0.82	0.81	-1.20
Hf	87	492.46	478.67	375.83	355.92	0.76	0.74	0.60

Figure 12.36
Cerium Field Duplicate Samples

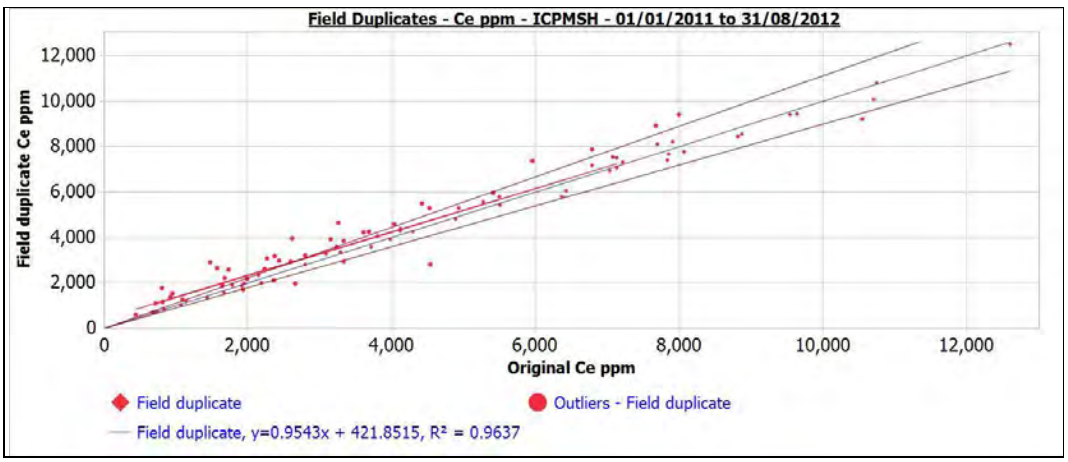


Figure 12.37
Dysprosium Field Duplicate Samples

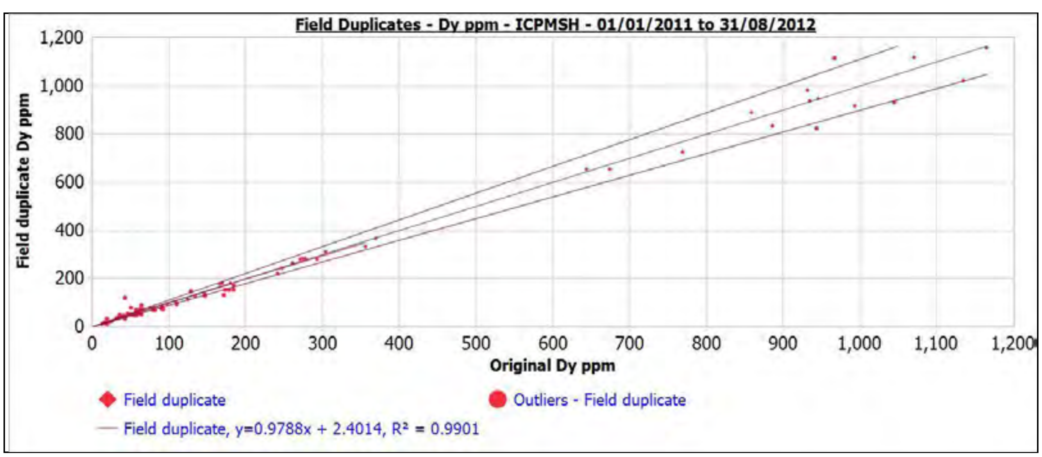


Figure 12.38
Europium Field Duplicate Samples

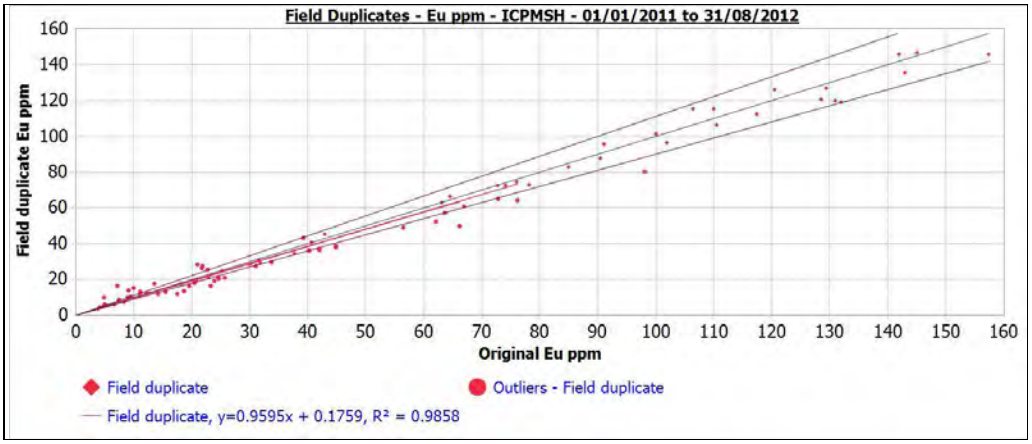


Figure 12.39
Niobium (ICP-MS) Field Duplicate Samples

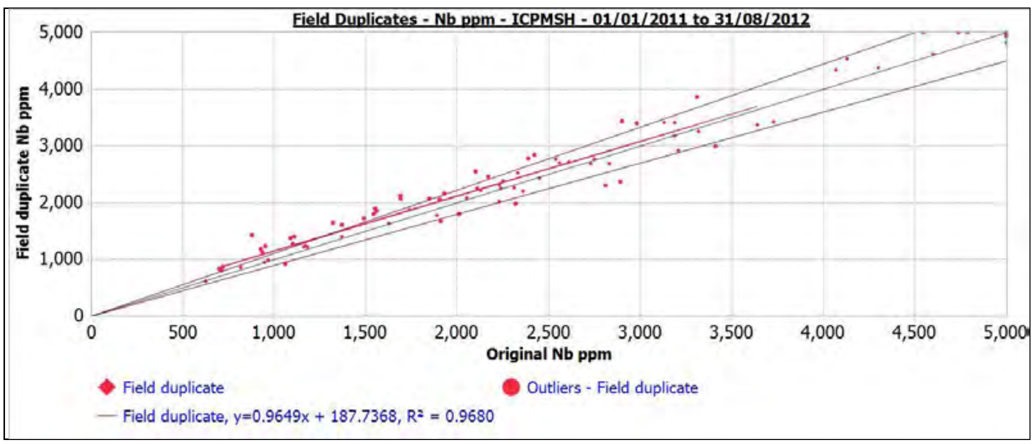


Figure 12.40
Niobium (XRF) Field Duplicate Samples

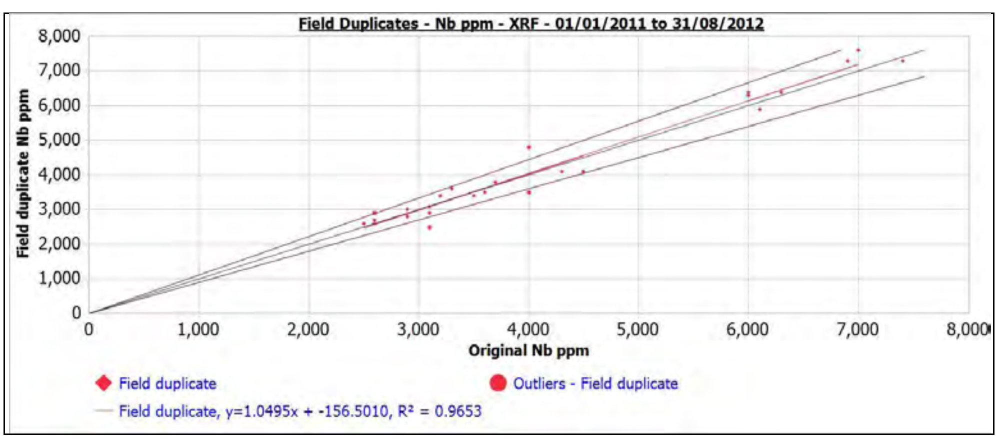


Figure 12.41
Neodymium Field Duplicate Samples

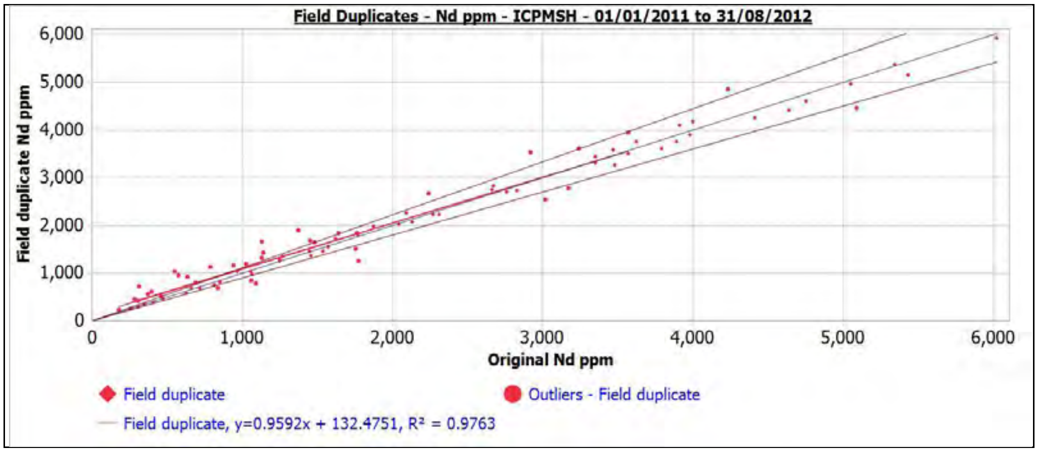


Figure 12.42
Zirconium (ICP-MS) Field Duplicate Samples

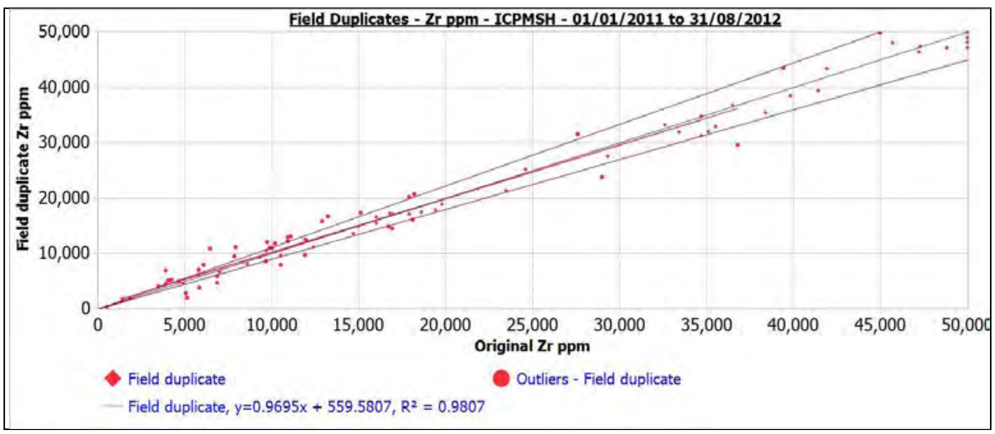
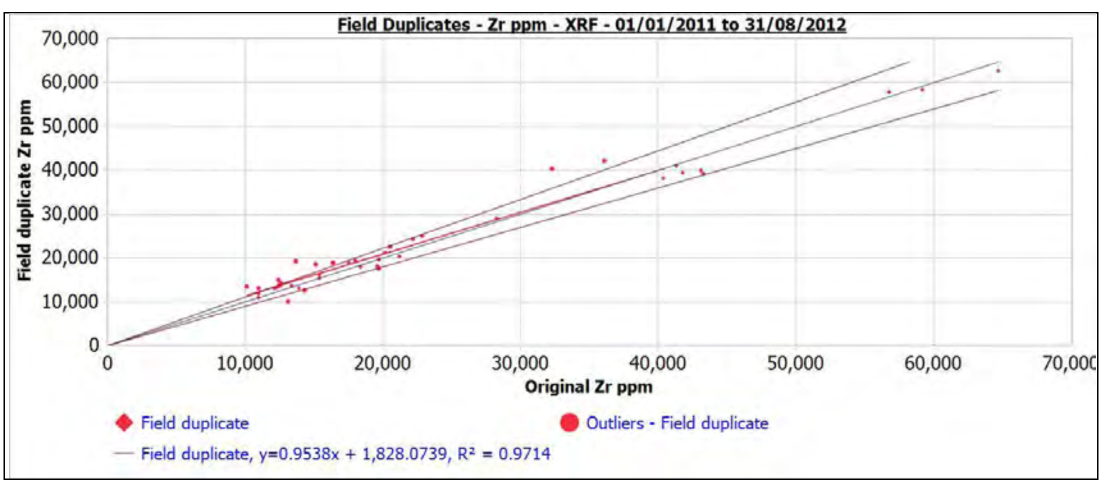


Figure 12.43
Zirconium (XRF) Field Duplicate Samples



12.2.2 General QA/QC Procedures

Avalon monitors the results of its internal standards during routine analysis of drill core. Due to the large number of elements involved, being 15 rare earth elements and three rare metals (niobium, tantalum and zirconium), it would be impractical to apply a normal logic table of failures where an analysis batch is failed on the basis of issues with one element. In addition, all core is analyzed and so about two-thirds of the samples that have chemical data are not significant in terms of the economics of the deposit. As a result, Avalon followed the following procedure for assessing analytical data:

- Batches were not failed if the samples analyzed were clearly far below any economic levels (not mineralized), unless the standards results were very grossly out of the acceptable range.

- The results of the standards were reviewed to see how many elements were out of the acceptable range as recommended in the standard certification, and if four elements were out of range (greater than three standard deviations), but two high and two low, and the remaining 14 elements were in range, the batch was accepted.

- If five or more elements were out of the acceptable range (greater than three standard deviations), and all in the same direction, either biased all high or all low, then the batch was re-analysed.

- The focus on evaluating the performance with respect to individual elements focused on those contributing the highest value to the deposit – neodymium, europium, terbium, dysprosium and niobium.

There were a few cases of blanks being out of the acceptable range. However, on close examination of the results, these were almost invariably clearly associated with sample switching, and it was clear that a mineralized sample and the blank had been switched. A few cases of standards being switched with core samples were also identified.

There was a noticeable reduction in the number of cases of standards being out of range in the case of ALS method MS81h compared to MS81.

The overall conclusions of the QA/QC work completed are as follows:

- Standard deviations of duplicate analyses on standards indicate that the precision of the laboratories is satisfactory, both for ALS and Acme.

- Duplicate analyses of standards and the duplicate reject analyses indicate that there is little systematic bias between ALS and Acme.

- ALS and Acme indicate means for the four standards utilized by Avalon within 5% of the accepted values for most rare earth elements, niobium and tantalum. Zirconium shows more deviation, with ALS MS81h and Acme Method 4A/4B being higher, as are the XRF analyses completed at ALS.

- Although there may be systematic differences between ALS and Acme for individual rare earths, in general, the TREE and HREE indicate that this difference is about 1% for TREE and 1.5% for HREE.

- Given the general agreement between laboratories on the mean of the standards, and the low standard deviation of duplicate analyses, Avalon considers that the standards are acceptable for QA/QC monitoring of the drill core analyses.

- In conclusion, the drill core analyses are considered sufficiently reliable for resource estimation purposes, with the caveat that analyses for zirconium require further investigation to establish the cause of the difference between the various methods.

- However, Avalon and RPA do not consider that the variation in zirconium analyses is material to the resource given the low proportionate value of zirconium relative to all other metals (rare earths and niobium).

13.0 MINERAL PROCESSING AND METALLURGICAL TESTING

Extensive metallurgical testwork has been completed at a number of different laboratories over the last four years. A large number of testwork reports have been issued to summarize this work, however, the feasibility study only presents that work used to generate final process design information (although all test reports are included in the appendices to the feasibility study).

The 15 lanthanide elements are divided into two groups. The ‘light’ rare earth elements (LREE) are those with atomic numbers 57 through 62 (lanthanum to samarium) and the ‘heavy’ rare earth elements (HREE) from 63 to 71 (europium to lutetium). Yttrium has properties more similar to the heavy lanthanides and is included within this group.

13.1 METALLURGICAL SAMPLES

Much of the pertinent metallurgical and mineralogical development studies have been undertaken using bulk composite samples that represent the Nechalacho deposit mineralization spatially and in terms of lithology. These selected composite samples tended to be selected to represent mineralization at different depths in the deposit in terms of elevation. The composites designated UZ were from Upper Zone mineralization and BZ were from Basal Zone mineralization.

Since 2010, Avalon has completed the following series of flotation pilot plant tests:

- 2010: Two mini-pilot plant tests (MPP) at Xstrata Process Support (XPS) using about 1 t of sample each.

- August 2011: Pilot plant at SGS Minerals, Lakefield, Ontario (SGS-M) using about 3.7 t of sample comprising one UZ and one BZ composite.

- February 2012: Pilot plant test program at SGS-M using approximately 40 t of sample, two UZ and two BZ composites.

- October 2012: MPP at XPS using about 2 t of sample from the BZ (two composites from deepest areas of the BZ zone).

All of these pilot plants were conducted using bulk samples sourced from drill core.

The 40 t pilot plant sample selected for the 2012 pilot plant operated by SGS-M comprised four main composites designated UZ-LG, UZ-AG, BZ-AG and BZ-MP. The UZ-LG and UZ-AG composites are from similar levels within the deposit, with the UZ-LG being low grade and the UZ-AG being similar to the average resource grade. The BZ-AG composite represents the upper half of the BZ and the BZ-MP composite the lower half of the BZ.

Subsequent testing undertaken during October, 2012 used samples designated as BZ-Q3and BZ-Q4, which originated from the deepest areas of the BZ-MP zone, where BZ-Q3 and Q4 are the upper and lower quarters, respectively, of the deeper half of the BZ.

13.2 MINERALOGY

The mineralogy of the Nechalacho deposit has been studied by SGS-M using QEMSCAN®, a scanning electron microscope (SEM) and an electron microprobe (EMP). A second major body of work continues to be conducted by a team under Professor Anthony Williams-Jones of McGill University, Montreal.

Nechalacho mineralization is hosted in nepheline syenite that has been extensively hydrothermally altered in areas of mineralization. Thus, the gangue mineralogy of the deposit comprises the nepheline syenite minerals along with alteration products, many of which are fine grained.

The payable elements of the Nechalacho deposit are typically hosted in a number of minerals, summarized as follows:

- LREEs dominantly occur in bastnaesite, synchisite, monazite and allanite.
- HREEs dominantly occur in zircon, fergusonite and rare xenotime.
- Zirconium (Zr), along with HREE, niobium (Nb) and tantalum (Ta) occurs in zircon and other zircono-silicates (eudialyte).
- Nb and Ta occur in columbite and ferrocolumbite, fergusonite and zircon.

The mineralogy of the Nechalacho ore is complex and guides metallurgical development and performance. Consequently, it is critical to have a comprehensive understanding of the minerals, their textures and compositions relative to position within the deposit.

The payable elements are zoned within the deposit with heavy rare earth oxides (HREO), Nb and Ta, relative to the light rare earth oxide (LREO) content, tending to increase with lower depths in the deposit. In the BZ, an almost continuous layer of mineralization at the base of the deposit, typically HREO makes up more than 15% of total rare earth oxides (TREO). The lower parts of the BZ have gradually increasing HREO contents; two individual 2-m interval drill core samples achieved HREO levels of over 50% of TREO

Above the BZ, the UZ comprises a zone of less continuous mineralization, with lens-like layers of higher grade TREO-Zr-Nb-Ta contained within an anomalous background rare earth content. The HREO content can be as low as 6% of TREO in the highest levels of the deposit.

Relative distribution of ore minerals shows very clear trends. The zircon content increases towards the bottom of the mineralization, in the UZ-AG, BZ-AG and BZ-MP. Fergusonite is the other ore mineral that shows the greatest relative increase from UZ to BZ, more than doubling in relative abundance. The columbite abundance drops in sympathy, presumably indicating a partial shift in the host for Nb and Ta from columbite in the UZ to fergusonite in the BZ. The LREE-bearing minerals also show clear trends, as allanite and monazite decrease from top to bottom in the deposit and bastnaesite increasing. Synchysite does not show a clear trend.

Detailed investigations into the textures of the zircon and zircon-silicates show particularly complex textures that suggest complex multistage origins.

Extensive mineralogical investigation of the deposit using QEMSCAN® and EMPA has been completed to aid in predicting or explaining metallurgical performance during the programs of flowsheet development testwork. The QEMSCAN® examination of the pilot plant composites demonstrated little change in overall grain size with depth. There was no statistically significant trend between the four composites, with the median grain diameters of about 12 and 11 micrometres for gangue and ore, respectively.

QEMSCAN® grain liberation analysis completed on the 40 t pilot plant composites suggested a general trend of steadily decreasing proportion of liberate and free (lib+free) minerals with depth, the UZ samples tended to show superior lib+free and complex grain characteristics than the BZ samples. However, there were increases in the proportion of lib+free mineral particles in the BZ-MP compared to the BZ-AG for bastnaesite, fergusonite, columbite, allanite and synchysite.

13.3 TESTWORK – MINERAL PROCESSING

Metallurgical analyses traced the performance of the cerium (Ce), yttrium (Y), Nb and Zr with the attention focused on the Zr as representing a valid indicator of the overall metallurgy. LREEs were tracked using Ce and La while the HREEs were tracked using Y. Mass balances were calculated using Bilmat™ (a mass balance program) to make minor adjustments to the actual assays while attributing balanced masses to each stream.

Results from the numerous bench scale tests (at SGS-M) were used to develop a flowsheet which was tested at a pilot plant scale.

Flotation results from the August, 2011 MPP program undertaken at SGS-M using about 3.7 t of sample, comprising one UZ and one BZ composite, have been used as the basis for the feasibility study flotation process design criteria. The material treated during this program of testwork was considered to be more representative of the total BZ than other pilot work which tested a variety of different materials. It is the BZ mineralization which will be mined and fed to the concentrator.

Following the 3.7 t MPP, a further 40 t pilot plant campaign was undertaken also at SGS-M, specifically to generate a bulk concentrate for down-stream hydrometallurgical process development. Some data generated in this program has also been included in the process design criteria.

It is noted that further bench scale flotation testwork has since been performed in an attempt to optimize flotation reagent addition. The results from this work (conducted by XPS) resulted in a further pilot plant campaign at XPS in October, 2012 to test this revised reagent circuit. Whilst successful in reducing reagent requirements, the final recoveries did not match those achieved in the 3.7 t program.

Additional investigations are currently progressing with the potential to remove the need for both the de-sliming and gravity circuits (both are sources of REE losses to tailings). These results look very encouraging, but further work is required before they can be incorporated into a final flowsheet.

13.3.1 Comminution

The crushing and grinding circuits have been selected and designed using test data generated during the 3.7 t and 40 t pilot test programs. The grinding and concentrator has been designed to process 2,000 t/d initially, with future expansion capability to 4,000 t/d.

A rod/ball mill combination was selected on the basis that this circuit offered the best opportunity to minimize fines production and to produce the narrowest flotation feed size distribution (flotation testwork had indicated the need for slimes removal ahead of flotation hence slimes production needs to be minimized).

For the feasibility study it was decided to install a rod mill sized for the maximum (expansion) throughput (4,000 t/d) and the ball mill would be sized for the 2,000 t/d rate, with the intention to install a second, identical unit in parallel for the up-grade to 4,000 t/d.

In November, 2010, FLSmidth (FLS) completed a comminution study which designed and sized the crushing, milling and classification equipment. In February, 2011, FLS issued a further report describing the de-sliming cyclone circuit, and then early in 2012, they issued a process design package including a full set of process flow diagrams (PFDs) for the entire mill and flotation processing facility.

The crushing plant was designed to produce a rod mill 80% passing feed size (F_{80}) of 16 mm (100% minus 25 mm) and the target ball mill circuit 80% passing product size (P_{80}) of 36 µm. In the initial simulations, FLS used a Bond rod mill work index (RWi) of 16.5 kWh/t and a Bond ball mill work index (BWi) of 15.7 kWh/t (as used in the prefeasibility study). The BWi was later updated to 17.5 kWh/t, which was based on recent grindability tests that used a more project applicable and finer closing size.

The FLS recommendation was to install a 3.4 m diameter by 5.5 m long rod mill with a 1,000 HP motor plus a 4.3 m diameter by 8.5 m long ball mill with a 3,500 HP motor. A duplicate parallel ball mill would be installed for the expansion.

The standard Bond rod mill work index determinations for the four composite samples used during the 40 t pilot plant test ranged from 13.9 to 14.8 kWh/t with the index tending to increase with depth. The two samples tested (BZ-Q3 and BZ-Q4) during September 2012 gave results of 15.8 and 15.5 kWh/t. Based on these results, the index used for the design (16.5 kWh/t) appears to be reasonable.

The standard Bond ball mill indices determined for the two composite samples selected for the August, 2011 3.7 t pilot plant and four composites used for the February, 2012 40 t pilot plant varied between 14.7 and 15.9 kWh/t. These tests used a ball mill closing test size of 75 µm. The two recent tests (October, 2012 test program) using a closing size of 53 µm gave results of 17.6 and 17.0 kWh/t. The selection of 17.5 kWh/t for the design appears to be reasonable and appropriate.

Standard Bond abrasion index test results on two 3.7 t, four 40 t and two XPS 2012 pilot plant test samples ranged from 0.428 and 0.607. These samples can be considered relatively abrasive.

13.3.2 Magnetic Separation

The purpose of the magnetic separation of the flotation feed (by low intensity magnetic separation (LIMS)) is to remove iron. This will reduce flotation reagent consumption and flotation feed volume as well as the iron in the feed to the hydrometallurgical plant.

Numerous tests using a drum magnet separator at a low magnetic field strength of approximately 650 gauss have been undertaken on different batch test and pilot plant samples. The results of these tests showed that the final mass pull to magnetics varies depending on feed material but was typically in the 7 to 10% range with metal losses generally around 3 to 4%.

13.3.3 De-sliming

Comparative flotation testwork using the feasibility study proposed reagent suite showed that the introduction of a de-sliming process ahead of flotation is essential in order to achieve acceptable recoveries at optimum concentrate grades and mass pull. A target cut point, 80% passing size (K_{80}), in the range of 7 to 10 µm was selected.

Various tests undertaken on different batch test and pilot plant samples indicated typical mass of slimes removed of around 8-15% with losses of REEs ranging from 4% to 15%. In the feasibility study the mass of slimes removed was stated as 9.3%.

FLS-Krebbs conducted de-sliming cyclone simulations and developed material balances for the de-sliming process area. A three-stage cyclone system was selected for the feasibility study.

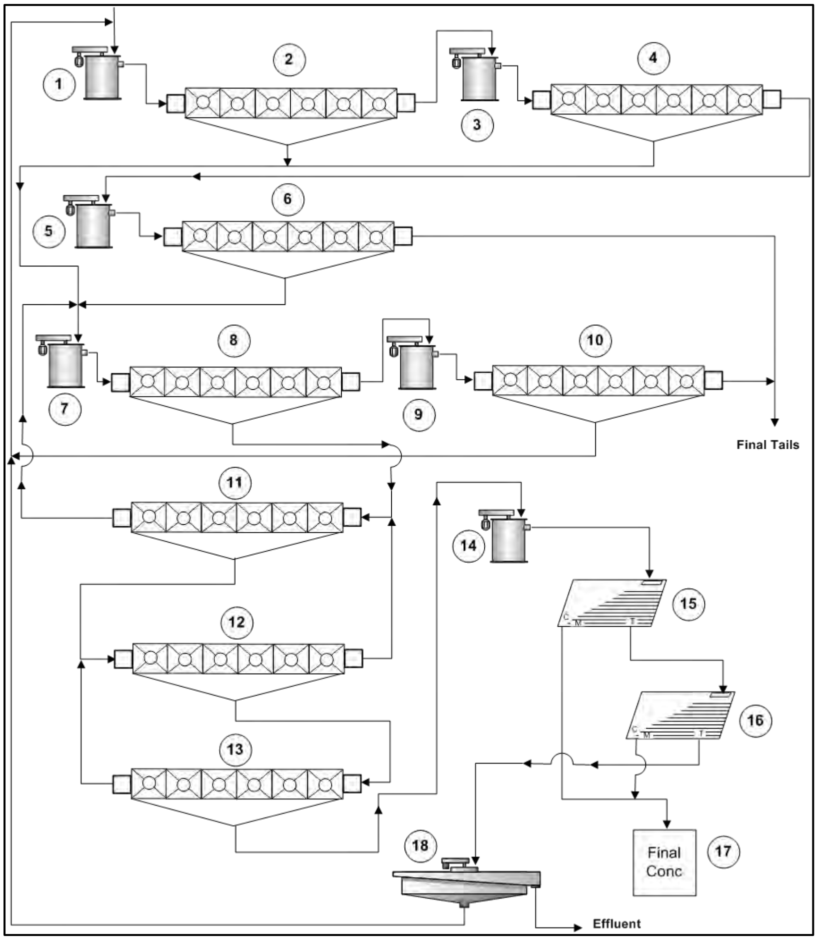
Work currently in progress evaluating alternative flotation reagent suites suggests it may be possible to eliminate this de-sliming process without impacting negatively on flotation performance. This work is expected to be completed in the second quarter of 2013.

13.3.4 Flotation

Numerous open circuit and locked cycle flotation tests (LCT) have been completed since 2008 by SGS-M under the direction of SBM Mineral Processing and Engineering Services Ltd. (SBM). This development work was undertaken using a variety of samples representing the various ore-types of the deposit. The flotation circuit developed by SGS-M was tested during the various pilot plant test runs described earlier.

The feasibility study uses the test conditions including the reagent consumptions and metallurgical results from two LCTs (F4 and F6) that were undertaken as part of the 3.7t pilot plant program at SGS-M. These tests were considered to be the most reliable representation of flotation performance for the project based on the composite samples used compared to the life-of-mine (LOM) plan, the flowsheet and the test conditions adopted. A schematic flowsheet used for the flotation and downstream gravity LCT and pilot plant tests is presented in Figure 13.1.

Figure 13.1
SGS-M Flotation and Gravity Flowsheet



Ref: SGS-M Test Report "Small Continuous Pilot-Scale Program on Thor Lake Ore", December, 2012.

1. Primary rougher conditioner.
2. Primary rougher.
3. Secondary rougher conditioner.
4. Secondary rougher.
5. Rougher scavenger conditioner.
6. Rougher scavenger.
7. 1st cleaner conditioner.
8. 1st cleaner.
9. 1st cleaner scavenger conditioner.
10. 1st cleaner scavenger.
11. 2nd cleaner.
12. 3nd cleaner.
13. 4nd cleaner.
14. Neutralizing conditioner.
15. Primary gravity table.
16. Secondary gravity table.
17. Concentrate filter.
18. Gravity tailings thickener.

The average results from LCT F4 and F6 are presented in Table 13.1.

Table 13.1
Average Flotation LCT F4 and F6 Results

Sample	Wt%	Grades (%)						Distribution (%)[1]					
		Nb_2O_5	Ce_2O_3	La_2O_3	Nd_2O_3	Y_2O_3	ZrO_2	Nb_2O_5	Ce_2O_3	La_2O_3	Nd_2O_3	Y_2O_3	ZrO_2
Feed	100	0.39	0.53	0.24	0.26	0.21	3.21	100	100	100	100	100	100
Final concentrate	18.0	1.61	2.28	1.05	1.13	0.94	14.93	73.9	76.6	78.35	77.2	80.55	83.75

[1] Distribution recoveries to final concentrate account for losses to slimes as well as flotation tailings.

The 3.7 t pilot plant program that immediately followed the LCT tests successfully produced over 200 kg of on-specification final concentrate which was used for hydrometallurgical testing. The metallurgical recoveries and reagent dosage rates achieved during the pilot plant testing were not optimized due to a lack of operating time to properly optimize the system.

The primary objective of the SGS-M 40 t pilot plant was to produce a bulk concentrate for use in the further development of the down-stream hydrometallurgical process. However, tests were conducted using a number of different feed materials and the impact of a number of operating parameters on performance was also investigated. The flowsheet adopted was very similar to that of the 3.7 t pilot test (see Figure 13.1) including de-sliming and gravity enrichment of the flotation concentrate.

13.3.5 Gravity Separation

SGS-M was contracted to conduct gravity separation testwork on cleaner flotation concentrate from the 40 t pilot plant. The objective of this work was to upgrade the final concentrator product by the removal of lighter, gangue material. The results of these tests suggested a concentrate upgrading factor of between 2 and 3.

13.3.6 Solid-Liquid Separation

Thickening testwork was conducted by SGS-M and Outotec on several products from the various pilot plant campaigns and various ore types. The streams investigated were:

- First Cleaner scavenger concentrate.
- Gravity tailings.
- Gravity concentrate (final product).
- Combined tailings.

The results from these tests were used for equipment selection and sizing for the feasibility study.

13.4 TESTWORK - HYDROMETALLURGY

Hydrometallurgical testwork has been conducted since late 2008. Tests have been conducted on both the bench and pilot scale on flotation concentrate. The feasibility study final process design criteria have been largely developed from the operating conditions used in the final (PP6) pilot plant campaign completed by SGS-M in September-October, 2012.

Hydrometallurgical plant processing consists of the following major unit processes.

- Pre-leaching (PL).
- Acid Baking (AB).
- Water Leach (WL).
- Neutralization and Impurity Removal (IR).
- Impurity Removal Re-dissolution.
- RE Precipitation (RP).
- Waste Water Treatment.
- Tailings Neutralization (TN).

A simplified process block flow diagram is provided in Figure 13.2.

13.4.1 Bench Scale Testwork

Extensive static bench work, mainly undertaken by SGS-M has been completed using samples representing the Nechalacho deposit. The main objectives of the bench testwork were to:

- Determine the best hydrometallurgical methods to extract REEs from the concentrate while minimizing co-extraction of contaminants.

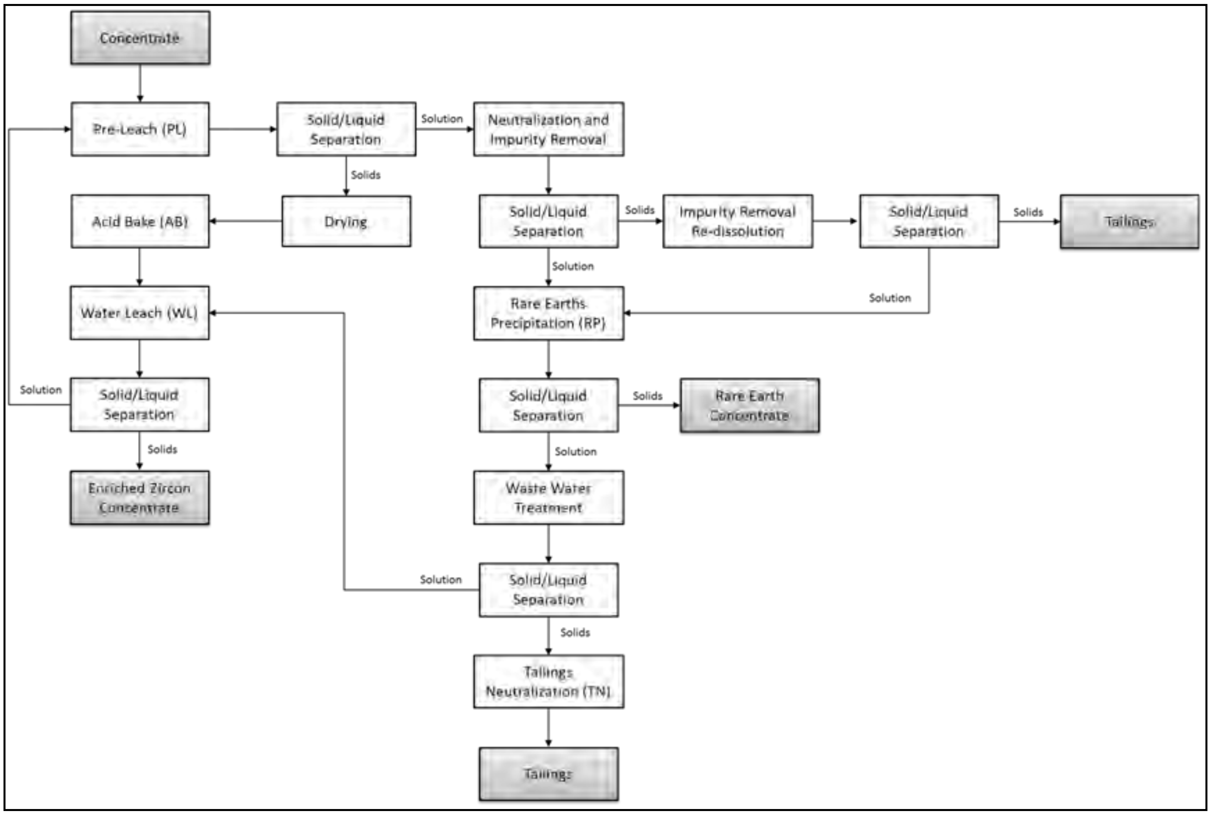
- Investigate and optimize key operating parameters such as temperature, reagent concentrations, pH, reaction time etc.

- Characterize chemical reactions (especially for acid bake) by determining reaction kinetics, conversion profiles and fractional conversion of REEs.

- Optimize reagent dosages.

- Provide input for subsequent continuous scale pilot campaigns.

Figure 13.2
Hydrometallurgical Process Block Flow Diagram



The main reagents used in the flowsheet are:

- Concentrated sulphuric acid, which is used in the acid bake process and the impurity removal and re-dissolution circuit.
- Dolomite, hydrogen peroxide and magnesia, which are used in the neutralization and impurity removal circuit.
- Lime, which is used in the rare earth precipitation circuit and for waste water treatment.

13.4.2 Mini Pilot Plant Testwork

Seven mini-pilot plant campaigns were conducted at SGS-M in order to investigate the acid bake operating parameters and to determine the optimal conditions leading to the best REE recoveries. These were conducted between January 25 and May 14, 2012, using concentrate derived from BZ (basal zone) and UZ (upper zone) type mineralization.

13.4.3 Pilot Plant Campaigns

Six hydrometallurgical pilot plant campaigns (PP1 to PP6) were conducted at SGS-M between June and October, 2012. The main objectives of these campaigns were to:

- Test a continuous version of the hydrometallurgical flow-sheet and demonstrate steady state operation.
- Optimize REE extraction in the pregnant solution (through the pre-leach, acid bake and water leach circuits) and achieve recoveries comparable to previously conducted bench tests.
- Remove target contaminants (iron, uranium and thorium (Fe, U, Th)) in the iron removal circuit while ensuring minimal REE losses.
- Ensure the final mixed rare earth precipitate (RP) product had an acceptable grade of rare earths higher than that in the input concentrate while minimizing the U and Th content to a level lower than 500 ppm as imposed by the Transportation of Dangerous Goods Act.
- Ensure the concentrations of species in the filtrate from the tailings circuit met target environmental levels determined via hydrological modelling (especially targets for Mg (800 mg/L) and SO_4 (500 mg/L), respectively).

The hydrometallurgical pilot plant campaigns operated at a throughput rate of around 2 kg/h of concentrate.

The final pilot plant campaign (PP6) operated between September 24 and October 5, 2012, and was the most successful. The results from this campaign proved the technical viability of the process and provided crucial input for the final hydrometallurgical flowsheet, process design criteria and process engineering adopted for the feasibility study. A picture showing the pilot plant set-up at SGS-M is presented in Figure 13.3.

Figure 13.3
Hydrometallurgical Pilot Plant at SGS-M

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The results for the pilot plant campaign PP6 are summarized in Table 13.2.

Table 13.2
Summary of Pilot Plant (PP6) Results

Element	Rare Earth Precipitate			EZC	
	Grade (%)	Recovery (%)[1]	Recovery (%)[2]	Grade (%)	Recovery (%)
La	2.40	79	83	0.16	17
Ce	5.36	78	81	0.41	18
Pr	0.66	78	82	0.06	19
Nd	2.66	72	75	0.28	25
Sm	0.44	62	62	0.07	34
Eu	0.065	61	66	0.011	37
Gd	0.465	61	65	0.076	38
Tb	0.073	59	64	0.013	40
Dy	0.376	56	60	0.077	43
Ho	0.061	53	54	0.016	46
Y	1.28	50	50	0.380	50
Er	0.128	43	45	0.047	56
Tm	0.014	37	37	0.007	62
Yb	0.066	31	31	0.053	68
Lu	0.007	25	25	0.008	74
Zr	0.27	1	1	12.48	97
Nb	<0.01	0	1	1.20	

Element	Rare Earth Precipitate			EZC	
	Grade (%)	Recovery (%)[1]	Recovery (%)[2]	Grade (%)	Recovery (%)
U+Th	0.040			0.019	

[1] The average recoveries across all runs during PP06 (including runs which experienced some problems).
[2] Average values measured during times when the pilot plant was operating smoothly under optimal conditions. The feasibility assumes average recoveries for initial operations and optimized higher values after the first year of operation.

One important accomplishment of PP6 was the reduced U and Th levels in the final rare earth product to below the 500 ppm limit governed by transportation regulations. PP6 was able to achieve a reduced net U and Th level by using hydrated lime instead of magnesium oxide as a precipitating reagent. Also, use of hydrated lime in the RP circuit resulted in Mg and sulphate (SO_4) concentrations in the tailings filtrate being consistently below target environmental levels as specified by hydrological modeling (800 mg/L for Mg and 4,500 mg/L for sulphate).

13.5 TESTWORK – RARE EARTH REFINERY

The refinery to be constructed at Geismar comprises two plants, the leaching and the separation plants. The leaching plant removes impurities from the hydrometallurgical rare earth precipitate in order to attain a purified rare earth element feed (+99.9%) to the separation plant where the individual rare earth products will be produced.

The leaching plant process consists of the following major unit processes:

- Acid leach.
- Selective precipitation.
- Solvent extraction, scrub and strip for impurity removal.
- Crud treatment.
- Rare earth precipitation.
- Hydrochloric acid digestion.
- Water treatment.

The separation plant process consists of the following major unit processes:

- Solvent extraction to separate LREEs from HREEs.
- Crud treatment.
- A series of solvent extraction systems for individual rare earth separation.
- Precipitation.
- Thickening.
- Filtration.
- Product drying.
- Product calcining.
- Product packaging.

A large number of testwork reports have been issued to summarize the testwork that has been undertaken at a number of different laboratories over the last few years. All relevant testwork has been completed using the rare earth precipitate produced during the hydrometallurgical pilot plant testwork program discussed in Section 13.4.3 above. All the solids from the six different hydrometallurgical pilot plants were combined to provide a total of 200 kg of rare earth precipitate for the leaching and separation plant testwork with average composition shown in Table 13.3.

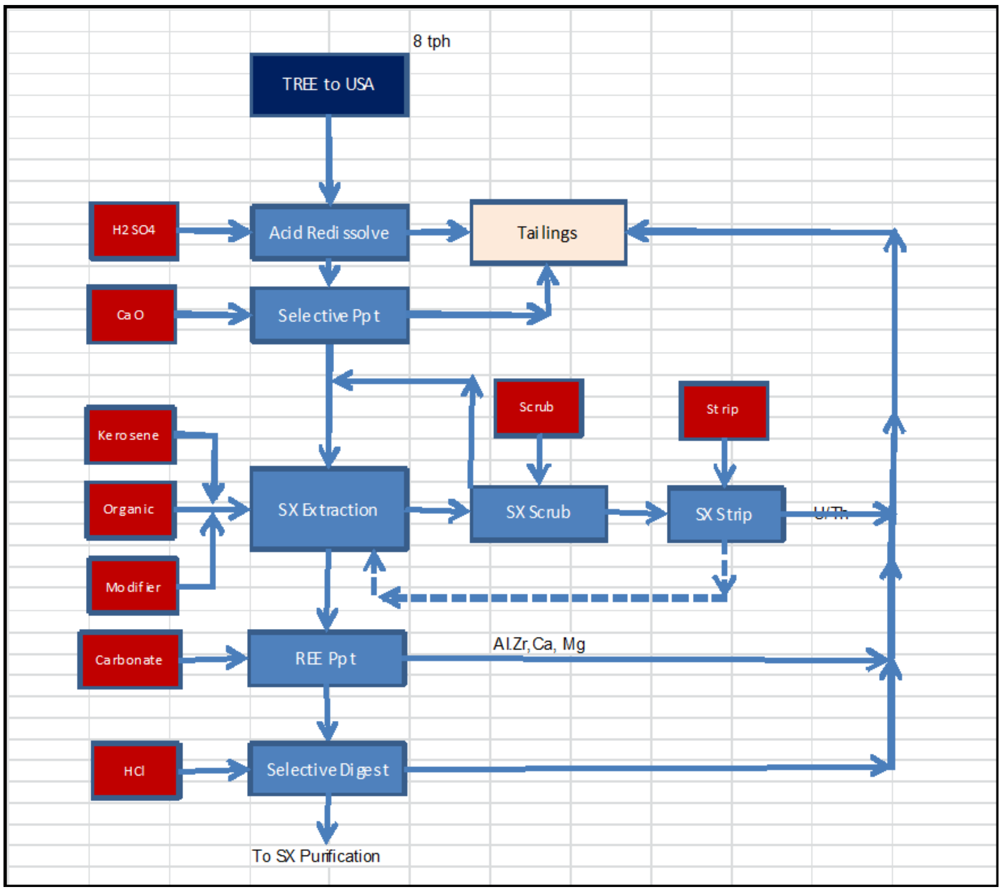
Table 13.3
Comparison of the Combined REE Precipitate Test Sample and Study Specifications

Element	Combined Actual REE Leach Plant Sample For Testwork (%)	Feasibility Study Process Design Criteria Leach Plant Feed (%)	Impurity Specification, Feed to Separation Plant (ppm)
La	2.69	1.765	
Ce	5.41	4.375	
Pr	0.66	0.571	
Nd	2.69	2.409	
Sm	0.54	0.429	
Eu	0.07	0.063	
Gd	0.49	0.485	
Tb	0.06	0.068	
Dy	0.33	0.338	
Ho	0.05	0.051	
Y	1.10	1.04	
Er	0.09	0.096	
Tm	0.01	0.012	
Yb	0.05	0.050	
Lu	0.005	0.006	
Zr	0.33	0.209	<10
Nb	0.02	0.004	
P	0.01	0.004	<10
Fe	0.76	0.554	<350
Si	0.56	0.216	<10
Na	0.08	0.004	1,378
K	0.02	0.004	
F	4.45	2.409	<200
Al	5.07	2.975	<200
Mg	7.41	-	<3,000
Ca	4.83	-	<7,000
S	6.79	-	
TREE Content	14	12	
LREE Content	12	10	
HREE Content	2	2	
U + Th (ppm)	610	500	<14.7

13.5.1 Leaching Plant

As mentioned previously, a large amount of testwork has been undertaken by a number of different laboratories over the last few years, including SGS-M and Mintek of South Africa. The most recent relevant testing has been conducted using the combined rare earth precipitate produced during the hydrometallurgical pilot plant testwork program. The resulting flowsheet developed from this work is presented in Figure 13.4.

Figure 13.4
Simplified Leach Plant Process Block Flow Diagram



SNC Feasibility Study Report, Section 22.

The results from the testwork programs completed to date have suggested that the specified impurity criteria shown in Table 13.3 for the feed to the rare earth separation plant can be achieved with a loss of rare earth to the residue of less than 2%.

13.5.2 Separation Plant

It is noted that the design of the rare earth separation plant is not based on testwork. It is based on the proven standard rare earth separation processing methods and equipment selections. The technology proposed for this plant is mainly based on Chinese technology.

The detailed design was provided by a Chinese expert who has many years of experience in the design and operation of similar plants in China. This detailed design was reviewed by Jerald (Jerry) Tabor, an independent consultant engaged by Avalon who has over 40 years of experience including in the design and operation of commercial production facilities for high purity rare earths such as lanthanum, cerium, praseodymium, neodymium, samarium, gadolinium and yttrium.

The separation plant extraction process for the feasibility study is based on experience and widely-accepted, well proven standard rare earth separation processing methods and equipment selections. It uses 16 solvent extraction systems (chloride solutions) to separate the individual rare earths from each other. The separation plant will produce 10 different pure rare earth oxides products in accordance with the specifications indicated in Table 13.4. The plant will also produce one separate stream containing four different rare earth carbonates for which there is limited market at the present time. This stream will be stockpiled initially and eventually disposed of if markets are not forthcoming.

Table 13.4
List of Products from the Separation Plant

Product	Design Plant Production (t/y)	Product Distribution (%)	Feasibility Study Specification[1]
La Oxide	1,583	16	3 N
Ce Oxide	3,572	36	3 N
Pr Oxide	451	4	3 N
Nd Oxide	1,783	18	3 N
Sm Oxide	391	4	2 N
Eu Oxide	49	0.5	4 N
Gd Oxide	371	4	3 N
Tb Oxide	54	0.5	4 N
Dy Oxide	271	3	4 N
Y Oxide	1,170	11	5 N
Lu Oxide	14	0.1	3 N
Er/Ho/Tm/Yb Carbonate Mix	292	3	2 N
Total	**10,000**	**100**	

[1] "N" stands for the number of nines purity produced as final product, for example 3 N = 99.9%.

14.0 MINERAL RESOURCE ESTIMATES

The mineral resource estimate for the Nechalacho project presented in the feasibility study based on the block model prepared by Avalon was audited originally by RPA on 21 November, 2012. Subsequent to this, Avalon updated the database and re-estimated the resource as of 3 May, 2013. The update included correction of some minor assay data entry errors and drill hole locations. The net effect of these changes is considered immaterial as the resource change was less than 1% in most individual parameters. The largest changes were for ZrO_2 grade, and the effect was an increase in grade in Measured and Indicated resources of between 0.1% and 3.2% of the overall grade in the various categories.

The current mineral resource estimated by Avalon and accepted by RPA for the Nechalacho deposit is summarized in Table 14.1. The mineral resource is reported at a cut-off value of US$320/t. The effective date of the mineral resource estimate is 3 May, 2013.

Table 14.1
Nechalacho Deposit Mineral Resource Estimate as at 3 May, 2013

Category	Zone	Tonnes (million)	TREO (%)	HREO (%)	ZrO_2 (%)	Nb_2O_5 (%)	Ta_2O_5 (%)
Measured	Basal	10.86	1.67	0.38	3.23	0.40	0.04
	Upper	-	-	-	-	-	-
Total Measured		**10.86**	**1.67**	**0.38**	**3.23**	**0.40**	**0.04**
Indicated	Basal	55.81	1.55	0.33	3.01	0.40	0.04
	Upper	54.59	1.42	0.14	1.96	0.28	0.02
Total Indicated		**110.40**	**1.49**	**0.24**	**2.49**	**0.34**	**0.03**
Measured and Indicated	Basal	66.67	1.57	0.34	3.05	0.40	0.04
	Upper	54.59	1.42	0.14	1.96	0.28	0.02
Total Measured and Indicated		**121.26**	**1.50**	**0.25**	**2.56**	**0.34**	**0.03**
Inferred	Basal	61.09	1.29	0.25	2.69	0.36	0.03
	Upper	122.28	1.26	0.12	2.21	0.32	0.02
Total Inferred		**183.37**	**1.27**	**0.17**	**2.37**	**0.33**	**0.02**

1. CIM definitions were followed for Mineral Resources.
2. Mineral Resources are estimated at a NMR cut-off value of US$320/t. NMR is defined as "Net Metal Return" or the in situ value of all payable metals, net of estimated metallurgical recoveries and off-site processing costs.
3. An exchange rate of US$1=CAD1.05 was used.
4. Heavy rare earth oxides (HREO) is the total concentration of: Y_2O_3, Eu_2O_3, Gd_2O_3, Tb_2O_3, Dy_2O_3, Ho_2O_3, Er_2O_3, Tm_2O_3, Yb_2O_3 and Lu_2O_3.
5. Total rare earth oxides (TREO) is HREO plus light rare earth oxides (LREO): La_2O_3, Ce_2O_3, Pr_2O_3, Nd_2O_3 and Sm_2O_3.
6. Rare earths were valued at an average net price of US$62.91/kg, ZrO_2 at US$3.77/kg, Nb_2O_5 at US$56/kg, and Ta_2O_5 at US$256/kg. Average REO price is net of metallurgical recovery and payable assumptions for contained rare earths, and will vary according to the proportions of individual rare earth elements present. In this case, the proportions of REO as final products were used to calculate the average price.
7. ZrO_2 refers to zirconium oxide, Nb_2O_5 refers to niobium oxide and Ta_2O_5 refers to tantalum oxide.

It should be noted that, unless otherwise stated, rare earth and yttrium oxides have been calculated and expressed herein as RE_2O_3 although oxides of cerium, praseodymium and terbium are often given as CeO_2, Pr_6O_{11} and Tb_4O_7, respectively.

14.1 MINERAL RESOURCE DATABASE

The database for the November 21, 2012 mineral resource estimate for the Nechalacho deposit contained 490 drill holes totalling 104,918.7 m. The database included 51 historic drill holes amounting to 5,588 m and 439 recent drill holes with a total length of 99,330.6 m. The estimate was based on 33,236 samples assayed for rare metals, rare earths, and other elements, from 450 drill holes, 48 historical and 402 recent. Samples from 41 historical drill holes have incomplete or no REE assays results. Only 21 of the historical drill holes sampled the Basal Zone, as it was not a target at that time.

Prior to the interpretation, data from the database was imported into MineSight software for geological modelling and resource estimation by Avalon geologists. The assay and lithological tables were merged, resulting in occasional splitting of assay samples at lithological contacts. The total number of samples in MineSight was 37,990. At the data import stage, the first density measurement encountered down drill hole was assigned to individual samples in the same drill hole until a new measurement was encountered. These samples form the basis of the resource estimate and are reflected in assay descriptive statistics tables.

14.2 DATABASE VALIDATION

RPA verified and validated the mineral resource database. The procedures are described in Section 12.0, Data Verification. RPA considered the database acceptable to use for resource estimation.

14.3 GEOLOGICAL INTERPRETATION

Avalon geologists initially recognized two domains of REO enrichment for the Nechalacho deposit, Upper and Basal Zones, differentiated by the relative concentration of HREO. The upper and lower contacts of the Basal Zone (BZ) and Upper Zone (UZ) were defined on the basis of significant changes in HREO% and TREO%, the BZ having a higher proportion of HREO than the UZ.

The UZ was defined as the volume between the top of the BZ and the base of the overburden or water. The contact with the Grace Lake granite or barren sodalite cumulate was used to limit the extent of the UZ and BZ to the south.

Increased drilling definition of the diabase dykes (see Section 7.1.2) has enabled improved modelling of the dyke geometry for the resource estimation purposes. The volume of dykes was subtracted from the UZ and BZ resource volumes such that the resources as quoted exclude the volume of dykes.

The Upper and Basal Zones of the deposit, and the drill pattern, are illustrated by the sections shown as Figure 14.1 and Figure 14.2.

Figure 14.1
Geological North-South Section UTM 416200E

Legend: Mineralized Zones
Upper Zone | Basal Zone – Top (BZ-T) | Basal Zone – Bottom (BZ-B)
TLS Thor Lake Syenite

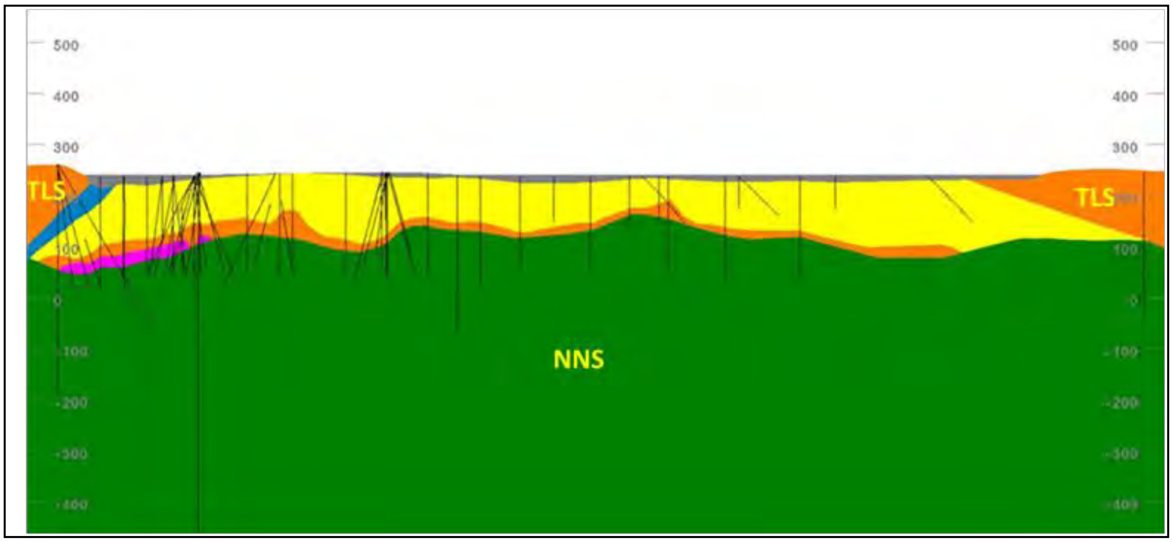
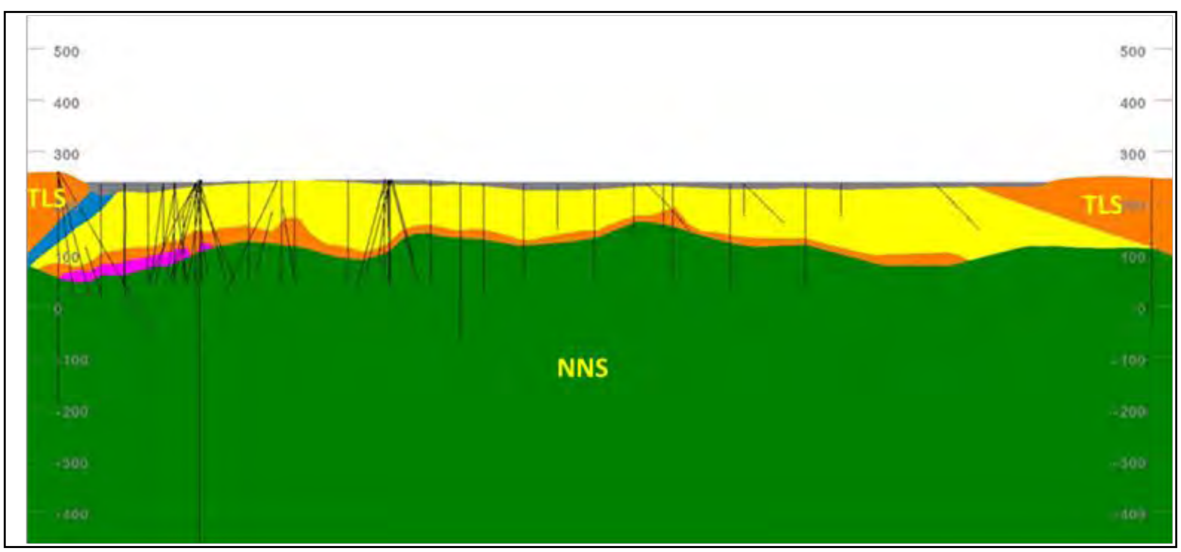


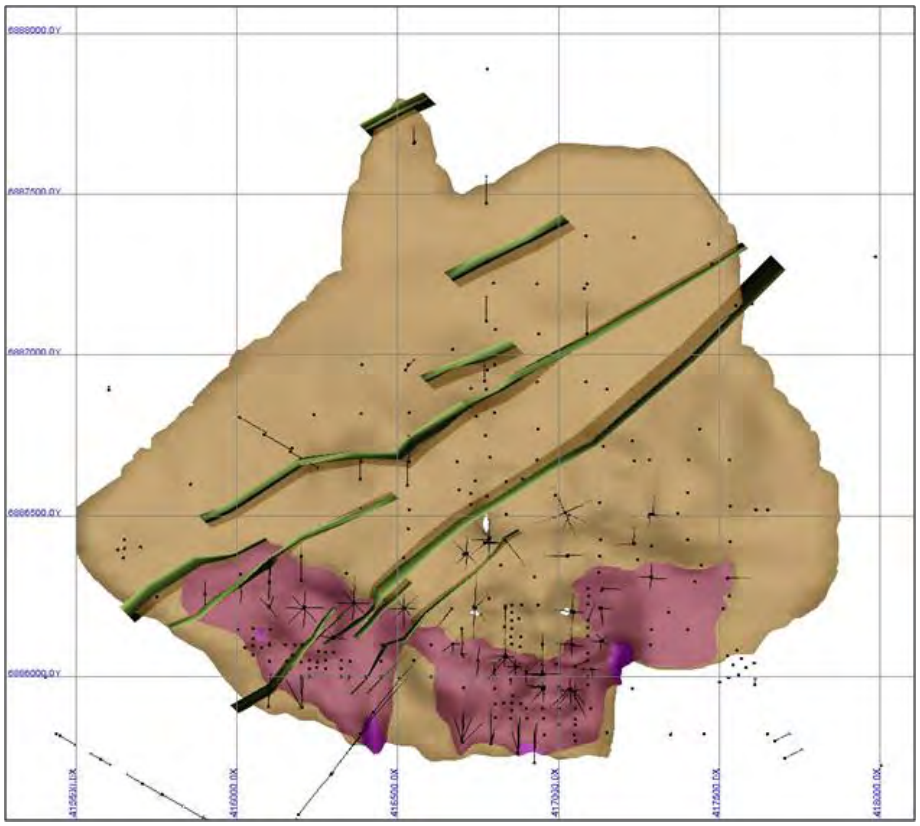
Figure 14.2
Geological North-South Section UTM 416800E

Legend: Mineralized Zones
Upper Zone | Basal Zone – Top (BZ-T) | Basal Zone – Bottom (BZ-B)
TLS Thor Lake Syenite



Subsequently, in the southern part of the deposit, the BZ was subdomained in Basal-Top and Basal-Bottom zones (see Figure 14.3).

Figure 14.3
Basal Top and Basal Bottom Zones



The base of the BZ is generally defined by a strong and sharp increase in the HREO% and TREO%. In some drill holes, the grade increase is not sharp and the contact is set at the start of the increase in HREO% and TREO%. The upper contact of the BZ is sharp or gradational. The BZ was split into two subdomains: the BZ-Bottom (BZ-B) consisting of material that consistently has TREO% greater than 1% and HREO% greater than 0.15% (domain averages 25% heavy-to-light ratio), and the BZ-Top (BZ-T) consisting of material that often has TREO% greater than 1% and HREO% higher than 0.15% (domain averages 17% heavy-to-light ratio).

14.4 BASIC STATISTICS AND CAPPING OF THE HIGH ASSAYS

The resource estimate is based on intercepts from 450 drill holes which intersected the mineralized zones. The descriptive statistics of the resource assays for BZ-B, BZ-T and UZ presented in Table 14.2 are based on the database available for the November, 2012 resource estimate. As noted above, for the May, 2013 estimate, minor corrections were applied to the assay database involving grades above the detection limit for Zr and one value corrected for Ce. A total of 257 over-limit assay results for Zr were updated resulting in an increase in average grade to 11,210 ppm for Zr in UZ, 23,974 ppm in the BZ-T and 36,814 ppm in the BZ-B. (In Table 14.2 and Table 14.3, H:T (%) refers to the ratio of HREO to TREO)

RPA investigated the necessity for capping of high grade assays. Top decile analysis performed for ZrO_2, Nb_2O_5 Dy_2O_3, Nd_2O_3 and Y_2O_3 showed that capping was not necessary. This was also indicated by low coefficients of variation. As more data become available, capping analysis should be reconsidered.

Table 14.2
Resource Assays Descriptive Statistics BZ-B, BZ-T, and UZ

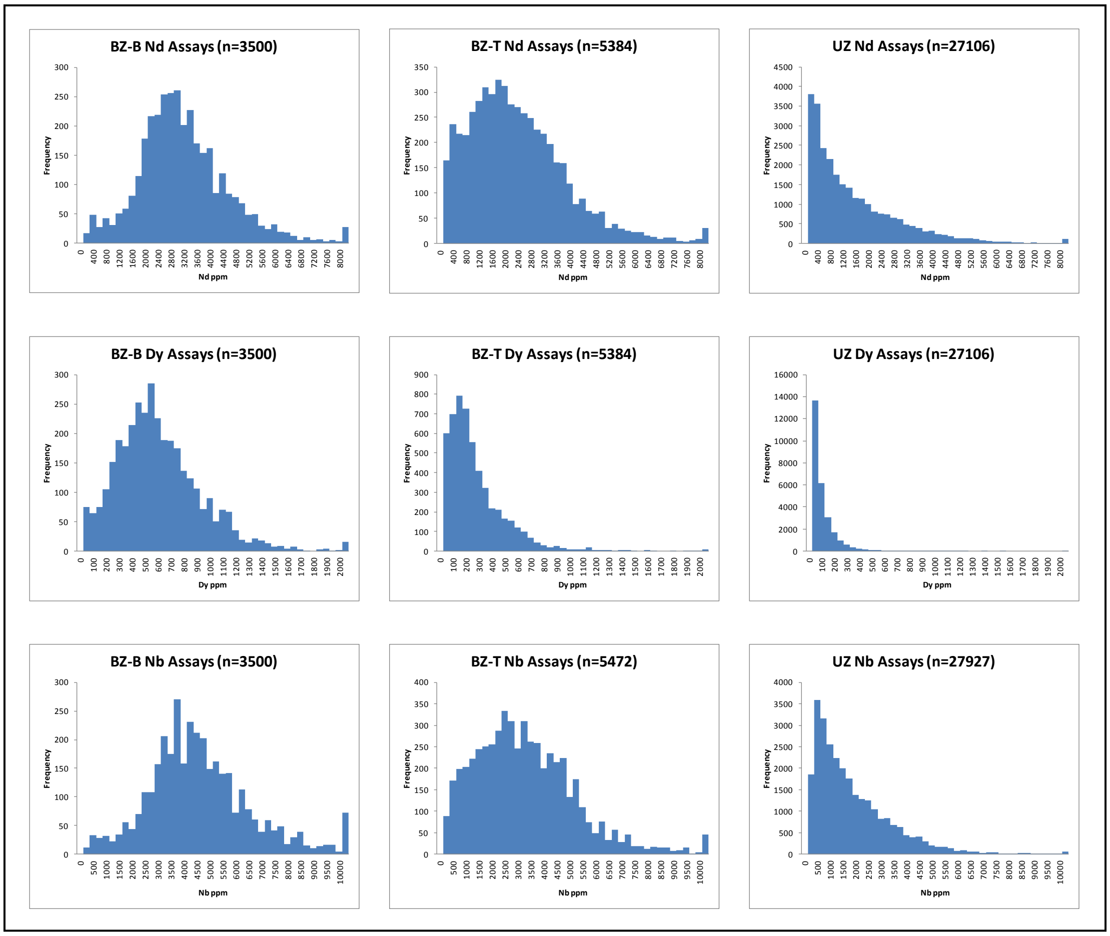
Basal Zone-Bottom Assays											
Oxide (ppm)	**La_2O_3**	**Ce_2O_2**	**Pr_2O_3**	**Nd_2O_3**	**Sm_2O_3**	**Eu_2O_3**	**Gd_2O_3**	**$T4_2O_3$**	**Dy_2O_3**	**Ho_2O_3**	**Er_2O_3**
Count	3500	3500	3500	3500	3500	3500	3500	3500	3500	3500	3500
Mean	2798	6214	783	3102	708	89	661	108	588	107	280
Median	2628	5843	743	2927	669	85	630	101	539	96	247
Std. Dev.	1375	2884	349	1451	318	40	300	57	352	70	190
Min	89	200	25	88	19	3	16	2	10	2	3
Max	39530	58550	5850	28684	3990	319	2629	707	4638	929	2132
CV	0.49	0.46	0.45	0.47	0.45	0.45	0.45	0.53	0.60	0.66	0.68
Oxide (ppm)	**Tm_2O_3**	**Yb_2O_3**	**Lu_2O_3**	**Y_2O_3**	**ZrO_2**	**Nb_2O_5**	**HfO_2**	**Ta_2O_5**	**Ga_2O_3**	**ThO_2**	**U_3O_8**
Count	3500	3500	3500	3500	3500	3500	3500	3500	2404	3500	3500
Mean	37	223	31	2622	36467	4517	721	477	139	110	25
Median	33	201	27	2375	34586	4250	689	455	137	76	21
Std. Dev.	25	148	21	1736	18059	2070	329	223	23	111	20
Min	0	2	0	47	378	82	7	2	36	3	1
Max	218	1040	155	25908	139153	18603	2464	1832	359	1514	373
CV	0.68	0.67	0.67	0.66	0.50	0.46	0.46	0.47	0.17	1.01	0.79
Item	**LREO (%)**	**HREO (%)**	**TREO (%)**	**H:T (%)**	**NMR (US$)**	**Density (g/cm^3)**					
Count	3500	3500	3500	3500	3500	3617					
Mean	1.36	0.47	1.84	25.01	849	2.90					
Median	1.28	0.43	1.72	26.55	791	2.88					
Std. Dev.	0.63	0.29	0.86	6.82	409	0.15					
Min	0.04	0.01	0.06	4.14	30	2.09					
Max	13.66	3.81	14.25	70.28	3730	3.80					
CV	0.47	0.60	0.47	0.27	0.48	0.05					

Basal Zone-Top Assays											
Oxide (ppm)	**La_2O_3**	**Ce_2O_3**	**Pr_2O_3**	**Nd_2O_3**	**Sm_2O_3**	**Eu_2O_3**	**Gd_2O_3**	**Tb_2O_3**	**Dy_2O_3**	**Ho_2O_3**	**Er_2O_3**
Count	5384	5384	5384	5384	5384	5384	5384	5384	5384	5384	5384
Mean	2114	4710	588	2297	478	57	398	54	258	43	104
Median	1900	4181	526	2044	428	51	358	45	191	28	68
Std. Dev.	1449	3260	399	1544	328	39	279	44	243	45	112
Min	11	24	3	13	4	1	4	1	2	0	0
Max	31661	58550	6840	13351	2817	304	2498	448	3134	602	1417
CV	0.69	0.69	0.68	0.67	0.69	0.69	0.70	0.80	0.94	1.06	1.08
Oxide (ppm)	**Tm_2O_3**	**Yb_2O_3**	**Lu_2O_3**	**Y_2O_3**	**ZrO_2**	**Nb_2O_5**	**HfO_2**	**Ta_2O_5**	**Ga_2O_3**	**ThO_2**	**U_3O_8**
Count	5384	5384	5384	5384	5346	5472	5346	5464	4610	5384	5395
Mean	13	80	11	1112	23688	3162	461	283	151	130	28
Median	9	54	7	764	20805	2905	426	252	147	96	23
Std. Dev.	15	89	13	1147	17201	1970	317	182	33	121	27
Min	0	0	0	10	37	5	1	0	28	2	1
Max	159	968	172	15621	158067	21367	2547	1542	396	1388	805
CV	1.11	1.11	1.14	1.03	0.73	0.62	0.69	0.64	0.22	0.93	0.98
Item	**LREO (%)**	**HREO (%)**	**TREO (%)**	**H:T (%)**	**NMR (US$)**	**Density (g/cm^3)**					
Count	5384	5384	5384	5384	5346	5302					
Mean	1.02	0.21	1.23	17.13	492	2.84					
Median	0.91	0.16	1.10	15.57	436	2.81					
Std. Dev.	0.69	0.19	0.83	7.70	335	0.17					
Min	0.01	0.00	0.01	4.17	5	1.91					
Max	11.15	2.38	11.78	92.09	2981	4.07					
CV	0.68	0.91	0.67	0.45	0.68	0.06					

Upper Zone Assays											
Oxide (ppm)	**La_2O_3**	**Ce_2O_3**	**Pr_2O_3**	**Nd_2O_3**	**Sm_2O_3**	**Eu_2O_3**	**Gd_2O_3**	**Tb_2O_3**	**Dy_2O_3**	**Ho_2O_3**	**Er_2O_3**
Count	27106	27105	27106	27106	27106	27106	27106	27106	27106	27106	27106
Mean	1382	3070	378	1459	273	29	197	21	84	12	29
Median	968	2096	257	984	179	19	125	13	50	7	17
Std. Dev.	1521	3247	383	1541	285	31	211	24	102	16	38
Min	7	16	3	9	2	0	1	0	1	0	0
Max	58650	106561	6143	51071	5800	633	5765	652	2279	375	842
CV	1.10	1.06	1.01	1.06	1.04	1.05	1.07	1.13	1.22	1.32	1.34
Oxide (ppm)	**Tm_2O_3**	**Yb_2O_3**	**Lu_2O_3**	**Y_2O_3**	**ZrO_2**	**Nb_2O_5**	**HfO_2**	**Ta_2O_5**	**Ga_2O_3**	**ThO_2**	**U_3O_8**
Count	27106	27106	27106	27106	26886	27927	26886	27891	24973	27106	27246
Mean	3	21	3	346	11111	1761	209	116	160	77	35
Median	2	11	1	194	7336	1315	140	89	154	50	23
Std. Dev.	5	30	4	458	11843	1553	209	103	44	86	51
Min	0	0	0	3	5	5	0	0	25	0	0
Max	103	721	110	11341	124022	22050	2726	1241	651	2504	1875
CV	1.41	1.44	1.48	1.33	1.07	0.88	1.00	0.89	0.27	1.12	1.48
Item	**LREO (%)**	**HREO (%)**	**TREO (%)**	**H:T (%)**	**NMR (US$)**	**Density (g/cm^3)**					
Count	27105	27106	27105	27105	26886	26691					
Mean	0.66	0.07	0.73	11.56	243	2.73					
Median	0.45	0.05	0.50	9.32	166	2.71					
Std. Dev.	0.69	0.09	0.76	7.32	239	0.14					
Min	0.00	0.00	0.01	1.67	2	2.13					
Max	17.99	1.63	19.62	79.32	5817	3.95					
CV	1.05	1.17	1.04	0.63	0.98	0.05					

Figure 14.4 shows resource assay histograms for Nb_2O_5 from rare metals group, Nd_2O_3 from light rare earths group, and Dy_2O_3 from heavy rare earths group, in BZ and UZ.

Figure 14.4
Assay Histograms BZ-B, BZ-T and UZ



The assays were composited in the three modelled mineralized domains, the BZ-B, BZ-T and UZ. The assays were composited within continuously sampled intercepts, in 2-m fixed length intervals starting from collar. Orphans less than one metre long were added to the last full composite. The assays were weighted by sample length. The composites were used for the mineral resource estimate.

Descriptive statistics of the composites in the BZ-B, BZ-T, and UZ are shown in Table 14.3. Histograms of the composites for Nb_2O_5, Nd_2O_3, and Dy_2O_3 in the three zones are shown in Figure 14.5. These elements are the highest contributors to the revenue from each of the three groups of elements: rare metals (Nb), LREO (Nd), and HREO (Dy).

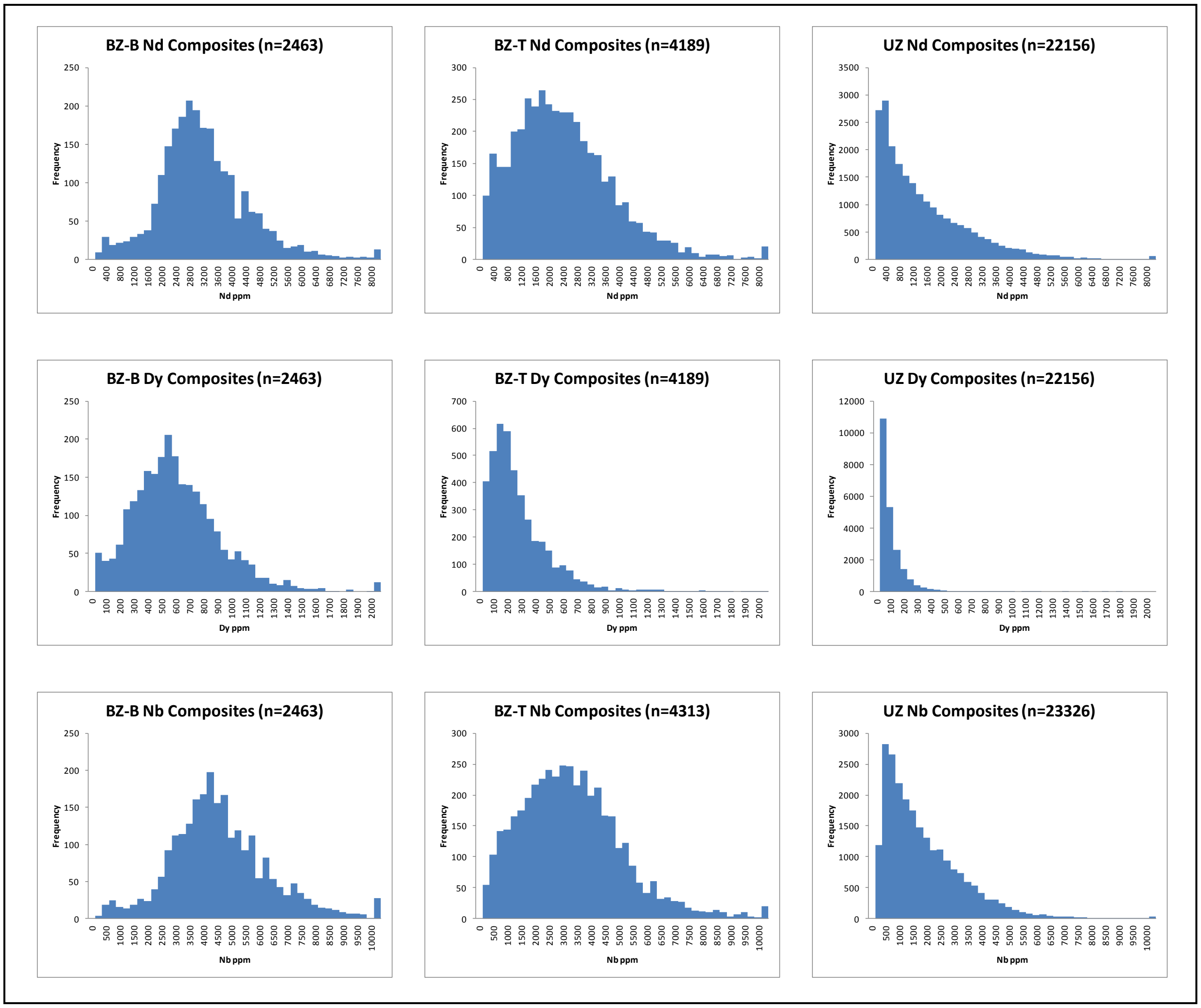
Table 14.3
Composites Descriptive Statistics BZ-B, BZ-T and UZ

Basal Zone-Bottom Composites											
Oxide (ppm)	**La_2O_3**	**Ce_2O_3**	**Pr_2O_3**	**Nd_2O_3**	**Sm_2O_3**	**Eu_2O_3**	**Gd_2O_3**	**Tb_2O_3**	**Dy_2O_3**	**Ho_2O_3**	**Er_2O_3**
Count	2462	2462	2462	2462	2462	2462	2462	2462	2462	2462	2462
Mean	2789	6199	780	3097	705	89	661	108	587	107	280
Median	2634	5860	747	2931	674	86	634	102	547	98	252
Std. Dev.	1251	2647	320	1333	292	37	276	53	332	67	180
Min	89	200	25	88	19	3	16	2	10	2	4
Max	22533	34603	3621	16755	2378	309	2234	644	4217	845	1938
CV	0.45	0.43	0.41	0.43	0.41	0.42	0.42	0.49	0.57	0.62	0.64
Oxide (ppm)	**Tm_2O_3**	**Yb_2O_3**	**Lu_2O_3**	**Y_2O_3**	**ZrO_2**	**Nb_2O_5**	**HfO_2**	**Ta_2O_5**	**Ga_2O_3**	**ThO_2**	**U_3O_8**
Count	2462	2462	2462	2462	2462	2462	2462	2462	1853	2462	2462
Mean	37	223	31	2614	35914	4486	717	474	138	113	25
Median	34	207	29	2411	34586	4289	694	455	136	80	21
Std. Dev.	24	139	19	1639	16364	1839	300	205	23	110	18
Min	0	2	0	50	655	217	20	8	67	3	3
Max	198	952	133	23510	139153	18603	2464	1832	355	1206	321
CV	0.64	0.63	0.63	0.63	0.46	0.41	0.42	0.43	0.16	0.97	0.70
Item	**LREO (%)**	**HREO (%)**	**TREO (%)**	**H:T (%)**	**NMR (US$)**	**Density (g/cm^3)**					
Count	2462	2462	2462	2462	2462	2571					
Mean	1.36	0.47	1.83	24.99	846	2.90					
Median	1.29	0.44	1.74	26.51	798	2.88					
Std. Dev.	0.58	0.27	0.79	6.59	380	0.14					
Min	0.04	0.01	0.06	4.90	32	2.09					
Max	7.98	3.47	8.35	70.28	3415	3.80					
CV	0.43	0.57	0.43	0.26	0.45	0.05					

Basal Zone-Top Composites										
Oxide (ppm)	**La_2O_3**	**Ce_2O_3**	**Pr_2O_3**	**Nd_2O_3**	**Sm_2O_3**	**Eu_2O_3**	**Gd_2O_3**	**Tb_2O_3**	**Dy_2O_3**	**Ho_2O_3**	**Er_2O_3**
Count	4189	4189	4189	4189	4189	4189	4189	4189	4189	4189	4189
Mean	2121	4730	590	2308	478	57	400	55	259	43	104
Median	1959	4289	539	2116	441	53	368	47	198	30	71
Std. Dev.	1309	2983	366	1439	305	36	262	41	227	42	106
Min	11	24	3	13	4	1	4	1	2	0	0
Max	14545	29626	3639	13351	2598	296	2444	403	2227	438	1093
CV	0.62	0.63	0.62	0.62	0.64	0.64	0.65	0.75	0.88	0.99	1.01
Oxide (ppm)	**Tm_2O_3**	**Yb_2O_3**	**Lu_2O_3**	**Y_2O_3**	**ZrO_2**	**Nb_2O_5**	**HfO_2**	**Ta_2O_5**	**Ga_2O_3**	**ThO_2**	**U_3O_8**
Count	4189	4189	4189	4189	4147	4313	4147	4301	3742	4189	4207
Mean	13	80	11	1115	23656	3189	463	284	151	132	29
Median	9	56	8	803	21481	3005	435	263	147	101	24
Std. Dev.	14	84	12	1070	16072	1843	300	173	32	115	27
Min	0	0	0	10	37	5	1	0	28	2	1
Max	153	918	132	11151	117605	15994	2510	1542	396	1252	805
CV	1.04	1.05	1.07	0.96	0.68	0.58	0.65	0.61	0.21	0.87	0.94
Item	**Length (m)**	**LREO (%)**	**HREO (%)**	**TREO (%)**	**H:T (%)**	**NMR (US$)**	**Density (g/cm^3)**				
Count	4189	4189	4189	4189	4189	4147	4173				
Mean	1.98	1.02	0.21	1.24	17.08	495	2.84				
Median	2.00	0.93	0.17	1.14	15.84	449	2.81				
Std. Dev.	0.18	0.64	0.18	0.77	7.31	313	0.17				
Min	0.20	0.01	0.00	0.01	4.18	5	1.91				
Max	2.95	6.37	1.90	6.88	70.53	2981	4.07				
CV	0.09	0.62	0.85	0.62	0.43	0.63	0.06				

Upper Zone Composites											
Oxide (ppm)	**La_2O_3**	**Ce_2O_3**	**Pr_2O_3**	**Nd_2O_3**	**Sm_2O_3**	**Eu_2O_3**	**Gd_2O_3**	**Tb_2O_3**	**Dy_2O_3**	**Ho_2O_3**	**Er_2O_3**
Count	22156	22156	22156	22156	22156	22156	22156	22156	22156	22156	22156
Mean	1356	3016	370	1429	267	29	192	21	81	12	28
Median	994	2166	266	1017	184	19	129	13	51	7	17
Std. Dev.	1373	3034	353	1397	261	28	193	22	91	14	34
Min	7	16	3	12	3	1	3	0	1	0	0
Max	55284	106561	5850	36991	4692	496	4370	479	1752	310	702
CV	1.01	1.01	0.95	0.98	0.98	0.98	1.00	1.05	1.12	1.22	1.24
Oxide (ppm)	**Tm_2O_3**	**Yb_2O_3**	**Lu_2O_3**	**Y_2O_3**	**ZrO_2**	**Nb_2O_5**	**HfO_2**	**Ta_2O_5**	**Ga_2O_3**	**ThO_2**	**U_3O_8**
Count	22156	22156	22156	22156	21866	23326	21866	23276	20777	22156	22353
Mean	3	20	3	334	10904	1765	207	115	160	76	34
Median	2	11	2	201	7712	1369	147	92	154	53	24
Std. Dev.	4	27	4	404	10908	1450	194	96	42	77	43
Min	0	0	0	3	6	5	0	0	25	0	0
Max	84	583	89	8169	102907	17468	1989	1102	628	2085	1312
CV	1.30	1.33	1.38	1.21	1.00	0.82	0.94	0.84	0.26	1.02	1.26
Item	**LREO (%)**	**HREO (%)**	**TREO (%)**	**H:T (%)**	**NMR (US$)**	**Density (g/cm^3)**					
Count	22156	22156	22156	22156	21866	22451					
Mean	0.64	0.07	0.72	11.49	238	2.73					
Median	0.46	0.05	0.51	9.43	170	2.71					
Std. Dev.	0.63	0.08	0.69	6.87	217	0.14					
Min	0.00	0.00	0.01	2.10	5	2.13					
Max	15.87	1.25	17.09	77.89	4564	3.95					
CV	0.98	1.08	0.97	0.60	0.91	0.05					

Figure 14.5
Composites Histograms BZ-B, BZ-T and UZ



14.5 CUT-OFF GRADE

RPA recognizes that both rare metals and rare earths contribute to the total revenue of the Nechalacho deposit.

An economic model was created, using metal prices that were updated from those used in the prefeasibility study, flotation and hydrometallurgical recoveries, the effects of payable percentages, and any payable Net Smelter Return (NSR) royalties (see Scott Wilson RPA, 2010). The payable percentages of elements contained within the enriched zirconium concentrate (EZC) were also included. The net revenue generated by this model is termed the NMR.

$$NMR = [(REO\%_1 \text{ x } R_1 \text{ x } P_1)+(REO\%_2 \text{ x } R_2 \text{ x } P_2)+ \ldots+(REO\%_n \text{ x } R_n \text{ x } P_n)]+\$EZC]$$

Where, for the rare earths, REO% is the element percent in the ore, R_1 is the total metallurgical recovery and P_1 is the price in US$/kg. $EZC is 35% of the estimated value per kilogram of payable elements contained within the EZC.

This resource estimate is based on the minimum NMR value being equal to an operating cost of US$320/t, a break-even cut-off value. This cut-off NMR was the estimate of the operating cost per tonne of ore. However, the final operating cost is different and is given in Section 21 of this report.

Table 14.4 lists the metal oxides, prices, recovery, and calculated metal values. The total NMR for each metal oxide is based on the sum of NMR values from the processes.

Table 14.4
Calculation of Metal Values

Oxide	Price (US$/kg)	Separated REE (US$/kg)	Acid Bake Residue (US$/kg)	Value for NMR (US$/kg)
Nb_2O_5	55.86	-	11.99	11.99
ZrO_2	3.77	-	1.07	1.07
Ta_2O_5	255.63	-	29.87	29.87
Y_2O_3	67.25	25.92	9.17	35.09
Eu_2O_3	1,392.57	652.49	147.24	799.73
Gd_2O_3	54.99	25.77	5.81	31.58
Tb_2O_3	1,055.70	486.67	114.56	601.23
Dy_2O_3	688.08	312.00	76.58	388.58
Ho_2O_3	66.35	28.58	7.94	36.52
Er_2O_3	48.92	17.75	7.08	24.82
Tm_2O_3	-	-	-	-
Yb_2O_3	-	-	-	-
Lu_2O_3	522.83	102.73	107.65	210.38
La_2O_3	8.75	6.08	0.19	6.28
Ce_2O_3	6.23	4.24	0.17	4.41
Pr_2O_3	75.20	49.44	2.72	52.16
Nd_2O_3	76.78	48.16	3.63	51.79
Sm_2O_3	6.75	3.47	0.60	4.07

Separated REE Values						
Oxide	Price (US$/kg)	Flotation Recovery (%)	Acid Bake Recovery (%)	Separation Recovery (%)	Separated REE Payable %	Separated REE (US$/kg)
Nb_2O_5	55.86	69	0	0	100	-
ZrO_2	3.77	90	0	0	100	-
Ta_2O_5	255.63	63	0	0	100	-
Y_2O_3	67.25	80	49	98	100	25.92
Eu_2O_3	1,392.57	80	60	98	100	652.49
Gd_2O_3	54.99	80	60	98	100	25.77
Tb_2O_3	1,055.70	80	59	98	100	486.67
Dy_2O_3	688.08	80	58	98	100	312.00
Ho_2O_3	66.35	80	55	98	100	28.58
Er_2O_3	48.92	80	47	98	100	17.75
Tm_2O_3	-	80	39	98	100	-

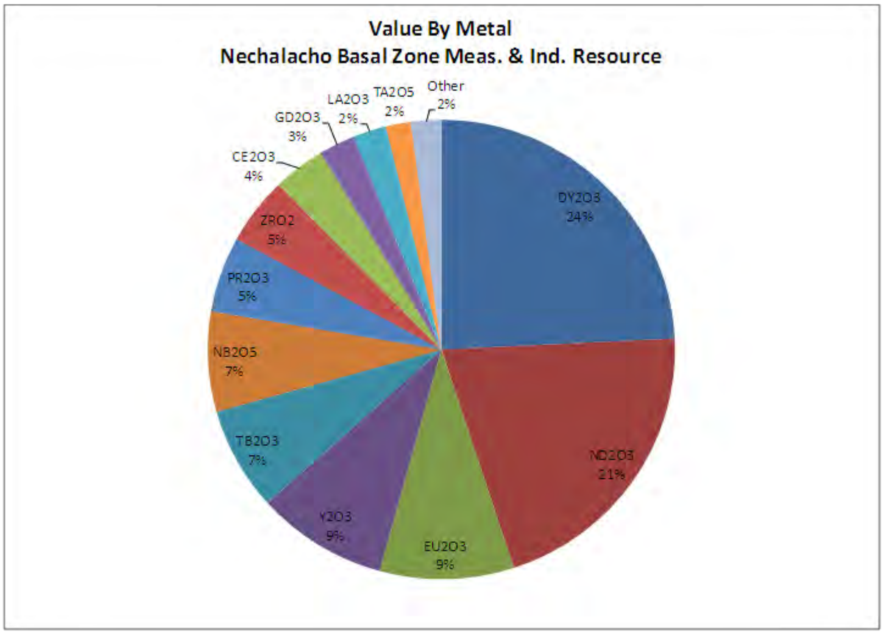
Separated REE Values						
Oxide	**Price (US$/kg)**	**Flotation Recovery (%)**	**Acid Bake Recovery (%)**	**Separation Recovery (%)**	**Separated REE Payable %**	**Separated REE (US$/kg)**
Yb_2O_3	-	80	31	98	100	-
Lu_2O_3	522.83	80	25	98	100	102.73
La_2O_3	8.75	80	89	98	100	6.08
Ce_2O_3	6.23	80	87	98	100	4.24
Pr_2O_3	75.20	80	84	98	100	49.44
Nd_2O_3	76.78	80	81	98	100	48.16
Sm_2O_3	6.75	80	66	98	100	3.47

Oxide	Price (US$/kg)	Flotation Recovery (%)	Acid Bake Residue (%)	Acid Bake Residue Payable (%)	Acid Bake Residue (US$/kg)
Nb_2O_5	55.86	69	89	35	11.99
ZrO_2	3.77	90	90	35	1.07
Ta_2O_5	255.63	63	53	35	29.87
Y_2O_3	67.25	80	49	35	9.17
Eu_2O_3	1,392.57	80	38	35	147.24
Gd_2O_3	54.99	80	38	35	5.81
Tb_2O_3	1,055.70	80	39	35	114.56
Dy_2O_3	688.08	80	40	35	76.58
Ho_2O_3	66.35	80	43	35	7.94
Er_2O_3	48.92	80	52	35	7.08
Tm_2O_3	-	80	60	35	-
Yb_2O_3	-	80	68	35	-
Lu_2O_3	522.83	80	74	35	107.65
La_2O_3	8.75	80	8	35	0.19
Ce_2O_3	6.23	80	10	35	0.17
Pr_2O_3	75.20	80	13	35	2.72
Nd_2O_3	76.78	80	17	35	3.63
Sm_2O_3	6.75	80	32	35	0.60

Prices for rare earths are typically gathered into an average price, referred to as a "basket price". The basket price is net of metallurgical recovery and payable assumptions for a mixed rare earth concentrate. Even with constant individual prices, recoveries and payables, the basket price varies according to the proportions of individual rare earth elements (in general, more heavy rare earths will raise the basket price). The basket price for the Measured and Indicated BZ resources is US$59/kg rare earth oxide, while the basket price for the entire resource is US$38/kg.

The value contribution of the payable metals in the resource is presented in Figure 14.6. Dysprosium, neodymium, europium and yttrium oxides account for approximately 63% of the revenue, based on the Measured and Indicated resource from the BZ.

Figure 14.6
In-Situ Value by Metal



14.6 VARIOGRAPHY AND TREND ANALYSIS

Omni-directional variograms were computed on BZ-T, BZ-B, and UZ composites for dysprosium (representative of the HREO group), neodymium (representative of the LREO group) and niobium (representative of the rare metals group). See Figure 14.7 through Figure 14.15.

In BZ-T, Dy shows ranges of 65 m, Nd of 50 m, and 50 m for Nb. In BZ-B, Dy displays a range of 160 m, Nd of 80 m, and 50 m for Nb.

The UZ omni-directional ranges are 100 m for Dy, 50 m for Nd, and 40 m for Nb.

Horizontal directional variograms were also computed by RPA for the BZ-B, BZ-T, and the whole BZ. Table 14.5 shows the orientation and length of maximum ranges identified.

Table 14.5
Maximum Ranges and Orientations in BZ Variograms

Domain	Dysprosium		Neodymium		Niobium	
	Max Range	Azimuth	Max Range	Azimuth	Max Range	Azimuth
BZ	160 m	multiple	80 m	65° to 105°	120 m	90° to 115°
BZ-T	120 m	175°	80 m	95°	90 m	100° to 145°
BZ-B	140 m	80°, 160°	120 m	0°, 95°	120 m	80° to 95°

Figure 14.7
BZ-T Composite Dysprosium Omni-directional Variogram

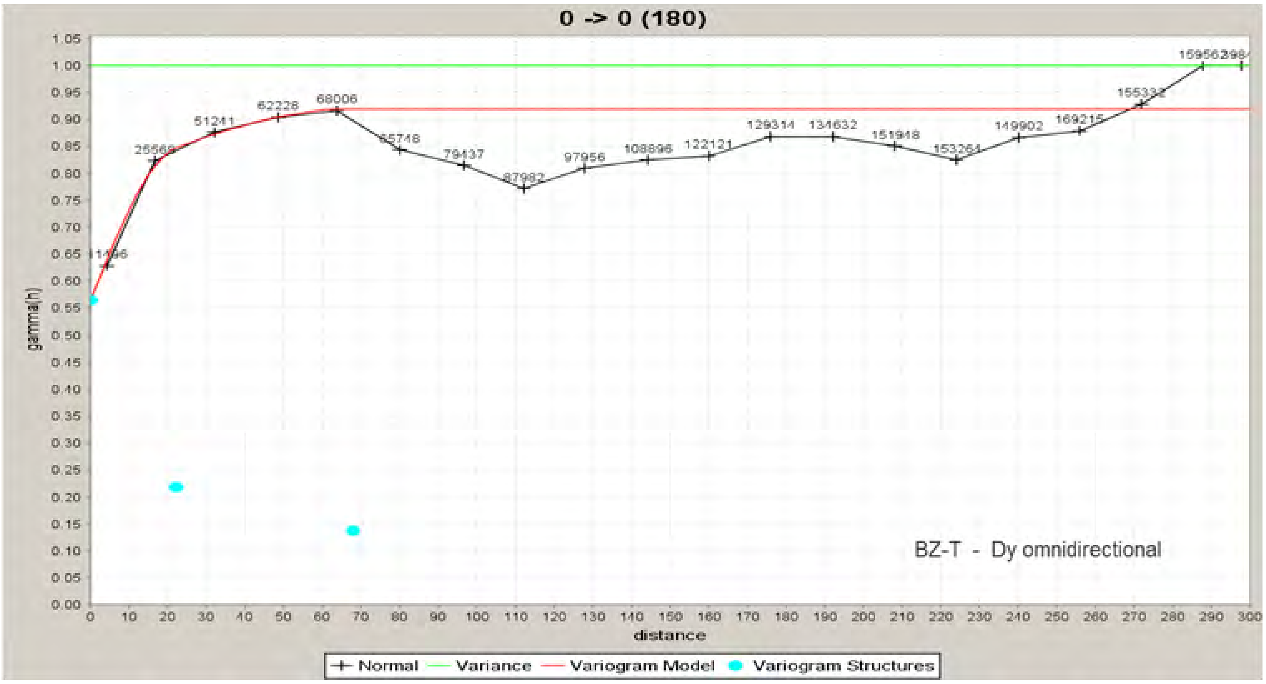


Figure 14.8
BZ-T Composite Neodymium Omni-directional Variogram

Figure 14.9
BZ-T Composite Niobium Omni-directional Variogram

Figure 14.10
BZ-B Composite Dysprosium Omni-directional Variogram

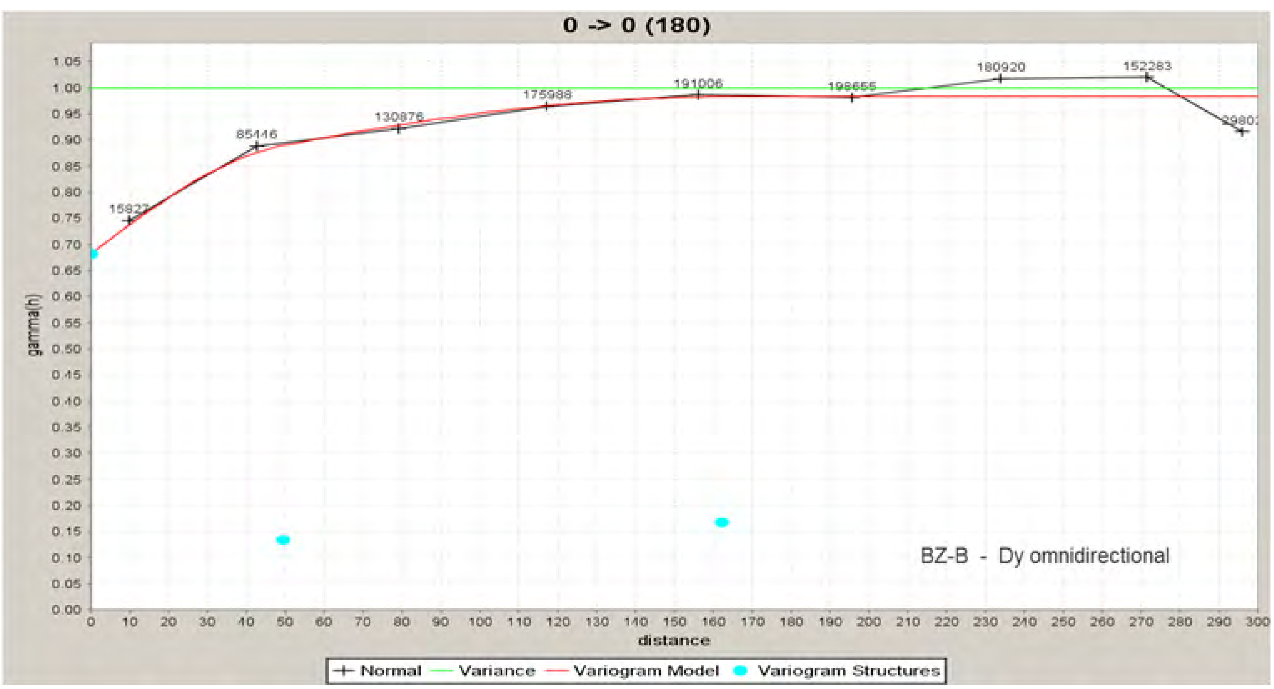


Figure 14.11
BZ-B Composite Neodymium Omni-directional Variogram

Figure 14.12
BZ-B Composite Niobium Omni-directional Variogram

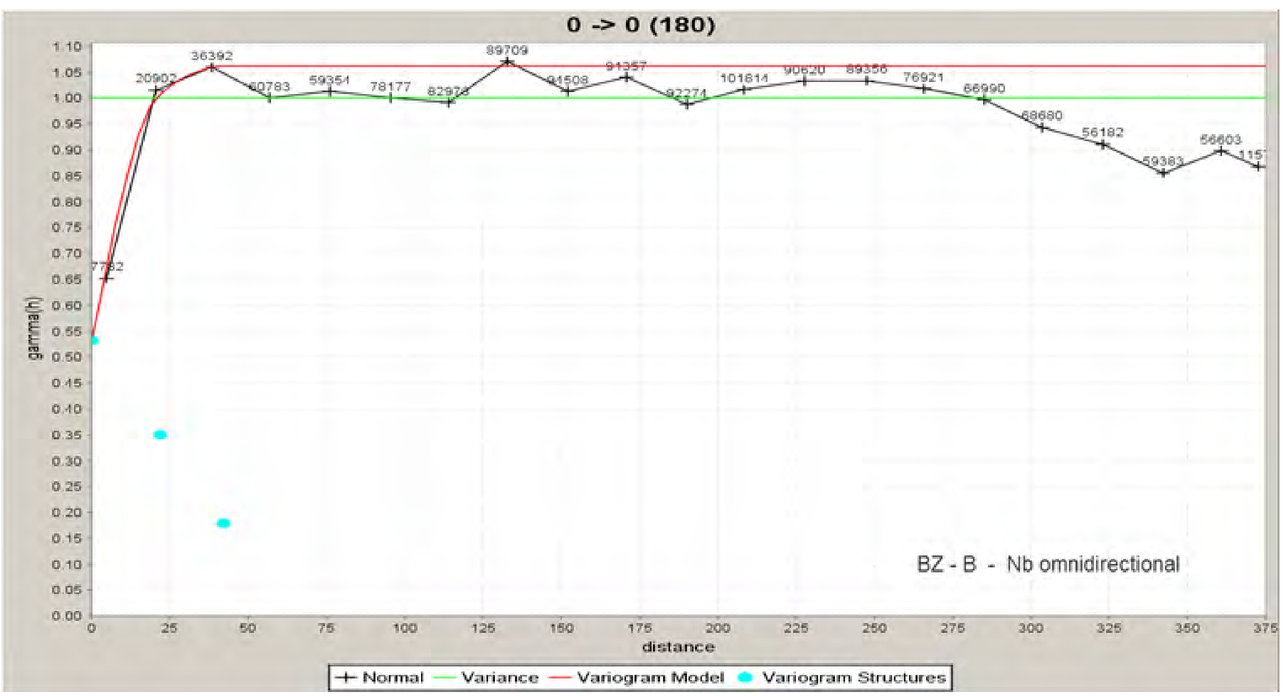


Figure 14.13
UZ Composite Dysprosium Omni-directional Variogram

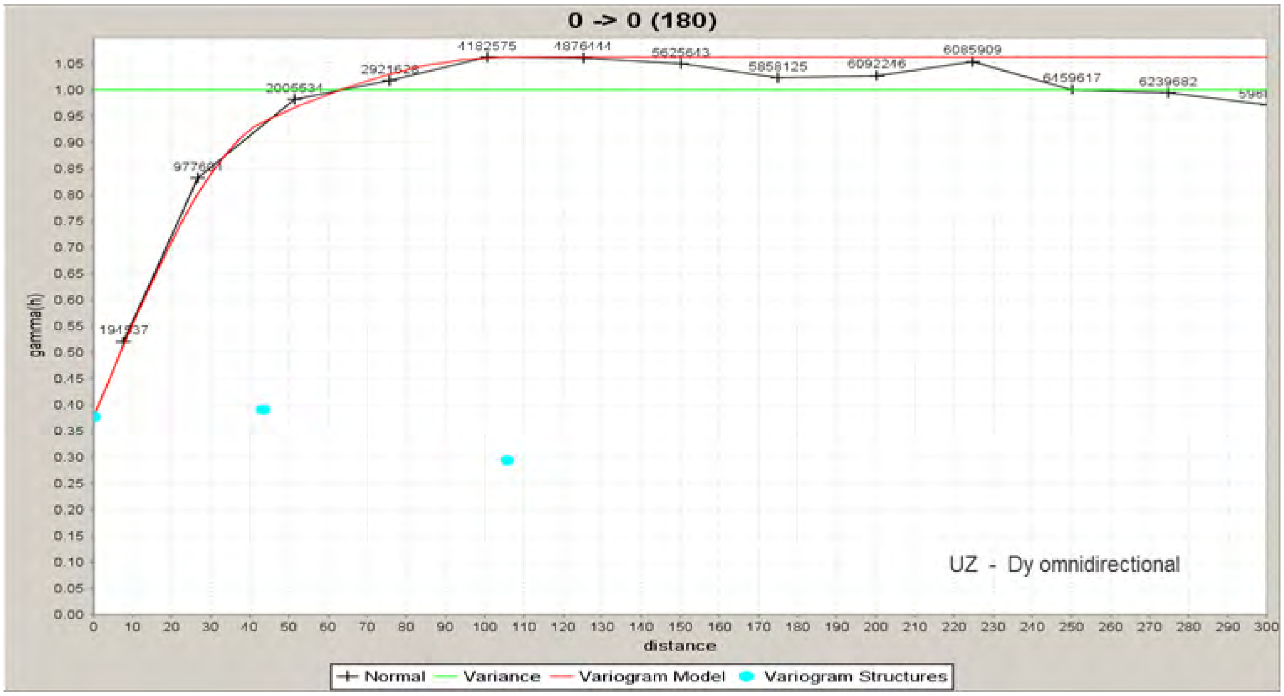


Figure 14.14
UZ Composite Neodymium Omni-directional Variogram

Figure 14.15
UZ Composite Niobium Omni-directional Variogram

The variography analysis indicated that a maximum distance of 60 m away from a drill hole was appropriate for classification into the Indicated category, and 120 m maximum distance for classification into the Inferred category. The shorter ranges obtained for the UZ suggest that supplementary conditions should be used for resource classification within the UZ.

14.7 BLOCK MODEL AND GRADE ESTIMATION

A block model, extending beyond the limits of the UZ, BZ-B and BZ-T, was created using MineSight software. The block sizes were 10 m east-west, 10 m north-south, and 5 m vertical. Table 14.6 lists the block model characteristics.

Table 14.6
Block Model Definition

Element	X (m)	Y (m)	Z (m)
Origin	415,000	6,885,000	300
Block size	10	10	5
Block count	300	400	80

The resource wireframes were used to identify the resource blocks and assign the percentage of each block inside the wireframe. For the blocks straddling the contacts between the UZ and BZ-T, and the BZ-T and BZ-B, a percent majority rule was used to assign blocks to either of the zones. Subsequently, based on the dyke wireframe, the percentage of dyke material was subtracted for the affected blocks.

The interpolation method used for the resource estimate was inverse distance squared, performed in three passes, with a horizontal search ellipse, separately for each of the mineralized zones. The first pass had the longest search radii and was used to populate blocks farthest from the drill holes, then subsequent passes overwrote blocks closer to drill holes. The third pass, with shortest ranges, was designed to preserve local grades. The composite selection strategy and the characteristics of the search ellipse for each pass are shown in Table 14.7.

Table 14.7
Grade Interpolation - Search Strategy Parameters

Zones	Ellipse	Anisotropy			Rotation About			Sample Selection		
		X (m)	Y (m)	Z (m)	Z	X	Z	Min Sample	Max Sample	Max per Drill Hole
All	Pass 1	240	240	120	0°	0°	0°	1	15	3
	Pass 2	60	60	30	0°	0°	0°	4	9	3
	Pass 3	20	20	10	0°	0°	0°	3	9	3

Individual oxide values in ppm were interpolated and then summed to obtain the sum of the light, heavy and total rare earth oxide values of each resource block.

The group of LREO comprises La_2O_3, Ce_2O_3, Pr_2O_3, Nd_2O_3 and Sm_2O_3. The group of HREO comprises Eu_2O_3, Gd_2O_3, Tb_2O_3, Dy_2O_3, Ho_2O_3, Er_2O_3, Tm_2O_3, Yb_2O_3 and Lu_2O_3, as well as Y_2O_3. The TREO value represents the sum of LREO and HREO.

The block NMR value was calculated by summing the contributions of all of the individual elements that generate revenue. Oxide formulae and price per kilogram are presented in Table 14.4, above.

The details on the collection of specific gravity data are presented in Section 11.0. Specific gravity data were handled in a similar manner to the grades. The specific gravity values were composited, and then interpolated using the same search criteria and interpolation method.

14.8 BLOCK MODEL VALIDATION

The block model provided by Avalon was imported in Gemcom GEMS by RPA for validation. The interpolated block grades were visually compared with the grades of the composites on vertical sections. Block domain flagging, percent assignment, density, and classification were also inspected. Calculation of LREO, HREO, TREO and NMR was verified on random blocks, and the resource report replicated in GEMS.

As part of the block model validation process, RPA developed an alternate, flattened block model using the “unwrinkle” procedure in GEMS. The base of the BZ was used as reference.

The flattened model contained only the BZ, as a single domain, in an attempt to diminish the influence of the undulating nature of the mineralized domain. The composites generated by Avalon for the BZ-B and BZ-T were used, and the same search strategy and interpolation method were applied.

Compared to the Avalon block model, the RPA flattened model displayed a smoother vertical and lateral grade distribution following the undulations of the BZ base. Local grade differences were noted along the contact of BZ-B and BZ-T and at the lateral limits of the BZ-B due to the different approach of the two models: a single domain in the RPA model versus two domains with hard boundaries in the Avalon model. The RPA model generated resource grades similar to those based on the Avalon block model.

In the opinion of RPA, the Avalon block model is a reasonable representation of the rare earths and rare metals mineralization of the Nechalacho deposit.

14.9 RESOURCE CLASSIFICATION

The criteria used for the mineral resource classification of the Nechalacho deposit include drilling density, availability of full set of assay data, and the mineralized zone. The BZ-T and BZ-B are considered a single unit for classification and reporting purposes.

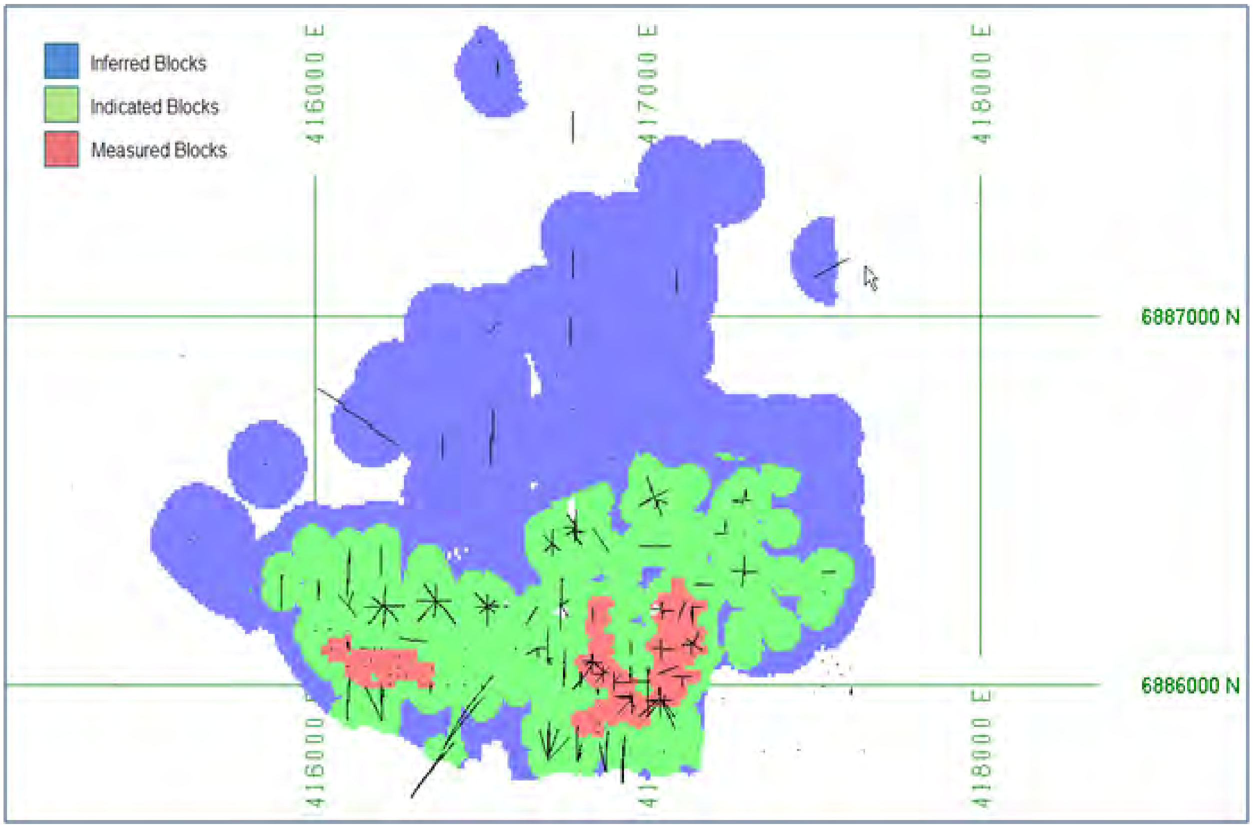
For all domains, blocks populated in the first pass and up to 120 m away from a drill hole were classified as Inferred.

Within the UZ, blocks populated in the second pass (a 60 m search ellipse and a minimum of two drill holes) were classified as Indicated. A manually-digitized contour was used to select and reclassify isolated Indicated blocks or patches of blocks into the Inferred category.

For the BZ, blocks within 60 m of a drill hole were classified as Indicated. A manually-digitized contour was used to select and reclassify isolated Indicated blocks or patches of blocks to the Inferred category.

In the BZ, two contiguous areas supported by Avalon drill holes spaced at approximately 20 m to 30 m apart were manually digitized to define the Measured blocks (see Figure 14.16).

Figure 14.16
BZ Block Resource Classification



The current mineral resource estimate (at 21 November, 2012) at a cut-off NMR value of US$320/t is shown in Table 14.1, above. The grades of individual rare earth oxides are presented in Table 14.8.

Table 14.8
Mineral Resource Estimate Grades of Individual Rare Earth Oxides and Specific Gravity

Category	Zone	Tonnes (million)	La_2O_3 (ppm)	Ce_2O_3 (ppm)	Pr_2O_3 (ppm)	Nd_2O_3 (ppm)	Sm_2O_3 (ppm)	Eu_2O_3 (ppm)	Gd_2O_3 (ppm)	Tb_2O_3 (ppm)
Measured	Basal	10.86	2,629	5,878	745	2,928	652	82	594	91
	Upper	-	-	-	-	-	-	-	-	-
Total Measured		**10.86**	**2,629**	**5,878**	**745**	**2,928**	**652**	**82**	**594**	**91**
Indicated	Basal	55.81	2,522	5,605	701	2,761	596	73	529	80
	Upper	54.59	2,686	5,970	740	2,853	539	58	387	42
Total Indicated		**110.40**	**2,603**	**5,786**	**720**	**2,806**	**568**	**66**	**459**	**61**
Measured and Indicated	Basal	66.67	2,539	5,649	708	2,788	605	75	539	82
	Upper	54.59	2,686	5,970	740	2,853	539	58	387	42
Total Measured and Indicated		**121.26**	**2,605**	**5,794**	**723**	**2,817**	**575**	**67**	**471**	**64**
Inferred	Basal	61.09	2,110	4,760	608	2,390	487	60	439	63

	Upper	122.28	2,312	5,367	661	2,576	465	51	340	35
Total Inferred		**183.37**	**2,245**	**5,165**	**643**	**2,514**	**473**	**54**	**373**	**44**

Category	Zone	Tonnes (million)	Dy_2O_3 (ppm)	Ho_2O_3 (ppm)	Er_2O_3 (ppm)	Tm_2O_3 (ppm)	Yb_2O_3 (ppm)	Lu_2O_3 (ppm)	Y_2O_3 (ppm)	SG
Measured	Basal	10.86	471	84	221	29	174	24	2,061	2.85
	Upper	-	-	-	-	-	-	-	-	-
Total Measured		**10.86**	**471**	**84**	**221**	**29**	**174**	**24**	**2,061**	**2.85**
Indicated	Basal	55.81	413	72	182	24	141	20	1,813	2.88
	Upper	54.59	160	23	54	6	39	5	649	2.80
Total Indicated		**110.40**	**288**	**48**	**119**	**15**	**91**	**13**	**1,237**	**2.84**
Measured and Indicated	Basal	66.67	422	74	189	25	147	20	1,853	2.88
	Upper	54.59	160	23	54	6	39	5	649	2.80
Total Measured and Indicated		**121.26**	**304**	**51**	**128**	**16**	**98**	**14**	**1,311**	**2.84**
Inferred	Basal	61.09	315	55	132	18	106	15	1,327	2.83
	Upper	122.28	137	20	46	6	40	6	560	2.81
Total Inferred		**183.37**	**196**	**32**	**75**	**10**	**62**	**9**	**816**	**2.82**

1. CIM definitions were followed for Mineral Resources.
2. Mineral Resources are estimated at a NMR cut-off value of US$320/t. NMR is defined as "Net Metal Return" or the in situ value of all payable metals, net of estimated metallurgical recoveries and off-site processing costs.
3. An exchange rate of US$1=CAD1.05 was used.
4. Heavy rare earth oxides (HREO) is the total concentration of: Y_2O_3, Eu_2O_3, Gd_2O_3, Tb_2O_3, Dy_2O_3, Ho_2O_3, Er_2O_3, Tm_2O_3, Yb_2O_3 and Lu_2O_3.
5. Total rare earth oxides (TREO) is HREO plus light rare earth oxides (LREO): La_2O_3, Ce_2O_3, Pr_2O_3, Nd_2O_3 and Sm_2O_3.
6. Rare earths were valued at an average net price of US$62.91/kg, ZrO_2 at US$3.77/kg, Nb_2O_5 at US$56/kg, and Ta_2O_5 at US$256/kg. Average REO price is net of metallurgical recovery and payable assumptions for contained rare earths, and will vary according to the proportions of individual rare earth elements present. The proportions are based on the actual planned production from the Nechalacho project.
7. ZrO_2 refers to zirconium oxide, Nb_2O_5 refers to niobium oxide, and Ta_2O_5 refers to tantalum oxide.

Table 14.9 lists the tonnage and TREO and rare metal oxide grades reported at various higher NMR cut-off values.

Table 14.9
Tonnes and TREO Grades at NMR Cut-off Values over US$320/t

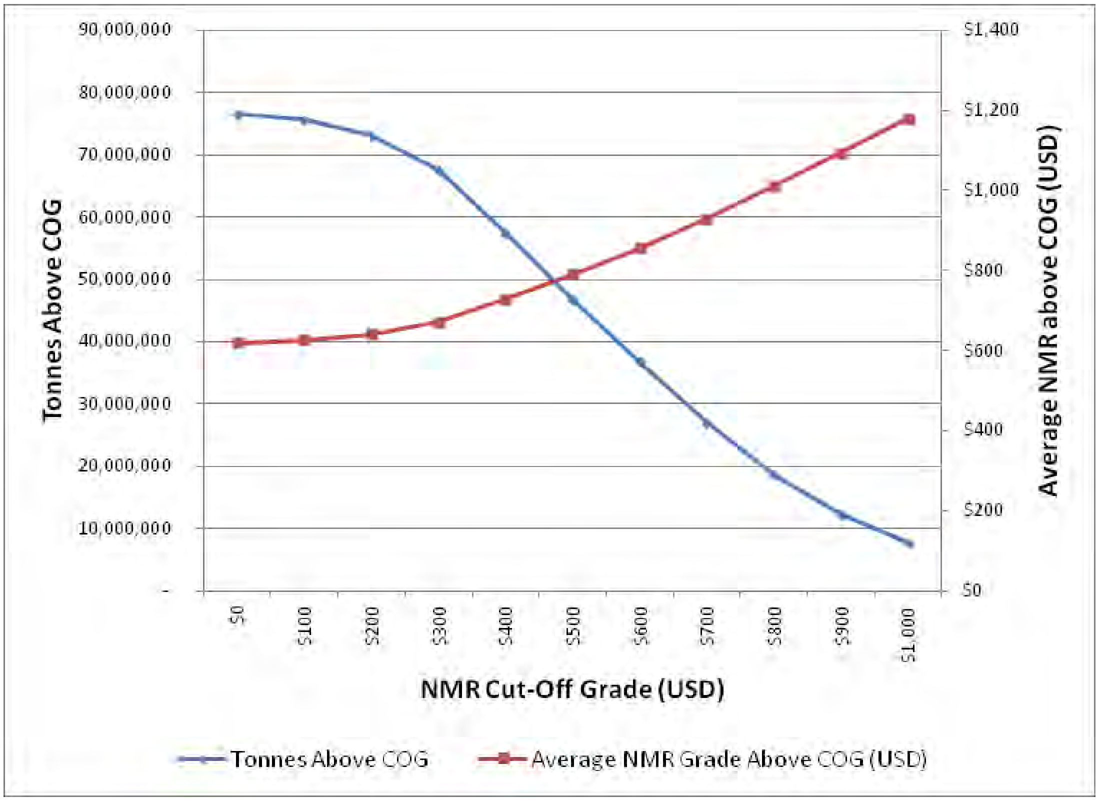
Zone	NMR Cut-off (US$/t)	Tonnes (Million)	TREO (%)	HREO (%)	HREO/TREO (%)	ZrO_2 (%)	Nb_2O_5 (%)	Ta_2O_5 (%)
Measured								
BASAL	≥320	10.86	1.67	0.38	22.26	3.23	0.40	0.04
BASAL	≥600	6.76	1.98	0.49	24.65	3.90	0.48	0.05
BASAL	≥800	4.08	2.22	0.59	26.60	4.37	0.53	0.06
BASAL	≥1,000	2.00	2.51	0.70	27.91	4.92	0.60	0.06
Indicated								
BASAL	≥320	55.81	1.55	0.33	20.99	3.01	0.40	0.04
BASAL	≥600	30.83	1.89	0.45	23.74	3.66	0.47	0.05
BASAL	≥800	15.08	2.18	0.56	25.75	4.19	0.53	0.06

BASAL	≥1,000	5.82	2.52	0.67	26.88	4.75	0.60	0.06
Measured plus Indicated								
BASAL	≥320	66.67	1.57	0.34	21.20	3.05	0.40	0.04
BASAL	≥600	37.59	1.91	0.45	23.90	3.70	0.47	0.05
BASAL	≥800	19.16	2.19	0.56	25.93	4.23	0.53	0.06
BASAL	≥1,000	7.81	2.52	0.68	27.14	4.79	0.60	0.06
Inferred								
BASAL	≥320	61.09	1.29	0.25	19.44	2.69	0.36	0.03
BASAL	≥600	19.30	1.69	0.37	22.08	3.33	0.45	0.04
BASAL	≥800	4.99	2.02	0.50	25.33	3.87	0.51	0.05
BASAL	≥1,000	1.08	2.42	0.62	25.79	4.23	0.56	0.06

1. CIM definitions were followed for Mineral Resources.
2. Mineral Resources are estimated at a NMR cut-off value of US$320/t. NMR is defined as "Net Metal Return" or the in situ value of all payable metals, net of estimated metallurgical recoveries and off-site processing costs.
3. An exchange rate of US$1=CAD1.05 was used.
4. Heavy rare earth oxides (HREO) is the total concentration of: Y_2O_3, Eu_2O_3, Gd_2O_3, Tb_2O_3, Dy_2O_3, Ho_2O_3, Er_2O_3, Tm_2O_3, Yb_2O_3 and Lu_2O_3.
5. Total rare earth oxides (TREO) is HREO plus light rare earth oxides (LREO): La_2O_3, Ce_2O_3, Pr_2O_3, Nd_2O_3 and Sm_2O_3.
6. Rare earths were valued at an average net price of US$62.91/kg, ZrO_2 at US$3.77/kg, Nb_2O_5 at US$56/kg, and Ta_2O_5 at US$256/kg. Average REO price is net of metallurgical recovery and payable assumptions for contained rare earths, and will vary according to the proportions of individual rare earth elements present. The proportions are based on the actual planned production from the Nechalacho project.
7. ZrO_2 refers to zirconium oxide, Nb_2O_5 refers to niobium oxide, and Ta_2O_5 refers to tantalum oxide.

Figure 14.17 presents the grade-tonnage curve for the Measured and Indicated resource in the BZ.

Figure 14.17
Grade-tonnage Curve for Measured + Indicated BZ



14.10 COMPARISON WITH PREVIOUS MINERAL ESTIMATE

The May, 2013 mineral resource estimate is compared with the January, 2011 estimate in Table 14.10.

Table 14.10
Mineral Resource Comparison - 2011 to 2013

Year	Resource	Tonnes (Million)	TREO (%)	HREO (%)	ZrO_2 (%)	Nb_2O_5 (%)	Ta_2O_5 (%)
2011	Basal Indicated	57.82	1.56	0.33	2.99	0.40	0.04
	Basal Inferred	107.26	1.35	0.26	2.83	0.37	0.04
	Upper Indicated	30.64	1.48	0.15	2.1	0.31	0.02
	Upper Inferred	115.98	1.27	0.12	2.37	0.34	0.02
2013	Basal Measured and Indicated	66.67	1.57	0.34	3.05	0.40	0.04
	Basal Inferred	61.09	1.29	0.25	2.69	0.36	0.03
	Upper Indicated	54.59	1.42	0.14	1.96	0.28	0.02
	Upper Inferred	122.28	1.26	0.12	2.21	0.32	0.02

For the BZ, the addition of new drilling led to the upgrade of Inferred material into the Indicated category, and upgrade of Indicated material into the Measured category.

Slight variations to the grades were also induced by changes to the metal prices used for NMR calculation and by the increased cut-off threshold.

The updated geological model changed the geometry of the mineralized zones and the dykes, while the maximum distance for the Inferred category in 2012 was shorter.

In the UZ, the tonnage of Indicated material almost doubled while the grades are slightly lower; the tonnage of Inferred material rose, while the grades are virtually unchanged. For the BZ, Indicated material increased in tonnage while the grades stayed the same; there is less Inferred material, mainly due to shorter radii used for this resource category.

The price for lutetium was changed between the November, 2012 mineral resource estimate and the final feasibility study (see Section 19.0), but the difference is insignificant and not material to the resource modelling.

15.0 MINERAL RESERVE ESTIMATES

The mineral reserve estimate for the Nechalacho project presented in the feasibility study was estimated from the block model prepared by Avalon and audited originally by RPA on 21 November, 2012. Subsequent to this, Avalon updated the database and re-estimated the resource as of 3 May, 2013. The update included correction of some minor assay data entry errors and drill hole locations. The net effect of these changes are considered immaterial as the changes for the reserves were zero for tonnage and 0.001% to 0.168% of the grade of individual elements (that is, no change in the second decimal place of TREO, HREO or individual rare earth oxides). The mineral reserve estimate is derived from this block model by applying the appropriate technical and economic parameters to extraction of the REE with proven underground mining methods (see Section 16.0, Mining Methods).

The mineral reserve has been estimated based on conversion of the high grade mineral resources at a cut-off value greater than US$320/t NMR. Payable elements include the REE, zirconium, niobium and tantalum. No Inferred mineral resources were converted to mineral reserves. The high grade mineral resources are 34.7% and 14.7% of the total Measured and Indicated mineral resources, respectively.

15.1 MINERAL RESERVE ESTIMATION METHODOGY

15.1.1 Cut-off Grade

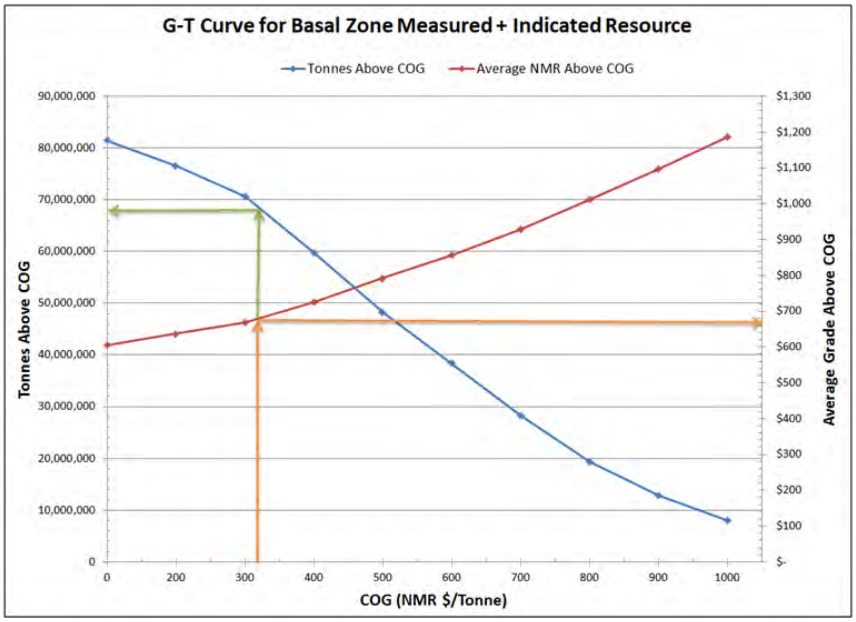
The cut-off grade for the design of the stopes was based upon a US$320/t NMR determined as described in Section 14.0. There are several payable products from the Nechalacho project, which include the individual REO products and the EZC which contains zirconium, niobium and tantalum as well as a portion of the HREO. The NMR/t was based upon the estimated product prices after allowance for recoveries at the concentrator, the hydrometallurgical plant and the rare earth refinery, and less the associated operating costs.

Figure 15.1 shows that at a cut-off value of US$320/t, there is about 68 Mt of Measured and Indicated mineral resources to sustain a mine life of over 90 years at a production rate of 2,000 t/d. The value of average grade material above cut-off grade is approximately US$670/t.

Details of the metal prices used are presented in Section 19.0. The price for lutetium was changed between the November, 2012 mineral resource estimate and the final feasibility study, but the difference is insignificant and not material to the resource modelling.

The cost parameters to determine the Proven and Probable mineral reserves are detailed in Section 21.0 of this technical report.

Figure 15.1
Cut-off Grade Determination Chart



15.1.2 Mine Design Parameters

The key design criteria set for the Nechalacho mine are:

- Initial design based on a 20-year life-of-mine (LOM) of high grade material.
- Mechanized cut and fill and long hole mining methods with paste backfill.
- Minimum mining thickness of 5 m.
- Extraction ratio of 94.2%.
- Internal dilution of 8.5%.
- External dilution of 5% applied to secondary stopes.
- Estimated total average dilution for the life of mine of approximately 11%.
- Production rate of 2,000 t/d ore (730,000 t/y).
- Ore bulk density of 2.91 t/m^3.

15.2 MINERAL RESERVE ESTIMATE

The mineral reserve estimate for the Nechalacho project shown in Table 15.1 has an effective date of 3 May, 2013.

The figures in the table are rounded to reflect that the numbers are estimates. The conversion of mineral resources to mineral reserves includes technical information that requires subsequent calculations or estimates to derive sub-totals, totals and weighted averages. Such calculations or estimations inherently involve a degree of rounding and consequently introduce a margin of error. Where these occur, Micon does not consider them to be material.

To the best of Micon's knowledge there are no legal, political, environmental, or other issues which would materially affect development of the Mineral Reserve estimated above.

Table 15.1
Mineral Reserve Estimate as at 3 May, 2013

Description	Mineral Reserve Category		
	Proven	Probable	Proven and Probable
Tonnage (Mt)	**3.68**	**10.93**	**14.61**
TREO (%)	**1.7160**	**1.6923**	**1.6980**
HREO (%)	**0.4681**	**0.4503**	**0.4548**
HREO/TREO	27.28%	26.61%	26.78%
La_2O_3	0.256%	0.256%	0.256%
Ce_2O_3	0.570%	0.567%	0.568%
Pr_2O_3	0.072%	0.071%	0.071%
Nd_2O_3	0.284%	0.283%	0.283%
Sm_2O_3	0.065%	0.065%	0.065%
Eu_2O_3	0.008%	0.008%	0.008%
Gd_2O_3	0.062%	0.061%	0.061%
Tb_2O_3	0.010%	0.010%	0.010%
Dy_2O_3	0.058%	0.056%	0.056%
Ho_2O_3	0.011%	0.010%	0.010%
Er_2O_3	0.029%	0.027%	0.028%
Tm_2O_3	0.004%	0.004%	0.004%
Yb_2O_3	0.023%	0.022%	0.022%
Lu_2O_3	0.003%	0.003%	0.003%
Y_2O_3	0.259%	0.249%	0.251%
ZrO_2	3.440%	3.309%	3.342%
Nb_2O_5	0.425%	0.413%	0.416%
Ta_2O_5	0.046%	0.045%	0.045%

1. CIM definitions were followed for Mineral Reserves.
2. Mineral Reserves are based on Mineral Resources published by Avalon in News Release dated November 26th, 2012 and audited by Roscoe Postle Associates Inc., and modified as of 3 May, 2013.
3. Mineral Reserves are estimated using price forecasts for 2016 for rare earth oxides given below.
4. HREO grade comprises Y_2O_3, Eu_2O_3, Gd_2O_3, Tb_2O_3, Dy_2O_3, Ho_2O_3, Er_2O_3, Tm_2O_3, Yb_2O_3, and Lu_2O_3. TREO grade comprises all HREO and La_2O_3, Ce_2O_3, Nd_2O_3, Pr_2O_3, and Sm_2O_3.
5. Mineral Reserves are estimated using a NMR cash cost cut-off value of US$320/t.
6. Rare earths were valued at an average net price of US$62.91/kg, ZrO2 at US$3.77/kg, Nb_2O_5 at US$56/kg, and Ta_2O_5 at US$256/kg. Average REO price is net of metallurgical recovery and payable assumptions for contained rare earths, and will vary according to the proportions of individual rare earth elements present. In this case, the proportions of REO as final products were used to calculate the average price.
7. Mineral reserves calculation includes an average internal dilution of 8.5% and external dilution of 5% on secondary stopes.

Micon believes the key assumptions, parameters and methods used to convert mineral resource to mineral reserve are appropriate. To the best of Micon's knowledge there are no known mining, metallurgical, infrastructure, permitting or other relevant factors that may materially affect the mineral reserve estimate.

16.0 MINING METHODS

16.1 INTRODUCTION

Underground mining of the Measured and Indicated mineral resource of the Basal Zone was investigated for the feasibility study. The majority of the mineral resource of the Basal Zone lies directly beneath and to the north of Long Lake, approximately 200 m below surface. Thus, the deposit is to be mined using underground mining methods.

The mine production rate is 2,000 t/d (730,000 t/y) of ore and the mine life is 20 years.

16.2 MINE DESIGN PARAMETERS

The key design criteria set for the Nechalacho project are:

- Mechanized cut and fill and long hole mining methods with paste backfill.
- Extraction ratio of 94.2%.
- Internal dilution of 8.5%.
- External dilution of 5% applied to all secondary stopes.
- Estimated total average dilution for the life of mine of approximately 11%.
- Ore bulk density of 2.91 t/m^3.
- Underground crushing with conveyor handling to surface.
- 24 h/d and 365 operating days per year.

Micon believes the mine design key assumptions, parameters and methods are appropriate for the proposed mining methods and the type of deposit.

16.2.1 Geotechnical Analysis

The available geotechnical information and analysis from the Nechalacho site has been reviewed by KPL (KPL, 2012a, 2012b) with the objective of providing rock mechanics comments and recommendations with respect to the underground mine plan. Geotechnical information for the mine design was based on geotechnical data collection completed in conjunction with Avalon's on-going exploration drill program.

The majority of the geotechnical data collection and analysis was completed in 2010 by Avalon and KPL, when eight geotechnical drill holes (2,153 m) were drilled to characterize the rock mass conditions in and around the proposed mine. The geotechnical work performed included packer testing, the installation of two thermistors and laboratory rock strength testing. In addition to the geotechnical drill holes, a total of 10,140 m of core from 41 drill holes was logged by Avalon's team. The selected holes covered a range of azimuths in order to reduce directional sampling bias. The geotechnical logging was focused on gathering the pertinent rock mass characteristics.

The results from the geotechnical analysis suggest that the rock masses encountered at the Nechalacho project site are generally good quality and that there is little variation with depth. General observations include the following:

- Drill core recovery was consistently close to 100%, suggesting that few zones of reduced rock mass quality were encountered.
- RQD values were generally in the 90% to 100% range.
- RMR values were generally between 60 and 80, typical of a good quality rock mass.
- Uniaxial compressive strength is 110-160 MPa.
- No major faults intersect the mineralization.
- The oriented geomechanical drill holes suggest that all the major rock units can be expected to have some vertical discontinuities, but that the dominant discontinuity orientation will be flat-lying. Typical discontinuity spacing is expected to be approximately 1.5 m for the flat discontinuities and approximately 3 to 4 m for the steep discontinuities.

Given the relatively high intact rock strength, rock mass performance is expected to be largely controlled by the orientation and characteristics of the observed discontinuities. Due to the relatively minor variations in rock mass quality that were observed and the importance of the rock masses in and around the orebody, the focus of most of the analyses was on the Hanging Wall, Basal Zone and Footwall domains.

16.2.1.1 Stope Dimensioning and Crown Pillar

The stope dimensioning and crown pillar stability analysis was performed by KPL. The analysis indicated that excavations 15 m wide, 5 m high and 100 m in length will be stable with the proper installation of ground support and mitigation strategies (KPL, 2012b).

16.2.2 Mining Dilution and Recovery

The deposit at the Nechalacho project is relatively flat lying and will be mined with a combination of longhole stoping, and cut and fill methods. The minimum thickness used in the development of the mineral reserve estimate was 5 m.

Internal dilution was estimated based upon the grades and densities within the resource model for the areas to be extracted and is approximately 8.5% in the form of material below the cut-off value of US$320/t (mostly in the hanging wall).

External dilution is estimated at 5% and will be incurred mainly from small failures that are expected to occur at the stope.

16.3 UNDERGROUND MINING

The lower grades of HREO relative to TREO near surface and recognition of the higher value of the Basal Zone due to higher HREO/TREO led to a mine plan utilizing underground mining methods focused on the Basal Zone of the deposit.

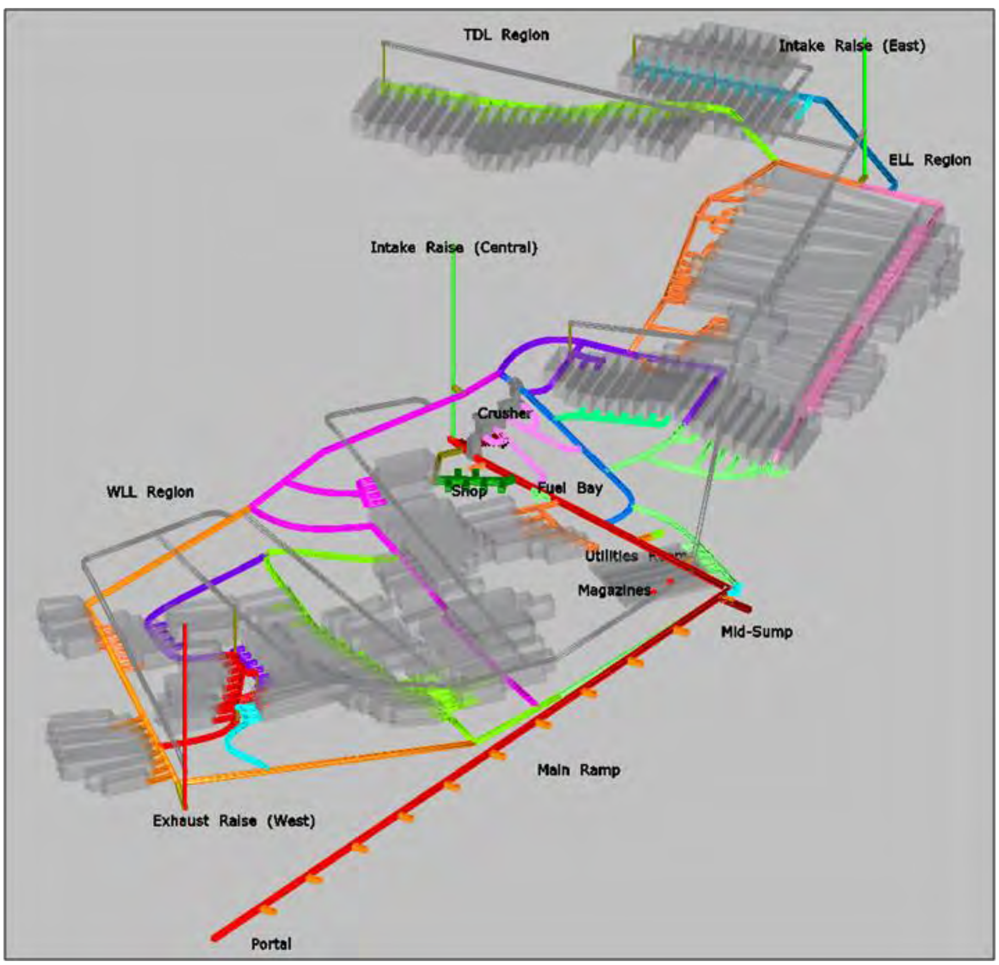
16.3.1 Mine Ramp

The mine will be accessed through a mine portal located near the concentrator. The dimensions of the 1,600 m main ramp will accommodate the overhead conveyor system and access to men and equipment.

16.3.2 Mining Method

The overall mine layout is shown in Figure 16.1.

Figure 16.1
Isometric View of Overall Mine Layout



Cut and fill and long hole stoping methods will be used with paste backfill to improve overall mine stability, reduce the surface footprint for the TMF and enable the extraction of secondary stopes for increased mining recovery. Skin pillars will be located between stopes to minimize the contamination of paste backfill material into the adjacent mining stopes.

Blasted material will be mucked and transported by mechanized underground rubber tyred equipment to the crusher station. The crushed ore will be transported to the surface by conveyor.

Zones less than 10 m thick will be mined by cut and fill methods in a primary and secondary mining sequence. Zones over 10 m thick will be mined with longhole stoping.

Secondary stopes will be mined after the adjoining primary stopes have been filled. The mining of the secondary stopes will be the same as the mining of the primary stope. Ore in the long hole stopes will be mucked remotely when the ore is located in open stopes, or when there are brow safety concerns.

16.3.3 Grade Control

Grade control within individual stopes will include sampling and analysis of drill cuttings collected from initial fan drilling in the central drift of the sill.

16.3.4 Ore and Waste Material Handling

Blasted ore will be hauled from the stopes by LHD or by truck directly to the truck dump bay feeding the run-of-mine coarse ore bin located above the underground crusher.

Until the crusher is in place and the conveyor is in service, all ore will be hauled to surface using low profile haul trucks. Development waste will also be hauled to surface in the same manner through this period.

After the crusher and conveyor are in place, the ore will be transported to surface by conveyor. Development waste will be hauled to a temporary waste stockpile on surface. Inert waste will be used for construction of site roads and the TMF.

16.3.5 Lateral and Vertical Development

Apart from the main ramp, there will be multiple underground lateral and vertical developments to support the mining activities.

The total annual mine development is presented in Table 16.1.

Table 16.1
Life-of-mine Development Schedule

Period	Total Lateral Development (m)	Total Raising (m)
Construction Year 1	1,876	0
Construction Year 2	2,587	436
Production Year 1	1,400	31
Production Year 2	3,055	0
Production Year 3	30	0.0
Production Year 4	4	6
Production Year 5 - 20	8,369	311
Total	17,320	784

16.3.6 Underground Mobile Equipment

A mine contractor will provide the underground mobile equipment and manpower for mine development. Avalon will take over operations at the commencement of commercial production, and will lease underground mobile equipment for the first five years of operation. The underground mining fleet will consist of the units for the mine development, production, and ancillary operations.

The underground equipment fleet is estimated based on an availability factor of 85% and comprises loaders, trucks, drills and support equipment.

Micon considers that the underground equipment fleet has been selected appropriately.

16.3.7 Underground Mine Personnel and Schedule

During the underground mining operations, a total of approximately 93 full-time mine employees will be required. An estimated 47 people will be on site for any given shift.

The mine labour and mine maintenance personnel will be divided into four crews. Each crew will work two 12 hours shifts per day. All the crews will be on a shift rotation of 14-day in and 14-day out routine.

Most mine administration, mine supervision, mine engineering and mine geology personnel will also work on a 14-14 schedule, on day shifts only. There will be two crews to cover the rotation schedule.

The management team will be on a rotation of 4-days in 3-days out (or 10-days in 4-days out) with a weekend rotation to ensure that one senior person is on site at weekends.

16.3.8 Mine Infrastructure

16.3.8.1 Crushing and Conveying

Ore will be crushed underground using a three stage crushing and screening facility and conveyed to the concentrator on surface.

Ore excavated during the mine development before commissioning of the underground crusher will be hauled and stockpiled on surface. Low grade ore encountered during development will be stockpiled near the portal and will be fed back into the system at a later date.

16.3.8.2 Electrical Distribution and Power Requirement

The electrical power for the site will be generated by diesel powered generators and distributed at 4,160 V. The feed to the mine will be by 4,160 V power cables installed in the ramp feeding load centres with 4,160:600 V transformers.

The estimated power consumption for the underground mining is 1.5 MW. This includes the mine ventilation system but excludes the crushing plant and conveyors which will account for a further approximately 1.5 MW of power.

16.3.8.3 Mine Ventilation and Heating

The mine ventilation design for the Nechalacho project was undertaken by Mine Ventilation Services Inc. (MVS) in 2102 (MVS, 2012). The mine fresh air intake will be through two raises located at the centroid of the deposit and the east end of the orebody. The central intake raise will also serve as a service raise for power lines and as an emergency escape way.

The mine ventilation air flow has been estimated based upon the mine equipment fleet, with an estimate of utilization and an additional allowance for losses and additional needs. The estimate of 198 m^3/s (415,000 cfm) exceeds the minimum requirements recommended by NWT Mine Health and Safety Act and Regulations

The feasibility study is based on the assumption that the emission of radon and thorium gas from the rock will not be an issue and will be appropriately diluted and exhausted with the mine air. Proper health and safety procedures for closing and sealing unused areas and for checking areas prior to reopening unventilated areas will be established to ensure that areas are suitably ventilated and that there are no noxious gases present before work commences in a new area or an area which has been closed for some time. The paste backfill will assist to seal most of the mined area and reduce the emission of these gases from the host rock.

The mine air heater will be operated in the winter months. The total estimated diesel fuel consumption for mine heating during the winter season is 4.5 ML.

16.3.8.4 Mine Dewatering

The mine is expected to have low groundwater inflow. Maximum inflow is estimated at 3.15 L/s (50 gpm) based upon the observations of the numerous core drill programs and observations from the test mine previously developed at the Nechalacho site.

Most of the mine water will be generated from the underground mining activities at an estimated flow of 16.2 L/s (257 gpm). All water will be diverted to the base of the ramp to the main sump.

16.3.9 Paste Backfill System

A feasibility study for paste backfill at the Nechalacho site was completed by Golder in 2012 (Golder, 2012).

The paste backfill plant will be located adjacent to concentrator. The paste fill distribution pipeline will be installed in the main ramp.

The tailings generated from the concentrator will be mixed with normal Portland cement to form the paste backfill material. Golder has estimated that a cement addition of 7% will be required for fill for the primary stopes with height greater than 18 m to obtain 0.59 MPa at 28 days curing period for 250 mm (10 in) slump. An estimated 2.5% of cement will be used for the secondary stopes to prevent liquefaction at 0.17 MPa for 28 days curing period for the same slump. The average cement content required on an annual basis will be between 4% and 5%. The daily Portland cement requirement is 76 t.

16.3.9.1 Mine Maintenance Facility

A maintenance facility and fuel station will be constructed underground to accommodate mobile equipment for minor repairs and fuelling. More complicated mechanical maintenance work will be completed at the mechanical shop located on surface within the concentrator building.

16.3.9.2 Mine Communication System

Mine communications will consist of telephone service to the main mine switchboard, as well as radio communications through a leaky feeder system. The communications system will also be used for monitoring and control of mine personnel, production equipment, ventilation systems, dewatering and backfill.

16.3.9.3 Mine Explosives Storage

Detonators, primers and explosives will be stored in separate approved explosives magazines which will be located underground. During the construction phase, temporary surface explosives magazines will be used.

The main explosive to be used at the Nechalacho site is ANFO, which will be prepared on a batch basis by Dyno Nobel in its proposed ANFO plant in Yellowknife. The explosive will be shipped to the project site in 450 kg totes and/or 25 kg bags. Dyno Nobel will supply all the explosives and detonators for the project.

The estimated average powder factor is 0.9 kg/t ANFO of blasted material and the annual ANFO consumption is estimated to be 285 t for the mine development. The annual consumption cartridge explosive is estimated to be 115 t for the longhole stopes.

16.3.9.4 Mine Sanitary System and Potable Water

Potable water for the underground mine will be provided in dedicated containers that will be resupplied regularly from the site potable water supply. Sanitary facilities underground will be approved self-contained units.

16.3.9.5 Refuge System

The main refuge station for the mine will be at the bottom of the ramp and will be equipped with potable water. The station can be sealed from smoke and can also supply a minimum of 12 h of fresh air.

As the mine footprint expands, additional refuge stations on the eastern and western flank of the mine will be constructed or mobile refuge chambers will be strategically located in these areas.

16.4 MINE PRODUCTION SCHEDULE

The Nechalacho production schedule is designed to extract high grade material located closest to the underground crusher station in the initial mining years to maximize present value.

Steady production at an average rate of 2,000 t/d ore is anticipated by the sixth month of operation.

The annual production schedule is presented in Table 16.2.

Table 16.2
Mine Production Schedule

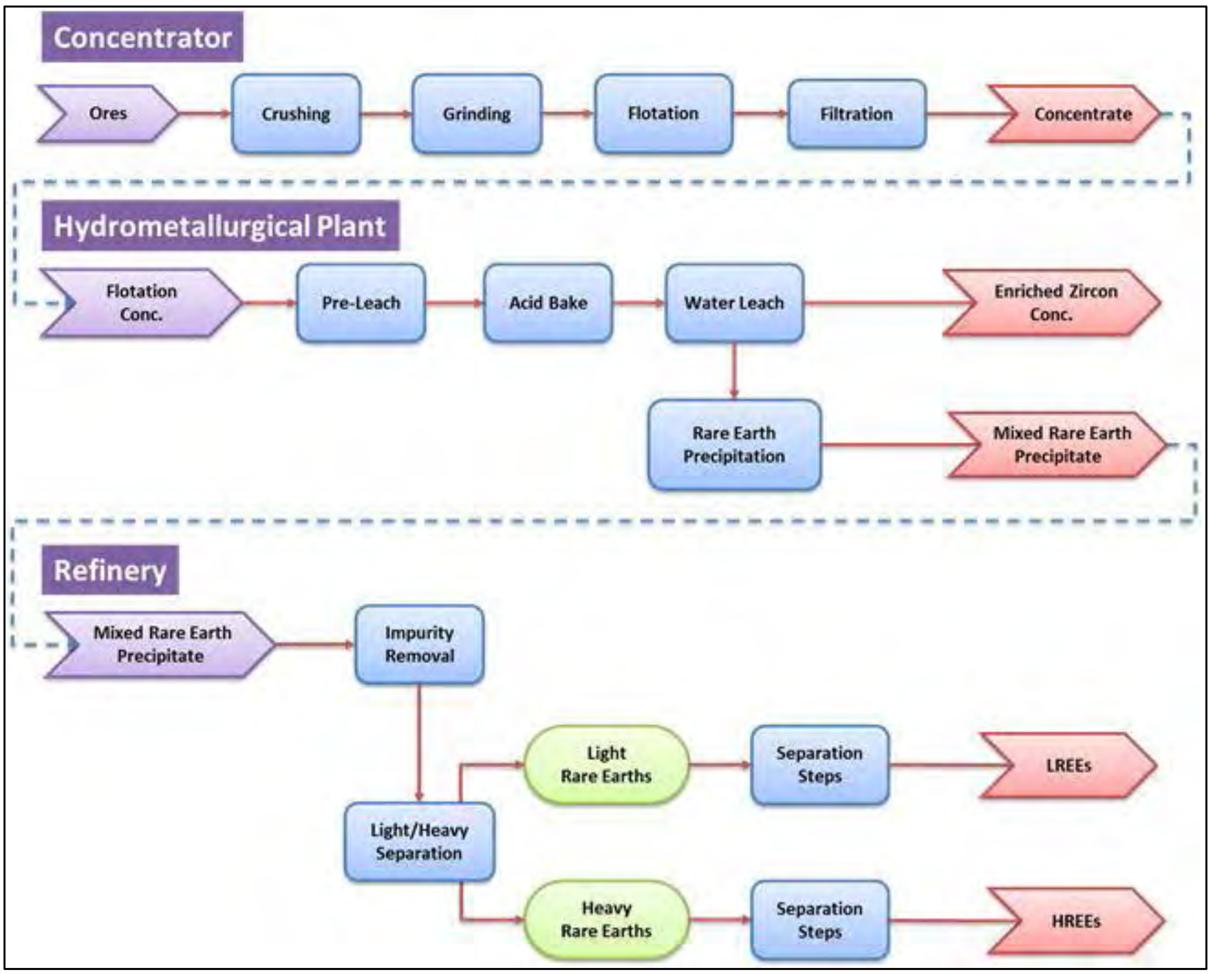
Year	Total	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
Mined Ore (Thousand t)																						
	14,600	103	725	730	730	730	730	730	730	730	730	730	730	730	730	730	730	730	730	730	730	632
Average Grade (%)																						
TREO	**1.70**	2.02	2.02	2.03	1.97	1.81	1.71	1.73	1.68	1.61	1.64	1.67	1.69	1.62	1.62	1.63	1.65	1.58	1.57	1.55	1.59	1.58
ZrO_2	**3.34**	4.07	4.18	4.31	4.08	3.75	3.42	3.37	3.29	3.11	3.14	3.15	3.23	2.93	2.68	2.74	2.83	2.97	3.36	3.31	3.47	3.35
Nb_2O_5	**0.42**	0.49	0.50	0.50	0.48	0.44	0.41	0.41	0.41	0.38	0.40	0.41	0.42	0.39	0.37	0.38	0.39	0.38	0.40	0.40	0.42	0.43
Ta_2O_5	**0.05**	0.05	0.05	0.05	0.05	0.05	0.05	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.05	0.04	0.04	0.04

17.0 RECOVERY METHODS

The processing facilities comprise three separate plants, the concentrator located at the Nechalacho site, the hydrometallurgical plant at the Pine Point site and the rare earth refinery at the Geismar site.

Extensive metallurgical testwork has been completed at a number of different laboratories over the last four years. This testwork is the basis for the flowsheets and the process design criteria developed for the three processing facilities. The relevant testwork programs are discussed in Section 13.0 of this report. A simplified flowsheet showing the three processes is presented in Figure 17.1.

Figure 17.1
Simplified Flowsheet for the Concentrator, the Hydrometallurgical Plant and the Refinery



17.1 CONCENTRATOR – NECHALACHO SITE

The concentrator has been designed to produce a rare earth concentrate feed for the hydrometallurgical plant at the Pine Point site. The design capacity of the concentrator is 2,000 t/d initially, with mining and plant layout provision for a potential future expanded capacity of 4,000 t/d.

The process design, including flowsheets, process design criteria and main equipment selection was developed by FLSmidth Inc. (FLS). The compilation of the process engineering and costing of the concentrator was completed by SLI.

17.1.1 Process Design Criteria

The process design criteria (PDC) for the project were initially developed by FLS. These criteria were updated for the feasibility study using data generated in 2012 from the pilot plant testwork, additional work indices tests and more recent de-watering testing. A summary of the key design criteria is shown in Table 17.1.

Table 17.1
Concentrator Key Process Design Criteria

Criterion	Units	Value
Nameplate feed capacity	t/d	2,000
Plant availability	%	91.3
Ore moisture (w/w)	%	5.26
Specific gravity		2.91
Final average concentrate mass pull	Wt%	18
Flotation feed size (80% passing – P_{80})	µm	38
Recovery – Ta_2O_5	%	74.3
Recovery – Nb_2O_5	%	73.9
Recovery – Ce_2O_3	%	76.6
Recovery – La_2O_3	%	78.4
Recovery – Nd_2O_3	%	77.2
Recovery – Y_2O_3	%	80.6
Recovery – ZrO_2	%	83.8

17.1.2 Process Description

The feasibility study includes an underground crushing system comprising a three-stage, open-circuit crushing facility, which was recommended by FLS. The run-of-mine (ROM) ore will be fed to the primary crusher from the underground ROM storage pocket, which will have a live capacity of 1,000 t. Fine crushed ore product (minus 25 mm) will be stored in a 2,000 t live capacity underground ore pocket then loaded onto the ore transport conveyor, which will be installed inside the mine access ramp. The ore will be delivered to a surface fine ore bin which will feed the grinding circuit. The crushing circuit has been designed to operate initially for 12 h/d with a potential increase to 24 h/d for an expanded capacity to 4,000 t/d.

The grinding circuit will be a conventional rod mill/ball mill operation. The rod mill will be operated in open circuit, and the ball mill in closed circuit with classifying hydrocyclones. The grinding circuit is designed to produce a product size of 80% passing size 38 µm.

The cyclone overflow will gravitate to two stages of magnetic separation, followed by a desliming circuit. The design of the desliming circuit was based on MetSim™ simulations performed by FLS. The circuit involved three stages of classification using hydrocyclones. The magnetics from the magnetic separation circuit and the fines from desliming will be routed to tailings. The deslimed slurry will feed the flotation circuit.

This flotation circuit design comprises three stages of bulk flotation, four stages of cleaner flotation and a single cleaner scavenger stage. Flotation concentrate will be pumped to a gravity separation circuit for further enrichment before being thickened and filtered to final product concentrate. The light material (gravity tailings) will be recycled to the bulk rougher flotation circuit.

The final concentrate will be stored in a bulk storage facility ready for loading into 40 t containers and barging across Great Slave Lake to Pine Point during the shipping window (July/September).

The tailings will be thickened, the overflow from which will be pumped to the process water tank although a portion will be fed to a water treatment plant to remove impurities. The tailings thickener underflow will be directed to either the tailings management facility (TMF) or the paste backfill plant. Water reclaimed from the paste backfill plant will be used as process water.

The paste backfill plant has been designed to produce 1,738 t/d of 250 mm (10 in slump) backfill using concentrator tailings. The backfill plant comprises vacuum disc filters, filtered tailings conditioning mixer, cement storage and feeding system, tailings/cement batch mixer and feed to the backfill distribution system.

17.2 HYDROMETALLURGICAL PLANT – PINE POINT SITE

The objective of the hydrometallurgical plant is two-fold:

- To condition the flotation concentrate for down-stream separation of REE at Geismar by removing certain impurities and producing a mixed rare earth precipitate.

- To separate a marketable enriched zirconium concentrate (EZC), (which also contains niobium and tantalum) for sale.

The simplified flowsheet is provided in Figure 17.1. The flowsheet and process design criteria are based on the testwork results that are discussed in Section 13.0.

17.2.1 Process Design Criteria

The key process design criteria developed for the design of the hydrometallurgical plant are provided in Table 17.2.

Table 17.2
Pine Point Hydrometallurgical Plant Key Process Design Criteria

Criterion	Units	Value
Nameplate feed capacity	dry t/d	360
Plant availability	%	89.9
Concentrate moisture (w/w)	%	10
Rare earth precipitate recoveries		
Recovery – La_2O_3	%	79.1
Recovery – Ce_2O_3	%	78.1
Recovery – Y_2O_3	%	49.5
Recovery – ZrO_2	%	1.0
ECZ recoveries		
Recovery – La_2O_3	%	16.5
Recovery – Ce_2O_3	%	18.4
Recovery – Y_2O_3	%	49.7
Recovery – ZrO_2	%	97.0

17.2.2 Process Description

The hydrometallurgical plant at Pine Point comprises the following process sections:

- Pre-leach.
- Acid bake.
- Water leach.
- Neutralization and impurity removal.
- Impurity removal re-dissolution.
- Rare earth precipitation.
- Waste water treatment.
- Tailings neutralization.

Concentrate will be barged across Great Slave Lake during the summer months in 40 t containers. On arrival at the hydrometallurgical plant, containers will be off-loaded into a bulk concentrate storage building sized to accommodate a year’s concentrate production from the Nechalacho site. The building will be insulated and heated in the winter to prevent the concentrate from freezing.

The concentrate will be fed to the pre-leach section of the plant where excess sulphuric acid produced in the water leach section is used to neutralize the base materials (e.g., carbonate and hydroxide minerals). This process reduces the overall plant reagent consumptions for both acid and dolomite. The product from the pre-leach circuit will be filtered and fed to the acid bake system while the filtrate will feed the iron reduction circuit.

The filter cake from the pre-leach circuit (or pre-leached concentrate) will be dried, mixed with concentrated sulphuric acid and fed into the acid baking rotary kiln where the REE in the concentrate will be converted to sulphates. The discharge from the acid bake kiln will be leached in water where approximately 80% of the LREEs and 50% of the HREEs will be leached. The solids containing the balance of the REEs, along with most of the zirconium, niobium and tantalum, will be filtered, washed, neutralized and dried to approximately 8% moisture. This dried product will be packaged and shipped to customers as EZC.

The rare earth filtrate from the water leach process will be cleaned through several neutralization and impurity removal steps. The majority of the iron and thorium will be precipitated using controlled pH adjustments with dolomite. The precipitate (solids) containing the impurities will be re-leached to recover the entrained rare earths which are recycled and filtered, and the remaining solids will be sent to the tailing facility for disposal. The filtrate (solution) from the filters will be transferred to the rare earth precipitation section.

The filtered solution from the impurities removal section will be precipitated using controlled pH adjustment to produce a mixed rare earth precipitate product. The RE slurry will be filtered and washed and the final rare earth precipitate will be dried to approximately 8% moisture. The rare earth product will be stored on site in preparation for shipping. The product will be loaded into containers, which are sealed, trucked to Hay River, loaded onto trains, and railed to the Geismar site.

A waste water treatment circuit will treat the clarified solution (mainly water containing sulphates) from the RE precipitation circuit, kiln off-gas scrubber overflow and solid residues from the neutralization and impurity removal circuit. The solutions will be neutralized with lime using pH control to reduce the magnesium to less than 800 ppm and sulphate to less than 4,500 ppm, thickened and filtered. The filtered solids will be washed and transferred to the Pine Point HTF.

The hydrometallurgical process requires a number of reagents, the main ones include:

- Concentrated sulphuric acid which will be produced on site in a sulphur burning sulphuric acid plant.
- Quicklime.
- Finely ground dolomite sourced primarily from local historic stockpiles containing high purity dolomite.

17.3 RARE EARTH REFINERY – GEISMAR SITE

The rare earth refinery facilities at Geismar consists of two key sections, the leaching plant refinery to remove impurities, and the separation plant where products are separated and refined to the quality required for the customers.

The feasibility study design of the refinery is based on extensive testwork, primarily undertaken at SGS-M and Mintek. However, additional testwork is currently underway at Mintek to investigate various optimization opportunities for better impurities removal that, if successful, will be incorporated into the next phase of project development.

The refinery design is based on experience and widely-accepted, well proven standard rare earth sulphate and chloride processing methods and equipment selections. The detailed design for the feasibility study was provided by a highly experienced processing expert based in China. Approximately 90% of the world's REE solvent extraction (SX) plants are located in China and the technology proposed for Avalon is based on this proven Chinese technology. Similar SX technologies are also in operation in Japan, France and the United States.

A simplified process flow diagram is provided in Figure 17.1. The relevant testwork for the refinery and impurity removal is discussed in Section 13.0.

17.3.1 Process Design Criteria

The basic design criterion for the refinery is to produce an estimated 10,000 t/y of saleable purified rare earth oxide/carbonate products from the mixed rare earth precipitate generated at Pine Point.

The key process design criteria developed for the design of the refinery facility are provided in Table 17.3.

Table 17.3
Geismar Refinery Key Process Design Criteria

Criterion	Units	Value
Annual production	dry t/y	10,000
Design production capacity	dry t/d	27.4
Utilization	%	91.3
Rare earth recovery	%	98.0

17.3.2 Process Description

The purpose of the refinery is to selectively remove impurities from the mixed rare earth precipitate produced at the Pine Point hydrometallurgical plant in order to produce a purified rare earth feed to the separation plant. The leach plant comprises a series of processes, including re-dissolution, precipitation, solvent extraction and selective precipitation. The main concerns with regard to impurities are the radioactive elements such as uranium and thorium. The main impurity removal steps include:

- Selective re-dissolution with sulphuric acid. The solid residue, which contains most of the calcium, niobium, tantalum, zirconium and silicon, will be neutralized, filtered and conveyed to the gypsum impoundment area.

- Selective precipitation to remove some of the impurities and obtain the pH required for the subsequent SX step.

- Once the remaining solution has been filtered and clarified to remove any entrained solids, solvent extraction will be used to remove impurities comprising mainly zirconium, aluminium, uranium, thorium and iron.

- The contaminants contained in the SX strip solution are recovered by selective precipitation. The precipitate will be filtered and transferred to the gypsum/residue impoundment area.

- The rare earths (both LREE and HREE) contained in the SX raffinate stream will be precipitated as carbonates using sodium carbonate. This precipitate will contain less than 1% impurities.

- The remaining light elements (Mg, Ca, K, Na) will remain in solution, which will be combined with the SX strip solution for treatment in the waste water treatment plant. After filtration, the waste slurry is pumped to the waste gypsum impoundment located adjacent to the separation plant.

- The precipitated rare earth carbonates will be leached using hydrochloric acid into an aqueous solution suitable to feed the downstream rare earth separation circuits.

The solvent extraction circuit at Geismar is based on the common Chinese configuration of stages and is divided into 16 extraction steps, each with a specific number of stages for loading, extraction, washing and stripping, and sized according to the feed composition. The design of entire extraction circuits comprises a total of approximately 1,000 mixer/settlers.

The separation plant will produce 10 different pure rare earth oxides products in accordance with the specifications indicated in Table 17.4. The plant will also produce one separate stream containing four different rare earth carbonates for which there is limited market at the present time. This stream will be stockpiled initially and eventually disposed of if markets are not forthcoming.

Table 17.4
List of Products from the Separation Plant

Product	Design Plant Production (t/y)	Product Distribution (%)	Feasibility Study Specification[1]
La Oxide	1,583	16	3 N
Ce Oxide	3,572	36	3 N
Pr Oxide	451	4	3 N
Nd Oxide	1,783	18	3 N
Sm Oxide	391	4	2 N
Eu Oxide	49	0.5	4 N
Gd Oxide	371	4	3 N
Tb Oxide	54	0.5	4 N
Dy Oxide	271	3	4 N
Y Oxide	1,170	11	5 N
Lu Oxide	14	0.1	3 N
Er/Ho/Tm/Yb Carbonate Mix	292	3	2 N
Total	**10,000**	**100**	

[1] "N" stands for the number of nines purity produced as final product, for example 3 N = 99.9%.

A kerosene mixture will be used as the extracting agent for most separations. Hydrochloric acid will be used as the stripping agent. Deionized water will be added in the washing and stripping stages to dilute and adjust the reagent concentrations.

The purified strip solution from the respective solvent extraction stage will feed their atmospheric precipitation tanks where soda ash or oxalic acid will be added in order to precipitate the pure REE as carbonates or oxalates, respectively. The slurry streams containing the precipitates will be thickened, filtered, dried and calcined to produce pure rare earth oxides. The filtrate will be forwarded to the water treatment facility. The mixed holmium, erbium, thulium, and ytterbium stream will be precipitated as carbonate and, hence, will not be calcined.

The dry rare earth oxide or carbonate products will be cooled and then packaged in drums ready for shipment to customers. The product storage facility will provide two weeks capacity, to interface between plant production and continuous product dispatch via rail, air or ocean transportation.

18.0 INFRASTRUCTURE

Avalon's rare earth project comprises the mine and concentrator at Nechalacho, a hydrometallurgical plant at Pine Point in the Northwest Territories, and a rare earth refinery at Geismar in Louisiana, United States.

The flotation concentrate produced at the Nechalacho site will be barged along the eastern side of Great Slave Lake to Pine Point where the flotation concentrate will be upgraded to a mixed rare earth precipitate and EZC. The rare earth precipitate will be shipped by rail to the Geismar site for leaching and separation of rare earths. The EZC will be shipped for sale to one or more third parties.

18.1 NECHALACHO SITE

The Nechalacho site infrastructure is designed to support the operation and maintenance of the mine and concentrator for its design capacity of 2,000 t/d of crushed ore. Future expansion of the concentrator to 4,000 t/d ore has also been accommodated, including space allowances for expansion of infrastructure facilities, such as the permanent accommodation camp and diesel fuel storage.

A site plan showing the surface infrastructure is presented in Figure 18.1

18.1.1 Access and Roads

There is no permanent road connecting the site with nearby communities and there are no plans to construct one. If necessary, ice road access along the northern shore of the Great Slave Lake will be utilized during construction. The site is accessible by barge during the summer months and year-round by aircraft using an existing airstrip. This airstrip will be extended to a length of 1,000 m as part of the Nechalacho project

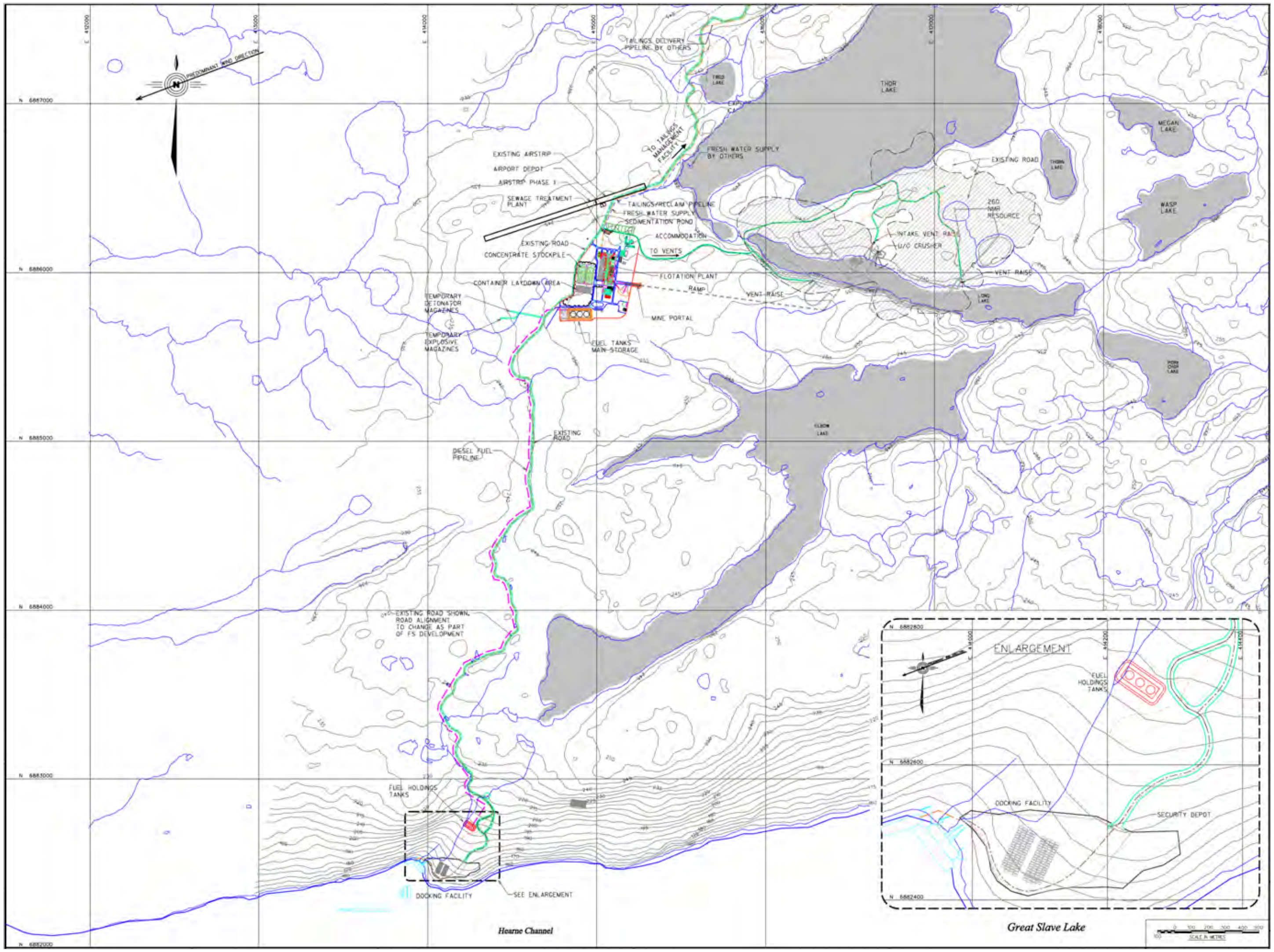
An existing 5 km long road connects the concentrator area with the planned dock site on the north shore of Great Slave Lake and will be upgraded during project construction to allow year-round operations. The road will be two-lane and 8 m wide. A lined berm on one side of the road will contain a pipeline for transfer of fuel from the barge dock holding tanks to the main tanks at the concentrator.

Roads within the concentrator area will generally be 6 m wide. Minor service roads will be 4.0 m wide. All roads within the site will have granular surfaces.

18.1.2 Dock Facility

The shoreline topography at Great Slave Lake south of the Nechalacho site is generally rocky and steep. The design of the permanent dock facility comprises a barge loading-unloading dock and a large marshalling yard.

Figure 18.1
Nechalacho Site Layout



A permanent sheet pile dock is not planned. A roll-in roll-out logistics method has been selected which is relatively inexpensive, is expected to reduce cycle times and to minimize potential environmental impacts.

The yard at the dock (approximately 2 ha in area) will be used to handle materials as they are offloaded and loaded, and for the transfer of containers to and from the Nechalacho site. The yard will include:

- 10 m to 30 m long ramp to access the barge deck for loading and unloading barges.
- 4.2 ML capacity lined and bermed fuel storage area.
- Two minimum 2,400 m^2 areas for full containers and for empty/returning containers.
- Parking area for intermodal freight and trucks.
- Diesel pumps and piping for transfer of fuel from barges to fuel storage tanks.
- A receiving/security/lunchroom facility.
- Diesel generation to power fuel pumps and site services.
- Small floating dock for the work boat.

All supplies, except diesel fuel, will arrive in standard shipping containers (those used for concentrate) or in full size containers.

18.1.3 Airstrip and Depot

The Nechalacho site has been serviced using float planes and ski planes, with helicopter support during freeze-up and break-up. In 2010, the airstrip was expanded to 30.4 m wide by 305 m long to provide year-round fixed-wing service to site.

The airstrip will be extended to a total length of 1,000 m for operations to allow year-round access for light- and medium-sized aircraft for light freight and personnel movement. It will not be lit for night operation and additional instrument controls will be installed to maximize safe operation and to minimize delays during the winter when daylight hours are short. The most common connection will be with Yellowknife.

An airstrip depot, located on the south side of the airstrip and west of the access road, will serve as communication centre, passenger waiting area and for security check in/check out.

18.1.4 Power

Power will be generated using diesel generators on the basis of a build-own-operate arrangement where a third party will provide complete operational facilities and provide the power to the Avalon facilities at an agreed cost.

The power plant will consist of five diesel generators rated 4.2 MW at 4.16 kV, 60 Hz, one generator will be in standby mode and one under maintenance. Nechalacho site peak power is estimated to be 13.6 MW (12.4 MW peak load + 10% spinning reserve). The power generation system will include a heat recovery system to heat facilities on site. Plant-wide primary power distribution will be at 4.16 kV and 4.16 kV/600 V unit substations, located in switchrooms, to step down the primary distribution voltage to 600 V. The 600 V/208 V-120 V transformers will be utilized to supply power to buildings services loads and other miscellaneous loads.

Any one diesel generator unit of the base load power plant will be utilized to supply backup power to critical process, electrical heat tracing and other loads during outages of the power plant due to major bus fault or any other fault causing total blackout. Power plant switchgear is designed with a bus sectional breaker to provide isolation of faulty bus section and to restore the backup supply from another bus section.

18.1.5 Water Supply, Storage and Distribution

Fresh water for the potable water treatment plant will be provided via a floating pump barge located in the southwestern portion of Thor Lake connected by pipeline (approximately 1,080 m long) to the concentrator site.

Fresh water for the potable water treatment, firewater and the concentrator will be stored in a tank with capacity to store a fire water reserve for three hours of fire fighting capability and one day of process water requirements. The fresh water distribution piping will be insulated and heat traced HDPE pipeline contained in utilidors around the site.

A portion of the water from the fresh water storage tank will be treated for potable use and stored in a tank with capacity for one day of requirements (135 m^3).

18.1.5.1 Grey Water and Sewage Treatment

Grey water and sewage from the concentrator site facilities will be collected and pumped to a rotating biological contactor (RBC) treatment plant. The effluent will be sterilized using ultra-violet sterilization and recirculated to the concentrator.

18.1.5.2 Waste Water Treatment

Waste water from the flotation process at will be treated using the following steps:

- Thickener to remove the solids reporting to the tailings management facility.
- Dissolved air flotation unit to remove fine suspended solids.
- Series of media filters to remove fine solids.
- Dissolved organics removal unit.
- Sludge filter presses.
- Control of pH of the water before release, as required.

18.1.6 Administration and Dry Facilities

The administration office will comprise open office space, private offices, lunch rooms, meeting/conference rooms, toilets and service rooms.

The metallurgical laboratory will be equipped with analytical facilities, offices and service rooms.

Dry facilities will include lockers and change facilities for both men and women.

18.1.7 Maintenance Shop

The maintenance shop will be located on the ground floor at the northeast corner of the concentrator building. Multiple bays accessed via large overhead doors will be used to service and maintain heavy mobile and process equipment. A welding area and offices for foremen will also be included.

18.1.8 Fuel Transportation and Storage

Diesel fuel for the Nechalacho site will be transported by barges in the summer months and temporarily stored in three holding tanks located close to the dock. Fuel will be pumped to the main fuel storage tank farm. No gasoline will be stored or dispensed at site.

The storage capacity is a nominal 30,000 m^3 for 12 months, of which 4,200 m^3 will be stored in the dock transfer tanks and the balance in the main fuel tanks located close to concentrator. These areas will be bermed.

18.1.9 Waste Handling Facilities

Waste handling facilities will comprise a sorting station and incinerator. The waste will be sorted on site and organic waste will be directed to the incinerator. Hazardous and recyclable wastes will be stored in sealed containers and shipped off-site on a regular and as-required basis to existing municipal handling facilities.

Solid waste will be recycled wherever possible. Non-recyclable waste will be incinerated. The ash from the incinerator and other approved inert construction waste will be deposited in the TMF. The solid waste incinerator has been sized on the basis of a minimum 2.5 kg per capita per day of domestic waste. Industrial waste generated during construction has been estimated at four times the domestic volume. The incinerator will be capable of handling 2 t/d of solid waste and will be a batch feed, two-stage unit. The units will be located in a heated and ventilated building and emissions will meet the limits stipulated in the codes and regulations for the Northwest Territories.

18.1.10 Parking

Parking for on-site surface service vehicles will be provided at the administration building and at the maintenance shop area and designated for operations and maintenance vehicles only. Dedicated vehicles, such as an ambulance, fire truck, waste truck and fuel truck will be parked close to the servicing facility.

18.1.11 Accommodation Complex

The accommodations complex is designed to provide accommodation, medical, dining and recreational requirements for 120 persons. Accommodation facilities are designed as two-storey prefabricated modules. These are configured as three wings which connect with a two-storey facility for common utilities and the service core building for recreational, kitchen, medical and similar facilities.

The accommodations complex will be built under a build-own-operate arrangement where a third party will provide complete operational facilities and provide accommodation and catering services during operations.

18.1.12 Security

In view of the remote nature of the Nechalacho site, there is little risk to the general public and little risk of public access to the site. Occasional summer visitors may come to the dock site by boat but will be informed that the dock and site are private.

There will not be a manned security station at any location on the site.

Where necessary, fencing will be installed to keep wildlife out of areas such as the waste storage. The use of containers for storage will minimize the requirement for such fencing.

18.1.13 Telecommunications

Proven, reliable, and state-of-the-art telecommunications systems will be provided for permanent operations and maintenance at the Nechalacho site facilities.

The project facilities, encompassing the Nechalacho and Pine Point sites in Canada, and the Geismar site in the United States will be connected via a combination of satellite land-station terminals and terrestrial communication links to provide access to voice, data and video services.

18.2 PINE POINT SITE

The hydrometallurgical plant will treat concentrate from the Nechalacho site to produce a rare earth precipitate and EZC.

The site layout is shown in Figure 18.2.

18.2.1 Access and Roads

The site is accessible year-round from Hay River via Highways 5 and 6 and an access road from Highway 6 to the site. A network of local gravel roads was developed for past mining operations and provides alternative access to the site.

An existing 13.8 km long gravel road connecting site with the dock located on the south shore of Great Slave Lake will be used for concentrate shipping during the summer months. In general, this road is in good condition with the exception of sections closer to the dock. The road will be designed for approximately 3,300 return trips per summer with 40-t concentrate container loads inbound to the hydrometallurgical plant.

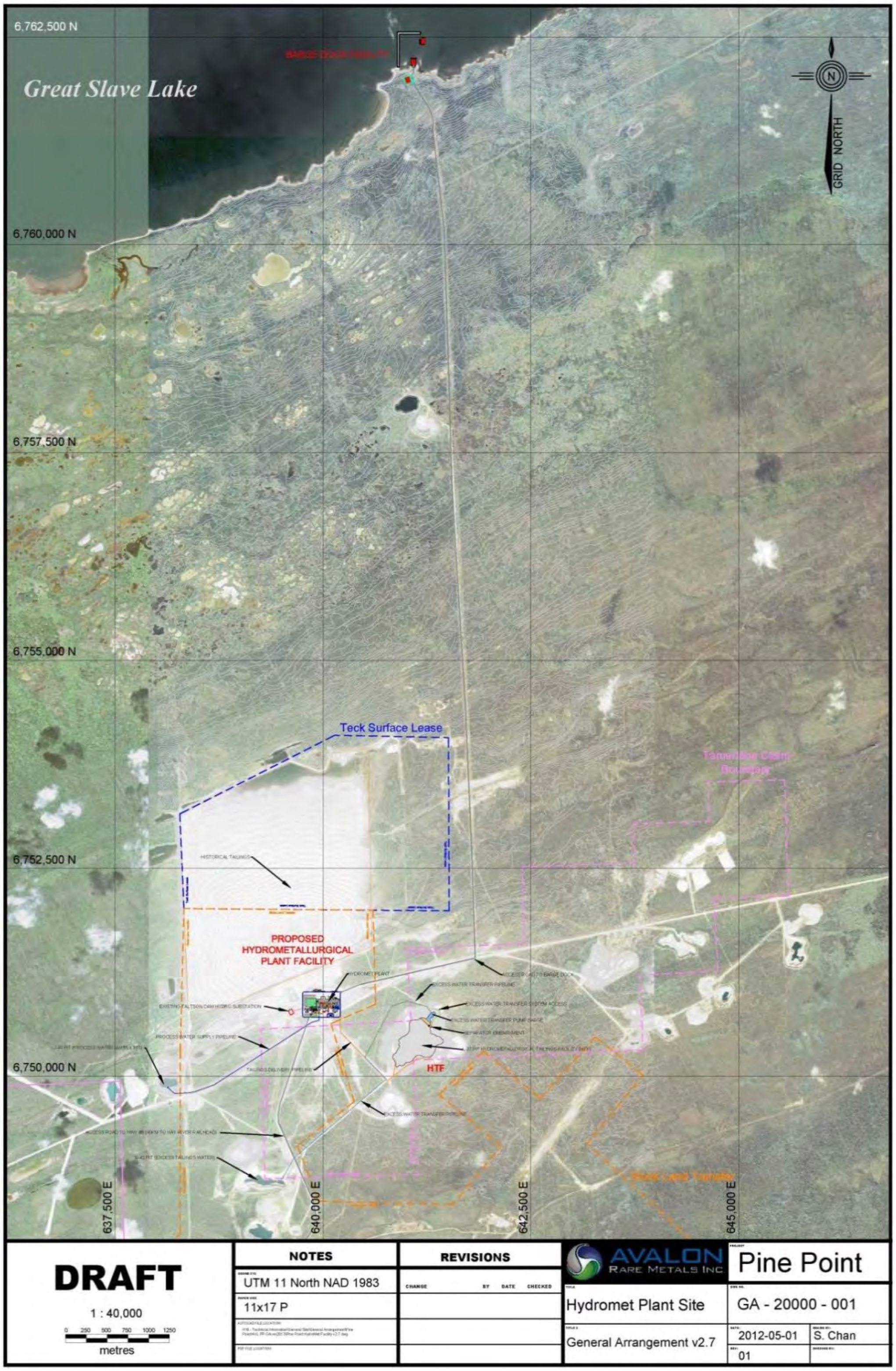
Site roads will generally two-lane and 8 m wide.

18.2.2 Dock Facility

At the location of the Pine Point docking facility, Great Slave Lake is very shallow with water depths of ±1 m within 150-200 m of the shoreline. Due to the shallow depth, dredging is planned to eliminate the need for spacer barges and to reduce restrictions on vessel draughts. Access to the docked barge will be via steel ramp or suitable gravel pushout ramp from the Great Slave Lake yard.

The dock yard will be used as a lay down area for concentrate containers that will be shipped from the concentrator at the Nechalacho site, as well as for returning empty containers. Approximately 3,300 containers will be shipped during a shipping season. The yard has been designed to accommodate a total of approximately 200 containers.

Figure 18.2
Pine Point Site Layout



18.2.3 Power

Primary power for Pine Point site will be provided via the existing NWT Hydro Corporation (NTHC) substation located at the former Pine Point mine site. A 12.5 kV overhead line from the NTPC 6 MW capacity substation to the hydrometallurgical site will be provided by NTHC. Power will be distributed throughout the plant from the 12.5 kV switchgear in the main substation, via distribution feeders.

A 250 kW, 600 V diesel generator with a 600 V MCC will be provided to meet the power requirement at the dock facilities.

Standby power requirements for critical loads will be supplied by a 2,600 kW, 12.5 kV rated diesel generator.

18.2.4 Concentrate and Reagent Handling

Two concentrate storage buildings are sized to store 11 months of production.

Prilled-sulphur feedstock for the sulphuric acid plant will be trucked from Hay River. Storage will be sufficient for two months of production.

Sulphuric acid (98.5%) will be stored in four product acid tanks with dedicated heating and circulating systems to prevent freezing. It will be built as a build-own-operate arrangement with a third party.

In addition to prilled sulphur, the principal bulk reagents will be dolomite, magnesium hydroxide, slaked lime, hydrogen peroxide and flocculants, all of which will be delivered by truck.

18.2.5 Water Supply and Distribution

Raw water for process operations will be pumped via a raw/fresh water pipeline over a distance of 2.2 km from the J-44 historic open pit, to a fresh water storage tank. Raw/fresh water will then be distributed from this storage tank to the potable water treatment plant, the fire water system distribution system, the boiler water treatment, the gland water storage tank and the process water storage tank.

Freshwater for the potable water treatment plant will be provided from the raw/fresh water tank. The system is sized for 160 L/person/d, for 66 people.

18.2.5.1 Sewage Treatment

Sewage will be treated using RBC treatment plants which are sized for 66 people. After treatment of the sewage, the effluent will be directed to the HTF.

18.2.6 Diesel Fuel

The main fuel storage and dispensing facility will consist of one 300 m^3 storage tank which will meet the fuel demand of all site operations for an average of four days to coincide with the time required to clear the road from Hay River during the winter months.

A fuel line will be installed from the storage tank to the sulphuric acid plant, the kiln and process boilers at the hydrometallurgical plant, and the maintenance shop. All storage tanks will be installed within an HDPE-lined earth bund sized to contain 120% of the volume of the largest tank.

18.2.6.1 Diesel Supply for Building Heating Systems

Diesel fuel will be the main source of heat energy for heating the east side of the hydrometallurgical plant, the administration building and domestic hot water heating.

Diesel fuel will be the secondary source of heat energy to supplement the heating system for the west side of the hydrometallurgical plant process area which will utilize waste heat generated by the process acid cooler to supply for the low temperature glycol-water heating system. It will also be the secondary source of heat energy to supplement the sulphuric acid plant, boiler building and turbine generator and auxiliary boiler building heating systems which will utilize steam generated by the waste steam boiler.

18.2.7 Waste Management

Solid waste will be collected at the point of origin in colour-coded containers. These containers will then be emptied into appropriate dumpsters located in a fenced area, from where they will be transported via a contractor to an appropriate municipal or private collection/treatment facility.

18.2.8 Administration Building

A two-storey administration building will have a footprint of approximately 610 m^2 and will be connected to the plant building. The ground floor will house security, first aid, change facilities and a laboratory. The second floor will house administrative offices, training room, conference room, printer/copier area, storage room and associated facilities.

18.2.9 Maintenance Facility

A maintenance area of approximately 550 m^2 will be provided within the hydrometallurgical plant building with service bays, offices and maintenance equipment.

18.2.10 Telecommunications

Proven, reliable and state-of-the-art telecommunications systems will be provided for permanent operations and maintenance at the Pine Point site facilities.

18.3 GEISMAR SITE

A site at Geismar, Louisiana, was selected as the location for leaching and separation of rare earth products from the mixed rare earth precipitate produced at the Pine Point metallurgical plant.

The Geismar site layout is provided in Figure 18.3.

18.3.1 Access and Roads

A two-lane access road connecting the Geismar site to the existing Ashland Road will be constructed over the right-of-the way with a length less than 610 m (2,000 ft). Roads within the refinery area will generally be 8 m wide, two-lane roads. An emergency access road will connect the western side of the site with the River Road and will provide alternate access as needed.

18.3.2 Rail Spur

A new rail spur, approximately 1.2 km in length, will be constructed from the existing CN railway line, located northeast of the Geismar site, to the product storage area. The line will facilitate the delivery of concentrate and regents and haulage of product from the site.

18.3.3 Water Supply and Distribution

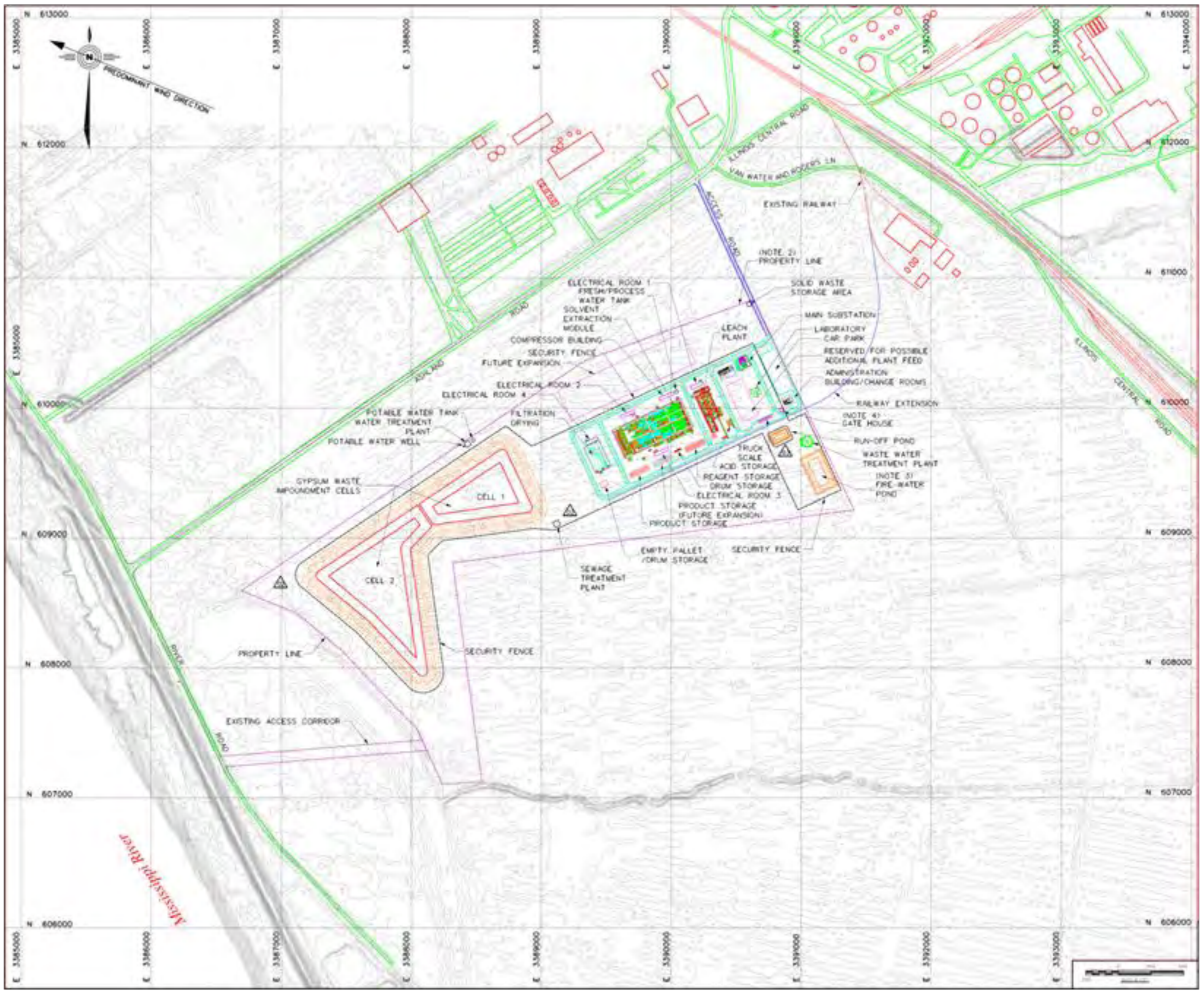
18.3.3.1 Fresh Water Supply

Fresh water for the packaged potable water treatment plant will be provided from a well equipped with a submersible pump. From the well, water will be pumped to a potable water treatment plant where the water will be treated and stored in a potable water storage tank. The system is sized for 160 L/person/day for 130 people. Water from the potable water storage tank will be pumped to all the facilities requiring potable water.

Fresh water for process and fire water will be provided via a HDPE pipeline from the adjacent chlor-alkali plant.

Fire water will be stored in a dedicated pond located in the southeast corner of the site and designed to hold the equivalent of three hours of fire fighting capability plus evaporative and contingency demands.

Figure 18.3
Geismar Site Layout



Nechalacho Project Feasibility Study, April, 2013.

18.3.3.2 Site Drainage and Run-off Pond

Drainage for the site is divided into two areas:

- Clean run-off from the undisturbed areas and water from roofs will be diverted and conveyed away from the site for discharge to the environment.

- Contaminated run-off from the contact areas within the site will be collected by ditches and conveyed to a run-off pond. This water will be pumped to a waste water treatment plant where it will be treated to meet effluent discharge criteria and released into the Mississippi River. Water from the stockpile will also be pumped to the waste water treatment plant where it will be treated and released.

The run-off pond is designed to store a one in 25 year, 24 h precipitation event. The pond will be considered a solid waste facility and, as such, it will be lined and equipped with two pumps in duty/standby mode.

18.3.3.3 Waste Water Treatment Facility

A waste water treatment plant will be provided to treat run-off water from the contaminated areas of the site and gypsum impoundment, as well as the process waste water stream prior to discharge to the Mississippi River. The plant will be designed to treat 87.7 m^3/h. Water treatment will include filtration and pH adjustment.

18.3.3.4 Sewage Treatment

Sewage will be treated using RBC treatment plants which have been sized on the basis of 160 L/person/day. After treatment of the sewage, the effluent will be sterilized using ultra-violet sterilization and discharged to the Mississippi River.

18.3.4 Administration Building

The administration building will be a stand-alone building located outside of the refinery site fenced area in order to minimize traffic in the process areas of the site. This will be a 959 m^2 single storey building, designed to accommodate 35 people, comprising reception area, offices, office working areas equipped with cubicles, conference rooms, lunch room, washrooms, print rooms and storage facilities.

18.3.5 Laboratory Building

The laboratory building will be a stand-alone building located inside the refinery fenced area. It will be a 390 m^2 single storey building comprising laboratory working areas, offices, lunch room, washrooms and utility rooms.

Laboratory working areas include sample preparation, metallurgical laboratory, assay laboratory and analytical areas. The HVAC system, including the laboratory control system, will be capable of maintaining negative pressure relative to corridors and adjacent non-laboratory areas, in addition to providing heated or air-conditioned outdoor makeup air to replace fume hood exhausts, dust collection exhausts and miscellaneous laboratory equipment exhausts.

18.3.6 Shops/Warehouse Building

The shops/warehouse building will be 892 m^2 in area, a stand-alone building located within the site fenced area and will comprise the staff dry and offices, and shops and warehouse sections.

18.3.7 Material Receiving, Storage and Handling

Rare earth precipitate from the Pine Point site will be shipped by rail to the refinery. The rare earth storage capacity will be sufficient to interface with the upstream feed processing for 24 h of operation.

18.3.8 Reagent Handling

The reagents required for the process include sulphuric acid, hydrochloric acid, sodium hydroxide, soda ash, oxalic acid, slaked lime, naphthenic acid, P507, DEPHA, and kerosene. Reagents will be transported to site by rail, ship or pipeline and stored on-site.

18.3.9 Power

A new 34.5 kV overhead line will be installed by Entergy Corporation to supply the main step-down substation. The new step-down substation and a 13.2 kV overhead line to supply power to the refinery will also be provided by Entergy.

For identified critical loads, an emergency diesel generator will be connected to the 13.2 kV main switchgear to provide power for short periods in the case of interruption to utility power supply.

18.3.10 Security

All personnel accessing the site will go through security control located at the gate house. Access to the separation plant and certain areas of the refinery will be restricted to authorized personnel only and controlled by an electronic access system.

Supplies and product off-site shipping procedures will be developed to ensure that all personnel are recorded and accounted for in a case of emergency.

18.3.11 Shipping of Product

Product will be stored in sealed containers and placed in shipping containers that will be used as storage facilities. Both road and railroad access will be available for product off-site shipping.

18.3.12 Telecommunications

Proven, reliable and state-of-the-art telecommunications systems will be provided for permanent operations and maintenance at the Geismar site facilities.

All telecommunications equipment will be to International Telecommunications Union standard and other applicable international and local industry standards.

An independent uninterruptible power supply system will supply power for 60 minutes of operation of the entire telecommunication system upon loss of the normal 120 VAC 60Hz power supply.

19.0 MARKET STUDIES AND CONTRACTS

The Pine Point hydrometallurgical plant will recover two products, a mixed rare earth precipitate and an enriched zirconium concentrate (EZC). The mixed rare earth preciptate will be shipped to Avalon's rare earth refinery at Geismar, Louisiana and the EZC will be sold to one or more third parties.

The plant at Geismar will recover a suite of separated rare earth products.

In preparing its analysis of markets for the Nechalacho project feasibility study, Avalon has collected historical price information, supply/demand analysis, and forecasts from different sources. The sources of price information include the websites of Metal-Pages™ and Asian Metal™, reports by BCC Research LLC (BCC), Roskill Information Services (Roskill) and Canadian Imperial Bank of Commerce (CIBC), verbal communication with Kaz Machida, a metal trader in the Japanese market, and private reports to Avalon by Industrial Minerals Company of Australia Pty Ltd (IMCOA).

Micon has reviewed the material presented below and has discussed the basis of Avalon's research and conclusions with representatives of the company. Micon considers that the results of the analysis and the price projections used in the economic model are reasonable.

19.1 AVALON MARKETING EXPERIENCE AND STRATEGY

Avalon's marketing team is led by Pierre Neatby, Vice President Sales and Marketing, who has over 19 years of experience in metals and industrial minerals marketing. The company is supported by market development specialists in key areas including China, the Republic of Korea and Japan.

Avalon has secured memoranda of understanding (MOUs) for the sale of its rare earth products and enriched zirconium concentrate (EZC), the details of which are confidential.

19.2 RARE EARTHS

The term "rare earths" refers to the 15 elements in the periodic table with atomic numbers 57 to 71. Also known as the lanthanide elements, they are neither particularly rare in nature, nor earthy in character but were so-called because of the difficulty early researchers had in separating them from each other and because they could be recovered as oxides, or "earths". The element, yttrium, often occurs with the rare earths and has some similar physical and chemical properties, although it is not a lanthanide.

The rare earths are divided on the basis of atomic weight into the more common 'light' rare earth elements (LREE) and the less common 'heavy' rare earth elements (HREE). In spite of its low atomic weight (39), yttrium has properties more similar to the HREE and is often included within this group. The rare earth element content of ores and products is generally expressed in terms of the rare earth oxide equivalent (REO) or total rare earth oxide content (TREO).

19.2.1 Rare Earth Supply

Rare earths are found in more than 200 minerals, of which about a third contain significant concentrations. Only a few are of commercial interest. The most important source minerals are carbonates (bastnaesite) and the phosphates (monazite and xenotime). Apatite (a calcium fluor-phosphate) may also contain rare earths. Heavy rare earths are more commonly found in minerals in granitic and alkaline igneous rocks and in ionic, or ion adsorption, clays. The main geological environments for occurrence of rare earths are listed below (Harben, 2002):

- Bastnaesite-bearing carbonatites (Mountain Pass deposit, California; Kola Peninsula, Russia; Sichuan, China).
- Monazite and xenotime-bearing mineral sands (west coast of Australia; east coast of India).
- Iron-bastnaesite deposits (Bayan Obo deposit, Inner Mongolia, China; Olympic Dam deposit, Australia).
- Ion adsorption clays (southern China).
- Loparite and eudialyte in alkaline intrusives (Kola Peninsula, Russia; Dubbo deposit, Australia).
- Pegmatites, hydrothermal quartz and fluorite veins (Northern Territories, Australia; Karonge deposit, Burundi; Naboomspruit deposit, South Africa).

By far the most important current sources are the Bayan Obo iron-rare earth deposits near Baotou, Inner Mongolia, the bastnaesite deposits in Sichuan, China and the ion adsorption clay deposits in southern China. China is the dominant source of all rare earths, accounting for approximately 90% of world production in 2012. Light rare earths are primarily produced in northern China (Inner Mongolia) and southwestern China (Sichuan). The heavy rare earths are primarily produced in southern China (principally in Guangdong, Jiangxi and Fujian) from ion adsorption clays.

In 2012, Molycorp Inc. (Molycorp) resumed mining operations at Mountain Pass, California, where a separation plant had been running on stockpiled material for a number of years. In Australia, Lynas Corporation Ltd. (Lynas) commissioned its mining operation at Mt. Weld, Western Australia, in 2011 and the separation plant in Malaysia (the Lynas Advanced Materials Plant, or LAMP) received a temporary operating permit in September, 2012.

There are distinct differences in the elemental composition of various rare earth sources, as illustrated in Table 19.1.

Table 19.1
Distribution of Rare Earth Elements from Deposits in China, United States and Australia
(%)

Source	**Baotou, Inner Mongolia, China**	**Sichuan, China**	**Guangdong, China[1]**	**Longnan, Jiangxi, China[1]**	**Mountain Pass, California**	**Mt. Weld, Western Australia[2]**
Ore Type	**Bastnaesite Concentrate**	**Bastnaesite Concentrate**	**High-Eu clay**	**High-Y clay**	**Bastnaesite**	**Monazite**
TREO in Concentrate	50	50	92	95		
Oxide						
La	23	29.2	30.4	2.1	33.2	23.88
Ce	50.1	50.3	1.9	0.2	49.1	47.55
Pr	5	4.6	6.6	0.8	4.34	5.16
Nd	18	13	24.4	4.5	12	18.13
Sm	1.6	1.5	5.2	5	0.789	2.44
Eu	0.2	0.2	0.7	0.1	0.118	0.53
Gd	0.8	0.5	4.8	7.2	0.166	1.09
Tb	0.3	0	0.6	1	0.0159	0.09
Dy	0	0.2	3.6	7.2	0.0312	0.25
Er	0	0	1.8	4	0.0035	0.06
Y	0.2	0.5	20	62	0.0913	0.76
Ho-Tm-Yb-Lu	0.8	0	0	5.9	0.0067	trace
Total REO	**100**	**100**	**100**	**100**	**99.9**	**100**

[1] TREO contents of Chinese ion adsorption clays represent relative distribution in concentrate.
[2] Central Lanthanide Zone distribution in mineral resources.
Neo Material Technologies Inc., Harben, 2002, Lynas Corporation Ltd.

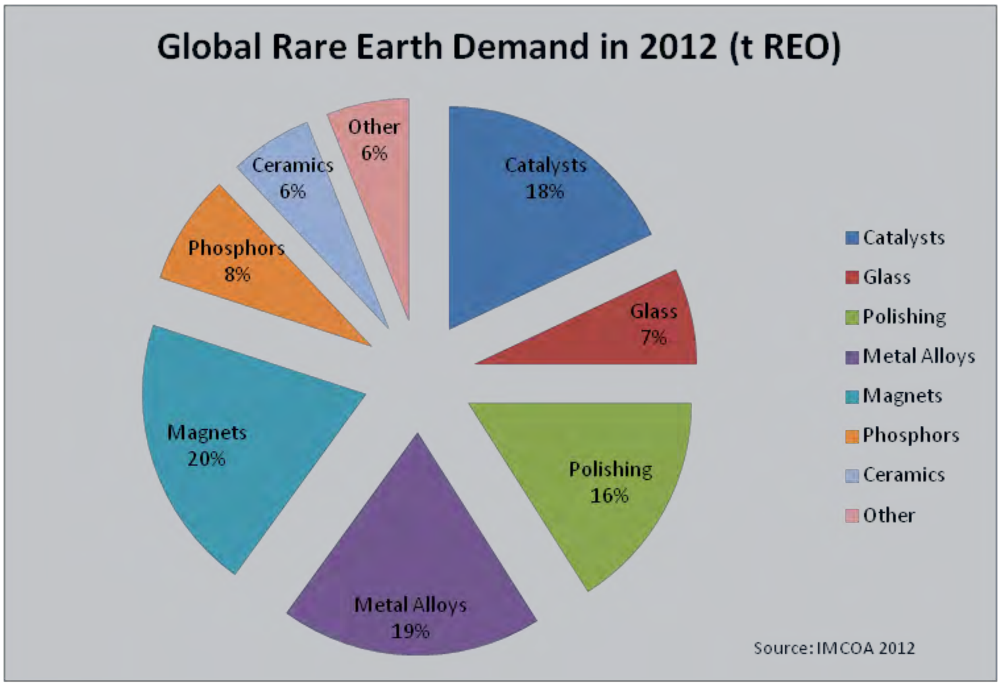
There is a disparity between the relative occurrence of the rare earth elements, as illustrated in Table 19.1, and the demand for individual oxides since, to a large extent, they each have specific market applications. At any given time, recovery and separation of the elements results in some being in excess supply and others in deficit. Overall production in terms of TREO is generally greater than the aggregate demand for the individual elements.

The Chinese government has actively supported the consolidation of the rare earth industry and the control of crude ore supply from illegal mining. Important, also, has been the imposition of export quotas and taxes on rare earth products reflecting Chinese government policy objectives of securing supplies for domestic manufacturing industry and maximizing added value. Overall, Avalon's analysis indicates that supply of rare earths from China will be constrained between 2014 and 2016 and that increased demand, particularly for the HREE, will have to be met primarily from non-Chinese sources.

19.2.2 Rare Earth Demand

Total demand for rare earths in 2012 was estimated at 115,000 TREO by IMCOA and the breakdown by end use is detailed in Figure 19.1.

Figure 19.1
Rare Earth Demand by End-Use



IMCOA, 2012.

Projected rare earth demand in 2016, broken down by region and by application, is shown in Table 19.2.

Table 19.2
Projected Rare Earth Demand by Region and Application in 2016
(Tonnes REO)

Application	China	Japan and Northeast Asia	United States	Others	Total
Catalysts	14,500	2,500	6,500	1,500	25,000
Glass	6,000	1,000	1,000	1,000	9,000
Polishing	19,000	2,000	3,000	1,000	25,000
Metal Alloys	20,000	2,500	2,000	1,500	26,000
Magnets	28,000	4,500	2,000	1,500	36,000
Phosphors	9,000	2,000	1,000	500	12,500
Ceramics	4,000	2,000	2,000	1,000	9,000
Other	6,500	3,500	8,000	2,000	20,000
Total	**107,000**	**20,000**	**25,500**	**10,000**	**162,500**

IMCOA, 2012 (estimates ±20% accuracy).

By 2016, total demand is expected to have increased by approximately 40%. Production in China may account for just over 70% of demand and the balance will be filled by output in the rest of the world. While overall supply of rare earths is expected to exceed total demand, it is expected that supply of some of the rare earth elements will be in deficit.

The Nechalacho Project is anticipated to produce 7,000 t to 10,000 t TREO/y and to meet demand for key rare earth elements including neodymium and HREE.

19.2.3 Outlook

Supply projections include provision for new, non-Chinese suppliers from a number of advanced projects, some of which are shown in Table 19.3.

Table 19.3
Advanced Rare Earth Projects

Company	Operation/ Project	Location	Target Start Date	Capacity (t/y REO)	Product Focus
Molycorp Inc.	Mountain Pass	United States	2013	20,000	LREE
Lynas Corporation	Mt. Weld	Australia	2013	11,000	LREE
Toyota Tsusho Corp./Indian Rare Earths Limited	Orissa	India	2013	7,000	LREE
Sumitomo/Kazatomprom[1]	Kazakhstan	Kazakhstan	2013	?	LREE
Great Western Minerals Group Ltd.	Steenkampskraal	South Africa	2015	5,000	LREE
Alkane Resources Ltd.	Dubbo	Australia	2016	1,605[2]	Zr, Y, Nd, Pr
Avalon	Nechalacho	Canada/United States	2016	10,000	Nd, Eu, Tb, Dy, Y

[1] By-product from processing of uranium tailings.
[2] Base case production rate.
Avalon and corporate reports.

Based on the predominance of LREE in the anticipated product mix of new supply sources outside China, IMCOA considers it likely that a supply shortfall in many of the more critical rare earths is likely to emerge by 2016, as detailed in Table 19.4. This is expected to result in firmer prices for neodymium, europium, gadolinium, terbium, erbium and yttrium, shown in red.

Table 19.4
Supply and Demand for Rare Earths, 2016

Rare Earth Oxide	Supply (t REO)	Demand (t REO)	Surplus/Shortage (t REO)
Lanthanum	49,500	34,300	15,200
Cerium	77,750	70,500	7,250
Praseodymium	8,650	7,525	1,125
Neodymium	28,000	30,025	-2,025
Samarium	3,275	1,150	2,125
Europium	450	500	-50

Rare Earth Oxide	Supply (t REO)	Demand (t REO)	Surplus/Shortage (t REO)
Gadolinium	2,175	2,650	-475
Terbium	250	500	-250
Dysprosium	1,100	900	200
Erbium	550	1,050	-500
Yttrium	7,300	13,600	-6,300
Ho-Tm-Yb-Lu	1,000	250	750
Total	**180,000**	**162,500**	

IMCOA, 2012.

19.2.4 Rare Earth Element Pricing

The market for rare earths and yttrium products is small, and reliable, publicly available pricing information, forecasts and refining terms are not readily accessible. The pricing methodology for the feasibility study is based on that provided in the August, 2011NI 43-101 Technical Report (RPA, 2011) with certain prices updated as described below.

Prices for lanthanum, cerium and samarium were reduced by 50% from the August, 2011 forecast since planned production from Molycorp and Lynas is expected to have a downward influence on pricing for these elements. The other LREE, neodymium and praseodymium, are used in magnet applications that are expected to expand as the availability of these elements increases and, therefore, prices in the August, 2011 forecast were retained for the feasibility study, with the exception of lutetium which is discussed in more detail below. The prices used in the economic analysis for the feasibility study are shown in Table 19.5.

Table 19.5
Avalon Forecast Prices for Rare Earth Oxides

Rare Earth Oxide	Feasibility Study Forecast 2016 Forecast (US$/kg, FOB China)
LREE	
La_2O_3	8.75
CeO_2	6.23
Pr_6O_{11}	75.20
Nd_2O_3	76.78
Sm_2O_3	6.75
HREE	
Eu_2O_3	1,392.57
Gd_2O_3	54.99
Tb_4O_7	1,055.70
Dy_2O_3	688.08
Ho_2O_3	66.35
Er_2O_3	48.92
Lu_2O_3	1,313.60
Y_2O_3	67.25

Avalon increased its forecast price for the heavy rare earth, lutetium, compared with the August, 2011 projection since its price movement has been different from that of the other HREEs. Other heavy rare earth prices increased early in 2011, peaked in mid-2011 and then began to fall immediately thereafter to reach a more stable level in the latter part of 2012, as illustrated by the trend of europium prices shown in Figure 19.2.

Figure 19.2
Price Trend for Europium Oxide
(99% minimum, FOB China, RMB/kg)

Asian Metal™.

Prices for lutetium also peaked in mid-2011 but reached a higher peak in late 2011 and have since levelled-off in late 2012 and early 2013 to a range that is well above the mid-2011 peak, as shown in Figure 19.3.

The price trend shown in Figure 19.3 is based on domestic prices for lutetium published by Asian Metal™. There is little trade in lutetium outside China and, in order to derive an estimated FOB price for the purpose of the feasibility study, the 28-month average price for lutetium in China from November, 2010 to February, 2013 was increased by 25% to account for the export tax and converted to US dollars at a rate of RMB 6.2339:US$1.00. In order to check that the resulting US$1,313.60/kg is conservative, Avalon compared it with European and Hong Kong trader quotes available as of February, 2013.

Figure 19.3
Price Trend for Lutetium Oxide
(99.9% minimum, FOB China, RMB/kg)

Asian Metal™.

19.2.4.1 Rare Earth Price Projections Beyond 2016

Prices for rare earths beyond 2016 have been held constant at 2016 levels for the purposes of the feasibility study. Demand is expected to continue to grow steadily and could be significantly higher if one or more of the potentially important end-use sectors are adopted widely, for example, electric/hybrid electric vehicles and wind turbines. On the supply side, there is insufficient information to predict accurately when, or if, new non-Chinese producers, less advanced than those shown in Table 19.3, will come onstream. Avalon believes that it is both reasonable and conservative to hold prices constant from 2016 onwards. Micon agrees with this approach.

19.3 ZIRCONIUM

As described in Section 17.0, a unique zirconium product will be produced in the hydrometallurgical plant at Pine Point. Referred to as enriched zirconium concentrate, or EZC, it will carry the niobium and tantalum values recovered from the Nechalacho project and is expected to have the following approximate composition:

ZrO_2	15-19%
Nb_2O_5	1.5-2.0%
Ta_2O_5	0.1-0.2%
LREO	1.0-1.25%
HREO	0.8-1.25%

At present, it is only in China that zircon is "cracked" to produce intermediate zirconium products, such as zirconium oxychloride (ZOC) and zirconium basic sulphate (ZBS), which are then upgraded or converted to a range of zirconium chemicals or to zirconium metal. Zirconia and zirconium chemicals account for just under 20% of total zircon demand. Zircon chemicals are used in a wide variety of applications including refractories, ceramic pigments, advanced ceramics and catalysts, abrasives and electronics.

The feedstock for the zirconium chemicals industry in China is zircon sand imported, principally, from Australia and South Africa. Given the zirconium chemicals industry in China, Avalon considers that the logical destination for EZC from the Nechalacho project is China. In January, 2013 the company announced the signing of a Memorandum of Understanding (MOU) with an Asian third party (see Avalon press release dated 29 January, 2013).

Projected revenue from EZC in the feasibility study is based on the terms of the MOU, net of the freight cost to China. Given the small number of players in this sector of the zirconium industry, Avalon considers pricing for EZC to be sensitive and proprietary.

19.4 CONTRACTS

Avalon has entered into several MOUs for the sale of its rare earth products. Negotiations to convert these MOUs into binding agreements are ongoing. As noted above, Avalon entered into an MOU for the sale of EZC in January, 2013. In Avalon's opinion, these MOUs show that there is significant interest in its EZC product.

Micon concurs that the MOUs in hand for both rare earth products and EZC demonstrate the potential for firm off-take agreements to be completed.

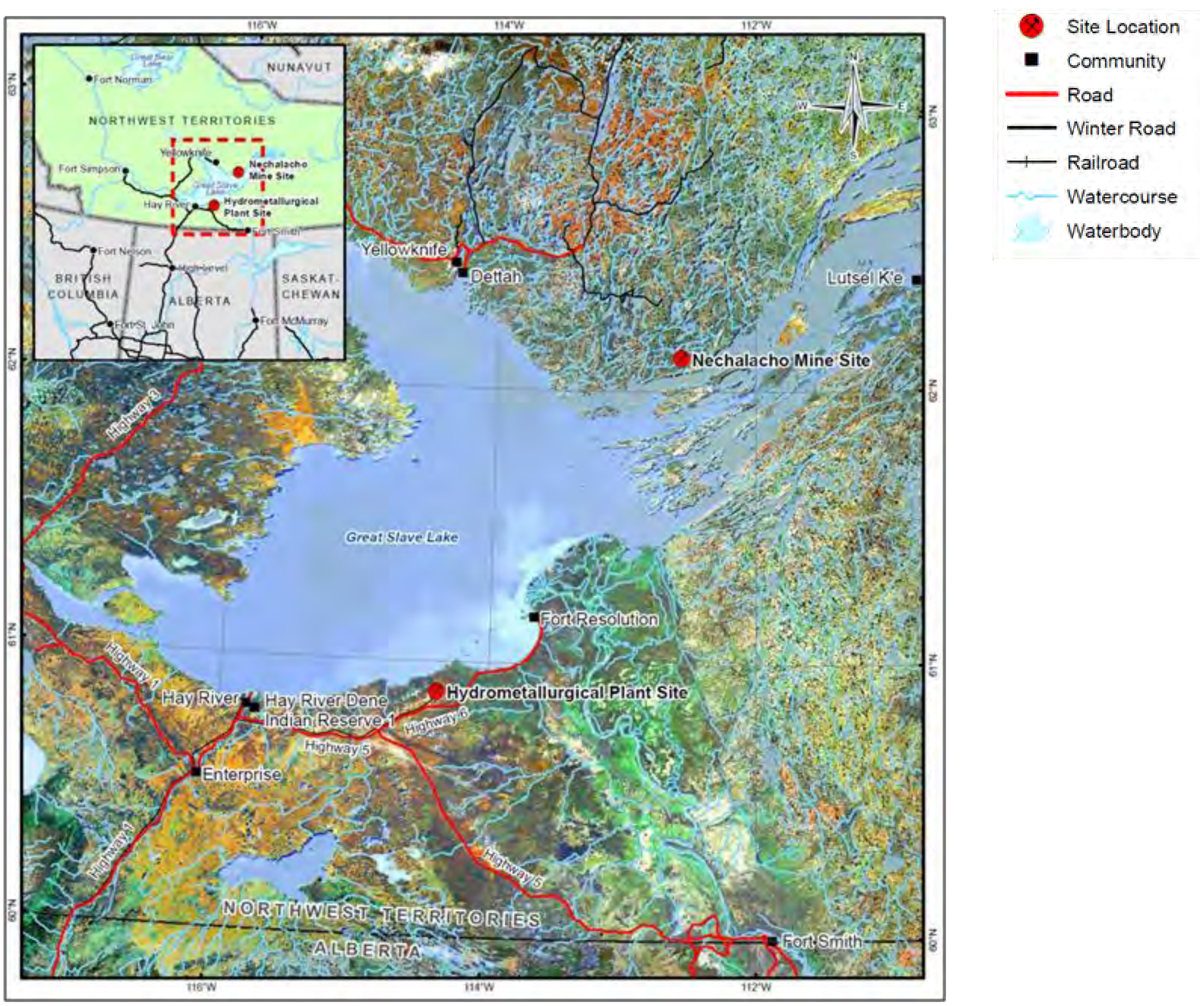
20.0 ENVIRONMENTAL STUDIES, PERMITTING AND SOCIAL OR COMMUNITY IMPACT

20.1 OVERVIEW, NECHALACHO AND PINE POINT SITES, NORTHWEST TERRITORIES

A complete description of the environmental setting and existing baseline conditions is provided in the Thor Lake Project Developers Assessment Report (DAR) prepared by EBA Engineering Consultants Ltd. (EBA) on behalf of Avalon in 2011 (EBA, 2011a), as well as in various environmental baseline studies and recent fieldwork by Stantec Inc., EBA Engineering Consultants Ltd. (EBA) and Knight Piésold Ltd. (KPL).

The Nechalacho and Pine Point site locations are shown in Figure 20.1.

Figure 20.1
Nechalacho and Pine Point Site Locations, Physical and Infrastructure Facilities



EBA prepared a summary description of the environment and permitting framework for the Nechalacho site for the purpose of the project feasibility study encompassing the following (see EBA, 2010):

- Existing environmental conditions.
- Potential environmental issues and associated mitigation measures.
- Long term (or future) reclamation and closure activities.
- Permitting requirements for the Nechalacho site.

20.1.1 Permitting

Subsequent to the acquisition of the property, and completion of community engagement meetings, Avalon applied to the Mackenzie Valley Land and Water Board (MVLWB) for an exploration permit. Two-year permits were granted effective July, 2007 and July, 2009. In 2011, a new exploration permit was granted for a five-year period.

The Mackenzie Valley Land and Water Resources Act (MVRMA) and Regulations establish a three-part environmental assessment process comprising:

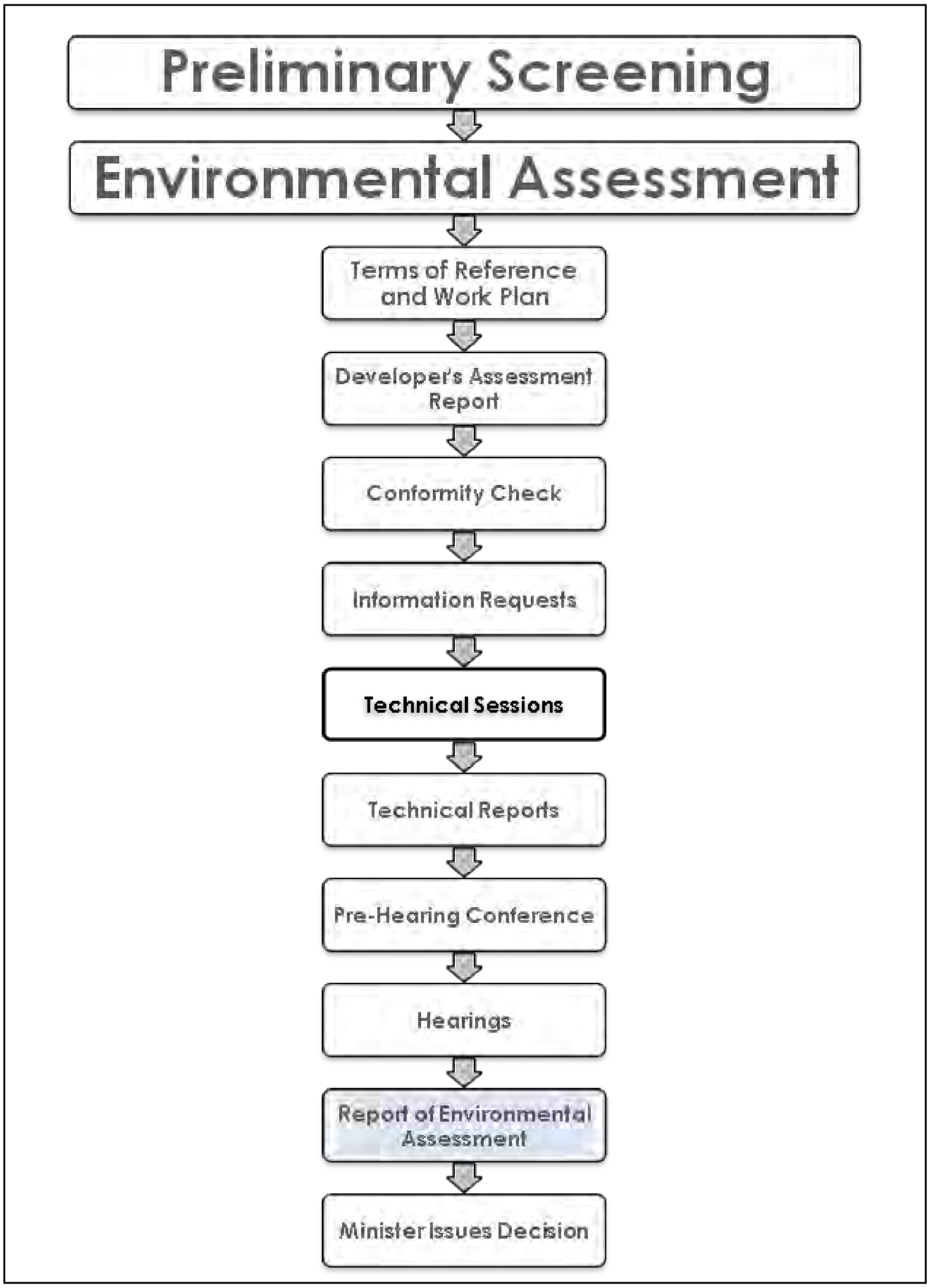
- Preliminary screening.
- Environmental assessment.
- Environmental impact review (panel review, if necessary).

On April 23, 2010, Avalon filed a detailed Project Description Report (PDR) with the MVLWB as the first step in its application for a Type A Water Licence and a Type A Land Use Permit (Avalon, 2010). On June 11, 2010, Avalon was advised by the Mackenzie Valley Environmental Impact Review Board (MVEIRB) that, as expected, the permit application would be referred for environmental impact assessment. Avalon submitted a DAR to the MVEIRB in May, 2011 and completed the conformity review and information request stages. The Technical Sessions were held during the period August 14-17, 2012 and the Public Hearings were held during the period February 18-26, 2013.

Preparation and issuance of the MVEIRB Report of Environmental Assessment is expected by the summer of 2013. This report will then be submitted to the Minister of Aboriginal Affairs and Northern Development. The MVEIRB environmental assessment process is shown in Figure 20.2. At the time of writing, the Nechalacho project is at the Report of Environmental Assessment stage.

Following Ministerial approval, the process will revert back to the MVLWB to determine the conditions for the Type A Water Licence(s) and Type A Land Use Permit(s) that are anticipated to be required for the Nechalacho and Pine Point sites. The MVLWB process for the Type A Water Licence(s) is anticipated to take 8 to 12 months to obtain and will include a public hearing process. The Type A Land Use Permit(s) will be processed concurrently.

Figure 20.2
MVEIRB Environmental Assessment Process



Other environmental permits/approvals anticipated to be required for the project include:

- A Navigable Waters Protection Act (NWPA) approval for the seasonal docking facilities at the two Nechalacho project sites and an Order in Council Proclamation of Exemption under Section 23 of the NWPA for the use of water bodies in the TMF.

- A Fisheries Act Letter of Advice from the Federal Department of Fisheries and Oceans (DFO).

20.1.2 Summary of Potential Residual Effects

The major potential environmental effects pathways include air, water and terrestrial resources. The main components of the Nechalacho and Pine Point sites are expected to have limited effects on the local receiving environment. Any identified environmental effects are expected to be generally limited to the immediate footprint and local study area of the Nechalacho site and associated infrastructure and, with the exception of the infilling of the small non-fish-bearing water bodies located within the TMF footprint, are reversible once activities ceased. Overall, the environmental assessment determined that, with the application of the proposed mitigation measures, the residual environmental effects for all valued environmental components (VECs) in the Nechalacho project area are anticipated to be negligible and insignificant.

Avalon and EBA is confident that with the application of the sound engineering, environmental planning and best management practices described in the feasibility study, as well as compliance with anticipated permits, licences, approvals, existing Federal and Territorial environmental regulations and guidelines, that the environmental issues associated with the development and operation of the project can be effectively addressed and managed (EBA, 2011a).

20.1.3 Valued Components

Environmental valued components (VCs), or key indicator species, were identified through consultation, environmental assessment process, knowledge and understanding of the biophysical aspects, applicable scientific principles and consideration of the following:

- Species currently listed under the Federal Species at Risk Act (SARA), assessed under the Committee on the Status of Endangered Wildlife in Canada (COSEWIC), or ranked under the Northwest Territories Wildlife Act or the Northwest Territories Species at Risk Act.

- Species or species groups considered of cultural importance.

- Species or species groups considered to be particularly sensitive to disturbance.

- Species or species groups dependent upon particular environmental features.

Environmental VCs considered in the evaluation of potential project effects include the following:

- Indicators of air/noise quality.
- Indicators of surface/groundwater quality.
- Fish and fish habitat.
- Indicators of terrain and permafrost integrity.
- Ecosystem and plant species including sensitive ecosystems, traditional use plants, and species at risk.
- Wildlife species at risk and species of cultural importance including:
    - Barren-ground caribou.
    - Woodland caribou.
    - Moose.
    - Wood bison.
    - Black bear.
    - Other furbearers.
    - Peregrine falcon.
    - Short-eared owl.
    - Common nighthawk.
    - Olive-sided flycatcher.
    - Rusty blackbird.
    - Yellow rail.
    - Horned grebe.
    - Whooping crane.

20.1.4 Seismicity

The central region of the Northwest Territories is historically a quiet seismic zone. Seismic hazard assessments for the Nechalacho and Pine Point sites were completed using probabilistic calculations based on design tables from the 2005 National Building Code of Canada (NRC, 2011).

The maximum acceleration for both project sites ranged from 0.007 g for a 1 in 100 year return period to 0.16 g for a 1 in 10,000 year return period. A review of the TMF consequence classification following the Canadian Dam Association 2007 Dam Safety Guidelines results in the TMF classification of ‘Significant’ (CDA, 2007). The resulting earthquake design ground motion is a 1 in 1,000 year event which corresponds to a maximum acceleration of 0.035 g for the Nechalacho and Pine Point sites.

20.1.5 Site Potential Effects and Site Monitoring

An environmental and socio-economic assessment for the Project was prepared in accordance with the requirements of the MVEIRB (MVEIRB, 2004). The application of sound engineering, environmental planning and best management practices, including compliance with existing environmental regulations and guidelines, will ensure that all of the environmental issues associated with the development and operation of the Project can be effectively addressed.

During operations, monitoring programs will be implemented in order to identify and mitigate potential effects on the Nechalacho and Pine Point site environments in the following general areas:

- Air quality.
- Noise.
- Surface hydrology
- Surface water quality.
- Hydrogeology.
- Fish and fish habitat.
- Surficial geology and soils.
- Ecosystems and vegetation.
- Wildlife and wildlife habitat.

After closure, monitoring will evaluate the success of the re-vegetation effort in Year 1 and Year 5 post-closure at a minimum.

Post-closure monitoring of the TMFs will include an annual inspection for a prescribed period to confirm that the completed closure measures are meeting expectations.

20.1.6 Nechalacho and Pine Point Site Closure and Reclamation Plans

The conceptual closure and reclamation plan for the project is based on a "design for closure" approach. The closure and reclamation plan will be a living document that is updated throughout the life of the Project to adapt to and reflect any changes that may arise. The closure and reclamation of all Project site facilities will be conducted in accordance with the terms and conditions of the future MVLWB Land Use Permit and Water Licence and accepted mine reclamation practices in the Northwest Territories and with engagement of local Aboriginal groups as per the Accommodation Agreements.

Reclamation and closure of all site facilities will be based on the Mine Site Reclamation Policy for the Northwest Territories, (Indian and Northern Affairs Canada (INAC, 2002) and the Mine Site Reclamation Guidelines for the Northwest Territories (INAC, 2007).

The objectives of the closure and reclamation plan are:

- To protect public health and safety.

- To minimize the effects of mining on the environment.

- To establish conditions that lead to acceptable long-term physical and chemical stability in reclaimed areas.

- To establish conditions that are appropriate for the surrounding environment and identified end land uses.

- To provide the public and government with a clear understanding of reclamation expectations.

The following components will be incorporated into the overall mine design, where possible, and into the closure and reclamation plans for relevant components of both the Nechalacho and Pine Point sites:

- Buildings and site infrastructure will be located, to the degree practical, on previously disturbed land.

- Organic and mineral soils collected from the Nechalacho site (the Pine Point site has no organics) will be salvaged and stored for future reapplication during reclamation of the site.

- Post-closure terrestrial monitoring will be limited to evaluating the success of the re-vegetation effort. Post-closure monitoring for re-vegetation success is envisioned to be conducted at Year 1 and Year 5 post-closure.

- Appropriate and approved seed mixes and plant material will form the basis of the re-vegetation strategy. Native species will be used where possible. All seed sources, whether native or agronomic will be certified weed-free.

- Following removal of surface facilities, the remaining fill embankments, borrow pits, access roads, and development footprint will be scarified and re-contoured as required to ensure surface stability, blend with the surrounding landscape, and facilitate the re-establishment of native vegetation.

- Test plots will be developed during operations in different ecosystem types that will require restoration to evaluate the applicability and suitability of reclamation techniques.

Monitoring during operation of the TMF will provide important input for performance evaluation and refinement of operating practices.

Costs associated with closure and ongoing monitoring of the Nechalacho and Pine Point sites are provided in Section 21, Capital and Operating Costs.

20.1.7 Community and Social Engagement

The Nechalacho and Pine Point sites are within the Akaitcho Dene asserted traditional territory and are subject to the comprehensive land claim negotiation between the Akaitcho Dene First Nations, i.e., Yellowknives Dene First Nation (Dettah), Yellowknives Dene First Nation (Ndilo), Lutsel K'e Dene First Nation (LKDFN), and Deninu Kue First Nation (DKFN), and the Federal Government. The Northwest Territory Métis Nation (NWTMN) is also negotiating a land claim in the South Slave region which encompasses a membership in Fort Resolution, Fort Smith, and Hay River.

The socio-economic study area for the Nechalacho site consists of those communities that are more likely to experience effects due to their relatively close proximity to the proposed site, as well as their possible contributions to the project workforce. On this basis, the primary study communities related to the Nechalacho site include Yellowknife, Ndilo, Dettah, Lutsel K'e and Fort Resolution.

Present non-traditional land uses near the Nechalacho site include mineral exploration, harvesting, recreation and tourism. Traditional harvesting continues to be an important part of Aboriginal cultures in the Northwest Territories.

The socio-economic study area for the Pine Point site consists of the communities of Hay River, Hay River Reserve, Fort Resolution and Fort Smith.

20.1.8 Negotiation Agreements

The Negotiation Agreements outline broad principles for co-operation and provides the basis for the negotiation of an Accommodation Agreement.

Avalon has entered into Negotiation Agreements with the Lutsel K'e Dene First Nation (LKDFN), Yellowknives Dene First Nation (YKDFN) and the Deninu Kue First Nation (DKFN). As of March 2013, Avalon has finalized Accommodation Agreements with the DKFN. Negotiations are ongoing with the LKDFN and the YKDFN. In the Pine Point area, negotiation agreements have been signed with the NWTMN.

20.1.9 Land Use

Present land uses near the Nechalacho and Pine Point sites include mineral exploration, harvesting, recreation and tourism.

The area surrounding Great Slave Lake offers a variety of year-round recreational opportunities. Blatchford Lake Lodge is an all season eco lodge located approximately 8 km north of the Nechalacho site.

Recreational and subsistence hunting and fishing also occur regularly in the North, South and East Slave regions (EBA, 2011a, b, c, d).

Traditional knowledge studies were conducted with the Yellowknives Dene First Nation, Lutsel K'e Dene First Nation, Deninu Kue First Nation, and Fort Resolution Métis Council for both the Nechalacho and Pine Point sites. The studies confirmed that hunting and trapping of mammals, birds and fish historically occurred near the sites (EBA, 2011b, c, d).

Participants of the studies noted that groups of people historically lived along the shoreline of Great Slave Lake. It was also noted that there are important historic and current travel routes on Great Slave Lake between Yellowknife, Fort Reliance, and Lutsel K'e. Culturally significant sites including graves, historic villages, treaty sites and spiritual sites, have been identified in the general areas near the Nechalacho site (EBA, 2011a, b, c, d).

Culturally significant sites have been identified in the general areas near the Pine Point site, but not within the proposed development sites.

Archaeological studies of the Nechalacho and Pine Point sites were conducted in 2011 and 2012 by Points West Heritage Consulting Ltd. (Points West) (see Points West, 2012 and 2013).

An archaeological protection plan is under development and will be finalized in the second quarter of 2013. No impacts on planned production areas are anticipated.

20.2 NECHALACHO SITE

20.2.1 Summary of Baseline Conditions

The Nechalacho site facilities include the mine, concentrator, tailings management facility (TMF) and associated road and dock facilities. The project is located in the Great Slave Upland High Boreal (HB) Ecoregion, which is a subdivision of the larger Taiga Shield HB Ecoregion. The land is dominated by low topography and fractured bedrock plains. Black spruce, jack pine, paper birch, and trembling aspen form discontinuous forested patches that are mixed in with exposed rock. Wetlands and peat plateaus commonly form around the margins of shallow lakes, as well as in depressions and lowlands. (See EBA, 2010).

In general, waters in the proposed Nechalacho site area have relatively high alkalinity, hardness and calcium. The lakes in the area tend to be clear with low suspended sediment levels, and low nutrient and metals concentrations. Three studies carried out in the vicinity of the Nechalacho site provide information to characterize and quantify the fisheries and aquatic resources that may be affected by proposed mining activities, Melville et al. (1989) and Stantec (2011a, b). These collectively investigated fish, fish habitat, water and sediment quality, and aquatic resources (phytoplankton, zooplankton, benthic invertebrates) in numerous lakes and streams within the potential effect footprint of Avalon's proposed mining activities. The water quality sampling program is ongoing and an aquatic invertebrate baseline report for the more detailed 2012 study is pending.

Eighteen lakes within the Nechalacho site area have been sampled for fish. Of these, A, Drizzle, Elbow, Fred, Great Slave, Kinnikinnick, Long, Murky, Redemption, and Thor lakes are considered fish-bearing, though some only for short periods each year due to their shallow nature; they freeze to the lake bed which prevents year-long habitat. The most common species found were the large bodied northern pike, lake whitefish and lake cisco and small bodied slimy sculpin and ninespine stickleback.

Both boreal and tundra animals frequent the area. Approximately 26 species of mammals may frequent this region. Tundra species, such as barren-ground caribou, are typically found within this ecoregion during the winter months only, spending the summers on the tundra. Other species, such as moose, gray wolf, grizzly bear, and wolverine are residents of both tundra and boreal forest. Boreal species such as mink and beaver reach their northern limit at this latitude and are seldom found beyond the tree line.

The Taiga Shield Ecoregion is also home to about 150 species of birds, the majority of which are seasonal migrants. The lakes and wetlands of the area provide habitat for a wide variety of waterbirds and shorebirds. A number of birds of prey (raptors) use this region, either as residents or migrants.

20.2.2 Hydrogeology

A hydrogeological study was carried out by Stantec from 2008 to 2010 to provide preliminary baseline information (Stantec, 2010). KPL conducted additional packer testing in 2010 and installed seven groundwater monitoring wells in 2011. That work consisted primarily of drilling and installing wells, hydraulic testing, determination of groundwater levels, and the analysis of groundwater chemistry (KPL 2011a, b). The groundwater sampling program is ongoing.

The conceptual hydrogeological model for the area between Thor Lake and Long Lake consists of shallow and deep aquifers separated by bedrock and discontinuous permafrost (KPL 2010b, c). It is unlikely that there is any significant connection between these two aquifers.

Near-surface groundwater samples reflected an oxidizing environment, compared with the reducing environment from which deep aquifer samples were collected. Water quality results were consistent and few seasonal trends were apparent. Chemical analyses identified exceedances of guideline levels for aluminum, cadmium, copper, iron, lead and silver. These analyses represent natural background geologic and hydrogeologic conditions due to the undeveloped nature of the project area.

20.2.2.1 Surface Hydrology

The Thor Lake watershed area drains into a larger watershed area before flowing into Great Slave Lake. Sub-catchment areas within the Thor Lake watershed have been identified based on available imagery and mapping. Existing hydrological information for the Nechalacho site area is described in Melville et al. (1989), Stantec (2011c) and Knight Piésold (2009a, 2012a).

20.2.2.2 Permafrost and Soils

The Nechalacho mine and concentrator site is located within the discontinuous permafrost zone, which results in the spatial distribution of permafrost being highly dependent on local factors (e.g., thin snow cover, northern exposure, presence of fine-grained sediments like silt and clay). Within the project area, landforms associated with permafrost consist mainly of frost-shattered bedrock and frost-heaved sediments that form small raised peat plateaus. Areas represented by permafrost degradation include thermokarst pits, collapsed fens and bogs, and thaw lakes. In general, the thickness of the active layer within the project area ranges between 40 and 200 cm, with variability being attributed to local terrain features.

20.2.2.3 Ecosystems and Vegetation

Descriptions of the ecosystems and vegetation present within the Nechalacho site were developed using a combination of ecosystem mapping, field surveys, and the compilation of historical information. Ecosystem maps were developed for a local study area, which covered approximately 1,797 ha and a regional study area, which covered approximately 44,030 ha. Digital air photo interpretation and satellite image classification techniques were used to map ecosystems within the local and regional study areas, respectively. A total of 163 field inspections were completed within the project area overall. A rare plant study was also initiated in the local study area in 2010. Specific details are provided in the DAR (EBA, 2011a).

20.2.3 Tailings Management Facility

20.2.3.1 Introduction

KPL completed a feasibility level design for the TMF at the Nechalacho site. The feasibility design has been completed to provide permanent and secure tailings storage, in addition to water storage and control to ensure protection of the environment during operations and in the long-term, post-closure.

20.2.3.2 Site Selection

In 2009, a site selection study was completed to identify the preferred option for tailings management at the Nechalacho site. As part of this study, seven potential sites were identified (KPL, 2009b). After an evaluation of the various options, discussions with

Aboriginal groups and a desktop review of available information, the preferred layout option was identified.

The TMF is located approximately 3 km northeast of the concentrator within the Ring and Buck Lake basins, in the headwaters of the northern watershed area ultimately reporting to Thor Lake. The TMF occupies approximately 150.5 ha. The overall site layout and final configuration of the TMF is shown in Figure 20.3.

20.2.3.3 Overview of Design Basis

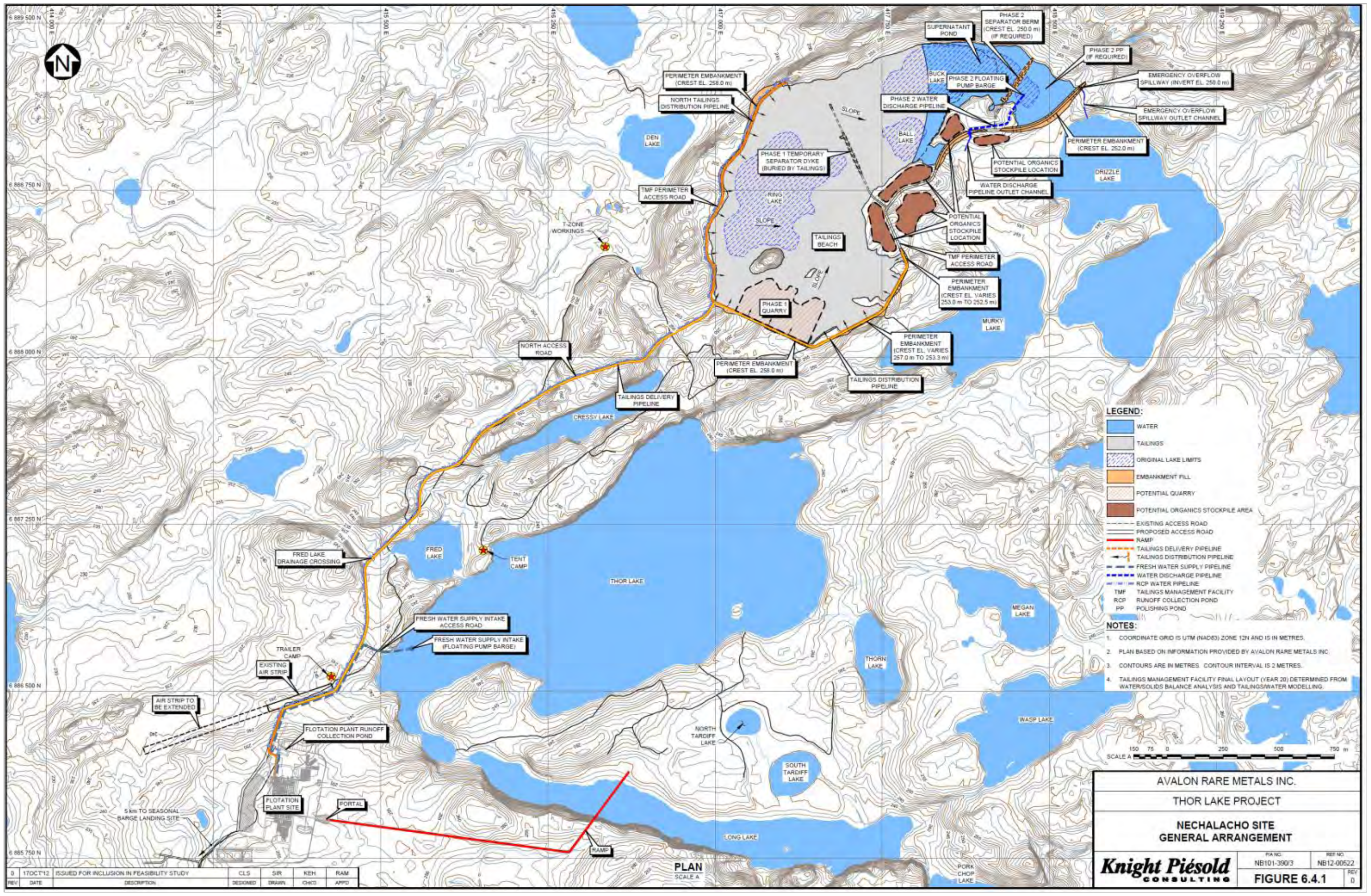
The principle objective of the TMF design is to provide tailings and water storage, ensure protection of the environment during operation and in the long term, and to achieve effective reclamation at mine closure. The feasibility design of the TMF has taken into account the following requirements:

- Permanent, secure and total confinement of all tailings solids within an engineered facility.
- Control, collection and decant of free draining liquids from the tailings during operations.
- The inclusion of monitoring features for all aspects of the facility to ensure performance goals are achieved and design criteria and assumptions are met over the mine life.

Embankment stability and design storm events chosen for the various infrastructure items are based on relevant criteria including the guidelines of the Canadian Dam Association (CDA, 2007), including the following:

- Emergency overflow spillway for IDF: 1 in 1,000 year, 24-hour storm event.
- Maximum design earthquake: 1 in 1,000 year (0.035 g).
- Perimeter embankment freeboard:
    - Containment of the Environmental Design Storm (EDS): 1 in 25 year, 24-hour storm event (46 mm).
    - Minimum spillway freeboard allowance: IDF plus 1.0 m for wave run-up.
- Tailings Delivery and Distribution Systems
    - Average process flow: 1586 dry t/d at 65% solids.
    - Design process flow: 73.2 m^3/h of slurry.

Figure 20.3
Nechalacho Site General Arrangement



The feasibility study design of the Nechalacho TMF is based on the preferred option location using a projected 20-y mine life at a processing rate of approximately 2,000 dry t/d. The life of mine storage capacity is 4.38 Mt.

20.2.3.4 Tailings Physical Properties

Flotation tailings samples were produced by a pilot plant in 2011 and 2012 and subjected to geotechnical testing by SGS Canada Inc. (SGS, 2012) (KPL, 2012b). Based on this testwork, the following tailings properties have been adopted for the feasibility design:

- Specific gravity: 2.83.
- Approximate grain size: 87% silt, 11% clay, 2% sand (98% passing 75 µm).
- Estimated settled dry density: 1.3 t/m^3.

Additional rheological testing was completed to provide input into the design of pumps and pipelines (KPL, 2012b).

20.2.3.5 Tailings Geochemical Properties

Under the direction of Avalon, environmental testing has been completed on selected tailings samples by SGS-M in 2010 and 2011 and from pilot plant work in 2012. Testwork included shake flask extraction (SFE), modified acid base accounting (ABA) and net acid generating (NAG) tests. The SFE testing indicated that the tailings samples were relatively 'clean' with no parameters exceeding the Federal Metal Mining Effluent Regulations. The ABA testing (2010 and 2011 results combined) indicated that some of the samples were potentially acid neutralizing, some samples were uncertain and one sample was potentially acid generating. All of the NAG testwork indicated that the tailings are highly unlikely to be acid generating.

The Phase 2 testing of the tailings and water completed by SGS in 2011 also included analysis of the fresh and aged pilot plant tailings decant water to determine the quality of the supernatant water over time. The results indicated that all controlled parameters were at concentrations below the MMER limits, and generally the aged water samples had decreased concentrations of most metals compared to the fresh water samples.

Water quality, water treatment and bioassay testing was completed by SGS on water from pilot plants in 2012. Treated samples blended with Drizzle Lake water in appropriate ratios were found to be non-toxic and non-sub-lethally toxic in standard testing. Treated water also met all metal mining effluent regulations in the effluent. Treated water blended with Drizzle Lake water met all Canadian Council of Ministers of Environment (CCME) guidelines for the protection of aquatic life. As there are no CCME guidelines for rare earth elements, utilizing the process utilized for the setting of the CCME guidelines, Avalon developed CCME-like guidelines for these elements. Treated and blended effluent also met these CCME-like guidelines. It is noted that some natural background water quality parameters in Drizzle and other lakes in the area do not meet the CCME guidelines. Low levels of uranium and thorium will be present in the tailing. This is not anticipated to have any impact on regulatory requirements.

Testing of water quality (acute and sub-lethal toxicity testing) has been completed on treated pilot plant water. Treated effluent quality passes all regulatory standards and toxicity testing. Avalon has committed to meet the CCME guidelines for the protection of aquatic life.

20.2.3.6 Foundation Conditions

Surface soils overlying the bedrock are intermittent but generally sparse. Where present, the surficial soils generally consist of tills, variable-textured glaciolacustrine sediments and glaciofluvial materials typically as thin, discontinuous deposits between rock outcrops over much of the area. Fine and coarse grained till deposits have been observed at the Nechalacho site with peat plateaus and organic soils occurring in wet depressions.

The bedrock in the vicinity of the TMF is generally comprised of good to very good quality syenite or diabase bedrock, with generally few discontinuities. Rock mass characteristics were collected for the bedrock to support foundation design and hydrogeological considerations. Based on the results, including Packer permeability testing, the bedrock is judged to have very low permeability.

Generally, discontinuous permafrost conditions have been observed in the Nechalacho site area in both bedrock and surface soil deposits.

20.2.3.7 Tailings Management

The TMF will include the tailings basin for primary settlement of solids and supernatant water storage and a polishing pond (PP) to allow for water clarification. The tailings and polishing pond will be established by the construction of several lined perimeter embankments and an internal separator berm/dyke to maintain the PP.

Tailings disposal will include spigotting and/or end of pipe discharge (seasonally dependent) from the constructed embankments and natural bedrock topography, generally on the west, southwest and northwest sides of the facility. This will develop a west–to-east sloping tailings beach and establish the supernatant pond on the eastern side of the facility.

During the first year of operations, all tailings will be sent to the TMF. From Year 2 of operations, through to the end of mine life, approximately 65% of the tailings produced will be used to generate paste backfill for the underground mine workings with the remaining 35% sent to the TMF. At any point in time, all tailings produced will either be sent to the paste backfill plant, or to the TMF (i.e., the tailings stream will not be split).

The TMF has been sized to permanently store 3.37 Mm^3 of tailings with an estimated settled dry density of 1.3 t/m^3.

20.2.3.8 Water Management

Water management will be a critical part of the TMF operations and will be carried out in close conjunction with the tailings deposition plan. One of the key aspects will be to maintain consistently acceptable water quality at the discharge point to the environment, as well as maintaining pond levels that are within the design limitations of the facility. The water balance was developed for design purposes and will become a key tool for determining water management strategies within the tailings facility during operations.

The water management strategy for the Nechalacho project consists of a ‘semi-closed loop’ system to minimize impact to the existing natural hydrologic flow patterns within the Thor Lake watershed. All fresh water required for the Nechalacho site will be taken directly from Thor Lake. The portion of the tailings solids and process fluids not sent underground as paste backfill, as well as all water from the runoff collection pond (includes treated underground mine water and impacted plant site runoff), will be sent to the TMF. There will be no reclaim of water from the TMF to the concentrator for use in the plant.

Ultimately, all excess water from the facility will be pumped from the polishing pond to the Drizzle Lake/Murky Lake drainage system using a floating pump barge, with flows eventually reporting to Thor Lake through natural drainage. A floating pump barge has been selected as it will provide operational flexibility over a decant system. Although treatment of the water is not expected to be necessary based on preliminary testing, a future permanent treatment system could be established at the discharge location to Drizzle Lake if required.

At the end of operations, in addition to the deposited tailings solids, the facility will have the capacity to maintain a minimum operating water volume of 155,000 m^3 plus additional capacity for temporary storm water containment. Water storage capacity will be significantly higher during the early years of operation, but will decline as tailings solids decrease the available capacity. In all cases, the freeboard available above the maximum operating level will exceed the runoff produced from the 1 in 25 year, 24 hour Environmental Design Storm (EDS) event (46 mm). An emergency overflow spillway has been included to convey the Inflow Design Flood (IDF) to the environment. The spillway has been designed with a minimum depth of 2 m, including 1 m for routing of the IDF and 1 m for wave run-up.

20.2.3.9 Facility Construction

The perimeter embankment cross-section will consist of a downstream rockfill zone, a transition zone on the upstream side of the rockfill and a bituminous geomembrane installed between layers of liner bedding/protection material on the upstream slope and tied into the foundation at the toe of the embankment using a concrete anchor plinth to reduce seepage. The entire basin is not lined, however, confinement will be provided by the good quality bedrock underlying the basin. A layer of riprap will be placed on top of the liner protection material to prevent damage to the liner due to erosion and ice.

The Phase 1 temporary separator dyke and Phase 2 separator berm, used to separate tailings solids from the polishing pond, will be constructed of homogeneous waste rock and/or processed fill.

Design analyses completed for the dams, included seepage and stability analyses.

20.2.3.10 Monitoring

Monitoring of TMF operating parameters will be carried out daily, monthly and annually to ensure optimum performance and allow refinement of the operation, as necessary. The achieved density of the tailings will be dependent on the tailings deposition practices and supernatant pond volume. Field information on the in situ tailings densities will be collected during the initial months of operations for comparison with the design values. This information will allow further optimization of the tailings deposition strategy and water balance model, adjustments to the pond operating levels and, if necessary refinement to ongoing operations and future design.

20.2.4 Waste Rock

Waste rock will be stockpiled temporarily during construction and incorporated into the construction of facilities and infrastructure at the site. During operations, however, waste rock generated will remain underground and will be placed in excavated openings. (See Section 16.0, Mining Methods).

20.2.5 Archaeological Studies

Culturally significant sites have been identified in the general areas near the Nechalacho site. An archaeology study of the Thor Lake area was initially conducted in 1988 as part of the Thor Lake Environmental Baseline Survey (Melville et al. 1989). The archaeological study was conducted to determine the cultural resources present at or within the Project. A subsequent study was conducted in 2011, under a Class 2 permit, by Points West for the Nechalacho site area (Points West, 2012). Further studies were conducted in 2012 (Points West, 2013).

The 2012 archaeological investigations of the mine area were to fill in the gaps that remained from the 2011 examinations where the extent of project components were revised or boundaries were not previously accurately identifiable. All mine related facilities proposed on the north side of Great Slave Lake were examined by pedestrian transects sufficient to provide good visual coverage. The focus was to cover all high and moderate potential ground and in that process sizeable areas of low potential ground were crossed and viewed. Existing ground exposures (bare ground) were frequent and were inspected.

The presently proposed project component locations, including the plant site and airstrip extension are considered fully assessed for the presence of archaeological resources and are considered unlikely to encounter any unrecorded extensive or intensive archaeological deposits.

20.2.6 Potential Socio-economic Effects and Mitigation Measures

Potential socio-economic effects and mitigation measures are described in the Avalon DAR, (EBA, 2011a).

20.3 PINE POINT SITE

20.3.1 Summary of Baseline Conditions

The Pine Point site is located in an existing brownfields area of the former Pine Point mine site and is located within the Great Slave Lowlands Mid-Boreal (MB) Ecoregion of the Taiga Plains Ecozone.

Vegetation is characterized by medium to tall closed stands of jack pine and trembling aspen. Poorly drained fens and bogs are covered with low open stands of larch, black spruce and various shrubs.

There are no streams or significant natural water bodies and, thus, no fisheries values or habitats at the Pine Point site.

Moose, woodland caribou and wood bison are the main ungulates found in the area, although none are considered common. Bird life is typical of the boreal forest and the south shore of the Great Slave Lake is considered to be an important concentration area for many bird species during annual migration.

The main components of the Pine Point site are expected to have limited effects on the local receiving environment. Any identified environmental effects are expected to be limited to the immediate footprints and local study area of the Pine Point site and associated infrastructure, and most will be reversible once activities ceased. With the application of the proposed mitigation measures, the residual environmental effects for all environmental VCs of the project at the Pine Point site are anticipated to be negligible and insignificant. Overall environmental benefits are anticipated as a result of a reduction in limonite waste stockpiles in the area through their utilization in the hydrometallurgical process, and restoration and rehabilitation of an abandoned open pit as a result of operations.

20.3.2 Hydrogeology

The hydrogeology for the Pine Point region has been previously studied and reported on by Geologic Testing Consultants (GTC, 1983) and Stevenson International Groundwater Consultants (Stevenson 1983, 1984). Six groundwater monitoring wells were installed in 2010 and seven monitoring wells were installed in 2011 by KPL (KPL, 2011c). The groundwater sampling program is ongoing.

The south side of Great Slave Lake comprises a lowland area and is considered a major groundwater discharge area.

20.3.2.1 Surface Hydrology

No streams are present in the proposed Pine Point site or the haul road alignment. The area of interest is flat to gently sloping and a considerable portion of the area is covered by poorly drained muskeg ranging up to 3 m deep. Extensive wetland areas and small lakes are located in the area of the haul road.

20.3.2.2 Groundwater

Local groundwater recharge to the bedrock aquifer at the Pine Point site is likely to be variable and largely controlled by the overburden geology. High rates of recharge are expected in areas where sinkholes are present but, in general, recharge will be limited by the presence of till overburden.

Local surface water/groundwater flows through the till, then downwards through fractured bedrock towards the water table. Several seepage points observed in historic pit walls indicate that there is some lateral flow within the unsaturated bedrock. This seepage is thought to be due to local infiltration being directed horizontally along bedding planes.

The bedrock units that represent the most productive aquifers are the Presqu'ile Formation and the Pine Point Formation. The Presqu'ile and Pine Point Formations form the main aquifer consisting of highly porous, well fractured dolomite.

The permeability of the Presqu'ile aquifer formation is very high with transmissivities in the order of $1x10^{-2}$ m^2/s (GTC 1983). The groundwater gradient in the Pine Point area is generally northwards towards Great Slave Lake while to the south of the Pine Point area, the groundwater gradient trends from west to east (Stevenson, 1983).

Although the Presqu'ile aquifer has a high permeability, the flow through it is thought to be slow due to the low gradient in the Pine Point area. Based on a preliminary calculation taking into account the documented transmissivity values, gradients and geological sections, the flow velocity is estimated to be less than 1 m per day.

20.3.3 Hydrometallurgical Tailings Facility

20.3.3.1 Introduction

KPL completed the feasibility level design for the hydrometallurgical tailings facility (HTF) at the Pine Point site (KPL, 2013). The feasibility design has been completed to provide permanent and secure tailings storage, in addition to water storage and control to ensure protection of the environment during operations and in the long-term (after closure).

20.3.3.2 Site Selection

The selected tailings disposal site is within the historic L-37 open pit previously operated by Pine Point Mines Ltd., subsidiary of Cominco Ltd. The site selection resulted from KPL's site visit in June, 2010 and consideration of tailings disposal alternatives (KPL, 2010a).

The HTF is located approximately 1 km southeast of the hydrometallurgical plant. The historic pit has an average depth of approximately 25 to 30 m and is approximately 300 m by 650 m in size, elongated northeast-southwest. There is a main access ramp located at the southwest end of the pit. An alternate historic ramp was developed to the north although it has been blocked by waste rock piles over most of its area. The final configuration of the facility is shown on Figure 20.4 with respect to the overall site layout.

20.3.3.3 Overview of Design Basis

The principle objective of the HTF design is to provide tailings storage, ensure protection of the environment during operations and in the long-term (after closure), and achieve effective reclamation at mine closure. The feasibility design of the HTF takes into account the following requirements:

- Permanent, secure and total confinement of all tailings solids within an historic open pit with engineered preparation and management strategies.

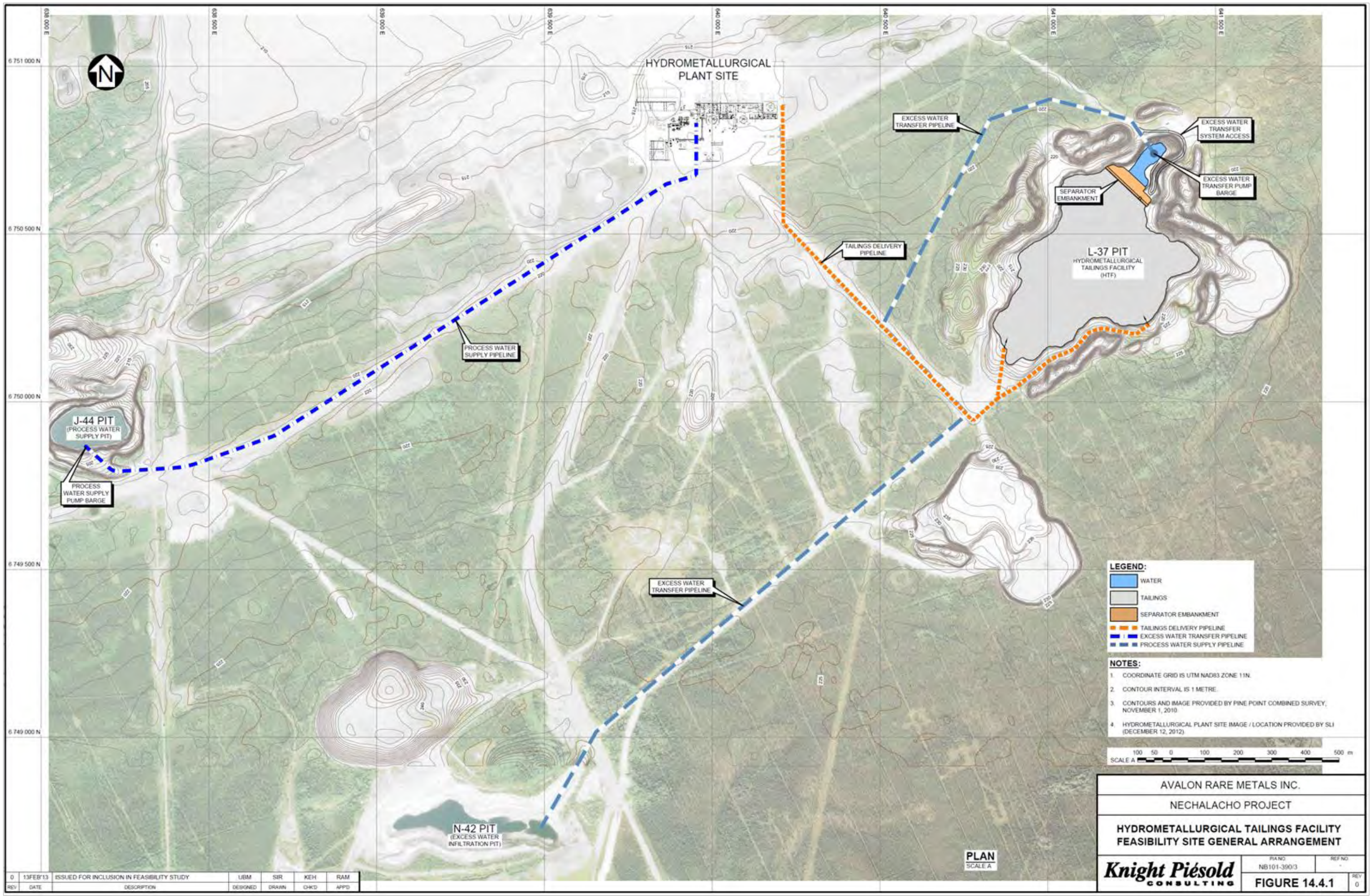
- Control, collection and removal of free draining liquids from the tailings during operations (to the maximum practical extent).

- The inclusion of monitoring features for all aspects of the facility to ensure performance goals are achieved and design criteria and assumptions are met.

The feasibility level design for the Pine Point HTF is based on the projected 20-year mine life at the processing rate for the hydrometallurgical plant. This translates to a total of approximately 1.34 Mt (2.58 Mm^3) of tailings. The general design basis for the HTF at the Pine Point site is summarized below.

- Tailings Production
  - Year 1 (includes 8-month ramp-up period): 47,870 dry t.
  - Year 2 to 20: 68,000 dry t/y.
- Tailings Delivery and Distribution Systems
  - Average process flow: 186 dry t/d at 30% solids.
  - Design process flow: 73.5 m^3/h of slurry

Figure 20.4
Pine Point Site General Arrangement



20.3.3.4 Tailings Physical Properties

Tailings samples were produced by a pilot plant in 2012 and subjected to geotechnical testing by SGS Canada Inc. and summarized in the feasibility design report (KPL, 2013). Based on these results, the following tailings properties have been adopted for the feasibility design:

- Specific gravity: 2.9.
- Approximate grain size: 0.4% sand, 99.6% silt.
- Estimated settled dry density: 0.52 t/m³.

In addition to the very low achievable settled density based on the settled density tests, the tailings exhibit a Bingham Plastic rheological behaviour which is characterized by yield stress. The yield stress of a slurry is the minimum force needed to cause the slurry to flow. The critical solids density (CSD) value was determined by plotting the yield stress versus solids density and was determined to be 30.5%. The rheological properties for the slurry at 30.4% solids were measured as follows:

- Yield stress: 16 Pa.
- Plastic viscosity: 2 mPa.s.

20.3.3.5 Tailings Geochemical Properties

Environmental testing was completed by SGS on selected tailings samples under the direction of Avalon. The testwork included the following:

- Shake Flask Extraction (SFE),
- Modified Acid Base Accounting (ABA).
- Net Acid Generating (NAG) tests.

20.3.3.6 Existing Pit Conditions

The walls of the L-37 Pit are primarily comprised of exposed dolostone bedrock overlain by natural till deposits ranging from 4 to 10 m in depth. Around the majority of the outside perimeter of the pit, stockpiles of till, waste rock and organics are present, with the bulk of these placed at the angle of repose. The exposed bedrock slopes are typically in the range of 60° to greater than 75°, with some localized areas approaching vertical. The natural till slopes typically range from 1.3H:1V and 1.5H:1V and show significant evidence of erosion. In many areas along the base of the pit slope, talus rock and eroded till materials have been deposited and sit at the angle of repose. The exposed dolostone bedrock forming the majority of the pit slopes is highly fractured which is suspected to be partially the result of blasting disturbance and subsequent weathering processes of the exposed slopes. The presence of some vugs was also noted during inspection of the pit walls, with some smaller voids present, possibly also the result of blasting disturbance and weathering.

The pit bottom is generally above the water table, however there are four small areas where the pit was mined deeper and where open water is now permanently exposed. The elevation of the water corresponds with the local groundwater table in the area (approximately 191 masl) which fluctuates only slightly over the year based on runoff conditions.

There is a significant amount of waste rock deposited in the bottom of the pit in small piles indicating the material was simply end-dumped. The piles are in various locations with the majority located along the SE side.

20.3.3.7 Tailings Management

The tailings and water management strategy has been developed to maximize use of previously disturbed areas within the existing brownfield site and to minimize impacts to the natural environment. All tailings solids and water will report to the HTF from the hydrometallurgical plant via a tailings delivery pipeline. Deposition will be accomplished by end-of-pipe discharge at select locations around the HTF. A separator embankment will be constructed across the facility to separate the tailings from the supernatant pond at the east end of the pit. The embankment is intended to promote drainage of the tailings, which will increase consolidation, increase densities and, ultimately, provide improved closure opportunities by having a less saturated surface on which to work.

The tailings delivery system will consist of a pump train and pipeline which will transport slurry from a tailings storage tank located at the plant site to the HTF. The tank will be sized such that the tailings delivery system will be operated two to three times per day.

20.3.3.8 Water Management

Water management systems associated with the HTF include the following components:

- Excess water transfer system.
- Excess water infiltration pit.
- Process water supply system.

The excess water transfer system has been designed for removal of excess supernatant water from the HTF. The excess water will be transferred by pipeline and discharged to the excess water infiltration pit (the historic N-42 pit), located southwest of the HTF. The majority of the transfer will occur during the summer months. Ultimately, all excess water from the HTF will infiltrate into the surrounding bedrock and back into the Presqu'ile aquifer, through seepage from the HTF and/or excess water infiltration pit.

A feasibility level water balance analysis was completed for the HTF. The monthly water balance analyses were completed for an eight month ramp-up period and projected 20-y mine life. The water balance indicates that the average transfer rate of water from the HTF to the excess water infiltration pit is approximately 108,300 m^3/y. The HTF has sufficient storage capacity to contain all the tailings solids and water and at all times maintain sufficient freeboard to safely accommodate the EDS event (corresponding to the PMP storm event) plus wave run-up.

All process water will be drawn from the historic J-44 pit, located approximately 2 km southwest of the hydrometallurgical plant. (See Figure 20.4, above).

A groundwater model was previously developed as part of the project pre-feasibility study to estimate the effect of the project operations (process water withdrawal and excess supernatant water infiltration) on the natural groundwater aquifer (KPL, 2011d). In 2012, EBA augmented this model with the addition of a conservative contaminant transport component to predict potential impacts on the downstream receiver. The results indicated that the project activities are not expected to have any significant impact on the groundwater regime, due to the very large volume of the Presqu'ile aquifer and its relatively shallow gradient. The impact on Great Slave Lake is predicted to be undetectable within the natural variation of the aquifer (EBA, 2012). Additional water quality monitoring close to the L-37 pit will be undertaken to verify the model predictions or to allow further mitigation measures to be taken, if required.

20.3.3.9 Construction

The amount of pit preparation required prior to start-up will depend on a final inspection of the pit walls and floor prior to construction. The primary goal of any pit preparation will be to facilitate HTF construction, ensure efficient operation and, ultimately, to ensure total containment of all tailings solids within the confines of the HTF. Inspections completed during the 2011 site investigations noted only minor issues that could potentially require work during pit preparation, including treating voids in pit walls, backfilling exposed water where tailings will be deposited, backfilling historic exploration drill holes and some general grading of the pit floor.

The L-37 pit has adequate capacity and freeboard to contain all tailings without the construction of perimeter embankments based on the current design.

The separator embankment will be constructed of waste rock material from historic mining activities currently piled in the pit bottom. A filter zone is included on the upstream side of the embankment which will consist of processed waste rock with a maximum particle size of 300 mm. Following construction of the initial embankment, additional raise(s) will be determined and scheduled based on the observed beach slopes and rate of rise at the location of the Separator Embankment. The Separator Embankment will have an upstream side slope of 1.3H:1V and a downstream side slope of 1.75H:1V with a 6 m wide crest (KPL, 2013). Each raise will include a swale excavated through the rockfill at the crest of the Separator Embankment to convey storm surge flows over the embankment to the water pond.

20.3.3.10 Monitoring

Monitoring of HTF operating parameters will be carried out daily, monthly and annually to ensure optimum performance and allow refinement of the operation, as necessary. The achieved density of the tailings will be partially be dependent on the tailings deposition practices and supernatant pond levels. Field information on the in situ tailings densities will be collected during the initial months of operations for comparison with the design values. This information will allow further optimization of the tailings deposition strategy and water balance model, adjustments to the pond operating levels and, if necessary refinement to ongoing operations and future design.

20.4 GEISMAR SITE

URS Corporation prepared the section of the project feasibility study dealing with environmental and permitting issues at the Geismar site.

20.4.1 Legal Requirements and Other Obligations (Permitting)

The 60.7 ha (150-acre) rare metal separation plant would require the construction of numerous facilities that include but are not limited to rail spur extensions, onsite infrastructure, wastewater collection, treatment and discharge facilities and solid waste impoundments. The construction and operation of the Geismar site will require the following environmental permits issued by the Louisiana Department of Environmental Quality:

- Air permit for air emissions. It is anticipated that the site will be a “small source” (minor source of criteria pollutants) and a minor source of hazardous air pollutants (HAPs) for air permitting.

- Solid waste permit for the management of gypsum waste produced in the manufacturing process and management of contact stormwater runoff prior to discharge. Waste gypsum and contact stormwater will be managed in surface impoundments. The construction of an earthen berm based solid waste disposal unit and the tankage for water treatment facilities will require compliance with the Louisiana Administrative Code for solid waste facilities (LAC 33 Chapter 7).

- Surface water discharge permit under the Louisiana Pollution Discharge Elimination System (LPDES) for the discharge of process and stormwater from the site. Process wastewater will be discharged to the Mississippi River through a permitted outfall. Non-contact stormwater will be discharged to surface drainage through monitored outfalls.

A “404 Permit” issued by the United States Army Corps of Engineers (USACE) is required for disturbance of wetlands at the site, if wetlands are disturbed. Avalon will obtain a 404/Section 10 Permit for the site, the access road and the LPDES discharge location to the Mississippi River.

Potable water is available from local utilities. Installation of wells for potable water will require a permit from the Louisiana Department of Health and Hospitals.

Authorization from the Pontchartrain Levee District is required for subsurface work (such as drilling or excavation) conducted within 1,500 ft of the Mississippi River levee. Portions of the site are within 1,500 ft of the levee.

Compliance with the Historical Preservation Act Section 106 will require a Phase I Cultural Resources Survey prior to construction.

20.4.2 Summary of Baseline Conditions

The rare earth leaching and separation plant site is located in Ascension Parish within the Mississippi River Delta Plain Region of the Coastal Plain Province of North America. The typical climate of this region is humid and subtropical with high average rainfall, high relative humidity, and high average temperatures. The major ecological regions in coastal Louisiana are palustrine forests, and coastal marshes.

The local physiography is characterized by terrace uplands, natural levees of the Mississippi River, and back swamps which formed in depressional areas between natural levees and terrace uplands. The area is predominately historic farmland replanted with mixed hardwood species, deciduous palustrine forested wetlands and bodies of fresh water. Small portions of the historic farmland have been developed into industrial sites along the natural levees of the Mississippi River.

Major waterbodies near the area include the Mississippi River, the New River and Bayou Conway. There are three small waterbodies within the Geismar site area which connect to Bayou Conway and the Mississippi River.

The wide variety of flora and fauna in the area is typical of humid subtropical regions nourished by abundant rainfall and normally mild winters. The back swamp and hardwood forest is an active ecosystem providing both soft and hard mast, as well as cover and water for birds, mammals, reptiles and amphibians. At the present time, there are no rare or endangered species listed by the U.S. Department of Interior Fish and Wildlife Services as indigenous to the area.

20.4.2.1 Air Quality

The proposed site for Avalon’s new plant is located in Geismar, Ascension Parish, Louisiana. The air quality in Ascension Parish is classified as non-attainment for ozone (volatile organic compounds and nitrogen oxides), and in attainment for all other criteria pollutants (carbon monoxide, sulphur dioxide, and particulate matter). Major sources of nitrogen oxides and volatile organic compounds will require stringent control requirements known as Lowest Achievable Emission Rates and offsets may be required for increased levels of these pollutants. Major sources of carbon monoxide, sulphur dioxide or particulate matter will require less stringent control requirements or Best Available Control Technology. The proposed plant is expected to be a minor source of criteria pollutants, and therefore, is not expected to have any significant impact on the current air quality in Ascension Parish.

20.4.3 Summary of Potential Residual Effects

The main components of the Geismar site are expected to have limited effects on the local receiving environment. Any identified environmental effects are expected to be limited to the immediate footprints and local study area of the leaching and separation plant site and associated infrastructure, and most will be temporary or reversible once construction activities cease. There are no streams or waterbodies that will be impacted by the development of the site. Therefore, the project is not expected to have any direct impact on fisheries, aquatic fauna, or aquatic habitats.

Overall, the assessment determined that with the application of the proposed mitigation measures, the residual environmental effects for all environmental VCs at the refinery site are anticipated to be negligible and insignificant, including the discharge of non-storm waters to the Mississippi River. Avalon is confident that with the application of sound engineering, environmental planning and best management practices, as well as compliance with anticipated permits, licences, approvals, existing Federal and State environmental regulations and guidelines, that the environmental issues associated with the development and operation of the Project can be effectively addressed and managed.

20.4.4 Valued Components

Environmental VCs were identified through consultation, environmental assessment process, knowledge and understanding of the biophysical aspects, applicable scientific principles and consideration of the following:

- Species currently listed under the Endangered Species Act of 1973.
- Species currently protected under the Migratory Bird Treaty Act of 1918.
- Species currently protected under the Bald and Golden Eagle Protection Act.
- Species or species groups considered of cultural importance.
- Species or species groups considered to be particularly sensitive to disturbance.
- Species or species groups dependent upon particular environmental features.

Environmental VCs considered in the evaluation of potential project effects are the same as those considered for the Nechalacho and Pine Point sites.

Wildlife Species at Risk, species with special conservation status, and species of cultural importance include the following:

- Eastern Spotted Skunk.
- Manatee.

- Bald Eagle.
- Four-toed Salamander.
- Gulf Sturgeon.
- Pallid Sturgeon.
- Inflated Heelsplitter.

20.4.5 Hydrogeology

The area along the eastern edge of the meander belt of the Mississippi River is a growing deltaic plain. Natural levees, which were formed during periods of overbank flooding, slope from the river's edge and grade into back swamp deposits at lower elevations. The deposition of sand, silt and clay forms the top stratum of the younger deltaic deposits with point-bar deposits within bends of the river channel.

The top stratum is underlain by older deltaic deposits and alluvium to depths of 245 m to 300 m within the Pleistocene channel of the Mississippi River. The deposits consist of interbedded clay, sand and gravel layers. The sand and gravel stratigraphic units form the principal aquifers of the area. The Plaquemine Aquifer, consisting of alluvial deposits, is the principal aquifer in Ascension Parish. However, in most of the region, the alluvial deposits which extends eastward beyond the limits of the river channel are of relatively minor importance, and the principal aquifers occur in the older deltaic deposits. The Plaquemine Aquifer and aquifers in the alluvial and older deltaic deposits apparently form an interconnected artesian aquifer system, at depths of about 45 m to 215 m. The deposits below a depth of about 215 m are mostly clay containing several sand layers that correlate with the "400- to 600-ft" (120 m to 183 m) aquifer system of the Baton Rouge area.

20.4.5.1 Surface Hydrology

The New River and the Mississippi River are the only surface water bodies in the vicinity of the proposed facility. New River is a small river with only local recreational use by area residents and is not a source of drinking water. It is a major tributary to Blind River which leads into Lake Maurepas.

The Mississippi River in the Geismar area is used primarily for transportation and industrial purposes. The Mississippi is also used for municipal purposes by cities along the river and serves as a discharge area for industrial processes water through National Pollutant Discharge Elimination System (NPDES) permitted outfalls.

20.4.6 Geismar Site Closure and Reclamation Plan

As described in Section 20.1.6, the conceptual closure and reclamation plan for the Geismar site is based on a "design for closure" approach and will be updated throughout the life of the project.

It is recognized that the refinery may continue to operate long after the Nechalacho and Pine Point site have been closed by treating rare earth feedstocks from other sources and third parties. Given the high level of industrial development in the area it is anticipated that during closure, when the separation plant and ancillary facilities will be removed, a portion of the site (not including the waste management facility) can be severed and resold for alternate development. Under these conditions, the site will be rehabilitated to the extent necessary for the subsequent development. The following is prepared for the entire site and assumes the unlikely event that resale is not achievable.

Where practical, the following components will be incorporated into the overall design and into the closure and reclamation plans for relevant components of the separation plant:

- Buildings and site infrastructure will be located, to the extent practical, on previously disturbed land. (Since the site is a former plantation, the entire area has been previously disturbed).

- Organic and mineral soils will be salvaged and stored for future reapplication during reclamation.

- Post-closure monitoring will be carried out to meet the regulatory requirements to assess the adequacy of the site clean-up and assure the stability of the gypsum waste disposal facility for 30 years.

- Appropriate and approved seed mixes and plant material will form the basis of the re-vegetation strategy. Native species will be used where practical and available. All seed sources, whether native or agronomic will be certified weed-free.

- Following removal of surface facilities, the plant and any ancillary building and development footprint will be scarified and re-contoured as required to ensure surface stability, blend with the surrounding landscape, and facilitate the re-establishment of native vegetation.

- Access roads and railroad access are planned to remain as assets for potential future development.

- Test plots will be developed during operations and Cell 1 of the waste management facility will be progressively rehabilitated during operations.

- Reasonable efforts to sell for reuse plant equipment and materials will be made (but no credit for this is included in the project economics). Scrap metal will be recycled to the extent practical, and waste will be disposed of in approved facilities as per local regulatory requirements.

Costs associated with closure and ongoing monitoring of the Geismar site are provided in Section 21.0.

21.0 CAPITAL AND OPERATING COSTS

21.1 CAPITAL COSTS

A summary of the feasibility study capital cost estimate for the project is presented in Table 21.1.

Table 21.1
Project Capital Cost Summary

Cost Category	NWT [1] ($ million)	LA [2] ($ million)	Total ($ million)
Mine development	81.58	-	81.58
Main process facilities	351.24	192.51	543.75
Infrastructure	150.68	78.82	229.50
EPCM	119.27	38.57	157.84
Indirect construction costs	175.56	27.25	202.81
Owner's costs	36.76	18.95	55.71
Contingency	120.91	44.90	165.81
Closing costs / bond	13.00	3.16	16.16
Pre-production capital cost	**1,049.00**	**404.16**	**1,453.16**
Sustaining capital	102.72	19.12	121.84
Total capital cost	**1,151.72**	**423.28**	**1,575.00**

[1] NWT – Costs applicable to the Nechalacho and Pine Point sites in the Northwest Territories.
[2] LA – Costs applicable to Geismar, Louisiana.

The capital cost estimate was prepared by SLI. The scope of the estimate encompasses the engineering, administration, procurement services, construction, pre-commissioning and commissioning of the Project. The estimate was completed to a level consistent with an AACEI (Association of Advanced Cost Engineering International) Class 3 estimate with target accuracy level of ±15%, based on second quarter 2012 prices, excluding escalation.

The total estimated pre-production capital cost is $1,453 million. The life-of-mine sustaining capital is estimated at $122 million.

21.1.1 Basis of Estimate

SLI uses its internal PM+ Estimating module to develop capital costs which includes a comprehensive evaluation from initial pricing to specific guidelines and allowances. The following documents were used to develop the capital cost estimate:

- Master project schedule.
- Process flow diagrams.
- Design criteria.
- Piping material selection.
- Electrical and mechanical equipment lists.
- Data sheets/specifications.

- Equipment and material quotes.
- Site plans.
- Preliminary geotechnical memo.
- General arrangement drawings.
- Concrete sketches and calculations (for major structures).
- Structural steel design sketches and calculations (for major structures).
- Architectural sketches and accommodation complex specification.
- Engineering material take offs (MTOs).
- Piping and instrumentation drawings (P&IDs).
- Electrical single line diagrams.
- Electrical equipment layout and tray routing sketches.
- Hazardous area classification design brief.
- Temporary facilities requirements.
- Construction packages' budgetary quotes.
- Current contractor crew rates and productivity information from SLI in-house experience.
- Current SLI in-house bulk material and minor equipment pricing.
- Scope of work as executed by various contractors and consultants.
- Owner's costs.

The MTOs prepared by SLI engineers were based on historical experience, and included appropriate overbuy/waste allowances and factors for material costing.

A design allowance was applied based on the degree of engineering completed and a comparison of quantities to historical experience. The design allowance was intended to account for the standard growth in quantities from ongoing interdisciplinary reviews and other design criteria that arise as engineering proceeds to final completion.

The foreign currency exchange rates used by SLI are shown below:

Country	Converted to	Rate of Exchange
USA (US$)	Canadian ($)	1.0285
Britain (GBP£)	Canadian ($)	1.5460
Australia (AUD$)	Canadian ($)	1.0500
EURO(EUR)	Canadian ($)	1.3357

All in labour rates were developed for the project based on local applicable published labour rates. Typical North America installation hours were used to estimate labour man-hours for all work items with the exception of major earthworks which were received from contractors. Productivity factors were applied to allow for productivity loss due to long working hours, remote site location and winter conditions.

21.1.2 Nechalacho Mine and Concentrator

A summary of the estimated capital costs for the Nechalacho site located on the northern side of Great Slave Lake is provided in Table 21.2. The scope of the Nechalacho site includes the underground mine, concentrator, paste backfill plant as well as the mine site utilities and infrastructure including power plant, accommodation, airstrip and dock facilities.

Table 21.2
Nechalacho Site Capital Cost Summary

Cost Category	Capital Cost Total ($ thousand)	% of Total
General site	9,960	1.8
Mine development	81,575	14.9
Mine paste plant and distribution system	25,791	4.7
Onsite infrastructure	13,682	2.5
Ancillary support facilities	9,629	1.8
Concentrator	147,741	27.0
Tailings management facility	20,947	3.8
Offsite infrastructure	2,616	0.5
Total direct costs	**311,941**	**56.9**
EPCM services	60,930	11.1
Other indirect costs	94,750	17.3
Total indirect costs	**155,679**	**28.4**
Owner's costs	21,115	3.9
Contingency	59,265	10.8
Total	**548,000**	**100.0**

The closure costs for the Nechalacho site are estimated to be $7 million.

21.1.3 Pine Point Hydrometallurgical Plant

A summary of the estimated capital costs for the Pine Point site located on the southern side of Great Slave Lake is provided in Table 21.3. The scope of the Nechalacho site includes the hydrometallurgical plant, tailings management facility and site utilities and infrastructure including dock facilities.

Table 21.3
Pine Point Site Capital Cost Summary

Cost Category	Capital Cost Total ($ thousand)	% of Total
General site	7,815	1.6
Onsite infrastructure	16,821	3.4
Ancillary support facilities	1,826	0.4
Sulphuric acid plant	2,533	0.5
Hydrometallurgical plant	235,637	48.3
Tailings management facility	6,483	1.3
Offsite infrastructure	435	0.1

Cost Category	Capital Cost Total ($ thousand)	% of Total
Total direct costs	**271,549**	**55.6**
EPCM services	58,339	12.0
Other indirect costs	80,814	16.6
Total indirect costs	**139,153**	**28.5**
Owner’s costs	15,652	3.2
Contingency	61,646	12.6
Total	**488,000**	**100.0**

The closure costs for the Pine Point site are estimated to be $6 million.

21.1.4 Geismar Rare Earth Refinery

A summary of the estimated capital costs for the rare earth refinery site located at Geismar, Louisiana, is provided in Table 21.4. The scope of the Geismar site includes the leaching and separation plants, residue storage facility and site utilities and infrastructure.

Table 21.4
Geismar Site Capital Cost Summary

Cost Category	Capital Cost Total ($ thousand)	% of Total
General site	8,088	2.0
Refinery	34,217	8.5
Solvent extraction plant	121,500	30.3
RE precipitation plant	3,528	0.9
Filtration, drying, calcining and packaging	29,508	7.4
Reagent storage	3,760	0.9
Utilities	58,489	14.6
Ancillary support facilities	12,244	3.1
Total direct costs	**271,334**	**67.7**
EPCM services	38,570	9.6
Other indirect costs	27,250	6.8
Total indirect costs	**65,820**	**16.4**
Owner’s costs	18,946	4.7
Contingency	44,900	11.2
Total	**401,000**	**100.0**

The closure costs for the Geismar site are estimated to be $3.2 million.

21.1.5 Sustaining Capital

The total estimated life-of-mine sustaining capital amounts to $121.8 million comprising $99.8 million for mine equipment, $0.7 million for Nechalacho tailings dam expansion, $2.3 million for Pine Point tailings dam expansion and $19.1 million for the expansion of the gypsum residue containment facility at Geismar.

The sustaining capital allocated to the Nechalacho mine was provided to cover the cost of major equipment purchase and replacement. For the first six years of the project the equipment will be leased and these costs will be carried in the operating costs. In Year 5 of development, the Owner's mine fleet will be ordered and then delivered in Year 6 of operation, and utilized thereafter. Major equipment rebuilds were scheduled after three years of operation and equipment replacement was scheduled after the fifth year from the time of initial purchase.

21.2 OPERATING COSTS

A summary showing the average annual and life-of-mine unit operating costs by project cost area is presented in Table 21.5.

Table 21.5
LOM Average Operating Costs per Project Cost Area

Project Section	Average Annual Costs ($ million)	LOM Average Unit Costs ($/t milled)
Reagents & Consumables	97.2	133.20
Fuel & power	50.7	69.48
Labour	36.7	50.26
Freight	29.4	40.31
G&A	26.8	36.74
Other	23.7	32.29
Project total	**264.5**	**362.28**

The feasibility study operating cost estimates have been prepared on an annual basis for the life of the mine by SLI. The operating cost estimate has been prepared with an estimated level of accuracy of ±15% based on the design of the plant and facilities as described in detail in the feasibility study. Operating cost estimates were based on prices as of fourth quarter 2012.

Salary and wages rates were generated by Avalon for the feasibility study and are based on wages and salary surveys. The basis for the data is supported by the Pricewaterhouse Coopers 2011 salary database for similar positions within the mining industry.

The operating cost estimates for the Northwest Territory operations are based upon low sulphur diesel fuel base price of $0.908/L FOB Hay River (prices from Midnight Petroleum, as of 9 April, 2013). An additional $0.05/L was added for the freight cost from Hay River to the Nechalacho site.

Mine costs are based on a combination of quotations for certain mine supplies, and experience from similar operations for the estimate of mine manpower and maintenance required for the mobile equipment fleet. Mine costs include all of the underground mining costs except for crusher operation which is included in the mill operating costs estimate.

Operating costs for the concentrator, hydrometallurgical plant and refinery were estimated from first principles. The concentrator operating costs were estimated from first principles. Consumables usage rates were derived from the metallurgical testwork and the unit prices from supplier quotes.

22.0 ECONOMIC ANALYSIS

22.1 BASIS OF EVALUATION

The objective of the feasibility study was to determine the viability of the project, comprising facilities located at three separate sites: an underground mine and concentrator to be located at Nechalacho, Thor Lake, 100 km southeast of Yellowknife, Northwest Territories, a hydrometallurgical plant to be located at Pine Point, 85 km east of Hay River, Northwest Territories, and a rare earth refinery to be located in Geismar, Louisiana, which will produce high purity separated REE oxides and carbonates.

Micon has prepared its assessment of the project on the basis of a discounted cash flow model, from which net present value (NPV), internal rate of return (IRR), payback and other measures of project viability can be determined. Assessments of NPV are generally accepted within the mining industry as representing the economic value of a project after allowing for the cost of capital invested. The sensitivity of this NPV to changes in the base case assumptions is then examined.

22.2 MACRO-ECONOMIC ASSUMPTIONS

22.2.1 Exchange Rate and Inflation

In this report, all financial results are expressed in Canadian dollars.

By convention, rare earth product prices are stated in US dollars (US$). Sales revenues and US dollar-denominated costs are converted to Canadian currency ($, or CAD) at the following rates:

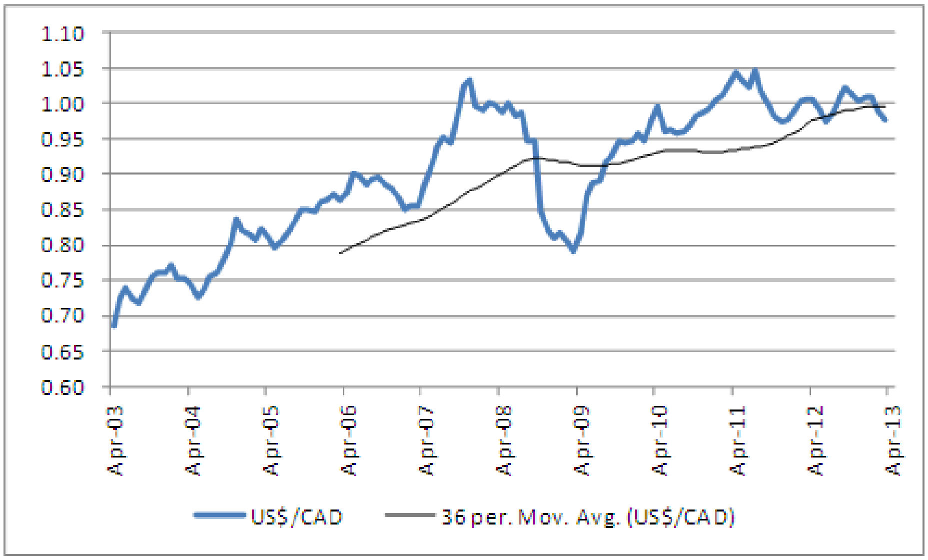
To end of 2015	US$0.97/CAD
2016	US$0.96/CAD
2017 and thereafter	US$0.95/CAD

Compared to current exchange rates, the forecast rates assume a modest weakening of Canadian currency versus the US dollar over the next four years, remaining within the range of forecasts published by major financial institutions and close to the trailing average rates observed in the past five years.

Historical monthly and trailing average rates are shown in Figure 22.1.

Cost estimates and other inputs to the cash flow model for the project have been prepared using constant money terms, i.e., without provision for escalation or inflation.

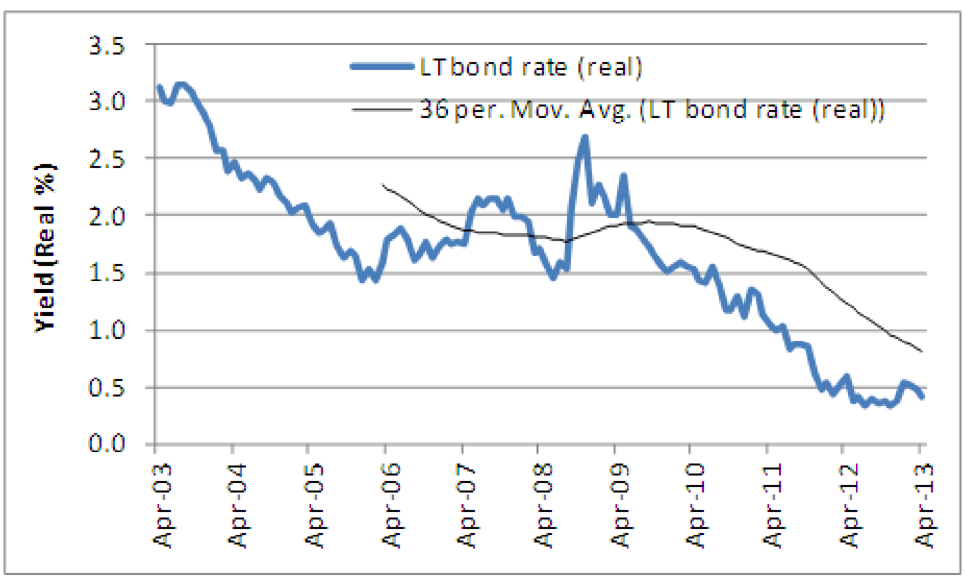
Figure 22.1
10-Year Historical Exchange Rate, US$/CAD



22.2.2 Weighted Average Cost of Capital

The project has been evaluated on an all-equity basis, in which case its weighted average cost of capital (WACC) is equal to its cost of equity, as calculated using the capital asset pricing model (CAPM). In order to calculate the cost of equity, Micon has taken the real return on Canadian long bonds as a proxy for the risk-free rate, as shown in Figure 22.2.

Figure 22.2
Historical Real Return on Canadian Long Bonds



The real rate of return has fallen from 3.0% 10 years ago to less than 0.5% in 2013. Micon has used a value of 1.5% in its base case, close to the average over the past 10 years, with high and low sensitivity cases 0.5% above and below this rate, respectively. CAPM also requires an estimate of beta. In the absence of a direct, REE-industry peer group, Micon has assumed that beta will be in the range 1.4 to 2.0, applicable to other multi-commodity and base metal mining producers. The resulting estimates of the cost of equity (and hence WACC, in the all-equity base case) range from 8% to 12%, as shown in Table 22.1. Micon has selected a value of 10%, being in the middle of this range, for evaluation of the base case.

Table 22.1
Suggested Range for Discount Rate

	Low	Base Case	High
Risk-Free Rate (%)	1.0	1.5	2.0
Beta	1.4	1.7	2.0
Risk Premium for Equity	5.0	5.0	5.0
Cost of Equity	8.0	10.0	12.0

22.2.3 Expected Product Prices

The basis for the price forecast used in evaluating the base case is discussed in Section 19.0 of this report. The cash flow projection uses a constant price forecast for each of the rare earth elements recovered from the rare earth precipitate and contained in the EZC, respectively, as given in Table 22.2. Rare earths separated and refined from rare earth precipitate are sold FOB Geismar, Louisiana. EZC revenues are based on a fixed discount being applied to the market value of the payable rare earth elements contained in the concentrate, FOB Pine Point, and are calculated in accordance with the terms of a MOU agreed with a potential customer in Asia.

Table 22.2
Forecast REO Prices

Rare Earth Oxide	Forecast Market Price (US$/kg)
La_2O_3	8.75
CeO_2	6.23
Pr_6O_{11}	75.2
Nd_2O_3	76.78
Sm_2O_3	6.75
Eu_2O_3	1,392.57
Gd_2O_3	54.99
Tb_4O_7	1,055.7
Dy_2O_3	688.08
Ho_2O_3	66.35
Er_2O_3	48.92
Tm_2O_3	-
Yb_2O_3	-
Lu_2O_3	1313.6
Y_2O_3	67.25

22.2.4 Royalties

There is one third-party royalty on the property which Avalon has a contractual right to repurchase for approximately $1.3 million (subject to escalation at prime). The repurchase of this royalty prior to the commencement of production has been provided for in Micon's evaluation.

22.2.5 Taxation Regime

Taxation has been computed for each jurisdiction within which the project will operate. For this purpose, arms-length transfer prices have been established between the three sites, i.e., the Nechalacho mine/concentrator, Pine Point hydrometallurgical plant, and the Geismar refinery. The basis of the transfer pricing is an allocation of revenues for each product pro-rata with the aggregate operating and capital costs for its production, averaged over the project life.

Taxes considered in the economic evaluation include the following.

22.2.5.1 Statutory Royalty (Mining Tax)

Canada's Northwest Territories collects a royalty of up to 13% on the net value of mine production. Product shipped from the Nechalacho mine to Pine Point is subject to this royalty, after deducting operating and product transport costs, and allowances for depreciation, development and processing.

22.2.5.2 Canadian Federal and Territorial Income Taxes

Rare earth product sales from Geismar and EZC sales from Pine Point are subject to Canadian Federal and Northwest Territorial income taxes at the rate of 15% and 11.5% respectively, after appropriate deductions for refining and transport costs, statutory royalties, capital cost allowances and non-capital losses brought forward. The project is eligible for accelerated capital cost allowances.

22.2.5.3 US Federal and Louisiana State Taxes

Refining charges at Geismar assume a 25% profit margin over operating costs. Hence, the refinery would be subject to US federal and Louisiana state income taxes at the rate of 35% and 8%, respectively, after appropriate allowances for depreciation.

22.2.6 Diesel Fuel Cost

An average diesel fuel price of US$ 0.908/L has been used in the study, including delivery to Hay River. Excise taxes have been added where appropriate. Delivery costs are then calculated separately for each site depending on volume and logistics.

22.2.7 Electrical Power Cost

Electrical power costs will be different for each of the three project sites:

- At the Nechalacho site, a diesel power station will be installed and operated under a BOO contract. Unit cost is forecast to be $0.345/kWh.

- Pine Point operations will be supplied from a hydroelectric power scheme installed when another company operated a mine and concentrator at this location. Unit cost is forecast to be $0.080/kWh.

- At Geismar, power is available from the grid. Unit cost is forecast to be $0.046/kWh.

22.3 TECHNICAL ASSUMPTIONS

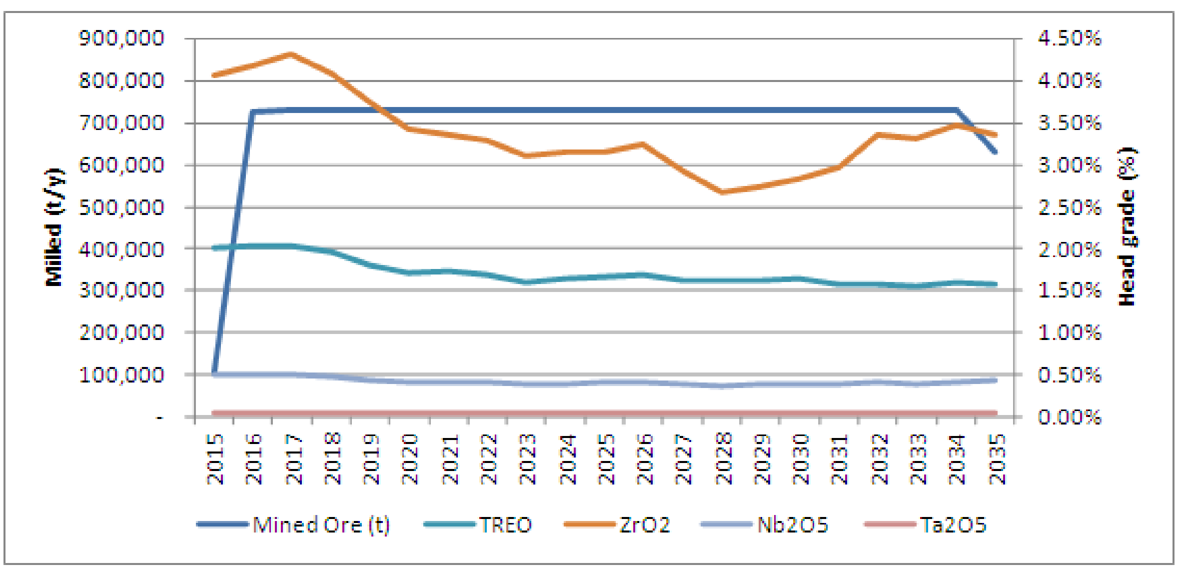
The technical parameters, production forecasts and estimates described elsewhere in this report are reflected in the base case cash flow model. These inputs to the model are summarized below. The measures used in the study are metric tonnes and kilograms.

22.3.1 Production and Sales Schedule

22.3.1.1 Nechalacho Mine

Figure 22.3 shows the tonnage and grade profile for material treated at the Nechalacho mine over the LOM period.

Figure 22.3
Nechalacho Production Schedule



The monthly production schedule provides for a linear ramp up over a six month period to a steady state throughput of 2,000 t/d mined and milled for a total of 730,000 t/y. This rate is maintained through the LOM period until production ceases in Year 20 (2035). A total of 14,600,000 t ore is scheduled for processing during this initial 20-year period.

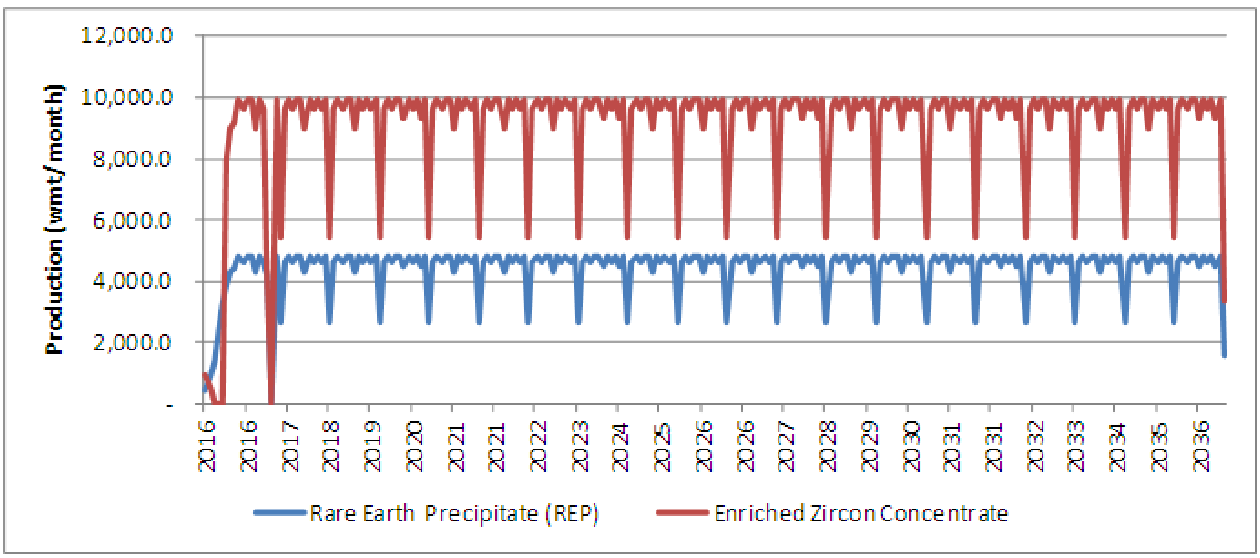
During the first three years, average grades are close to 2.0% TREO, falling thereafter to an average of 1.7% TREO over the LOM.

22.3.1.2 Pine Point Hydrometallurgical Plant

The hydrometallurgical plant at Pine Point will process the flotation concentrate received from Nechalacho and will make two products. The first is a mixed rare earth precipitate which is shipped to the refinery at Geismar, Louisiana for separation. The second product is an EZC which is shipped to Asia for sale.

Figure 22.4 shows monthly production of both products, reflecting an annual shutdown of the hydrometallurgical plant for routine maintenance.

Figure 22.4
Products Shipped from Pine Point



22.3.1.3 Product Sales Revenue

EZC produced at the hydrometallurgical plant is sold FOB Pine Point whereas the rare earth precipitate is separated into its constituent REE for sale FOB Geismar, Louisiana. Figure 22.5 shows the annual revenues derived from each. EZC sales are shown prior to the deduction shipping costs, which are included in operating expenses. Over the LOM period, annual revenues average $456.5 million from rare earth product sales and $189.3 million from EZC, equating to an average of $884.65/t milled.

Figure 22.5
Annual Sales Schedule

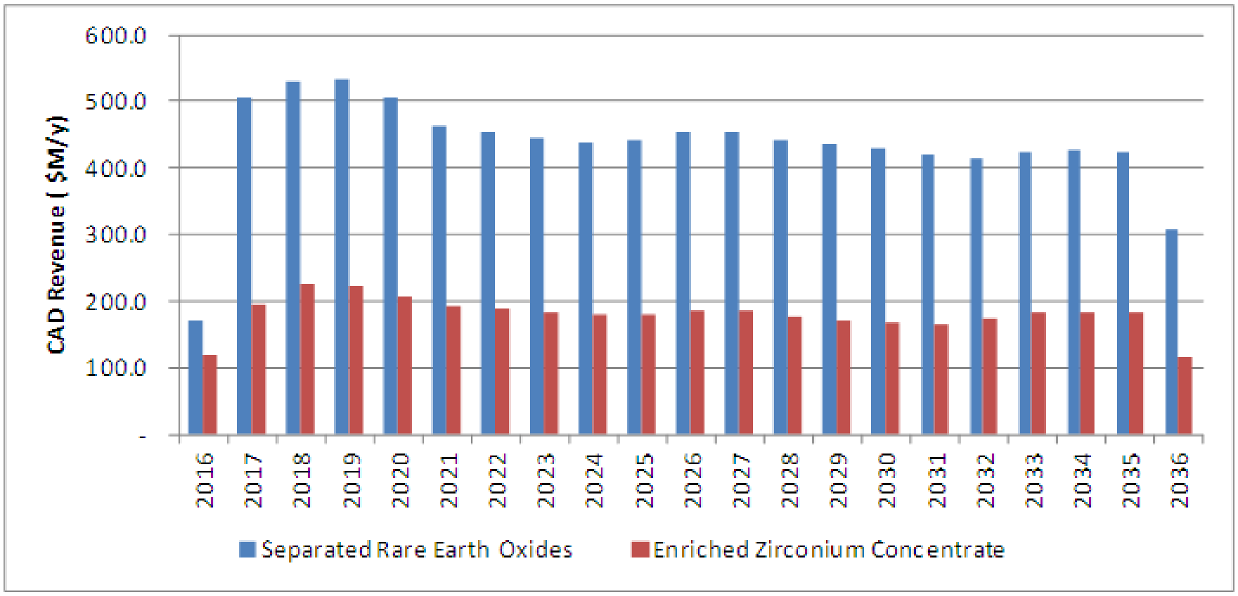


Table 22.3 shows the expected annual average production of separated REE in oxides and REE carbonate products (stated as oxide equivalents) over the 20-year initial mine life.

Table 22.3
LOM Annual Production Averages

Oxide	Contained in EZC (t/y)	Separated REO (t/y)	Total (t/y)
La_2O_3	193.2	1,187.6	1,380.8
Ce_2O_3	502.6	2,521.6	3,024.2
Pr_2O_3	58.8	322.3	381.1
Nd_2O_3	349.0	1,179.9	1,528.9
Sm_2O_3	122.2	220.9	343.1
Eu_2O_3	16.1	31.5	47.6
Gd_2O_3	121.2	231.5	352.7
Tb_2O_3	21.0	37.6	58.6
Dy_2O_3	128.6	195.8	324.4
Ho_2O_3	27.9	32.5	60.4
Er_2O_3	89.2	71.6	160.8
Tm_2O_3	13.8	8.0	21.8
Yb_2O_3	89.9	40.0	129.9
Lu_2O_3	13.5	4.6	18.1
Y_2O_3	728.8	724.9	1,453.7
Sub-total	**2,475.8**	**6,810.3**	**9,286.1**
ZrO_2	19,763.3	-	19,763.3
Nb_2O_5	2,230.5	-	2,230.5
Ta_2O_5	243.0	-	243.0
Total	**24,712.6**	**6,810.3**	**31,522.9**

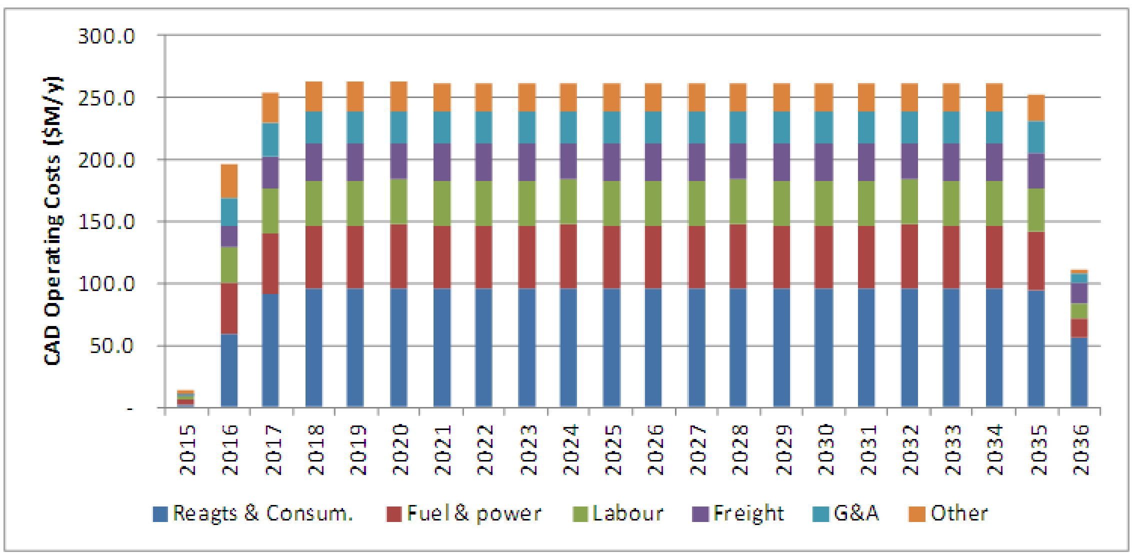
22.3.2 Operating Costs

Table 22.4 shows a summary of annual and unit operating costs for the project. Figure 22.6 shows these annual costs over the LOM period.

Table 22.4
Operating Cost Summary

Item	Average Annual Cost ($M/y)	Average Unit Cost ($/t)
Reagents and consumables	97.23	133.20
Fuel and power	50.72	69.48
Labour	36.69	50.26
Freight	29.43	40.31
G&A	26.82	36.74
Other	23.58	32.30
Total	**264.47**	**362.28**

Figure 22.6
Annual Operating Costs



22.3.3 Capital Expenditure

Initial capital costs to construct the facilities at Nechalacho, Pine Point and Geismar are shown in Table 22.5, together with aggregate sustaining capital over the remainder of the LOM period for each of these sites.

In the cash flow projection, bonds covering the full value of closure costs included in the above estimate are assumed to be required prior to the commencement of operations.

Table 22.5
Capital Expenditure Summary

Cost Category	NWT[1] ($ million)	LA[2] ($ million)	Total ($ million)
Mine development	81.58	-	81.58
Main process facilities	351.24	192.51	543.75
Infrastructure	150.68	78.82	229.50
EPCM	119.27	38.57	157.84
Indirect construction costs	175.56	27.25	202.81
Owner's costs	36.76	18.95	55.71
Contingency	120.91	44.90	165.81
Closure costs/bond	13.00	3.16	16.16
Pre-production capital cost	**1,049.00**	**404.16**	**1,453.16**
Sustaining capital	102.72	19.12	121.84
Total capital cost	**1,151.72**	**423.28**	**1,575.00**

[1] NWT – Costs applicable to the Nechalacho and Pine Point sites in the Northwest Territories.
[2] LA – Costs applicable to Geismar, Louisiana.

22.3.4 Working Capital

Working capital has been provided for in the cash flow model. Initial fills, parts and consumable stores are included in initial capital costs. Provision has been made for accounts payable on 45 days of non-labour working costs, and 60 days of ongoing capital costs. Accounts receivable equivalent to 15 days revenue on sales of separated rare earth products and EZC to third parties have also been provided for. The resulting net working capital provision in most operating periods is within the range of +$5 to -$5 million.

22.4 BASE CASE EVALUATION

Table 22.6 shows the annual cash flows for the project over the initial 20-year LOM period. Figure 22.7 presents a summary of the LOM cash flows, and the cumulative discounted and undiscounted cash flows.

Figure 22.7
LOM Cash Flow

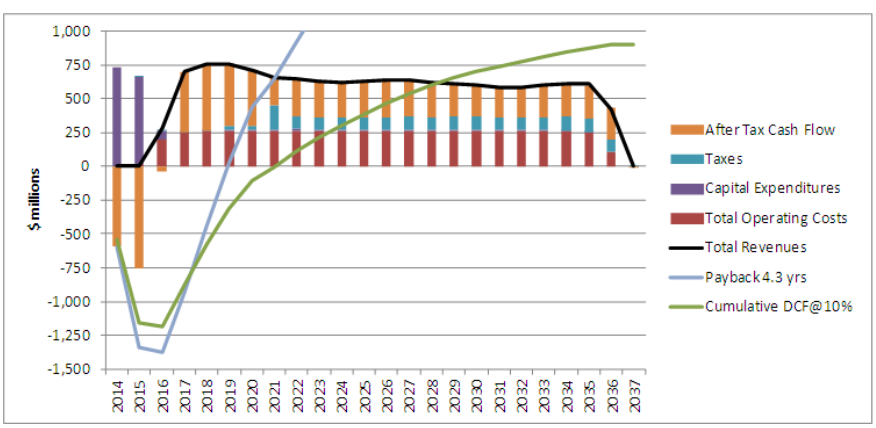


Table 22.6
LOM Annual Cash Flow Projection

Year			LOM	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Year (Calendar)			Total/Avg.	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037
Mined Ore (t)			14,600,000		103,417	724,583	730,000	730,000	730,000	730,000	730,000	730,000	730,000	730,000	730,000	730,000	730,000	730,000	730,000	730,000	730,000	730,000	730,000	730,000	632,000	-	-
Grade (%)																											
	TREO		1.70%		2.02%	2.02%	2.03%	1.97%	1.81%	1.71%	1.73%	1.68%	1.61%	1.64%	1.67%	1.69%	1.62%	1.62%	1.63%	1.65%	1.58%	1.57%	1.55%	1.59%	1.58%	-	-
	ZrO2		3.34%		4.07%	4.18%	4.31%	4.08%	3.75%	3.42%	3.37%	3.29%	3.11%	3.14%	3.15%	3.23%	2.93%	2.68%	2.74%	2.83%	2.97%	3.36%	3.31%	3.47%	3.35%	-	-
	Nb2O5		0.42%		0.49%	0.50%	0.50%	0.48%	0.44%	0.41%	0.41%	0.41%	0.38%	0.40%	0.41%	0.42%	0.39%	0.37%	0.38%	0.39%	0.38%	0.40%	0.40%	0.42%	0.43%	-	-
	Ta2O5		0.05%		0.05%	0.05%	0.05%	0.05%	0.05%	0.05%	0.04%	0.04%	0.04%	0.04%	0.04%	0.04%	0.04%	0.04%	0.04%	0.04%	0.04%	0.05%	0.04%	0.04%	0.04%	-	-
Payable content (t)	TREO		146,223	-	-	4,511	7,544	8,695	8,299	7,608	7,386	7,443	7,109	6,929	7,057	7,247	7,148	6,927	6,898	7,059	7,019	6,797	6,679	6,742	6,799	4,328	-
	ZrO2		395,266	-	-	12,218	21,674	25,206	23,735	21,651	20,140	19,953	19,113	18,522	18,494	18,907	18,769	16,909	15,871	16,465	16,703	18,399	19,671	19,961	20,374	12,531	-
Revenues ($ million)		$/t milled																									
Separated Rare Earth Oxides		625.33	9,129.8	-	-	170.6	507.5	531.3	532.1	506.1	464.0	455.4	445.9	438.2	443.1	454.6	455.7	443.3	436.7	428.9	420.5	414.4	423.6	426.5	423.5	308.1	-
Enriched Zirconium Concentrate		259.33	3,786.2	-	-	118.0	196.0	225.4	222.2	208.2	191.5	187.9	183.8	180.9	181.6	186.0	185.3	176.6	170.7	168.8	165.6	173.3	182.2	183.7	183.5	115.0	-
Total Revenues		884.66	12,916.0	-	-	288.6	703.5	756.8	754.3	714.3	655.5	643.3	629.7	619.1	624.6	640.6	641.0	619.9	607.3	597.7	586.0	587.6	605.8	610.2	607.0	423.2	-
Operating Costs ($ million)		$/t milled																									
	Reagts & Consum.	133.20	1,944.7	-	2.0	59.5	92.1	96.5	96.5	96.6	96.5	96.5	96.5	96.6	96.5	96.5	96.5	96.6	96.5	96.5	96.5	96.6	96.5	96.5	94.8	55.9	-
	Fuel & power	69.48	1,014.4	-	3.5	40.6	48.4	50.5	50.5	50.5	50.5	50.5	50.5	50.5	50.5	50.5	50.5	50.5	50.5	50.5	50.5	50.5	50.5	50.5	47.1	16.5	-
	Labour	50.26	733.7	-	3.0	29.2	36.4	36.4	36.4	36.5	36.4	36.4	36.4	36.5	36.4	36.4	36.4	36.5	36.4	36.4	36.4	36.5	36.4	36.4	34.7	10.9	-
	Freight	40.31	588.6	-	0.5	17.4	26.0	29.3	29.3	29.3	29.3	29.3	29.3	29.3	29.3	29.3	29.3	29.3	29.3	29.3	29.3	29.3	29.3	29.3	29.3	17.3	-
	G&A	36.74	536.4	-	2.2	21.7	26.6	26.6	26.6	26.7	26.6	26.6	26.6	26.7	26.6	26.6	26.6	26.7	26.6	26.6	26.6	26.7	26.6	26.6	25.3	7.6	-
	Other	32.30	471.5	-	2.6	27.9	24.2	24.2	24.2	24.2	22.8	22.8	22.8	22.8	22.8	22.8	22.8	22.8	22.8	22.8	22.8	22.8	22.8	22.8	21.3	3.3	-
Total Operating Costs		362.28	5,289.3	-	13.8	196.3	253.8	263.5	263.5	263.7	262.2	262.2	262.2	262.4	262.2	262.2	262.2	262.4	262.2	262.2	262.2	262.4	262.2	262.2	252.4	111.5	-
EBITDA ($ million)			7,626.7	-	(13.8)	92.3	449.7	493.3	490.8	450.6	393.4	381.1	367.5	356.7	362.5	378.5	378.9	357.5	345.1	335.5	323.9	325.3	343.6	348.0	354.6	311.7	-
Taxes	Property taxes		37.5	-	0.3	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	3.9	3.6	3.3	3.0	2.8	2.5	2.2	2.0	1.7	1.4	1.1	0.8	0.0	-
	NWT Royalty		105.0	-	-	-	-	-	-	-	0.7	8.4	5.8	7.3	7.1	6.5	7.3	6.1	6.8	6.3	4.7	4.0	3.1	5.7	6.8	7.9	10.4
	Income Taxes		1,528.5	-	-	-	-	-	26.1	28.3	174.3	85.5	84.6	80.5	80.4	83.1	89.6	81.9	90.3	94.7	85.5	85.3	85.5	99.2	90.8	83.0	0.0
Total Taxes ($ million)			1,671.0	-	0.3	1.1	1.1	1.1	27.2	29.4	176.1	95.0	91.5	91.7	91.1	92.9	100.0	90.8	99.6	103.2	92.1	91.0	90.0	106.1	98.5	90.9	10.4
After Tax Cashflow before CAPEX and NWC			5,955.7	-	(14.1)	91.2	448.6	492.1	463.7	421.2	217.3	286.1	276.0	265.0	271.4	285.6	278.9	266.7	245.6	232.3	231.8	234.3	253.6	242.0	256.1	220.8	(10.4)
Change in Working Capital			(0.0)	-	(6.0)	8.7	2.6	(0.0)	(0.5)	(3.5)	(0.8)	(0.1)	(1.2)	0.4	0.2	1.2	(1.5)	(0.2)	(0.9)	0.2	(1.2)	1.6	(0.1)	0.5	7.4	(6.8)	-
Capital Expenditures																											
	Northwest Territories		1,151.7	516.6	475.2	57.2	-	0.1	1.9	7.4	4.7	15.3	6.3	4.3	8.2	5.2	11.8	4.7	2.0	7.7	5.2	11.8	4.7	0.7	0.6	-	-
	Louisiana		423.3	76.9	267.0	60.2	0.1	0.4	0.6	0.6	1.9	2.8	2.8	2.8	1.5	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.3	(0.0)	-
Total Capital Expenditures			1,575.0	593.5	742.2	117.4	0.1	0.5	2.5	8.0	6.6	18.1	9.1	7.0	9.7	5.9	12.4	5.3	2.6	8.3	5.9	12.4	5.3	1.4	0.9	(0.0)	(0.0)
Pre-Tax Cash Flow	($ million)		6,051.7	(593.5)	(750.0)	(33.8)	447.0	492.8	488.8	446.0	387.5	363.1	359.7	349.2	352.6	371.4	368.0	352.4	343.4	327.1	319.3	311.2	338.4	346.2	346.3	318.5	-
Cumulative Pre-Tax Cash Flow				(593.5)	(1,343.5)	(1,377.3)	(930.2)	(437.4)	51.4	497.4	884.9	1,248.1	1,607.7	1,957.0	2,309.5	2,680.9	3,048.9	3,401.3	3,744.7	4,071.8	4,391.1	4,702.3	5,040.7	5,386.9	5,733.2	6,051.7	6,051.7
After Tax Cash Flow	($ million)		4,380.7	(593.5)	(750.3)	(34.8)	445.9	491.7	461.6	416.6	211.4	268.1	268.2	257.5	261.4	278.5	268.0	261.6	243.8	223.9	227.2	220.3	248.4	240.1	247.8	227.6	(10.4)
Cumulative After Tax Cash Flow				(593.5)	(1,343.8)	(1,378.6)	(932.7)	(441.0)	20.6	437.2	648.7	916.7	1,184.9	1,442.4	1,703.9	1,982.4	2,250.4	2,512.0	2,755.8	2,979.7	3,206.8	3,427.1	3,675.5	3,915.6	4,163.5	4,391.1	4,380.7

Table 22.7 shows the results of the economic evaluation of the base case cash flows.

Table 22.7
Economic Evaluation of Base Case

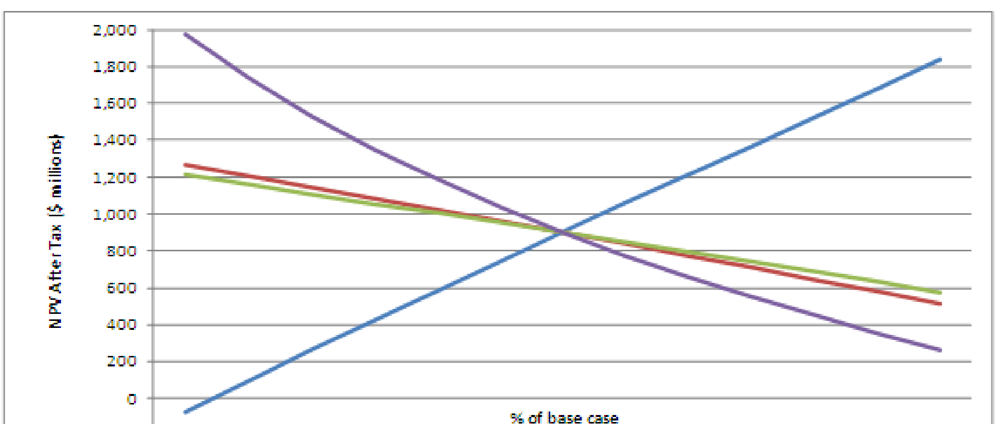
	Discount rate (%/y)	Pre-Tax $ million	After Tax $ million	Payback (y)
Undiscounted Cash Flow		6,052	4,381	4.3
Net Present Value	8	1,833	1,262	5.5
	10	1,351	900	6.3
	12	981	620	7.2
Internal Rate of Return		22.5%	19.6%	

The project provides a payback of 4.3 years on the undiscounted cash flow, or 6.3 years on the cash flow discounted at 10%, leaving a considerable reserve "tail". The cash operating margin is seen to remain positive over the whole LOM period, and is particularly strong in the first four years of full production.

22.5 SENSITIVITY STUDY

Micon has analysed the sensitivity of the NPV at a discount rate of 10% of the project's after-tax cash flow to changes in key parameters. Factors impacting revenue (price and overall recovery), operating costs, capital expenditure and exchange rate of Canadian to US dollars were each varied over a range of 30% above and below base case values. Figure 22.8 presents the results of this analysis.

Figure 22.8
Revenue, Capital and Operating Cost Sensitivity

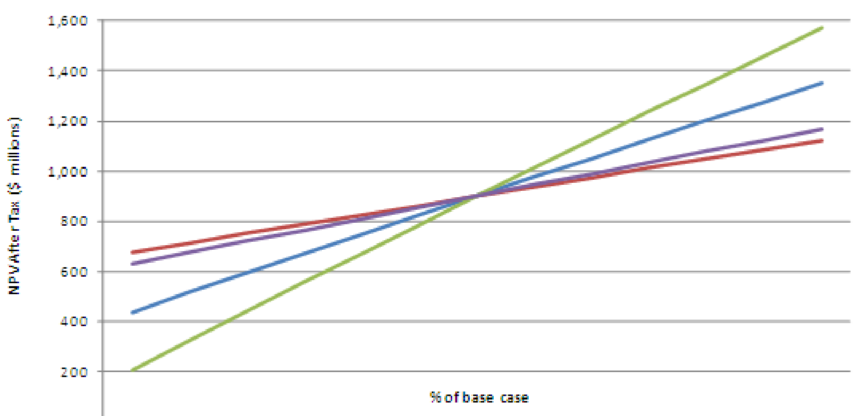


	70%	75%	80%	85%	90%	95%	100%	105%	110%	115%	120%	125%	130%
Revenue	-75	93	258	421	583	742	900	1058	1212	1368	1524	1680	1834
OPEX	1264	1205	1145	1085	1025	963	900	837	774	711	646	579	511
CAPEX	1211	1159	1108	1057	1006	953	900	846	793	740	685	631	576
USD/CAD	1976	1742	1535	1353	1188	1038	900	774	659	552	452	357	259

It will be seen that revenue factors and exchange rate have the greatest impact on project returns, but that an adverse change of more than 25% in project revenues are required to reduce NPV to near zero. The project is less sensitive to capital and operating costs, with the project being only slightly more sensitive to operating costs than to capital expenditure.

Figure 22.9 shows a detailed analysis of the sensitivity of the project to changes in price of total, heavy and light rare earth oxides (TREO, HREO and LREO, respectively) and of EZC.

Figure 22.9
Product Price Sensitivity

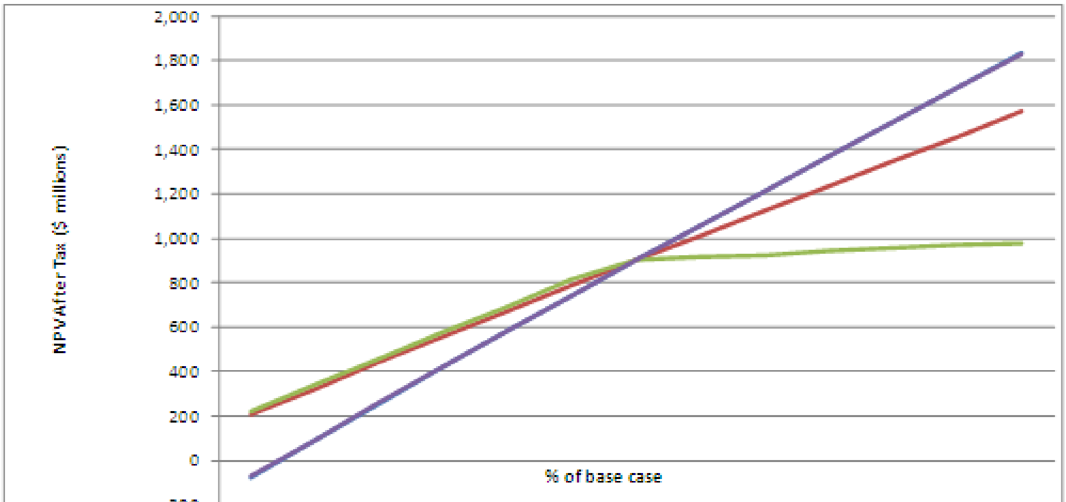


	70%	75%	80%	85%	90%	95%	100%	105%	110%	115%	120%	125%	130%
HREO	439	517	595	671	748	823	900	976	1052	1127	1201	1276	1351
LREO	677	714	751	788	825	863	900	937	975	1011	1048	1085	1121
TREO	208	326	443	559	673	786	900	1013	1125	1236	1347	1459	1571
EZC	632	677	721	766	810	855	900	945	989	1034	1078	1122	1165

The project is seen to be more sensitive to revenues from TREO than EZC, and that the project is more sensitive to HREO prices than to LREO prices.

Figure 22.10 shows an analysis of sensitivity to throughput and process recoveries in the concentrator, hydrometallurgical plant and refinery. Since only the first 20 years of production has been considered, a reduced throughput results in a reduction in the tonnage of ore processed over the LOM, causing an apparent increase in sensitivity versus the comparatively low sensitivity to an increase in throughput. However, sensitivity to recovery in the Nechalacho concentrator is seen to be greater than it is to throughput or recovery at either of the two downstream plants. Nevertheless, an adverse change of more than 25% is required to reduce NPV to near zero.

Figure 22.10
Sensitivity to Throughput and Process Recoveries



	70%	75%	80%	85%	90%	95%	100%	105%	110%	115%	120%	125%	130%
Hydromet Rec%	-75	93	258	421	583	742	900	1058	1212	1368	1524	1680	1834
Separation Rec%	208	326	443	559	673	786	900	1013	1125	1236	1347	1459	1571
Throughput	218	340	460	577	693	818	900	914	926	946	959	968	978
Flotation Rec%	-72	96	260	423	584	743	900	1057	1211	1366	1522	1677	1831

22.6 CONCLUSIONS

Micon concludes that the base case cash flow and sensitivity studies demonstrate robust economics that support the declaration of mineral reserves and warrant continued development of the Nechalacho REE project as described in this study.

23.0 ADJACENT PROPERTIES

At the time of writing, there are no mineral claims or leases adjacent to Avalon's Thor Lake leases and claims as reported in Section 4 of this report. All of the known rare metal deposits related to the Blatchford Lake Complex are owned by Avalon.

24.0 OTHER RELEVANT DATA AND INFORMATION

24.1 PROJECT IMPLEMENTATION

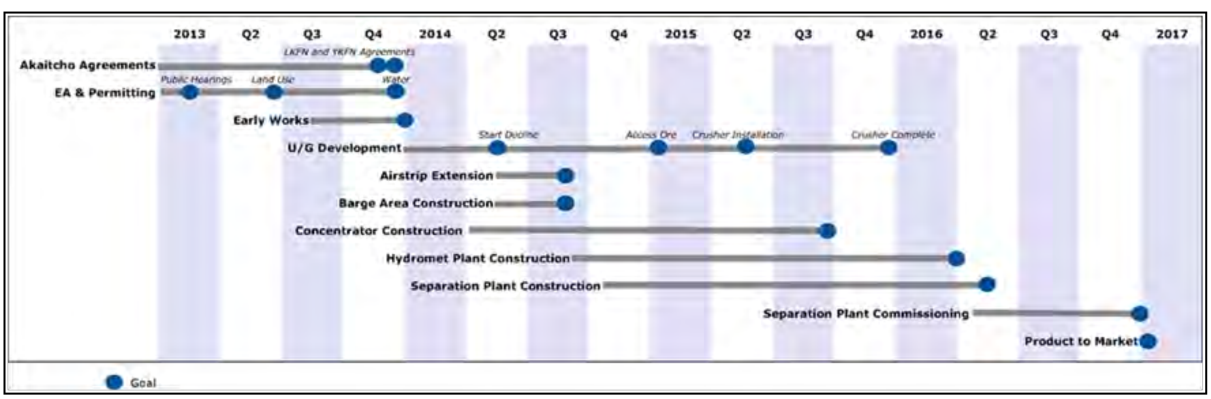
The project consists of a mine and three processing plants located at the Nechalacho, Pine Point and Geismar sites, as described elsewhere in this report.

Avalon will engage EPCM contractors at each site. These will have responsibility for design, procurement, contracting, and project and construction management of the plants and infrastructure facilities. Site specific project management will be the responsibility of each of the EPCM contractors, with overview by the Avalon Owner's team. Avalon has established a recruitment schedule and is working with project management companies to provide critical personnel and project support to supplement as required.

The EPCM contractors will be responsible for all the EPCM work in order to complete the project up to mechanical completion and hand-over to Avalon for commissioning. A management team from Avalon will facilitate the flow of information and approvals as the work progresses. During construction, temporary offices and support facilities will be established at each of the three project sites.

A project master schedule provided in Figure 24.1 shows the overall major activities and milestone dates. The schedule was developed based on Avalon's requirement to have the refinery at Geismar mechanically complete at the beginning May, 2016, and it takes into consideration equipment delivery and limitations related to Nechalacho Site access.

Figure 24.1
Project Master Schedule



The project has been developed based on the following requirements:

- Safety and protection of the environment cannot be compromised.
- The project must be delivered in accordance with the schedule, budget and quality requirements established for the site.

- Completion of the project in a timely manner is considered critical to meet market requirements.

- Training, hiring and purchasing from Northern and Aboriginal partners will be given priority.

The Nechalacho site is considered to be the most critical of the three sites from a schedule perspective, due primarily to the limited access. A detailed critical path schedule has been developed for each of the project sites.

24.2 HUMAN RESOURCES

Successful implementation of Avalon's human resources (HR) policies and procedures will enable Avalon to achieve its goals and objectives for its employees and contractors. Avalon's goal of increasing local participation rates will be facilitated through its training and development initiatives. Avalon is committed to worker training throughout the life of the project.

Avalon's HR Management Plan program will initially be designed to fill apprenticeship and technological occupations. In addition, all project contractors and sub-contractors will be required to adhere to Avalon's goal of maximizing Northern and Aboriginal employment.

Manpower recruitment will begin shortly after financing is established to bring onboard an Owner's team of experts to direct the project through the construction phase at each site. Avalon's HR department will work with the EPCM contractor to hire technical staff to work during construction and upon completion of construction those staff will roll into operations. Professional search firms will be utilized during peak recruitment periods and for positions where there is a recognized shortage of qualified candidates.

Avalon is committed to employee training. It recognizes that training initiatives are critical to achieving a healthy, safe, and productive workplace. Training initiatives will focus on site-based programs that train employees in specific skills, safety, environmental and administrative subjects. Half of all technical staff will start work during construction, a full three months prior to start-up. The prestart-up training will consist of site specific classroom and on-the-job practical instruction. Training areas include:

- Safety systems and safe work practices.
- First aid and emergency response.
- Environmental and waste management.
- Cross cultural training.
- Skill specific on-the-job training (supervisor and employee).
- Apprenticeship training programs (skilled employee).
- Administration functions.

24.2.1 Manpower Requirements

24.2.1.1 Nechalacho

It is anticipated that construction personnel will include up to 560 persons during construction of the Nechalacho mine, concentrator and surface infrastructure. The construction jobs will include employment generated through third-party contract opportunities needed to service the project.

During the underground mining and processing operations at Nechalacho, approximately 216 full-time employees will be required. Approximately 20 positions will be retained for the project closure and reclamation phase.

24.2.1.2 Pine Point

During construction of the hydrometallurgical facilities it is expected that some 460 construction personnel will be employed at site.

The hydrometallurgical processing operations at Pine Point will require approximately 77 full-time employees. Approximately 10 positions will be retained during the reclamation phase.

24.2.1.3 Geismar

Approximately 300 positions will be required at site at any given time during construction of the refinery and related facilities, with a peak of about 450 positions. The construction jobs will include employment generated through third-party contract opportunities needed to service the Project.

During processing operations at Geismar, approximately 133 full-time employees will be required. If the facility does not continue to operate as a custom-feed operation, about 10 employees will be retained for the closure and reclamation phase.

24.3 RISK ANALYSIS

Risk management for the project follows the principle of creating an environment and a context for dealing pro-actively with threats and opportunities, from identification, analysis and prioritization, through to mitigation and close-out of the risk. The risk management process is a continuous process to deal with potential threats and opportunities that could impact the expected outcome of the project.

During preparation of the feasibility study, two joint Owner and Contractor risk identification workshops were conducted to systematically explore all critical processes and project areas in identifying potential risks and establishing the initial risk register. The areas of project management, project controls, procurement, engineering, construction, mining, geotechnical, environmental, and health and safety were represented. Following the workshops, mitigation plans and actions were developed and carried out to the extent possible during the feasibility study phase. In addition, two risk reviews were conducted to update the status and reassess the risks identified in the earlier workshops and analyse any newly identified risks, including a site specific review of the Geismar site which had not been identified in earlier workshops. Going forward, a plan will be developed for the EPCM and execution phases of the project that would start with the risk register developed in the feasibility study and build on the basic process described below and followed throughout the feasibility study and into operations.

Ninety-seven project construction phase risks and 54 operating phase risks have been identified. Of the 97 project phase risks, 54 remain active and there are no high risks remaining following the implementation of mitigation plans to be carried out during project execution. Of the 54 operating risks identified, 38 remain active, all of which will be reduced to medium or lower once mitigation plans are carried out during project execution and operations.

The key risks identified for construction, and for which mitigations have been developed, include:

- Safety and health, mitigated by comprehensive health and safety (HS) plan.

- Project delay due to permitting mitigated by existing permits, robust project and well-resourced permitting group.

- Adequate work force mitigated by training and HR management plan.

- Logistics and procurements, particularly for the Nechalacho mine/concentrator sites mitigated by experienced personnel and detailed planning.

- Project delay due to Aboriginal negotiations mitigated by existing agreement and ongoing negotiations.

- Hurricane risk in Louisiana mitigated by site selection and design.

- Cold weather risk in the Northwest Territories mitigated by experience personnel and management plans.

- Tailing dam failure mitigated by robust design, small structures and management plans at Nechalacho, and no dams being required at Pine Point.

- Project delay associated with the perceived risk associated with uranium and thorium mitigated by low concentrations, good science and open and transparent consultation.

For operations, some of the above will continue, with additional risks associated with:

- Environmental risks causing cost or production delay mitigated by robust design and management and monitoring plans.

- Community risks causing cost or production delay mitigated by community consultation, well managed Aboriginal agreements and agreements with the Northwest Territories government.

- Risk of cost escalation/inflation mitigated by long term contracts, contingency and potential capital cost reductions.

- Foreign currency risk mitigated by investigation and implementation of a currency hedging strategy, if appropriate.

- REE price fluctuations mitigated by reliable supply and long term contracts with strategic partners and customers. There will be a long term agreement in place for EZC.

Hazard analysis (HAZAN) was also performed as part of work in the feasibility study phase and identified mitigations have been incorporated into the feasibility study design.

During detailed engineering, additional design reviews will be performed such as, but not necessarily limited to, hazard and operability (HAZOP) studies, and constructability reviews that will contribute to risk reductions during construction and operation phases.

24.4 HEALTH AND SAFETY

Avalon has confirmed that safety is a core value above all others in its operations and that it will implement a philosophy of zero accidents. Leadership and commitment to the Environment and safety policy will be demonstrated by management by active participation in setting goals and objectives, reviews of accidents and incidents and participation in reviews and audits.

As noted above, a risk management program has been implemented which will continue throughout the life of the operation. In addition to the many anticipated risks associate with mining and construction, site specific management plans will be developed. Management of change will be a key process.

Health and safety management plans will be developed for construction, commissioning and operations. These plans will detail the capital, operating and manpower health and safety resources required. Detailed risk management plans and standard operating procedures (SOPs) will be developed for all identified hazardous or high risk activities, with employee participation where practical and appropriate. Appropriate training will be completed as required by all employees, contractors and subcontractors, and the training will be documented. Accidents and incidents will be investigated and reported in a timely fashion, and communicated to ensure that they are not repeated. Regular internal and external inspections and audits will ensure compliance with the plans, and assist in measuring performance with the health and safety targets and objectives. Non-conformances will be identified and corrected. Performance against goals and objectives will be assessed, communicated and action items/implementation plans identified on a regularly scheduled basis or as required. All this will be regularly reviewed by management, with the goal of zero incidents/or accidents and continuous improvement in health and safety performance. In addition to the use of the traditional lagging indicators, leading indicators appropriate to the phase of work will also be developed and tracked. Avalon has committed to be ISO certified for Environmental, Health & Safety at all sites, and for Quality at the leaching and separation plants in Geismar by Year 3 of operations.

While the focus will be on prevention of accidents, emergency and crisis management plans will be developed.

25.0 INTERPRETATION AND CONCLUSIONS

SLI has completed a feasibility study for the Nechalacho rare earth element project on behalf of Avalon. The results of the feasibility study demonstrate that the project is technically feasible and economically robust. Table 25.1 presents the key project parameters, based on 100% equity financing.

Table 25.1
Key Project Parameters

Metric	Units	Quantity
Pre-tax economics		
IRR	%	22.5
NPV at 8%/y	$ million	1,833
NPV at 10%/y	$ million	1,351
NPV at 12%/y	$ million	981
After tax economics		
IRR	%	19.6
NPV at 8%/y	$ million	1,262
NPV at 10%/y	$ million	900
NPV at 12%/y	$ million	620
Payback period	y	4.3
Mining		
Mineral reserves	t	14,600,000
Production rate	t/y	730,000
Initial mine life	y	20
Total revenue	**$ million/y**	**645.8**
Rare earth oxides	$ million/y	456.5
EZC	$ million/y	189.3
Unit basis	$/t mined	884.65
Operating costs	**$ million/y**	**264.5**
Reagents	$ million/y	97.2
Fuel and power	$ million/y	50.7
Labour	$ million/y	36.7
Freight	$ million/y	29.4
General and Administration	$ million/y	26.8
Other	$ million/y	23.7
Unit basis	$/t mined	362.28

Revenues from HREE are in excess of 50% of total project revenue and revenue from neodymium, europium, terbium, dysprosium and yttrium account for over 82% of revenue for separated REO. Sales of lanthanum and cerium represent less than 4.5% of total revenue.

Sensitivity analysis demonstrates that revenue factors and exchange rate have the greatest impact on project returns, but that an adverse change of more than 25% in project revenues are required to reduce NPV to near zero. The project is less sensitive to capital and operating costs, with the project being only slightly more sensitive to operating costs than to capital expenditure.

The project is seen to be more sensitive to revenues from TREO than EZC, and that the project is more sensitive to HREO prices than to LREO prices.

Micon concludes that the base case cash flow and sensitivity studies demonstrate robust economics that support the declaration of mineral reserves and warrant continued development of the Nechalacho REE project as described in the feasibility study and in this technical report.

25.1 MINERAL RESOURCE ESTIMATE

The mineral resource which forms the basis of the Nechalacho project was estimated by Avalon and accepted by RPA, and is summarized in Table 25.2. The mineral resource is reported at a cut-off value of US$320/t. The effective date of the mineral resource estimate is 3 May, 2013.

Table 25.2
Nechalacho Deposit Mineral Resource Estimate as at 3 May, 2013

Category	Zone	Tonnes (million)	TREO (%)	HREO (%)	ZrO_2 (%)	Nb_2O_5 (%)	Ta_2O_5 (%)
Measured	Basal	10.86	1.67	0.38	3.23	0.40	0.04
	Upper	-	-	-	-	-	-
Total Measured		**10.86**	**1.67**	**0.38**	**3.23**	**0.40**	**0.04**
Indicated	Basal	55.81	1.55	0.33	3.01	0.40	0.04
	Upper	54.59	1.42	0.14	1.96	0.28	0.02
Total Indicated		**110.40**	**1.49**	**0.24**	**2.49**	**0.34**	**0.03**
Measured and Indicated	Basal	66.67	1.57	0.34	3.05	0.40	0.04
	Upper	54.59	1.42	0.14	1.96	0.28	0.02
Total Measured and Indicated		**121.26**	**1.50**	**0.25**	**2.56**	**0.34**	**0.03**
Inferred	Basal	61.09	1.29	0.25	2.69	0.36	0.03
	Upper	122.28	1.26	0.12	2.21	0.32	0.02
Total Inferred		**183.37**	**1.27**	**0.17**	**2.37**	**0.33**	**0.02**

1. CIM definitions were followed for Mineral Resources.
2. Mineral Resources are estimated at a NMR cut-off value of US$320/t. NMR is defined as “Net Metal Return” or the in situ value of all payable metals, net of estimated metallurgical recoveries and off-site processing costs.
3. An exchange rate of US$1=CAD1.05 was used.
4. Heavy Rare Earth Oxides (HREO) is the total concentration of: Y_2O_3, Eu_2O_3, Gd_2O_3, Tb_2O_3, Dy_2O_3, Ho_2O_3, Er_2O_3, Tm_2O_3, Yb_2O_3 and Lu_2O_3.
5. Total Rare Earth Oxides (TREO) is HREO plus Light Rare Earth Oxides (LREO): La_2O_3, Ce_2O_3, Pr_2O_3, Nd_2O_3 and Sm_2O_3.
6. Rare earths were valued at an average net price of US$62.91/kg, ZrO_2 at US$3.77/kg, Nb_2O_5 at US$56/kg, and Ta_2O_5 at US$256/kg. Average REO price is net of metallurgical recovery and payable assumptions for contained rare earths, and will vary according to the proportions of individual rare earth elements present. In this case, the proportions of REO as final products were used to calculate the average price.
7. ZrO_2 refers to zirconium oxide, Nb_2O_5 refers to niobium oxide and Ta_2O_5 refers to tantalum oxide.

25.2 METALLURGICAL TESTWORK

Development of the processing flowsheets for the concentrator, hydrometallurgical plant and rare earth refinery is supported by extensive metallurgical testwork that has been completed at a number of different laboratories over the last four years.

Much of the pertinent metallurgical and mineralogical development studies have been undertaken using bulk composite samples that represent the Nechalacho deposit mineralization spatially and in terms of lithology. These selected composite samples tended to be selected to represent mineralization at different depths in the deposit in terms of elevation. The composites designated UZ were from Upper Zone mineralization and BZ were from Basal Zone mineralization.

25.2.1 Concentrator

The 40 t pilot plant sample selected for the 2012 pilot plant operated by SGS-M comprised four main composites. Subsequent testing undertaken during October 2012 used samples which originated from the deepest areas of the BZ-MP zone.

Extensive mineralogical investigation of the deposit using QEMSCAN® and EMPA has been completed to aid in predicting or explaining metallurgical performance during the programs of flowsheet development testwork.

25.2.2 Hydrometallurgical Plant

Hydrometallurgical testwork has been conducted since late 2008. Tests have been conducted on both the bench and pilot scale on flotation concentrate. The feasibility study final process design criteria have been largely developed from the operating conditions used in the final (PP6) pilot plant campaign completed by SGS-M in September-October, 2012.

Seven mini-pilot plant campaigns were conducted at SGS-M in order to investigate the acid bake operating parameters and to determine the optimal conditions leading to the best REE recoveries. These were conducted between January 25 and May 14, 2012, using concentrate derived from BZ (basal zone) and UZ (upper zone) type mineralization.

25.2.3 Refinery

A large number of testwork reports have been issued to summarize the testwork that has been undertaken at a number of different laboratories over the last few years. All relevant testwork has been completed using the rare earth precipitate produced during the hydrometallurgical pilot plant testwork program.

The design of the rare earth separation plant is based on proven standard rare earth separation processing methods and equipment selections. The technology proposed for this plant is based on mainly Chinese technology and the detailed design was provided by a Chinese expert who has many years of experience in the design and operation of similar plants in China. The detailed design was reviewed by an independent consultant engaged by Avalon who has over 40 years of experience including the design and operation of commercial production facilities for high purity rare earths such as lanthanum, cerium, praseodymium, neodymium, samarium, gadolinium and yttrium.

The separation plant extraction process uses 16 solvent extraction systems to separate the individual rare earths from each other and will produce 10 different pure rare earth products.

25.3 MINERAL RESERVE ESTIMATE

As noted above, the mineral resource database was updated early in May, 2013. The mineral reserve estimate is derived from this block model by applying the appropriate technical and economic parameters to extraction of the REE with proven underground mining methods and is shown in Table 25.3.

Table 25.3
Mineral Reserve Estimate as at 3 May, 2013

Description	Mineral Reserve Category		
	Proven	Probable	Proven and Probable
Tonnage (Mt)	**3.68**	**10.93**	**14.61**
TREO (%)	**1.7160**	**1.6923**	**1.6980**
HREO (%)	**0.4681**	**0.4503**	**0.4548**
HREO/TREO	27.28%	26.61%	26.78%
La_2O_3	0.256%	0.256%	0.256%
Ce_2O_3	0.570%	0.567%	0.568%
Pr_2O_3	0.072%	0.071%	0.071%
Nd_2O_3	0.284%	0.283%	0.283%
Sm_2O_3	0.065%	0.065%	0.065%
Eu_2O_3	0.008%	0.008%	0.008%
Gd_2O_3	0.062%	0.061%	0.061%
Tb_2O_3	0.010%	0.010%	0.010%
Dy_2O_3	0.058%	0.056%	0.056%
Ho_2O_3	0.011%	0.010%	0.010%
Er_2O_3	0.029%	0.027%	0.028%
Tm_2O_3	0.004%	0.004%	0.004%
Yb_2O_3	0.023%	0.022%	0.022%
Lu_2O_3	0.003%	0.003%	0.003%
Y_2O_3	0.259%	0.249%	0.251%
ZrO_2	3.440%	3.309%	3.342%
Nb_2O_5	0.425%	0.413%	0.416%
Ta_2O_5	0.046%	0.045%	0.045%

1. CIM definitions were followed for Mineral Reserves.
2. Mineral Reserves are based on Mineral Resources published by Avalon in News Release dated November 26th, 2012 and audited by Roscoe Postle Associates Inc., and modified as of 3 May, 2013.
3. Mineral Reserves are estimated using price forecasts for 2016 for rare earth oxides given below.
4. HREO grade comprises Y_2O_3, Eu_2O_3, Gd_2O_3, Tb_2O_3, Dy_2O_3, Ho_2O_3, Er_2O_3, Tm_2O_3, Yb_2O_3, and Lu_2O_3. TREO grade comprises all HREO and La_2O_3, Ce_2O_3, Nd_2O_3, Pr_2O_3, and Sm_2O_3.
5. Mineral Reserves are estimated using a NMR cash cost cut-off value of US$320/t.

6. Rare earths were valued at an average net price of US$62.91/kg, ZrO2 at US$3.77/kg, Nb_2O_5 at US$56/kg, and Ta_2O_5 at US$256/kg. Average REO price is net of metallurgical recovery and payable assumptions for contained rare earths, and will vary according to the proportions of individual rare earth elements present. In this case, the proportions of REO as final products were used to calculate the average price.
7. Mineral reserves calculation includes an average internal dilution of 8.5% and external dilution of 5% on secondary stopes.

The mineral reserve has been estimated based on conversion of the high grade mineral resources at a cut-off value greater than US$320/t NMR. Payable elements include the REE, zirconium, niobium and tantalum. The mineral reserve estimate is derived from this block model by applying the appropriate technical and economic parameters to extraction of the REE with proven underground mining methods

No Inferred mineral resources were converted to mineral reserves. The high grade mineral resources are 34.7% and 14.7% of the total Measured and Indicated mineral resources, respectively.

25.4 MINING

Underground mining of the Measured and Indicated mineral resource of the Basal Zone was investigated for the feasibility study. The majority of the mineral resource of the Basal Zone lies directly beneath and to the north of Long Lake, approximately 200 m below surface. Thus, the deposit is to be mined using underground mining methods.

The mine production rate is 2,000 t/d (730,000 t/y) of ore and the mine life is 20 years.

25.4.1 Mine Design Parameters

The key design criteria set for the Nechalacho project are:

- Mechanized cut and fill and long hole mining methods with paste backfill.
- Extraction ratio of 94.2%.
- Internal dilution of 8.5%.
- External dilution of 5% applied to all secondary stopes.
- Estimated total average dilution for the life of mine of approximately 11%.
- Ore bulk density of 2.91 t/m^3.
- Underground crushing with conveyor handling to surface.
- 24 h/d and 365 operating days per year.

Ore will be crushed underground using a three stage crushing and screening facility and conveyed to the concentrator on surface.

Mined stopes will be backfilled using tailings generated from the concentrator mixed with normal Portland cement to form the paste backfill material.

25.5 PROCESSING

The processing facilities comprise three separate plants, the concentrator located at the Nechalacho site, the hydrometallurgical plant at the Pine Point site and the rare earth refinery at the Geismar site.

The design of the concentrator and hydrometallurgical plant is supported by extensive metallurgical testwork, including pilot scale studies. The feasibility study design of the refinery is also based on extensive testwork but additional testwork is currently underway to investigate various optimization opportunities for better impurities removal that, if successful, will be incorporated into the next phase of project development.

The refinery design is based on experience and widely-accepted, well proven standard rare earth sulphate and chloride processing methods and equipment selections. The detailed design for the feasibility study was provided by a highly experienced processing expert based in China. Approximately 90% of the world's REE SX plants are located in China and the technology proposed for Avalon is based on this proven Chinese technology. Similar SX technologies are also in operation in Japan, France and the United States.

25.6 INFRASTRUCTURE

The Nechalacho project is supported by infrastructure required for the three operating sites and the transportation of materials and product between the sites.

The flotation concentrate produced at the Nechalacho site will be barged along the eastern side of Great Slave Lake to the hydrometallurgical plant at Pine Point where the flotation concentrate will be upgraded to a mixed rare earth precipitate and EZC. The rare earth precipitate will be shipped by rail to the Geismar site for leaching and separation of rare earths. The EZC will be shipped for sale to one or more third parties.

25.7 MARKETING

Avalon has undertaken analysis of markets for rare earth products and zirconium to support its product marketing efforts and to provide input for the economic analysis of the Nechalacho project.

Avalon's market analysis has included assessment of rare earth supply and, based on the predominance of LREE in the anticipated product mix of new supply sources outside China, considers that a supply shortfall in many of the more critical rare earths is likely to emerge by 2016. This is expected to result in firmer prices for neodymium, europium, gadolinium, terbium, erbium and yttrium. The projection of prices is based on analysis of data from a number of published and private sources.

Avalon has entered into several MOUs for the sale of its rare earth products. Negotiations to convert these MOUs into binding agreements are ongoing. Avalon has also entered into an

MOU for the sale of EZC. It is considered that the MOUs in hand for both rare earth products and EZC demonstrate the potential for firm off-take agreements to be completed.

25.8 ENVIRONMENTAL STUDIES, PERMITTING AND SOCIAL OR COMMUNITY IMPACT

25.8.1 Northwest Territories

A complete description of the environmental setting and existing baseline conditions for the mine, concentrator and hydrometallurgical plant is provided in the Developers Assessment Report (DAR) as well as in various environmental baseline studies and recent fieldwork Avalon's consultants.

25.8.1.1 Permitting

In 2010, Avalon filed a detailed Project Description Report (PDR) with the MVLWB as the first step in its application for a Type A Water Licence and a Type A Land Use Permit and was advised by the MVEIRB that the permit application would be referred for environmental impact assessment. In 2011, the DAR was submitted to the MVEIRB and the conformity review and information request stages were completed. The Technical Sessions were held during the period August 14-17, 2012 and the Public Hearings were held during the period February 18-26, 2013.

Preparation and issuance of the MVEIRB Report of Environmental Assessment is expected by the summer of 2013. This report will then be submitted to the Minister of Aboriginal Affairs and Northern Development. Following Ministerial approval, the process will revert back to the MVLWB to determine the conditions for the Type A Water Licence(s) and Type A Land Use Permit(s) that are anticipated to be required for the Nechalacho and Pine Point sites. The MVLWB process for the Type A Water Licence(s) is anticipated to take eight to 12 months to obtain and will include a public hearing process. The Type A Land Use Permit(s) will be processed concurrently.

25.8.1.2 Reclamation and Closure

Reclamation and closure of all site facilities in the Northwest Territories will be based on the Mine Site Reclamation Policy for the Northwest Territories and the Mine Site Reclamation Guidelines for the Northwest Territories.

25.8.1.3 First Nations Agreements

Avalon has entered into Negotiation Agreements with the LKDFN, YKDFN and the DKFN First Nations. As of March 2013, Avalon has finalized Accommodation Agreements with DKFN. Negotiations are ongoing with LKDFN and YKDFN. In the Pine Point area, negotiation agreements have been signed with the NWTMN.

25.8.1.4 Tailings Disposal

Feasibility level designs have been completed for the TMFs at both the Nechalacho and Pine Point sites in order to provide permanent and secure tailings storage, in addition to water storage and control to ensure protection of the environment during operations and in the long-term, post-closure.

25.8.2 Louisiana

Baseline environmental studies have been completed for the Geismar site.

The construction and operation of the Geismar site will require environmental permits issued by the Louisiana Department of Environmental Quality.

A "404 Permit" issued by the USACE is required for disturbance of wetlands at the site, if wetlands are disturbed. Avalon will obtain a 404/Section 10 Permit for the site, the access road and the LPDES discharge location to the Mississippi River.

The conceptual closure and reclamation plan for the Geismar site is based on a "design for closure" approach and will be updated throughout the life of the project.

25.9 CAPITAL AND OPERATING COSTS

25.9.1 Capital Costs

The capital cost estimate was prepared by SLI. The scope of the estimate encompasses the engineering, administration, procurement services, construction, pre-commissioning and commissioning of the Project. The estimate was completed to a level consistent with an AACEI Class 3 estimate with target accuracy level of ±15%, based on second quarter 2012 prices, excluding escalation.

The total estimated pre-production capital cost is $1,453 million. The life-of-mine sustaining capital is estimated at $122 million.

The total estimated life-of-mine sustaining capital amounts to $121.8 million comprising $99.8 million for mine equipment, $0.7 million for Nechalacho tailings dam expansion, $2.3 million for Pine Point tailings dam expansion and $19.1 million for the expansion of the gypsum residue containment facility at Geismar.

25.9.2 Operating Costs

The feasibility study operating cost estimates have been prepared on an annual basis for the life of the mine by SLI. The operating cost estimate has been prepared with an estimated level of accuracy of ±15% based on the design of the plant and facilities as described in detail in the feasibility study.

Salary and wages rates were generated by Avalon for the feasibility study and are based on wages and salary surveys. The operating cost estimates for the Northwest Territory operations are based upon low sulphur diesel fuel base price of $0.908/L FOB Hay River. An additional $0.05/L was added for the freight cost from Hay River to the Nechalacho site.

Mine costs are based on a combination of quotations for certain mine supplies, and experience from similar operations for the estimate of mine manpower and maintenance required for the mobile equipment fleet. Mine costs include all of the underground mining costs except for crusher operation which is included in the mill operating costs estimate.

Operating costs for the concentrator, hydrometallurgical plant and refinery were estimated from first principles. Consumables usage rates were derived from the metallurgical testwork and the unit prices from supplier quotes.

25.10 ECONOMIC ANALYSIS

Micon has prepared its assessment of the project on the basis of a discounted cash flow model, from which NPV, IRR, payback and other measures of project viability can be determined. Assessments of NPV are generally accepted within the mining industry as representing the economic value of a project after allowing for the cost of capital invested. The sensitivity of this NPV to changes in the base case assumptions was then examined.

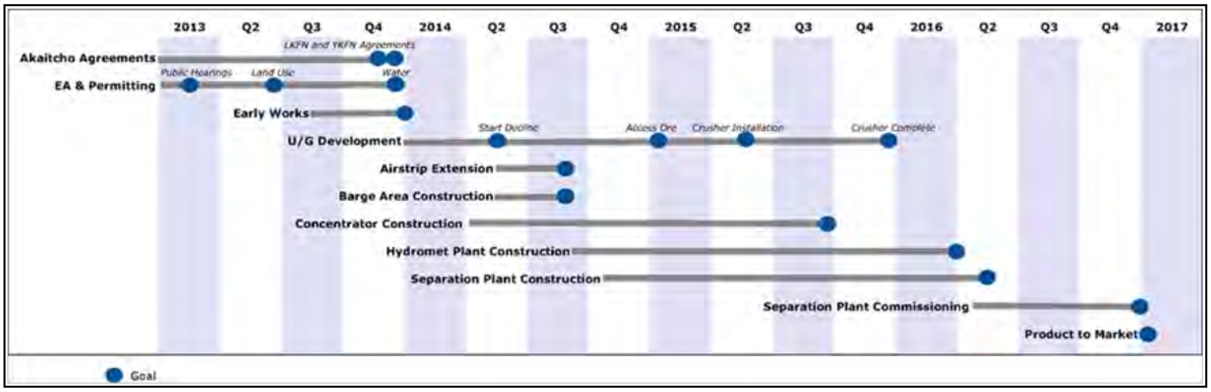
Micon concludes that the base case cash flow and sensitivity studies demonstrate robust economics that support the declaration of mineral reserves and warrant continued development of the Nechalacho REE project as described in this study.

25.11 PROJECT IMPLEMENTATION

Avalon will engage EPCM contractors at each of the three project sites. These will have responsibility for design, procurement, contracting, and project and construction management of the plants and infrastructure facilities. Site specific project management will be the responsibility of each of the EPCM contractors, with overview by the Avalon Owner's team.

A project master schedule provided in Figure 25.1 shows the overall major activities and milestone dates.

Figure 25.1
Project Master Schedule



The schedule was developed based on Avalon's requirement to have the refinery at Geismar mechanically complete at the beginning May, 2016, and it takes into consideration equipment delivery and limitations related to Nechalacho Site access.

The project has been developed based on the following requirements:

- Safety and protection of the environment cannot be compromised.
- The project must be delivered in accordance with the schedule, budget and quality requirements established for the site.
- Completion of the project in a timely manner is considered critical to meet market requirements.
- Training, hiring and purchasing from Aboriginal partners and the north will be given priority.

The Nechalacho site is considered to be the most critical of the three sites from a schedule perspective, due primarily to the limited access. A detailed critical path schedule has been developed for each of the project sites.

25.12 RISKS AND UNCERTAINTIES

During preparation of the feasibility study, two joint Owner and Contractor risk identification workshops were conducted to systematically explore all critical processes and project areas in identifying potential risks and establishing the initial risk register. The areas of project management, project controls, procurement, engineering, construction, mining, geotechnical, environmental, and health and safety were represented. Mitigation plans and actions were then developed and carried out to the extent possible during the feasibility study phase. In addition, two risk reviews were conducted to update the status and reassess the risks identified in the earlier workshops and analyse any newly identified risks, including a site specific review of the Geismar site which had not been identified in earlier workshops.

A plan will be developed for the EPCM and execution phases of the project that would start with the risk register developed in the feasibility study and would be followed throughout the feasibility study and into operations.

The key risks identified for construction, for which mitigations have been developed, and which are specific to the Nechalacho project, include:

- Logistics and procurements, particularly for the Nechalacho mine/concentrator sites mitigated by experienced personnel and detailed planning.
- Project delay due to aboriginal negotiations mitigated by existing agreement and ongoing negotiations.
- Hurricane risk in Louisiana mitigated by site selection and design.
- Cold weather risk in the Northwest Territories mitigated by experience personnel and management plans.
- Tailing dam failure mitigated by robust design, small structures and management plans at Nechalacho, and no dams being required at Pine Point.
- Project delay associated with the perceived risk associated with uranium and thorium mitigated by low concentrations, good science and open and transparent consultation.

For operations, some of the above will continue, with additional risks associated with:

- REE price fluctuations mitigated by reliable supply and long term contracts with strategic partners and customers. A long term agreement will be in place for EZC.

26.0 RECOMMENDATIONS

Given that the base case cash flow and sensitivity studies warrant continued development of the Nechalacho project as presented in the feasibility study, a number of recommendations are made for further work during project development.

26.1 GEOLOGY AND EXPLORATION

The Nechalacho deposit is not fully delineated in terms of the definition of the limits of the mineralization. The greatest potential for increase of tonnage of mineralization similar to the known mineralization is to the north of the present inferred resources. It is recommended that additional drilling be undertaken in order to define the northern limit of the Nechalacho deposit and to demonstrate that mineralization may exist under apparently un-mineralized Thor Lake Syenite.

26.2 MINING

It is recommended that:

- Optimization of the stoping sequence and stoping plans be undertaken, using specialized software, in order to determine whether further increases in the feed grades in the early years of the mine life can be achieved.

- The opportunity for higher mine production rates be assessed.

- The opportunity to increase mining of direct shipping ore be assessed.

- Additional geotechnical analysis and optimization analysis be undertaken relating to the proposed stope excavation spans and the extraction ratio in order to minimize and mitigate the associated rock mechanics risks and challenges of mining under large excavations and high extraction ratio. A numerical analysis is recommended to assess whether the location of pillars in ore or low grade/waste can be incorporated into the mine plan with a view to recovery at the end of mine life.

- Further paste backfill studies are undertaken in order to increase the effectiveness of paste fill contribution to mine stability. Cement or binder screening and optimizations are recommended to be performed at an early stage to reduce the overall operating cost for paste fill.

- A laboratory or on-site flow loop be completed to confirm the flow characteristics of the backfill paste to confirm that the positive displacement pumps located at the concentrator can service the entire mine without the need of a booster station.

Further mine design optimization analysis can be performed in the following areas to reduce the operating costs of the project:

- Stope optimization with mine scheduling and optimization software to increase the feed grade to the concentrator in early years of project life.

- Review of sill excavation from 15 m to 5 m wide and perform fan drilling especially in the long hole stopes to parallel holes drilling from a 15 m slashed sill.

- Investigation into the use of friction bolts on the side walls to reduce operating cost and increase efficiencies for the development of non-permanent excavations such as backfill drifts, headings in ore.

26.3 MINERAL PROCESSING

26.3.1 Concentrator

It is recommended that Avalon continues:

- With mineralogical studies to improve the understanding of compositional and textural variation of minerals laterally and vertically within the deposit, the chemical nature of the complex pseudomorphs within the lower parts of the BZ, and the relationship of mineral texture with metallurgical performance.

- To investigate means to improve process efficiency and reduce operating costs, including optimization of the comminution circuit, alternative flotation reagents and optimization of the cleaner flotation circuit.

26.3.2 Hydrometallurgical Plant

It is recommended that:

- Avalon continues with testwork to further optimize the hydrometallurgical flowsheet and identify potential improvements to the process that increase metallurgical efficiencies and/or reduce operating and capital costs including:

  - Optimization of the acid bake process.

  - Improvement of the HREE recoveries into the rare earth precipitate product using caustic cracking.

  - Optimization of the HREE recoveries by acid leaching at elevated pressures.

  - Direct treatment of high grade mineralization.

26.3.3 Rare Earth Refinery

It is recommended that Avalon continues:

- Further testwork with the primary objective of optimizing the impurity removal process.

- To work with Mintek to investigate enhancements to the process flowsheet which offer the potential to reduce capital and operating costs in the rare earth solvent extraction separation plant.

- To assess opportunities for toll processing of rare earth feedstocks for third parties.

- To assess opportunities to have a third party process output from the hydrometallurgical plant under a tolling agreement.

26.4 INFRASTRUCTURE AND TRANSPORTATION

It is recommended that Avalon carries out a detailed trade-off study to assess the merits of re-establishing the rail line between Hay River and Pine Point in order to reduce trucking costs.

26.5 COMMUNITY ENGAGEMENT

It is recommended that Avalon continues to engage with its Aboriginal partners and to utilize traditional knowledge with the objective of improving environmental performance.

26.6 MARKETING

It is recommended that Avalon continues to move forward to convert MOUs into binding contracts and continues to develop marketing and strategic relationships that will establish the company as a credible participant in the rare earth industry.

26.7 PROJECT DEVELOPMENT

Micon recommends that Avalon proceeds with continued development of the Nechalacho rare earth element project as described in the feasibility study prepared by SLI.

26.7.1 Potential Opportunities

Avalon has identified a number of opportunities for project optimization that may improve project economics, reduce technical risk, enhance metallurgical recoveries and improve operational efficiencies. These include:

- Optimization of the crushing and grinding circuit, plant layouts and materials of construction.

- Metallurgical testwork to further improve reagent selection and flotation recoveries.

- Improvements to the hydrometallurgical plant processes.
- Alternative impurity removal scenarios.
- Potential to separate lanthanum and cerium at the hydrometallurgical plant and stockpile for future sales.
- Potential to reintroduce cracking of zircon at a later date to increase direct production of HREE and separate the by-products from the EZC.
- Potential sales of magnetite by-product from the concentrator.
- Potential to defer construction of the refinery and toll process mixed rare earth concentrate through a refinery or refineries built and operated by others.
- Potential to use excess capacity in the refinery to toll process third party production and reduce operating costs.

These opportunities are under consideration and will continue to be investigated as the project proceeds. Avalon has a budget of $3 million to carry out this work over the next six to 12 months.

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28.0 DATE AND SIGNATURE PAGE

The effective date of this Technical Report is 17 April, 2013.

"Tudorel Ciuculescu" *{Signed and sealed}*

Tudorel Ciuculescu, M.Sc., P.Geo., Roscoe Postle Associates Inc.
Signing Date: 31 May, 2013

"Barnard Foo" *{Signed and sealed}*

Barnard Foo, P.Eng., Micon International Limited
Signing Date: 31 May, 2013

"Richard Gowans" *{Signed and sealed}*

Richard Gowans, P.Eng., Micon International Limited
Signing Date: 31 May, 2013

"Kevin Hawton" *{Signed and sealed}*

Kevin Hawton, P.Eng., Knight Piésold Ltd.
Signing Date: 31 May, 2013

"Christopher Jacobs" *{Signed and sealed}*

Christopher Jacobs, CEng, MIMMM, Micon International Limited
Signing Date: 31 May, 2013

"Jane Spooner" *{Signed and sealed}*

Jane Spooner, M.Sc., P.Geo., Micon International Limited
Signing Date: 31 May, 2013

29.0 CERTIFICATES

CERTIFICATE OF AUTHOR
Tudorel Ciuculescu

I, Tudorel Ciuculescu, M.Sc., P.Geo., as an author of this report entitled "Technical Report Disclosing the Results of the Feasibility Study on the Nechalacho Rare Earth Elements Project" prepared for Avalon Rare Earths Inc. and dated May 31, 2013, do hereby certify that:

1. I am Senior Geologist with Roscoe Postle Associates Inc. of Suite 501, 55 University Ave Toronto, ON, M5J 2H7.

2. I am a graduate of University of Bucharest with a B.Sc. degree in Geology in 2000 and University of Toronto with a M.Sc. degree in Geology in 2003.

3. I am registered as a Professional Geologist in the Province of Ontario (Reg. #1882). I have worked as a geologist for a total of 12 years since my graduation. My relevant experience for the purpose of the Technical Report is:
   - Preparation of Mineral Resource estimates.
   - Over 5 years of exploration experience in Canada and Chile.

4. I have read the definition of "qualified person" set out in National Instrument 43-101 (NI 43-101) and certify that by reason of my education, affiliation with a professional association (as defined in NI43-101) and past relevant work experience, I fulfill the requirements to be a "qualified person" for the purposes of NI 43-101.

5. I visited the Nechalacho Rare Earth Elements Project on 25th-27th of April, 2011 and 9th-11th of October, 2012.

6. I am responsible for Sections 6 through 12, 14 and 23 of the Technical Report.

7. I am independent of the Issuer applying the test set out in Section 1.5 of NI 43-101.

8. I have had no prior involvement with the property that is the subject of the Technical Report.

9. I have read NI 43-101, and the Technical Report has been prepared in compliance with NI 43-101 and Form 43-101F1.

10. At the effective date of the Technical Report, to the best of my knowledge, information, and belief, the sections for which I am responsible in the Technical Report contain all scientific and technical information that is required to be disclosed to make the Technical Report not misleading.

Dated this 31st day of May, 2013

"Tudorel Ciuculescu"   {Signed and sealed}

Tudorel Ciuculescu, M.Sc., P.Geo.

CERTIFICATE OF AUTHOR
Barnard Foo

As co-author of this report entitled "Technical Report Disclosing the Results of the Feasibility Study on the Nechalacho Rare Earth Project", dated 31 May, I, Barnard Foo, do hereby certify that:

1. I am a Senior Mining Engineer, employed by, and carried out this assignment for Micon International Limited, Suite 205, 700 West Pender, Vancouver, British Columbia, V6C 1GB.
   Tel: 1-604-647-6463 Fax: 1-604-647-6455
   E-mail: bfoo@micon-international.com

2. I hold the following academic qualifications:

Laurentian University, B.Eng., Mining Engineering	1998
University of British Columbia, M.Eng., Rock Mechanics	2007
University of Northern British Columbia, Executive MBA	2010

3. I am a registered Professional Engineers of Ontario (Membership # 100052925);

4. I have worked as a mining engineer in the minerals industry for 15 years;

5. I am familiar with NI 43-101 and, by reason of education, experience and professional registration; I fulfill the requirements of a Qualified Person as defined in NI 43-101. My work experience includes 4 years as an mining engineer in cassiterite, base and precious metal deposits, 5 years in underground and open pit geotechnical engineering and 6 years with in mine design and mining project evaluations for the mineral industry;

6. I have visited the Nechalacho project site on 13 and 14 March, 2013;

7. I am responsible for the preparation of Sections 15 and 16 of this Technical Report;

8. I am independent of Avalon Rare Earth Inc. according to the definition described in National Instrument 43-101 and the Companion Policy 43-101 CP;

9. I have had no prior involvement with the mineral property in question, other than providing consulting services.

10. I have read NI 43-101 and this Technical Report has been prepared in compliance with the Instrument;

11. As of the date of this certificate to the best of my knowledge, information and belief, the sections of this Technical Report for which I am responsible contain all scientific and technical information that is required to be disclosed to make this report not misleading.

Dated this 31 day of May, 2013.

"Barnard Foo" {signed and sealed}

Barnard Foo, M.Eng., P.Eng., MBA.
Senior Mining Engineer

CERTIFICATE OF AUTHOR
Richard Gowans

As the co-author of this report entitled "Technical Report Disclosing the Results of the Feasibility Study on the Nechalacho Rare Earth Project", dated 31 May, 2013, I, Richard Gowans do hereby certify that:

1. I am employed by, and carried out this assignment for Micon International Limited, Suite 900, 390 Bay Street Toronto, Ontario, M5H 2Y2 tel. (416) 362-5135 fax (416) 362-5763 e-mail: rgowans@micon-international.com

2. I hold the following academic qualifications:

   B.Sc. (Hons) Minerals Engineering, The University of Birmingham, U.K., 1980

3. I am a registered Professional Engineer of Ontario (membership number 90529389); as well, I am a member in good standing of the Canadian Institute of Mining, Metallurgy and Petroleum.

4. I have worked as an extractive metallurgist in the minerals industry for over 30 years.

5. I do, by reason of education, experience and professional registration, fulfill the requirements of a Qualified Person as defined in NI 43-101. My work experience includes the management of technical studies and design of numerous metallurgical testwork programs and metallurgical processing plants.

6. I visited the Nechalacho and Pine Point project site on 13 March, 2013.

7. I am responsible for the preparation of Sections 13, 17, 18 and 21 of this report entitled "Technical Report Disclosing the Results of the Feasibility Study on the Nechalacho Rare Earth Project", dated 31 May, 2013.

8. I am independent Avalon Rare Metals Inc, as defined in Section 1.5 of NI 43-101.

9. I have had no prior involvement with the mineral property in question.

10. I have read NI 43-101 and the portions of this report for which I am responsible have been prepared in compliance with the instrument.

11. As of the date of this certificate, to the best of my knowledge, information and belief, the sections of this Technical Report for which I am responsible contain all scientific and technical information that is required to be disclosed to make this report not misleading.

Signing Date: 31 May, 2013

"Richard Gowans" {signed and sealed}

Richard Gowans P.Eng.
President, Micon International Limited

CERTIFICATE OF AUTHOR
Kevin Hawton

As a co-author of this report entitled "Technical Report Disclosing the Results of the Feasibility Study on the Nechalacho Rare Earth Project", dated 31 May, 2013, I, Kevin Hawton, P.Eng., do hereby certify that:

1. I am employed by, and carried out this assignment for
   Knight Piésold Ltd
   1650 Main Street West
   North Bay, Ontario P1B 8G5
   tel. (705) 476-2165   fax (705) 474-8095
   e-mail: khawton@knightpiesold.com

2. I hold the following academic qualifications:

   B.Eng., Civil Engineering, Ryerson University, Toronto, ON. 1993

3. I am a licenced member of the Association of Professional Engineers of Ontario (Registration No. 90402694) and with The Association of Professional Engineers, Geologists and Geophysicists of the Northwest Territories (Registration No. L1733); as well, I am a member in good standing of the following organizations; The Canadian Geotechnical Society (CGS), Canadian Society for Civil Engineering (CSCE), Prospectors and Developers Association of Canada, International Society for Soil Mechanics and Geotechnical Engineering (ISSMGE) and International Permafrost Association (IPA).

4. I have worked as a consultant engineer servicing the mining industry for over 19 years.

5. I do, by reason of education, experience and professional registration, fulfill the requirements of a Qualified Person as defined in NI 43-101. My work experience includes geotechnical engineering for the mining industry and the planning, design and construction of tailings and water management facilities.

6. I have visited both project sites located in Canada during the month of August in 2011.

7. I am responsible for the preparation of Sections 20.2.3.1, 20.2.3.2, 20.2.3.3, 20.2.3.4, 20.2.3.6, 20.2.3.7, 20.2.3.8 20.2.3.9, 20.2.3.10, 20.2.4 and 20.3.3 of this report entitled "Technical Report Disclosing the Results of the Feasibility Study on the Nechalacho Rare Earth Project", dated 31 May, 2013.

8. I am independent of Avalon Rare Metals Inc., as described in Section 1.5 of NI 43-101.

9. I have had no prior involvement with the mineral property in question.

10. I have read NI 43-101 and the portions of this report for which I am responsible have been prepared in compliance with the instrument.

11. As of the date of this certificate, to the best of my knowledge, information and belief, the sections of this Technical Report for which I am responsible contain all scientific and technical information that is required to be disclosed to make this report not misleading.

Signing Date: 31 May, 2013

"Kevin Hawton" {signed and sealed}
Kevin Hawton, P.Eng.

CERTIFICATE OF AUTHOR
Christopher Jacobs

As co-author of this report entitled “Technical Report Disclosing the Results of the Feasibility Study on the Nechalacho Rare Earth Project”, dated 31 May, 2013, (the “Technical Report”), I, Christopher Jacobs, do hereby certify that:

1. I am employed by, and carried out this assignment for:
   Micon International Limited, Suite 900 – 390 Bay Street, Toronto, ON, M5H 2Y2
   tel. (416) 362-5135 email: cjacobs@micon-international.com
2. I hold the following academic qualifications:
   B.Sc. (Hons) Geochemistry, University of Reading, 1980;
   M.B.A., Gordon Institute of Business Science, University of Pretoria, 2004.
3. I am a Chartered Engineer registered with the Engineering Council of the U.K. (registration number 369178);

   Also, I am a professional member in good standing of: The Institute of Materials, Minerals and Mining; and The Canadian Institute of Mining, Metallurgy and Petroleum (Member);
4. I have worked in the minerals industry for 30 years; my work experience includes 10 years as an exploration and mining geologist on gold, platinum, copper/nickel and chromite deposits; 10 years as a technical/operations manager in both open-pit and underground mines; 3 years as strategic (mine) planning manager and the remainder as an independent consultant when I have worked on a variety of deposits including gold and silver;
5. I do, by reason of education, experience and professional registration, fulfill the requirements of a Qualified Person as defined in NI 43-101;
6. I have not visited the Nechalacho REE Property;
7. I am responsible for the preparation of Section 22.0 of the Technical Report,
8. I am independent of Avalon Rare Metals Inc., as defined in Section 1.5 of NI 43-101;
9. I have had no previous involvement with the Property that is the subject of this Technical Report;
10. I have read NI 43-101 and the portions of this report for which I am responsible have been prepared in compliance with the instrument;
11. As of the date of this certificate to the best of my knowledge, information and belief, the sections of this Technical Report for which I am responsible contain all scientific and technical information that is required to be disclosed to make this report not misleading.

Dated this 31st day of May, 2013

“Christopher Jacobs” {signed and sealed}

Christopher Jacobs, CEng, MIMMM

CERTIFICATE OF AUTHOR
Jane Spooner

As a co-author of this report entitled "Technical Report Disclosing the Results of the Feasibility Study on the Nechalacho Rare Earth Project", dated 31 May, 2013, I, Jane Spooner, P.Geo., do hereby certify that:

1. I am employed by, and carried out this assignment for
Micon International Limited
Suite 900, 390 Bay Street
Toronto, Ontario, M5H 2Y2
tel. (416) 362-5135 fax (416) 362-5763
e-mail: jspooner@micon-international.com

2. I hold the following academic qualifications:
B.Sc. (Hons) Geology, University of Manchester, U.K. 1972
M.Sc. Environmental Resources, University of Salford, U.K. 1973

3. I am a member of the Association of Professional Geoscientists of Ontario (membership number 0990); as well, I am a member in good standing of the Canadian Institute of Mining, Metallurgy and Petroleum.

4. I have worked as a specialist in mineral market analysis for over 30 years. I have managed consulting assignments on behalf of Micon, including those requiring independent Technical Reports under NI 43-101.

5. I do, by reason of education, experience and professional registration, fulfill the requirements of a Qualified Person as defined in NI 43-101. My work experience includes the analysis of markets for base and precious metals, industrial and specialty minerals (including rare earth elements), coal and uranium.

6. I have not visited the project site.

7. I am responsible for the preparation of Sections 1 through 5, 19, Section 20 with the exception of Sections 20.2.3.1 through 20.2.3.4, 20.2.3.6 through 20.2.3.10, 20.2.4 and 20.3.324, Section 25 and Section 26 of this report entitled "Technical Report Disclosing the Results of the Feasibility Study on the Nechalacho Rare Earth Project", dated 31 May, 2013.

8. I am independent of Avalon Rare Metals Inc., as described in Section 1.5 of NI 43-101.

9. I have had no prior involvement with the mineral property in question.

10. I have read NI 43-101 and the portions of this report for which I am responsible have been prepared in compliance with the instrument.

11. As of the date of this certificate, to the best of my knowledge, information and belief, the sections of this Technical Report for which I am responsible contain all scientific and technical information that is required to be disclosed to make this report not misleading.

Signing Date: 31 May, 2013

"Jane Spooner" {signed and sealed}
Jane Spooner, M.Sc., P.Geo.